

PART- D: TECHNICAL SPECIFICATIONS

CONSTRUCTION OF 122 NOS RESIDENTIAL FLATS FOR SBI OFFICERS AT DN NAGAR, ANDHERI WEST MUMBAI IN EPC MODE



CLIENT

S  Contractor

CONSULTANT

Pa 
PITHAVADI AND PARTNERS



STATE BANK OF INDIA

CONSTRUCTION OF 122 NOS RESIDENTIAL FLATS
AT DN NAGAR, ANDHERI WEST, MUMBAI IN EPC MODE

Tender No.: SBI/CC/2025-26/DNN/01

Last date & time of submission of tenders
08.10.2025 up to 15:00 Hrs.

Part-D
TECHNICAL SPECIFICATIONS

CLIENT:
STATE BANK OF INDIA
PREMISES DEPARTMENT, 9th Floor, SBI Bhavan,
CORPORATE CENTRE, Nariman Point,
MUMBAI- 400021.

CONSULTANTS:
PITHAVADIAN AND PARTNERS, CHENNAI
Architecture –Interior – Planning-Engineering-Management
10,Murugappa Road, Nawab Gardens, Kotturpuram, Chennai-600085.

TECHNICAL SPECIFICATION

GENERAL

GENERAL TECHNICAL SPECIFICATIONS**SECTION-1: BRIEF SCOPE OF WORK FOR CONSTRUCTION WORKS****1. Brief Scope of work is as follows:**

- i. The Contractor has to prepare and submit complete detailed designs and drawings based on the conceptual layouts and elevations for the proposed building project and obtain approval from Proof checking consultants suggested by the PMC/ SBI (for structural designs) and final approval from PMC/SBI (within 6 months from the date of Letter of Acceptance (LoA) issued by the PMC/ SBI).
 - ii. Obtain all necessary approvals required for commencement, execution and completion of the proposed building from the local authorities, including environmental clearance.
 - iii. Pre-construction activities such as clearance of site, site office, labour camp, site laboratory and other related activities to be carried out by the Contractor before the commencement of construction works at site.
 - iv. The Contractor shall execute all required works at his own cost, if left omitted in the scope, for making the building fit for occupation and functional use by the user department. Nothing extra shall be paid on this account.
 - v. Execution of the works and construction of buildings shall be as per approved drawings, design and plans as well as obtaining clearances required for occupation of the building from the local bodies.
 - vi. Technical Specifications for Civil, Plumbing and sanitary, Electrical (HT & LT), Lifts, Fire-fighting systems, STP and Landscaping works have been given in detail in the respective chapters & schedules of this bid document and same may be referred.
 - vii. All following works and allied works shall be part of the scope of the Contractor.
 - a) Civil Work
 - b) Plumbing & Sanitary Works
 - c) Internal & External Electrical Works (HT & LT)
 - d) Fire Fighting Systems
 - e) STP
 - f) Site Development
 - viii. Submission of Completion drawings
- 1.1. The Contractor should visit/revisit and examine the site of work and satisfy himself as to the nature of the existing roads, municipal drains, supply lines and other means of communication and other details pertaining to the work and local conditions and facilities for obtaining his own information on all matters affecting the execution of work. No extra charge made in consequence of any misunderstanding, incorrect information on any of these points or on the grounds of insufficient description, shall be allowed.
 - 1.2. The work shall be carried out in conformity with the approved drawings & designs and within the requirements of architectural, electrical, structural and other specialized services drawings.

The Contractor shall cooperate with all trades and agencies working on the site. He shall

make provision for hangers, sleeves, structural openings and other requirements well in advance to prevent hold up of progress of the construction schedule. All supports to the civil structure shall be provided with dash fasteners. On award of the work, Contractor shall submit a schedule of construction as per clause of the agreement for the approval of the Project Architect/Bank. All dates and time schedule agreed upon shall be strictly adhered to within the stipulated time of completion/ commissioning along with the specified phasing, if any.

On completion of work, Contractor shall submit six complete set of "As built drawings" along with soft copies (BIM Model), CAD Files, PDF, etc. to the PMC /SBI. These drawings shall have the following information.

- 1) Layout plans of each floors.
- 2) Location of all mechanical and electrical equipment with layout and plumbing and sanitary pipes.
- 3) Electrical wiring details.
- 4) Firefighting system details.
- 5) STP details.
- 6) All shop drawings shall be updated from time to time for the purpose of making completion drawings.
- 7) No completion certificate shall be issued unless the above drawings are submitted. Piping and drainage works shall be tested as specified under the relevant clauses of specifications.
- 8) Contractor or his associate agency engaged to do this work must hold a valid plumbing or any other as required license by the municipal authority or other competent authority under whose jurisdiction the work falls.

a) Development works

The work is to be carried out complete in all respect including services. The Contractor is required to connect all the external services like Water Supply, Sewerage, Drainage, Electric Supply, LAN, Telephone Lines etc. to the main lines of the authorities/service providers or any other agency and this shall be considered as integral part of Scope of work and deemed to be included in the quoted price of the agency. The Contractor shall obtain all mandatory statutory & Municipal approvals, "Occupation and Completion" or any other document required to use the building for residential purposes.

b) Roads and Pathways

- Approach and peripheral roads & pathways as per fire guidelines and functional requirements, to the building for Construction as per site requirements.
- The construction of guard rooms, SS gates, wicket gates, dustbins, sign boards, guide maps, location boards, direction boards and numbering etc. All complete as per the approved drawings of local bodies and direction of Project Architect/ Bank.

c) Storm water drains, sewer lines, rainwater harvesting

- Construction of storm water drains, sewer lines, rainwater harvesting in the external area around the building including connection to the trunk sewer line/STP etc. as per given schedule of Quantities.
- Execution of the roof top rainwater harvesting system for collection of rainwater including laying of pipelines and construction of substructures / superstructures as per given schedule of

Quantities, etc. as per the scope of work.

d) Internal water supply

- Execution of the Internal water supply system in all the building components.
- Providing and laying of internal water supply grid with external water supply grid network in all building components.

e) External water supply

- Execution of water supply system of the whole campus.
- Providing and laying water supply lines around the building and connection to Concerned Local Authority main water line which is to be connected to underground water tank.

xii) Specialized Civil, Electrical & Mechanical works:

The tenderer must associate himself with agencies of the appropriate eligibility for each of specialized nature of items/work listed in Tender Document. Such works shall be got executed only through associated agencies specialized in these fields. The tenderer whose tender is accepted shall indicate the name(s) of his associated specialized agencies those fulfilling the eligibility criteria laid down in Tender Document after award of work and at least 30 days before commencement of such items / work but within 90 days of Letter of Acceptance with their credentials whichever is earlier for the approval of the PMC /SBI of that component, whose decision shall be final and binding.

The Work shall be carried out as per the Specifications and drawings (Architectural, Structural and MEP). Any deviation, extra items & substitute Items shall be dealt as per Clause of General Conditions of Contract.

Note: All works must be executed as per specifications provided in the bid document and BOQ specifications. In case of getting more clarity on Tender Specifications or any missing or mismatch, CPWD, IS Codes, National Building Code Specifications shall be adopted, with due approval from the PMC/ SBI. No claim whatsoever may be entertained at later stage.

The scope of works & specifications is given in general, but they are not exhaustive i.e., does not mention all the incidental works required to be carried out for complete execution of the item of work. The work shall be carried out, all in accordance with true intent and meaning of the specifications and the drawings taken together, regardless of whether the same may or may not be particularly shown on the drawings and/ or described in the specifications, provided that the same can be reasonably inferred their form. There may be several incidental works, which are not mentioned in the contract document/specifications but will be necessary to complete the item in all respect. All these incidental works/ costs which are not mentioned but are necessary to complete the work shall be deemed to have been included in the overall amount quoted by the contractor for various components of work. No adjustment of rates shall be made for any variation in quantum of incidental works due to variation/change in actual working drawings. Also, no adjustment of rates shall be made due to any change in incidental works or any other deviation in such element of work (which is incidental to the items of work and are necessary to complete such items in all respects) on account of the directions of Project Architect/ Bank. Nothing extra shall be payable on this account.

All cost of providing and making buildings with services, landscape and horticulture works fully complete in all aspect unless specifically mentioned in the contract document and making buildings with services fully functional are included in the cost tendered for this work.

LIST OF MATERIALS OF APPROVED BRAND AND THEIR MANUFACTURES

The materials, accessories, fittings, etc. be used in the civil, Electrical, plumbing machinery, and lift installation work, shall be one of the following particular make or equivalent IS Make. The direction of selection of any particular make shall be rest with APMCF/SBI.

CIVIL		
No	Item	Approved make
1	Cement	Ultratech, Ambuja, ACC, JK
2	Sand	Locally available & as approved sample
3	Aggregates	Locally available & as approved sample
4	Bricks	As per approved sample by Client
5	AAC Block	Siporex, Aerocon, Ascolite, Godrej Tuff, Ultratech
6	Reinforcement bar	SAIL, TISCON, JSW, RINL.
7	Structural steel	SAIL, TATA Steel, Jindal, RINL- Vizag, Arcelor Mittal, Essar, JSW
8	Anti termite treatment	Bayer-Premise, PCI, Godrej HI-CARE
9	Waterproofing compound	Pidilite, Sikka- BASF, Fosroc, Asian
10	Weather sealant	Fosroc, Sikka- BASF, Dr. Fixit (Pidilite), Asian
11	Tile adhesive	Saint gobain - Weber, Ardex Endura, Kerakoll, Pidilite, Roff, MYK Laticrete, Dr Fixit Fevimate TL
12	Epoxy grouting	Saint gobain - Weber, Ardex Endura, Kerakoll, Pidilite, Roff, MYK Laticrete, Dr Fixit Fevimate TL
13	Paint & Putty	Asian, Berger, Nerolac, Birla Opus, JK, Akzonobel
14	Texture Paint	Asian, Luxture, Berger, Birla Opus
15	Italian Marble	As per approved sample
16	Granite	As per approved sample
17	Kotah Stone	As per approved sample
18	Vitrified tiles	Simpolo, Kajaria, Somany, Nitco, RAK
19	Ceramic Tiles	Simpolo, Kajaria, Somany, Nitco, RAK
20	Gypsum Plaster	Gyproc - Saint Gobain, Birla-white, JK, MYK
21	Thin Profile Glass Door	Kubik, Dorma, Saint Gobain, Ozone
22	Glass door hardware & fittings	Dorma, Haffle, Enox, Hettich, Godrej
23	Glass	Modiguard, Saint-Gobain, Asahi
24	Flush door	Greenly-door, Century- door, Archidply – door
25	Metal Door	Doorwin, JSW, Tata Pravesch, Jindal
26	Fire Rated Door	MP SWASTIK/ SHAKTI HORMANN/ SHREEJI WOODCRAFT/ BHAWANI FIRE PROTECTION
27	Ply (BWP - IS 710 & BWR 303)	Green ply, Century ply, Gurjan Ply
28	Laminate	Greenlam, Century, Merino, Timex, Royal touch, Formica,
29	Veneer	Greenlam, Century ply, Timex
30	Aluminum heavy duty section	Jindal, Hindalco, Maan, Vedanta
31	Polycarbonate sheet	Makrolon, Lexan, Bayer, Dunpalon, Sabic, Coxwell

32	Anchor fastener and bolts	Hilti, Fischer
33	Modular Glass Partition	Sonic, Kubik, Ozone
34	Hardware & fittings	Hettich, Haffle, Dorma, Godrej
35	Aluminium profile handles & frames	Olive, Hettich, Haffle, Godrej
36	Door hardware & accessories	Godrej/ Hettich/ Haffle/ Dorma/ Dorset
37	High Pressure Laminate	Fundermax, Royal Touche, Alstone, Greenlam, Century
38	Aluminium Fins	Metalium, Supersil, Hunter duglass
39	ACP (Aluminium Composite Panel)	Eurobond, Aludecor, Alstone, Alstrong
40	Paver Block	GK Concrete, Hindustan Tiles/Neptune Industries Ltd/PAVIT
41	Paver tiles 25 mm thick	Kajaria/Somany/Unistone/Pavit
42	FRP Door	Advance FRP/ Sancom Composites/ Arihant Enterprise
43	Concrete Admixtures	MCBAUCHEMIE, ROFFE, SIKA, FOSROC, BASF

PLUMBING		
Sr. No.	Description	Approved Make/ Approved Vendor
1	CP GRATING (P TRAP)	Astral, Finolex, Supreme
2	Sanitary vessels	Jaquar, Cera, Kohler, Grohe
3	Sanitary accessories	Jaquar, Cera, Kohler, Grohe
4	BALL VALVE	L&T/ KIRLOSKAR BROS./ GM ENGG/ HONEYWELL
5	GUN METAL WHEEL VALVE	L&T/ KIRLOSKAR BROS./ GM ENGG/ HONEYWELL
6	PRESSURE REDUCING VALVE	L&T/ KIRLOSKAR BROS./ GM ENGG/ HONEYWELL
7	BUTTERFLY VALVE	L&T/ KIRLOSKAR BROS./ GM ENGG/ HONEYWELL
8	NON-RETURN VALVE	L&T/ KIRLOSKAR BROS./ GM ENGG/ HONEYWELL
9	CAST IRON MANHOLE COVER	SJIF Group/ BIC/ Maa Sharda Casting
10	CAST IRON GRATING	SJIF Group/ BIC/ Maa Sharda Casting
11	UPVC PIPES/FITTINGS	ASTRAL / SUPREME / PRINCE/ FINOLEX
12	CPVC PIPES/FITTINGS	ASTRAL / SUPREME / PRINCE/ FINOLEX
13	PVC PIPES/FITTINGS	ASTRAL / SUPREME / FINOLEX / PRINCE
14	WATER METER	KRANTI/ DANESH /UPC/ FLOWTECH/ HONEYWELL
15	WATER LEVEL INDICATOR	KRANTI/ DANESH /UPC/ FLOWTECH/ HONEYWELL
16	ELECTRIC GEYSER	V-GUARD/A O SMITH/ BAJAJ / HAVELLS
17	BALL FLOAT VALVE	L &T/ KIRLOSKAR BROS./ GM ENGG/ HONEYWELL
18	SEWAGE TREATMENT PLANT	ION EXCHANGE/ TRIVENI ENGG/ EUREKA FORBES/KELVIN/ HYDROFLUX
19	DOMESTIC/ RAINWATER TREATMENT PLANT	ION EXCHANGE/ TRIVENI ENGG/ EUREKA FORBES/KELVIN/ HYDROFLUX

20	PUMPS	SIEMENS / CROMPTON / HAVELLS/ KIRLOSKAR
21	RCC HUME PIPES EXTERNAL MAIN UNDER GROUND PIPE	INDIAN HUME PIPE/ PRANALI/ ALCOCK
22	C.I. PIPE & FITTINGS	BENGAL IRON /ELECTROSTEEL/ TISCO/ JINDAL
23	G.I. PIPES	TISCO / JINDAL / ASIAN
24	G.I. PIPES FITTINGS WATER SUPPLY	TATA / JINDAL / ASIAN
25	GI TO GI JOINTS	TATA / JINDAL / ASIAN
26	PRESSURE GAUGE	H. Guru / Jack Tech / Forbes Marshall
27	AIR RELEASE VALVE	L &T/ KIRLOSKAR BROS./ GM ENGG/ HONEYWELL
28	GULLY TRAP	ASTRAL / SUPREME / FINOLEX / PRINCE
29	PVC WATER TANK	SINTEX/SUPREME/ASTRAL

Electrical		
Sr. No.	Description	Approved Make/ Approved Vendor
1	11 KV VCB	Schneider / ABB/ Lauritz Knudsen
2	11 KV HT Cable	Finolex / Polycab / KEI/Torrent
3	11 KV HT End Termination Kit	Raychem / 3M / RPG
4	Transformer/CSS	Power Lite / Kokila Transformers/Raychem/KV TRANSPower INDUSTRIES
5	DG Set	Alternator: -Kirloskar Green /Crompton /Stamford/LeroySomer. Engine: -Cummins / Caterpillar / Volvo Penta / Kirloskar/ Perkins
6	LT Panels/Feeder Pillar -Normal Fabricated	Authorized Panel Builder of Legrand / Schneider /Lauritz Knudsen - Panel Manufacturers: Adishwaram Corp/ AD Enterprise/Hi-Tech Engineers.
7	Distribution Boards	Legrand / Schneider /Panasonic/ Lauritz Knudsen
8	Low Voltage Cable	Havells/Finolex / KEI/ Polycab / RR Kabel /Torrent
9	Cable Tray (Ladder Type / Perforated)	OBO Bettermann / Legrand / Indiana / Profab
10	UPS	Eaton/ APC/ Numeric
11	SPD (Surge Arrester)	OBO / Legrand / Schneider/Lauritz Knudsen
12	LT Switchgear (ACB)	Legrand /Schneider/ Lauritz Knudsen
13	LT Switchgear (MCCB)	Legrand /Schneider/ Lauritz Knudsen
14	LT Switchgear (MCB)	Legrand /Schneider/ Lauritz Knudsen/ Anchor by Panasonic
15	LT Switchgear (Contactor, Relay, MPCB)	Legrand /Schneider/ Lauritz Knudsen/ Anchor by Panasonic
16	Auto Transfer Switch (ATS)	Legrand /Schneider/ Lauritz Knudsen
17	Busbar Trunking	Legrand /Schneider/ Lauritz Knudsen
18	Meters (Analog)	Schneider-Conserve/Trinity/Selec/Nippen
19	Meters/Energy Meters & Load Managers (Digital)	Schneider-Conserve/Trinity/Selec/Nippen

20	Indicating Lamps	Schneider / Lauritz Knudsen / Salzer/ GE/ ABB/ As Per OEM Standard
21	Electric Timer	Siemens / Legrand / Lauritz Knudsen / Legrand / MECO/ Siemens/ BCH / Anchor by Panasonic
22	Rotary Switch	Siemens / Keycee / Salzer / As Per OEM Standard
23	Push Button and Push Button Set	Siemens / Schneider electric / Lauritz Knudsen / Legrand / C&S / ABB/ GE
24	Selector Switch	Keycee / Salzer / MECO/ Siemens/ Lauritz Knudsen/ Schneider/ Abb/ As Per OEM Standard
25	APFC Relay	Enercon / Lauritz Knudsen / Trinity / Beluk/ Siemens/ EPCOS/ Abb/Schneider/Selec
26	LT Capacitors	Lauritz Knudsen / Legrand / Schneider/ Datar
27	Lugs	Dowell's / 3d / hEX Jainson / Comet / HMI/ CCI/
28	Bimetallic Lugs	Dowell's / 3d / hEX Jainson / Comet / HMI/ CCI/
29	Cable Gland	3d / Comet / HMI/ Grippwell/ Dowels/Polycab
30	PVC Conduits and Accessories	Precision / AKG/ Astral / Polycab/ Anchor By Panasonic
31	M.S. Conduit and Accessories	Akg / BEC / Steelcraft/ Anchor By Panasonic/ MK – Blenze Plus / Legrand-Arteor / Schnedier-Zencelo / Norisys /Panasonic- Europa/Cabtree-Athena/L &T- Englaze/ABB-IVIE & model shall be as approved as per architect.
32	Modular Switches, Sockets & Other Accessories	Contractor to provide minimum 3 no sample of different make for approval of end user & architect
33	Metal Clad Socket With MCB	Legrand / Hensel / Schneider/ Hager/ Lauritz Knudsen/ ABB / Anchor by Panasonic
34	PVC Tape	Steel Grip or Equivalent
35	PVC Junction Box	Hensel / Clipsal/ Sintex
36	Wires for Internal Wiring	Finolex/ Polycab / RR Kabel/ Anchor by Panasonic
37	Connectors (Colours as per Phase & Neutral)	WAGO / Phoenix Contact/ Connectwell
38	Control Transformer [PT/ CT]	Ashmor / Kappa / Elmex/ Ae/ Precise/ Pragati/ / ECS/ Kalpa
39	Paint	Nerolac / Asian Paints/Jotun
40	Ceiling Fan / Exhaust Fan	Crompton / Bajaj /Havells/ Usha/Almonard/Anchor by Panasonic
41	Floor Trunking	MK / Legrand / OBO Bettermann/ Schneider
42	Floor Junction box	Legrand / MK / OBO Bettermann/ Schneider
43	Fire Extinguisher	Firex/ Minimex / Safex / Ceasefire
44	Chemical Earthing	OBO Bettermann / JEF /JMV
45	Convectional LPS	OBO Bettermann / JEF /JMV
46	DC Battery Charger	Chhabi / HBL/Amararaja
47	SMF Battery	Exide / Rocket / Amaron-Amararaja/ Standard/ Prestolite
48	Solar PV Cells	Adani/Tata/Waaree/Bhel/Anchor by Panasonic
49	Grid TIE Inverter	Hitachi / SMA/ABB/EVVO/Waaree/Delta/Fimer
50	External light pole	Wipro/Havells/Philips/Panasonic

51	Light fixture	Wipro/Havells/Philips/Panasonic
52	BOREWELL PUMP	GRUNDFOS / XYLEM/ KBL / KSB
53	BOREWELL COLUMN PIPE	ASHIRVAD/ SUPREME/ ASTRAL
54	FLEXIBLE SUBMERSIBLE COPPER CABLE	FINOLEX/ POLYCAB/ RR KABEL
55	HYDRO PNEUMATIC SYSTEM	FRANKLIN/GRUNDFOS/XYLEM/KSB
56	SUBMERSIBLE PUMPS	GRUNDFOS / KIRLOSKAR / KSB / XYLEM
57	NON CLOG SUBMERSIBLE DE WATERING PUMP FOR RAIN WATER	GRUNDFOS / KIRLOSKAR / KSB / XYLEM
58	MUD PUMP FOR DRAINAGE	GRUNDFOS / KIRLOSKAR / KSB / XYLEM
59	LIFT	OTIS, Schindler, Orbis Elevator, Kone, Mitsubishi,

Fire Fighting		
SR. NO.	DESCRIPTION	APPROVED MAKE / APPROVED VENDOR
1	FIRE EXTINGUISHERS	KANEX, MINIMAX, SAFEX/SAFE PRO
2	FIRE FIGHTING PUMP - ELECTRICAL DRIVEN	KIRLOSKAR/LUBI/CROMPTON
3	FIRE FIGHTING PUMP - DIESEL DRIVEN	KIRLOSKAR/LUBI/CROMPTON
4	BATTERY	EXIDEs / AMCO / AMARON
5	JOCKEY PUMP	KIRLOSKAR/LUBI/CROMPTON
6	LT Panels/Feeder Pillar -Normal Fabricated	Authorized Panel Builder of Legrand / Schneider /Lauritz Knudsen - Panel Manufacturers : Adishwaram Corp/ AD Enterprise/Hi-Tech Engineers.
7	HYDRANT VALVE	SHAH BHOGILAL, NEWAGE, SWATI
8	SHORT BRANCH PIPE	SHAH BHOGILAL, NEWAGE, SWATI
9	FIRST AIDE-HOSEREEL & HOSE CABINET	SHAH BHOGILAL, NEWAGE, SWATI
10	RRL HOSE PIPE	SHAH BHOGILAL, NEWAGE, SWATI
11	FIRE BRIGADE CONNECTION	SHAH BHOGILAL, NEWAGE, SWATI
12	GI PIPES AND ACCESSORIES	JINDAL, ASIAN, SURYA PRAKASH
13	COATING & WRAPPING	IWL / RUSTEK / NEOTAPE
14	PAINTING	ASIAN / NEROLAC / BERGER
15	ANTICORROSIVE TAP	PIPEKOT / IWL / RUSTECH
16	Y STRAINER	ZOLOTO, KARTAR, SANT, NVR
17	AIR RELEASE VALVE	SANT / ZOLOTO / SHAH BHOGILAL/NVR VALVES
18	SPRINKLER	HD, TYCO, WIKING
19	FLOW SWITCHES	DANFOSS / FORBES / MARSHALL / SWITZER/HONEYWELL
20	ISNTALLATION CONTROL VALVE	HD, TYCO, NEWAGE
21	WATER CURTAIN NOZZLE	HD, SHAH BHOGILAL, NEW AGE
22	SOLENOID VALVE	HD, TYCO, NEW AGE
23	ROSSETE PLATE	HD, TYCO, WIKING

24	SPRINKLER FLEXIBLE PIPE	HD, TYCO, WIKING
25	PRESSURE SWITCH	HD, TYCO, WIKING
26	AUTO GLOW SIGNAGES	GLOWLITE / PROLITE / AUTOGLOW
27	EXIT SIGNAGES	GLOWLITE / PROLITE / AUTOGLOW
28	PUBLIC ADDRESS SYSTEM	Honeywell-X618/Aties-IDA8/Bosch-Prisario
29	FIRE FIGHTER TELEPHONE HANDSET	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
30	LT CABLES	Polycab, Avocab, R&M
31	LT TERMINATION	COMET OR EQUIVALENT
32	ADDRESSABLE FIRE ALARM PANEL	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
33	HEAT DETECTOR	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
34	SMOKE DETECTOR	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
35	MULTI CRITERIA DETECTOR	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
36	RESPONSE INDICATOR	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
37	MANUAL CALL POINT	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
38	HOOTER	Honeywell - Notifier, Bosch - UL, Johnson Control, Apollo-Panasonic
39	FIRE ALARM CABLE	Polycab, Avocab, R&M
40	PUMP ON OFF SWITCH	JAIN INSTRUMENTS / N.K.TECHNO FAB / BHAKTI ENTERPRISES
41	ELECTRIC METERS	SCHNEIDER / NIPPEN / Lauritz Knudsen / SECURE
42	STARTER / SWITCHES / MCB / MCCBS / etc.	LEGRAND / Lauritz Knudsen / SCHNEIDER
43	SWITCHGEAR / SFUS	LEGRAND / Lauritz Knudsen / SCHNEIDER
44	LEVEL INDICATOR	TECHNICA / WAAREE/ FLOWTECH
45	CLAMPS / FASTENERS	HILTI / FISCHER / MUPRO
46	TEST DRAIN ASSEMBLY WITH SIGHT GLASS	TYCO / VIKING / GRINNEL
47	AIR VESSEL	NEMA, ZENITH, AS PER CPWD SPECIFICATIONS TESTED UP TO 25KG/ SQM
48	ANTI-VIBRATING MOUNTING PADS, EXPANSION JOINTS	DULOP, RESISTOFLEX, EASY FLEX, FLEXIONICS, VIMPA

Notes: -

1) Note: - The above makes are approved for this particular work. However, in case of non-availability or discontinuation, besides the above makes, Bank/ Project Architect has the right to permit use of any equivalent brand / material matching the specified criteria / quality standards.

2) The contractor should obtain prior approval from Project Architect/ Bank before placing order for any specific materials. Project Architect/ Bank may / delete any of the makes or brands out of the above list.

- 3) All materials should conform to relevant standards and codes of BIS.
- 4) Materials with I.S.I. mark shall be used duly approved by the Project Architect/ Bank.
- 5) If any material is found to be not up to the mark, the contractor will have to produce original bills/certificate from the manufacturer or his authorized Distributor for authenticity and genuineness of the material for consideration and as per make approved by the Project Architect/ Bank. The same will not be considered for payment.
- 6) In exceptional case only, equivalent material's makes (Which are not mentioned in above list) shall be permitted that too with dual approval from Project Architect/ Bank. SBI reserves the right to accept or reject materials without assigning reason.

PROFORMAS/TABLES OF VARIOUS TESTS, MATERIALS, PACT, GUARANTEES

Sr. No	Particulars	Nos
1.	Record of Cement/Received/Used/Balance.	Table I
2.	Proforma of Paint/Lead/CICO Register	Table II
3.	Proforma for Reinforcement Bars Received	Table III
4.	Format of Receipt Of Materials At Site	Table IV
5.	Format of Monthly Progress Report	Table V
6.	Proforma for Bulkage Test of Sand Register	Table VI
7.	Proforma for Silt Test Register	Table VII
8.	Proforma for Sieve Analysis of Fine Aggregate Register	Table VIII
9.	Proforma for Sieve Analysis of Coarse Aggregate Register	Table IX
10.	Proforma for Slump Test Register	Table X
11.	Proforma of Cube Test Register	Table XI
12.	Proforma for Hindrance to Work	Table XII
13.	Proforma for Running A/c. Bill	Table XXI
14.	Account of Secured Advance if Admissible on Materials Held at Site by the Contractors	Table XIV
15.	Format for Memorandum For Payment	Table XV
16.	Format of Measurement Book	Table XVI
17.	Format of Site Order Book	Table XVII
18.	Format For Application By Contractor For Extension Of Time.	Table XVIII
19.	Details of Insurance Policies	Table XIX

Note : i) Contractor has to get the above record maintained in registers at site and to be kept securely at site.

TABLE-I**RECORD OF CEMENT RECEIVED / USED / BALANCE**

S. No.	Cement in stock Bags	Cement received (Bags)	Total Cement received (Bags)	Source from which received	Description of work where cement is used	Number of cement bags consumed	Balance in stock	Signature of Contractor & APMCF/ (Bank's Representative (Periodical)
1	2	3	4	5	6	7	8	9

TABLE-II
RECORD OF PAINT / LEAD / CICO REGISTER

Name of work :
 Name of the Contractor :
 Agreement No. :

Date of Receipt	Source Receipt with Ref. To S.O./Indnt	Qty. Received	Progressive Total	Item of work for which issued with approx. qty. work done in case of paint only	Date of issues	Quantity issued	Qty. returned at the end of the day	Total issued	Delay Balance at hand	Contractors initials	Site Engineers initials	Signature of APMCFC/ (Bank's Representative (Periodical))
1	2	3	4	5	6	7	8	9	10	11	12	13

Register for bitumen should be maintained. The format will be similar to that for cement.

TABLE-III**PROFORMA FOR REINFORCEMENT BARS RECEIVED (In KGS.)**

Truck No.	Challan No.	Name of Supplier	Binding Wire	6mm dia.	8mm dia.	12mm dia.	16mm dia.	20mm dia.	25mm dia.	Total Received
1	2	3	4	5	6	7	8	9	10	11

Number of diameters given is only illustrative. Open more columns for other diameters wherever needed.

TABLE-IV**FORMAT OF RECEIPT OF MATERIALS AT SITE**

Sr. No.	Description	Opening balance	Receipt during month	Consumption during month	Closing balance	Total Quantity received till date
1	Cement (M.T.)					
2	Mild steel (M.T.)					
3	Tor steel (M.T.)					
4	Coarse aggregate (cu.mt.)					
5	Fine aggregate (cu.mt.)					
6	Teak wood (cu.mt.)					
7	Bricks (Nos.)					
8	Tiles (Nos.)					

Sr. No.	Description of work	Date of Commencement	Due date of completion	Percentage progress achieved
1	General building work			
2	Security equipment work			
3	Pest control treatment work			
4	Sanitary & Plumbing work			
5	Electrical work			
	Fire Fighting Works			
7	Other work			

TABLE-V
FORMAT OF MONTHLY PROGRESS REPORT (Annexure X)

Name of work :

Progress report for the month :

Report No. :

Sr No.	Description	Approximate quantity executed (Till Previous Month)	Details of work location where work is done	Approximate quantity executed (Current Month)	Total Quantity Executed
A.	GENERAL BUILDING WORK:				
1	Foundation work				
2	Reinforcement fabrication				
3	Shuttering work				
4	Reinforced cement concrete				
5	Masonry work				
6	Wood work				
7	Plastering work				
8	Flooring work				
9	Glazing work				
10	Roof treatment work				
11	Painting work				
B	Pest control treatment				
C.	Security equipment work				
D.	Sanitary and plumbing work:				
1	Water supply				
2	Drainage work				
3	Fitting and fixtures				
E.	Electrical installation work				
F.	Fire Fighting Works				
G.	OTHER TRADES				

Signature & Seal of Contractor

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TABLE-VI**PROFORMA FOR BULKAGE TEST OF SAND REGISTER**

Sr. . No.	Date of Test	Volume of dust sand in Cylinder inundated & stirred	Volume inundated Sand in Cylinder	Percentage of Bulkage	Signature of Site Engineer	Signature of Contractor	Initial of APMCF / Bank's representative (Periodical)
1	2	3	4	5	6	7	8

TABLE-VII**PROFORMA OF SILT TEST REGISTER**

Sr. No .	Date of Test	Height of Sand in Cylinder inundated& stirred	Height of Silt	Max percentage of silt as specified	Percentage of silt obtained	Signature of Site Engineer	Signature of Contractor	Initial of APMCF / Bank's Representative (Periodical)
1	2	3	4	5	6	7	8	9

TABLE-VIII**PROFORMA SIEVE ANALYSIS OF FINE AGGREGATE REGISTER**

Sr. No	Date of Test	Wt. of Material to be tested	Sieve as per I.S. designation	Wt. of Sand retained in sieve	% retained in each sieve successively	Cumulative % retained in each sieve	F. M.	Signature of Site Engineer	Signature of Contractor	Signature of Architect/ Bank's representative (Periodical)

TABLE-IX**PROFORMA OF SIEVE ANALYSIS OF COARSE AGGREGATE REGISTER**

S. N o.	Date of Testi ng	Wt. of Mater ial to be tested	Nomin al size of Aggreg ate	I.S. Sieve designat ion	Standar d passing for graded aggreg ate. of nominal size	Test Res ult	Obtain ed passin g	Signat ure of Site Engine er	Signatu re of Contrac tor	Signature of APMCF /Bank's representative (Periodical)
1	2	3	4	5	6	7	8	9	10	11

TABLE-X
PROFORMA FOR SLUMP TEST REGISTER

Sr. No.	Date of Testing	Type of work for which slump taken	Specified slump		Slump Obtained		Signature of Site Engineer	Signature of Contractor	Signature of Architect/ Bank's representative (Periodical)
			When Vibrators are used	When Vibrators are not used	When Vibrators are used	When Vibrators are not used			
1	2	3	4	5	6	7	8	9	10

TABLE-XI
ANNEXURE-XI
PROFORMA OF CUBE TEST REGISTER

Date of taking Cube + Lime	Sample No.	No. of Cubes taken	Specific marking of Cubes	Proportion of mixture	Description of work carried out	Signature of Engineer taking sample	Signature of Contractor	7/28 Days Testing				Permissible Compressive strength of Concrete / 28 Days / 7 days		Remarks on Test Report and No.	Remarks of Architect/ Bank's representative Periodical
								Date of Test	Test Result Kg/ Sq. cm	Avg. Strength Kg. / Sq. cm.	Standard strength Kg / Sq. cm.	7 Days	28 Days		
1	2	3	4	5	6	7	8	9	10	11	12	13		14	15

**TABLE-XII
ANNEXURE-VIII.**

PROFORMA FOR HINDRANCE TO WORK

Name of Work : Date of Start of work :
 Name of Contractor : Period of Completion :
 Agreement No. : Dt. of Completion of work :

S.No .	Nature of Hindrance	Date of Occurrence of Hindrance	Date of which Hindrance was removed	Period of which Hindrance existed	Signature of Site Engineer	Signature of APMCF / Bank's Representative
1	2	3	4	5	6	7

TABLE XIII
PROFORMA FOR RUNNING A/C BILL

- i. Name of Contractor / Agency :
 ii. Name of Work :
 iii. Sl.No. of this Bill :
 iv. No. & Date of previous Bill :
 v. Reference to Agreement No. :
 vi. Date of Written order to commence :
 vii. Date of Completion as per Agreement :

S.No.	Item Description	Unit	Rate (Rs.)	As per Tender	
				Quantity	Amount (Rs.)
1	2	3	4	5	

Upto Previous R.A. Bill		Up Date (Gross		Present Bill		Remarks
Quantity	Amount (Rs.)	Quantity	Amount (Rs.)	Quantity	Amount (Rs.)	
6		7		8		9

Note: 1. If part rate is allowed for any items, it should be indicated with reasons for allowing such a rate.

 Net Value since previous bill

2. If ad-hoc payment is made, it should be mentioned specifically.

CERTIFICATE

The measurements on the basis of which the above entries for the Running Bill No. -----
----- were made have been taken jointly on ----- and are recorded at pages ---
----- to ----- of measurement book No. -----.

Signature and
date of Contractor

Signature and
date of Architects
Representative (Seal)

Signature and
date of Site Engineer

The work recorded in the above-mentioned measurements has been done at the site
satisfactorily as per tender drawings, conditions and specifications.

Signature & Date of Architect

Signature & Date of
Site Engineer

Signature & Date of
Bank's Engineer

TABLE - XIV
ACCOUNT OF SECURED ADVANCE, IF ADMISSIBLE ON MATERIALS HELD AT SITE BY THE CONTRACTOR

S.No.	Item	Quantity	Unit	Amount	Remarks
1	2	3	4	5	6

Total value of materials at Site.

Secured Advance @ ----- of above value -

B

CERTIFIED:

- (i) That the materials mentioned above have actually been brought by the Contractor to the site of the work and on advance on any quantity of any of this item is outstanding on their security.
- (ii) That the materials (are of imperishable nature) and are all required by the Contractor for use in the work in connection with the items for which rates of finished work have been agreed upon.

Dated Signature of
 Site Engineer
 Preparing the bill
 Rank -----

Date signature of
 Banks Architects-----
 (Name of the Architects)

 Dated Signature of the Contractor

TABLE - XV
MEMORANDUM FOR PAYMENT

R/A BILL NO.

1.	Total value of work done since previous bill (A)			Rs. -----
2.	Total amount of secured advance due since Previous Bill (B)			Rs. -----
3.	Total amount due since Previous Bill (C) (A+B)			Rs. -----
4.	PVA on account of declaration in price of Steel, Cement and other materials and labour as detailed in separate statements enclosed.			Rs. -----
5.	Total amount due to the Contractor			Rs. -----
	<u>OBJECTIONS:</u>			
i)	Secured Advance paid in the previous R/A		Rs. -----	
ii)	Retention money on value of works as per accepted tenders up to date amount Rs.		Rs. -----	
	Less already recovered		Rs. -----	
	Balance to be recovered		Rs. -----	
iii)	Mobilization Advance, if any			
(a)	Outstanding amount (principal + interest) as on date		Rs. -----	
(b)	To be recovered in this bill		Rs. -----	
iii.	Any other Departmental materials cost to be recovered as per contract, if any		Rs. -----	
iv.	Any other Departmental service charges to be recovered if any, as per contract (water, power etc.) enclose statement.		Rs. -----	
	Total Deduction as per contract (F)		Rs. -----	

Signature & Seal of Contractor

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	Adjustments, if any ----- Amount less received by Contractor in ----- R/A Bill (as per statement of Contractor)		Rs. -----	
	P.V.A.		Rs. -----	
	Total amount payable as per contract (E+F+G)		Rs. -----	
	(Rupees ----- in words)			

The bill amount to Rs. ----- (both figures and words) has been scrutinized by us after due checking of the measurements of work as required and is recommended for payment.

Date: -----

Signature of

Architect

with Seal

The bill amount to Rs. ----- certified by Consultants has been scrutinized by me after due test checking of measurements of works as required and is recommended for payment for an amount of Rs.

Date : -----

Signature of SBI Engineer

	<u>STATUTORY DEDUCTION:</u>			
i)	Total Amount due (E)		Rs. -----	
ii)	Less I.T. Payable		Rs. -----	
iii)	Less S.T. Payable		Rs. -----	
	Net Payable		Rs. -----	

These figures given in the Memorandum for payable has been verified and bill passed for payment --
----- (in words and figures)

Date: -----

Signature of the Premises Officer

Signature & Seal of Contractor

Table XVI**FORMAT OF MEASUREMENT BOOK (ANNEXURE- XX)****1ST Page:**

STATE BANK OF INDIA

.....office,

Measurement Book No.

(Pages 1 to.....)

This book is issued to Shri.....

Signature of A.G.M. (Premises & Estate)

Certified that this book contains..... pages

Signature of the official (to whom the book is issued)**MEASUREMENT BOOK PAGES NOS. 1 TO.....**

Item No.	Description	Unit	Measurement No.L B D/H	Quantity	Remarks

Site Engineer
Contractor
(Head of PMC)

Architect
(Head of Architect Consultant)

Checking/Test checking

Bank's Engineer

Date of checking/Test checking

NOTE :

Checking and test checking pertains to items wherever initialed.

Table XVII**FORMAT OF SITE ORDER BOOK (ANNEXURE- XVI)**

Name of the work_____

Date of Commencement_____

Sr. No	Remarks/ Instructions of the site Engineer/ Architect	Dated Initials of site Engineer/ Architect	Initials of the Contractor for having received the instructions	Action taken with date	Dated initials of the site Engineer	Remarks of the Architects APMCF/SBI Officials
1	2	3	4	5	6	7

Table XVIII**Format For Application By Contractor For Extension Of Time (Annexure- VII)**

1.	Name of the Contractor				
2.	Name of the work as given in the Agreement				
3.	Agreement WO				
4.	Tender amount				
5.	Date of commencement of work				
6.	Period allowed for completion as per agreement				
7.	Date of completion as per agreement				
8.	Period for which extension of time has been given				
			<u>Date</u>	<u>Month</u>	<u>Year</u>
	a)	1st extension vide Bank's Letter No.			
	b)	2nd extension vide Bank's Letter No.			
	c)	3rd extension vide Bank's Letter No.			
9.	Reasons for which extensions have been previously given (copies of the previous applications should be attached)				
10.	Period for which extension is applied for and the reasons thereof including hindrances, time for extra work assigned, if any etc.				

Signature of Contractor

Table XIX**DETAILS OF INSURANCE POLICIES (ANNEXURE XVII)**

Type of policies	Name of Insurance	Amount Rs.	Policy No.	Validity
CAR policy including 3rd party liability				
Workmen's Compensation				
Any other Policy				

Remarks :

- 1) This is only an 'on-account' payment and is not to be interpreted either as approval of work, materials brought or affixed at site or for that matter approval of any sort.
- 2) The quantum of work done and materials delivered at site have been certified by.....
- 3) should you wish to audit such work, kindly contact the undersigned and oblige.

APMCF/Architects

TECHNICAL SPECIFICATION

CIVIL, PLUMBING AND SANITARY

WORKS

Civil work
Mode of Measurements

1. EXCAVATION:

- 1.1. Footings: Area of excavation for footing shall be measured equal to area of lowest concrete course as shown on drawing. Depth shall be measured vertically from ground level to bottom of concrete course or dry rubble packing as the case may be.
- 1.2. Plinth beams: Depth of excavation for plinth beam shall be measured from ground level up to bottom of beam and width equal to width of beam.
- 1.3. Where excavation is made, in trenches, measurements for cutting depth shall be taken by means of tape and staff/ levels and the width of lowest concrete or rubble packing shall be considered as the width of excavation. When excavation is made for leveling the site, levels shall be taken before start and after completion of work and the total quantity of excavation in cutting computed from these levels.
- 1.4. Where soil including decomposed or soft rock and hard rock are mixed, hard rock after excavation shall be as per relevant IS codes applicable for measurement.
- 1.5. Width of excavation shall be considered as per BOQ Specific Items and as per preamble.
- 1.6. The unit of measure in all the above cases shall be in cubic meters or as specified in the Bill of Quantities.

2. EARTH FILLING:

Measurement for filling when it has been stipulated to be separately paid for, shall be, unless otherwise specified, as follows: -

- 2.1. In open spaces: Filling shall be measured from cross sections of embankments, before start of work and after completion of work by means of level taken at suitable places. When it is not possible to measure filling from cross section, it may be measured in loose stacks or lorry measurements with previous written permission of PMC and 20% deduction shall be made from measured quantity to arrive at payable quantity.
- 2.2. In plinth: Consolidated filling shall be measured without any deduction of voids.
- 2.3. The unit of measure in above cases shall be in cubic meters or as specified in the Bill of Quantities.

3. BRICK MASONRY:

- 3.1. Walls exceeding 150 mm brick/ thick shall be measured in unit of one cubic meter. Deductions for all openings, lintels, recesses shall be made except for the following:
 - i) When openings are less than 45 cm in both the directions or less than 45 cm dia on the surfaces.
 - ii) When beams & wall plates do not have bearing over entire thickness of wall.
 - iii) No extra will be paid for providing such openings, recesses etc..
- 3.2. Half brick walls: Net area over one surface shall be measured. Deductions for all openings, lintels, recesses shall be made as in 1.3.1.

4. CEMENT CONCRETE (Plain and Reinforced) :

- 4.1. Cement concrete items shall be measured exclusive of the steel reinforcement and plaster thickness but shall include necessary cost of shuttering, scaffolding, supporting, hacking, centering and curing. Items like R.C.C. precast jalis, R. C. pipes and other such articles which are normally manufactured in factories as well as those items which have been so specifically mentioned in schedule of quantities shall be measured inclusive of reinforcement. No deductions will be made when openings are less than 45 cm in both directions or 45 cm in diameter and no extra will be paid for providing such openings.
- 4.2. Foundation concrete: Will be measured in the unit of one cubic meter and to exact dimensions as shown on drawing or as actually laid as per instructions.
- 4.3. Footings, columns, beams, lintels, sills and bed blocks: Shall be measured in cubic meters. Portions of beams and columns embedded in slab shall be paid at the rate of slab. Only projecting rib of beam shall be paid for at the rate of beam. In case of junctions of columns and beams and/or lintels, columns shall be measured between slabs. In case of junction of columns and footings, the footing will be measured in full and the column above the footing.
- 4.4. Slabs and Chajjas : Shall be measured in cubic meters. Slab shall be measured full throughout. For chajja only projected portion shall be measured.
- 4.5. Parapet wall, apron wall and drop wall from chajja having drop exceeding 5 cm. Actual cubic contents for portion projecting over slab or beam shall be measured. If drop from chajja is 5 cm or less, the same shall be measured under chajja item.
- 4.6. Projected bands: Projection of 15 cm or less in breadth and thickness shall only be considered as band. The band shall be measured in cubic meters. Deductions will not be made on account of grooves, patties, bands, molds etc. nor will any extra be paid for forming such grooves or features.
- 4.7. Staircase: Measurements shall be per cubic meter comprising of step and soffit slab. All landings, and landing beams shall be paid separately under slab and beam measurements. In the case of soffit slab resting on beams, the portion of beam projecting below landing slab shall be measured and paid as beam. Side parapet walls, railings, finishing of risers and treads and plastering etc. shall be paid separately.
- 4.8. Reinforcement: Shall be measured on standard weight basis for the length and size of bars as shown in drawing. Wastage, rolling margin, spacers etc. required for construction purpose, and binding wire will not be measured. Lapping of bars shall be provided only as per instructions of the RCC Consultant, and the same shall be paid for. The rate will be inclusive of labour involved in cleaning, cutting, bending, binding, & erecting. Underweight reinforcement will not be accepted.
- 4.9. The rate will be inclusive of labour involved in cleaning, cutting, bending and erecting in position. The unit of measure shall be in metric tons.

5. STRUCTURAL STEEL WORK:

- 5.1. Weight of bolts, nuts, rivets, washers etc. used will not be considered for payment. Only the weight of the main members calculated on length basis at standard weight will be paid to nearest cm. No deduction shall be made for holes, bolts or rivets and wastage involved in cutting, for notching ends of sections or intermediate points for making connections. No additional payment shall be made for welding, riveting and bolting. The units of measure shall be in Kg. or MT as per Schedule of Quantities. Gusset plates shall be actually weighed and paid accordingly.

6. DOORS, WINDOWS, ROLLING SHUTTERS AND GATES:

6.1. These shall be measured in the unit of Square Meters.

- i) Teak wood doors, windows and ventilators: Clear area over one face exclusive of frame shall be measured. Hold fasts and portions embedded in masonry or flooring shall not be measured.
- ii) Steel doors, Fire doors, windows and louvers: Clear area over one face inclusive of frame shall be measured. Hold fasts or portions embedded in masonry or flooring shall not be measured.
- iii) Steel rolling shutters and rolling grills; Dimensions shall be the clear width between side jambs and clear height between floor and bottom of lintel or beam. Cover will not be measured separately.

7. FLOORING, SKIRTING AND DADO:

The net area covered shall be measured in sq. mts.

8. PLASTER:

Net area of surface plastered shall be measured in Sq. Mts. No deduction will be made for openings each less than 0.50 Sq.Mt. in area. No extras will be payable for any grooves, patties, bands, molds, (including drip molds) which are deemed to be included in the internal and external plastering items.

9. PAINTING AND COLOUR OR LIME WASH:

NOTE: All woodwork and steel items given in Schedule of Quantities are generally inclusive of painting, but if these are required to be measured separately, then the following procedure shall be adopted.

9.1. Net area of surface painted shall be measured in sq.mts. No deductions shall be made for unpainted surfaces or openings less than half sq.mts. each. The rates shall be inclusive of cleaning glasses and fittings.

- a) Walls: Net area of surface painted shall be measured. Extra for moldings, recesses and the like shall not be paid.

- b) WOODWORK:

Description	How measured	Multiplying factor
i) Paneled framed ledged braced and battened	Measured flat (not girthed) including frame; edges; chocks cleats etc shall be deemed to be included in the item.	1.30 (for each side)
ii) Flush	Measured flat (not girthed) excluding frame edges; chocks cleats etc shall be deemed to be included the item.	1.00 (for each side)

iii) Partly paneled & partly glazed / glazed	----- As above -----	1 (for each side)
iv) Fully glazed	----- As above -----	0.80 (for each side)
v) Guard bars balustrades gratings and railings	Measured flat over all, no deduction shall be made for opening. (Supporting members shall not be measured separately)	1 (for painting all over)

c) METAL WORK:

Description	How measured	Multiplying factor
i) Fully glazed or gauzed doors and windows / partitions	Measured flat over) frame. No deduction shall be made for openings. In case of gates supporting members like stays, guide rails, hinges shall not be measured	0.50 (for each side)
ii) Rolling Shutters	Measured flat (size of opening) overall; jamb guides, bottom rails and locking arrangements etc. shall be included in the item (top cover shall be measured separately)	1.10 (for each side)
iii) Collapsible gate	Measured flat (size of opening)	1.50 (for painting overall)
iv) R.C.C jali and Fencing	Measured flat overall No deductions shall be made for openings	1 (for each side)

MATERIALS MINIMUM SPECIFICATIONS/ REQUIREMENTS

- a. Material shall be of best approved quality obtaining and they shall comply with the respective Indian Standard Specification.
- b. Samples of all materials shall be got approved before placing order and the approved sample shall be deposited with the Architect/ Bank.
- c. In case of non-availability of materials in metric sizes the nearest size in FPS units shall be provided with prior approval of the Architect/Bank for which neither extra will be paid nor shall any rebates be recovered.
- d. The tests under the caption "Mandatory Tests" and listed in the Annexure to these specifications shall be carried out by the Contractor regularly and in routine fashion and if directed, the Contractor shall arrange for other tests also to be carried out in any approved Testing Laboratory. The test certificates in original shall be submitted to the Architects and all charges connected with such testing including charges for repeated tests, if ordered, shall be borne by the Contractor.
- e. It shall be obligatory for the Contractor to furnish certificate, if deemed by the Architect/ Bank, from manufacturer or the material supplier, that the work has been carried out by using their material and as per their recommendations.
- f. All materials supplied by the Employer / any other Specialist Firms shall be properly stored, and the Contractor shall be responsible for its safe custody until they are required on the works and till the completion of the work.
- g. All equipment and facilities for carrying out field tests on materials shall be provided by the Contractor without any extra cost.
- h. Unless otherwise shown on the Drawings or mentioned in the "Schedule of Quantities" or special specification, the quality of materials, workmanship, dimensions, etc., shall be as specified as hereunder.

1. Cement:

Cement shall comply in every respect with the requirements of the latest publications of IS: 269 and unless otherwise specified ordinary Portland Cement shall be used.

The weight of ordinary Portland Cement shall be taken as 1440 kg. per cu.m. (90 lbs.per Cu.ft.). Cement shall be measured by weight and in whole bags, and each undisturbed and sealed 50 kg. bag being considered equivalent to 35 liters (1.2 Cu.ft.) in volume care should be taken to see that each bag contains full quantity of cement. When part bag is required, cement shall be taken by weight or measured in measuring boxes.

No make of cement other than that approved by the Architect/ Bank will be allowed on works and the source of supply shall not be changed without approval of the Architect/ Bank in writing. Test certificates to show that cement is fully complying with the specifications shall be submitted to the Architects and notwithstanding this, the Architect/ Bank may at his discretion, order that the cement brought on site and which he may consider damaged or of doubtful quality or for any reason whatsoever, shall be retested by the Contractor at his own cost, in an approved testing laboratory and fresh certificates of its soundness shall be produced. Cement ordered for retesting shall not be used on any work pending results of retest.

Cement shall be stored in weather-proof shed with raised wooden plank flooring to prevent deterioration by dampness or intrusion of foreign matter. It shall be stored in such a way as to allow the removal and

use of cement in chronological order of receipt i.e., first received being used first. Cement deteriorated and or clotted shall not be used on the work but shall be removed at once from the site. However, allowing use of warehouse set cement shall be determined by the Architect/ Bank.

2. River Sand:

River sand shall confirm to IS: 383 and relevant portion of IS: 515. It shall pass through a I.S. sieve 4.75 mm. (3/16 B.S.) test sieve, leaving a residue not more than 5%. It shall be from natural source i.e. only river or crushed stone screenings, if allowed, chemically clean, sharp, hard durable, well graded and free from dust, pebbles, clay, shale, salt, organic matter, loam, mica or other deleterious matter. The sum percentages of all deleterious substances to acceptable limits. River sand shall not contain any trace of salt, and it shall be tested and river sand containing any trace of salt shall be rejected.

The fine aggregate i.e. river sand for concrete shall be graded within limits as specified in IS: 383 and the fineness Modules may range between 2.60 to 3.20.

The fine aggregate shall be stacked carefully on a clean hard dry surface so that it will not get mixed up with deleterious foreign materials. If such a surface is not available a platform of planks or corrugated iron sheets or brick floor or a thin layer of lean concrete shall be prepared.

3. Fine & Coarse Aggregate:

Shall consist of crushed or broken stone 95% of which shall be retained on 4.75 mm. IS tests sieve. It shall be obtained on crushing Granite, Quartzite, Trap, Basalt, or similar approved stones from approved quarry and shall confirm to IS:383 and IS 515. Fine & Coarse aggregate shall be chemically inert when mixed with cement and shall be cubical in shape and be free soft, friable, thin, porous, laminated or flaky pieces. It shall be free from dust and any other foreign matter.

Gravel / Shingle of desired grading may be permitted as a substitute in part or full in plain cement concrete if the Architect is otherwise satisfied about the quality of aggregate. For all the R.C.C. works the size of coarse aggregate shall be 20 to 25 mm. and fine aggregate shall be 10 to 15 mm.

4. Rust Free Reinforcement/ Epoxy coated reinforcement:

Reinforcement shall be of mild steel tested quality confirming to I.S.: 432-1966 and any other I.S. applicable or deformed bar confirming to IS:1786 and Is:1139 or hard drawn Fe 415 /500/550 (Tor Steel) steel wire fabric confirming to IS:1566;1967.

All finished bars shall be free from cracks, surface flaws, laminations, jagged and imperfect edges.

5. Bricks:

Bricks shall generally comply with IS:1077 except in size which shall be classified as 1st and 2nd class. 1st class bricks shall be the best quality locally available table moulded, well burnt but not over burnt, have plain rectangular faces with parallel sides and sharp right-angled edges, have a fine compact and uniform texture. The bricks shall be free from cracks, chips, flaws, stones or subsequent to soaking in water. It shall emit a clear ringing sound on being struck and shall not absorb water more than 20% by weight. Common building bricks shall have a compressive strength of 35 kg. / sqm unless otherwise specified for first class bricks.

6. Neeru:

Shall be made of Class "C" Lime (i.e. pre-fat lime) as mentioned in IS: 712. It shall be slaked with fresh water then sifted and reduced to a thick paste by grinding in a mill. Neeru thus prepared shall be kept moist until used and no more than that can be consumed in 15 days shall be prepared at time.

7. Surkhi :

Shall be made by grinding well burnt bricks, brick bats, burnt clay balls, etc., the brick etc., to be used shall be prepared from selected clay. The quality shall confirm to IS:1344.

Bricks bats, etc., shall be ground in mechanical disintegrator to a fine powder passing through IS Sieve No. 9 (2.36 mm.) with a residue not exceeding 10% by weight.

Surkhi for lime surkhi plaster shall be ground to fine powder in a mortar mill to pass through IS Sieve 150 micron (No. 100).

Surkhi shall be stored in a weather-proof shed on a brick pave platform.

8. Water:

Water for mixing cement / lime / surkhi mortar or concrete shall not be salty or brackish and shall be clean, reasonably clear and free from objectionable quantities of silt and traces of oil, acid and injurious alkali, salts, organic matter and other deleterious materials which will either weaken the mortar or concrete or cause efflorescence or attack the steel in reinforced cement concrete. Water shall be obtained from sources approved by the Architect. Potable water is generally considered satisfactory for mixing and curing concrete, mortar masonry, etc., where water other than main source is used this shall be tested in an approved testing laboratory to establish its suitability. All charges connected therewith shall be borne by the Contractor.

9. Timber:

Timber shall be well seasoned and of the best quality Indian Teak of specified species viz., Dandeli, Balarshah, Melabar, C.P.

Timber shall be considered as well seasoned, if its moisture content does not exceed the following limits.

- a) Timber for frames 14%
- b) Timber for planking, shutters, etc. 12%

The moisture content of timber shall be determined according to method described in paragraphs 4 of IS:287 for Maximum permissible moisture content of timber used for different purpose in different climatic zones.

In measuring cross-sectional dimensions of the frame pieces tolerance up to 1.5 mm. shall be allowed for each planed surface.

10. Superior quality Indian Teak Wood:

Superior quality Indian Teakwood means Dandeli, Balarshah, and Malabar Teak. It shall be of good quality and well-seasoned. It shall have uniform colour, reasonably straight grains, and shall be free from large. Loose, dead knots, cracks, shakes, warp, twists, bends, borer holes, sapwood or defects of any kind. No individual knot shall be more than 1 cm. in diameter and aggregate areas of all knots shall not exceed 1/2% of area of the piece. There shall not be less than 6 growth rings per 2.5 cm. width.

a) 1st Class Indian Teakwood:

1st Class Indian Teakwood means C.P. and Bular teak of good quality and well-seasoned. It shall have uniform colour, reasonably straight grains and shall be free from large. Loose dead knots, cracks, shakes, warp, twists, bends, sapwood or defects of any kind. No individual knot shall be more than 2.5 cm. in diameter and aggregate areas of all the knots exceed 1% areas of the piece. There shall not be less than 5 growth rings per 2.5 cm. width.

b) 2nd Class Indian Teakwood:

Shall be similar to first class Indian teak wood except that knot up to 4 cm. diameter and aggregate area of all knots up to 1 1/2% of the area of the piece shall be allowed. There shall not be sapwood up to 15% is allowed.

11. Flush Doors:

All flush doors shall be solid core exterior grade unless otherwise specified and it shall generally conform to IS:2202 and shall be fabricated as described under specification.

12. Steel Windows and Doors:

Steel windows and doors shall be fabricated of steel sections conforming to IS:226. They shall conform to IS 1038. Unless otherwise specified the details of construction etc., shall be as described under specification.

13. Floor Tiles:

Designer pre-cast concrete tiles and interlocking paver block, plain cement tiles, chequered tiles, mosaic tiles terrazzo tile shall conform to IS:1237. For neutral shade tiles grey cement shall be used. Tiles shall be compacted by mechanical vibration and hydraulically pressed. It shall be of choice shade and shall have desired pattern of chip distribution. The sizes of chips to cement in terrazzo or mosaic floor shall be as specified in IS:1237. The size and thickness of tiles shall be as approved by the Architect.

14. Ceramic / Vitrified Tiles:

White or coloured glazed tiles shall comply with IS:777 or relevant or latest I.S. code. It shall be from an approved manufacturer and shall be flat and true to shape. They shall be free cracks, crazing, spots, chipped edges and corners. The glazing and colour shall be uniform shade and unless otherwise specified the tiles shall be 6 mm. thick.

15. Marbles:

Marble slabs for flooring, dado veneering etc., shall be of kind specified in the item such as white or pink, Makrana, Chittor black, Bhanslana black, Jaisalmer yellow, Baroda green, Patiala (Pepsu) grey, etc., Marble from which slabs are made shall be selected quality, hard, sound dense and homogenous in texture and free from cracks, weathering, decay and flaws. Before starting the work, the contractor shall get the sample of Marble slabs approved by the Architect. The slabs shall be machine cut and machine polished.

16. Kotah / Shahbad / Cudappa / Granite:

Shall be of selected quality, hard, sound, dense, and of homogeneous texture, free from cracks decay, weathering and flaws. Stone slabs shall be of uniform colour as approved by the Architect. They shall be machine cut and machine polished where specified and shall confirm to the required size. Thickness shall be specified in the respective items.

17. Glazing:

Glass used for glazing shall be float glass of best quality, free from flaws, specks bubbles and shall be 5 mm. thick up to 0.60 x 0.60 mm. size and for larger size it shall be minimum 5 mm. thick unless otherwise specified in the Schedule of Quantities.

18. CPVC and Fittings

All pipes used for toilet internals, kitchen internal and external piping, ring main pipe shall be of chlorinated polyvinyl chloride (CPVC), made as per ASTM-D 2846 from 15 mm dia to 63mm dia. All CPVC pipes shall be of the approved make. They shall be sound with good surface finish, mechanical strength and capacity. They shall be of the diameter/ nominal bore, of the pipes for which they are intended and as specified in the item specification / as directed by the Project Architect/ Bank.

19. MPI. Sheets:

MPI. Sheets shall be of a gauge specified in the description of the item and shall conform to the IS:277. The sheets shall be free from cracks, spilt edges, twists, surface flaws, etc. They shall be clean bright and smooth. Galvanising shall be uninjured and the perfect condition. The sheet shall show no sign of rust or white powdery deposits on the surface. The corrugations shall be uniform in depth and pitch and parallel.

20. Paints:

Lime for lime wash, dry distemper, oil bound distemper cement primer, oil paint, enamel paint, flat oil paint, plastic emulsion paint, anti-corrosive primer, red lead, water-proof cement paint and exterior grade Acrylic Emulsion paint, cement paint, sand-tex matt shall be from an approved manufacturer and shall conform to the latest Indian Standard for various paints and shall be as per Green Building Norms **with low VOC content** withing prescribed limits. Ready mixed paints as received from the manufacturer

without any admixture shall be used, except for addition of thinner, if recommended by the manufacturer.

21. Mortar:

- a. Lime Surkhi Mortar: Lime and surkhi shall confirm to the specifications. It shall be composed of approved lime and surkhi in proportion of 1 lime to 2 surkhi mixed thoroughly. The ingredients shall be accurately gauged by measure and shall be well and evenly mixed together on a platform and water added to make it homogeneous. When large quantities are required, the mortar shall be mixed in a mechanical grinder.
- b. Cement Mortar: Cement mortar shall be of proportions specified for each type of work in the schedule. It shall be composed of Portland Cement and sand. The ingredients shall be accurately gauged by measure and shall well and evenly mixed together in a mechanical pan mixer, care being taken not to add more water than is required. No mortar that has begun to set shall be used. River sand shall be used unless otherwise specified.
- c. If hand mixing is allowed, then it shall be done on pucca water-proof platform. The gauged materials shall be put on the platform and mixed dry. Water will then be added and the whole mixed again until it is homogenous and of uniform colour. Not more than one bag of cement shall be mixed at one time and which can be consumed within half an hour of its mixing.
- d. Composite Lime, Cement, Sand Mortar: The mortar shall be of proportions specified for each type of work in the schedule of quantities. It shall comprise of Portland cement, lime and sand. Lime shall be measured in gauge boxes similar to one used for measuring cement and sand to the proportion specified and sufficient water then added to it to form a thick slurry thus obtained shall then be added to dry cement and sand mixture and thoroughly mixed to make a workable homogenous mortar of uniform colour by adding more water if necessary. Mechanical mixers shall generally be used for mixing such mortars. If hand mixing is allowed it shall be done on pucca platform.

22. AAC Blocks:

AAC (Autoclaved Aerated Concrete Blocks All units shall be sound and free of cracks or other defects which interfere with the proper placing of unit or impair the strength or performance of the construction. Minor chipping resulting from the customary methods of handling during delivery, shall not be deemed grounds for rejection.

Where units are to be used in exposed wall construction, the face or faces that are to be expelld shall be free of chips, cracks, or other imperfections, except that if not more than 5 percent of a consignment contains slight cracks or small chippings not larger than 25 mm, this shall not be deemed grounds for rejection.

The maximum variation in the length of the units shall not be more than 5 mm and the maximum variation in the height and width of unit, not more than ± 3 mm

The drying shrinkage shall be not more than 0.95 percent for Grade 1 blocks and @ 10 percent for Grade 2 blocks when tested as per table no 9.4 of IS 2185 -3(1984).

23. Windows/ UPVC windows

The cross section of the profile must confirm to the shape and dimensions of the manufacturer's specification and drawing maximum tolerance on outer surface shall not be more than ± 0.5 mm and glazing and seal grooves shall not deviate more than ± 0.3 mm.

The straightness of the profile as measured on the surface shall not deviate by more than 1.0 mm/meter.

The weight of the profile section per meter shall not be less than 3% of normal value.

24. RMC GUIDELINES

24.1. GENERAL REQUIREMENTS OF RMC

24.1.1. Basis of Supply: All concrete shall be supplied and invoiced in terms of cubic meters (full or part) of compacted fresh concrete as discharged from the transportation unit. The RMC shall be supplied in the quantity and having the quality in accordance with the requirements agreed by the producer or supplier and purchaser or user; however, the concrete supplied shall generally comply with the requirements of IS 456/ IS 457 and IS 4926. The volume of fresh concrete in a given batch shall be determined from the total mass of the batch divided by the density of the concrete. The total mass of the batch shall be determined as the mass of the concrete in the batch including the total mixing water. All Batching shall be carried out by mass, except water and admixture, which may be measured by volume.

24.1.2. Specifying Concrete – Ordering RMC shall be manufactured and supplied on either of the following basis.

1. Performance basis: Specified characteristic strength based on 28 days (or any other specific age) compressive strength in accordance with IS 456/ IS 457.
2. Prescriptive basis: Specified Mix Proportion.
3. When concrete is supplied on the basis of specified strength, responsibility for proportioning of mix rests with the producer/ seller whereas in case of specified mix proportion, the responsibility for proportioning of mix rest with the purchaser and the purchaser accepts the responsibility for concrete strength and its performance. Thus, it is desirable to place the supply order on specified strength basis. This system is based on performance parameters and is best way to order the RMC because RMC producer, who is expert in the field would design an economical mix with desired properties. In all cases the purchaser/user is required to furnish the following information for the guidance of the producer/ seller.
 - i) Type of cement to be used.
 - ii) The max. Size and type of aggregates to be used.
 - iii) The workability specified by slump or any other requirements.
 - iv) Following additional requirements should be specified to satisfy durability requirements:
 - a) Minimum and maximum cement content to be used in production of concrete.
 - b) Maximum water cement ratio to be kept.
 - c) Total chloride content in concrete: - Total chloride content should not exceed 0.15% by mass of cement in case of RCC work (IS:456) . For prestressed concrete work, total chloride content should not exceed 0.06% by mass of cement. (IS: 1343)
 - d) Total sulphate content: - It should not exceed by 4% by mass of cement.
 - e) Permeability test requirements. (if any)
 - f) Any special requirements such as pumpability or self-compacting concrete (SCC) etc.

24.1.3. Mixing of Concrete Ready-mix concrete is mixed and delivered to the point/site designated by the purchaser/user by means of one of the following combinations of operations.

- a. Central Mixed or Stationary mixed Concrete: Concrete that is mixed in a stationary mixer at plant that mixes the concrete completely before it is discharge into either in a truck agitator or truck mixer operating at agitating speed or in non-agitating equipment as agreed to by the purchaser/user. Central mix plants are sometimes referred to as wet batch or pre mix plants. When a truck mixer or agitator is used for transporting concrete which has been mixed before leaving the plant, the concrete shall agitated during transit and re-mixed at the site for at least 2 minutes so that the concrete is of the required uniformity.
- b. Shrink Mixed Concrete: Concrete that is first partially mixed in a stationary mixer and then mixed completely in a truck mixer. The time of partial mixing shall be minimum required to intermingle the ingredients. After transfer to the truck the amount of mixing at the designated mixing speed will be that necessary to meet the requirements for uniformity of concrete. Generally, it is two minutes of mixing in truck drum at mixing speed. This is not being practiced.

- c. Truck mixed concrete: Although, truck mixed concrete is also one of the methods of mixing of Ready Mixed Concrete, for the purpose of this chapter, truck mixed concrete shall not be allowed as RMC, as automatic record keeping arrangement such as digital computer slips etc. are not possible in such type of mixing. Regarding mixing whether in a stationary or central mixer it shall be ensured that it complies with performance criteria of mixing efficiency test as per IS 4634:1991. Mixing efficiency test shall be performed at least once in a year.

24.1.4. Information to be supplied by the purchaser/user of RMC plant The purchaser shall provide the details of the concrete mix or mixes required by him and all pertinent information on the use of the concrete and the specified requirements. Where the purchaser specifies a designed mix to be supplied it is essential that all relevant information is conveyed to the producer. In order to assist in this, the format given in IS 4926 Annex D may be completed and forwarded to the producer at the time of enquiry. The concrete mix shall be specified by its constituent materials and the properties or quantities of those constituent to produce a concrete with the required performance. The assessment of the mix proportions shall form an essential part of the compliance requirements. The purchaser shall provide the producer with all pertinent information on the use of the concrete and the specified requirements. In order to assist in this, the format given in IS 4926 Annex D may be followed with suitable modifications as applicable to prescribed mixes.

Purchaser responsibilities: The purchaser of Ready-mix concrete has the following responsibilities:

- i) When placing procedures can potentially alter the characteristics of fresh concrete, it is the responsibility of the purchaser to inform the producer of changes to the mixture requirements to accommodate these effects. An example is pumping concrete in place.
- ii) When a job uses more than one type of concrete mixture, it is the purchaser's responsibility to verify the mixture delivered and direct it to the correct placement location.
- iii) The purchaser should check and sign the delivery ticket and document any special occurrences on the ticket.
- iv) When strength tests are used for acceptance of concrete, the samples should be obtained at the point of discharge from the transportation unit.
- v) The purchaser or his representative should ensure that proper facilities are available for curing the test specimens at the jobsite and that standard practices are followed for subsequent curing and testing. Certified personnel should conduct the tests.
- vi) Test reports should be forwarded to the producer in a timely manner to ensure that deficiencies are rectified.

24.1.5. Information to be supplied by the producer/owner of RMC plant

Upon the request, the producer shall provide the purchaser with the following information before any concrete is supplied:

- a) Nature and source of each constituent material.
- b) Source of supply of cement, and
- c) Proposed proportions or quantity of each constituent/m³ of fresh concrete. d) When requested, the producer shall provide the purchaser the following information of admixtures:
 - i) Generic type(s) of the main active constituent(s) in the admixture.
 - ii) Whether or not the admixture contains chloride and if so, the chloride content of the admixture expressed as a percentage of chloride ion by mass of admixture.
 - iii) Where more than one admixture is used, confirmation of their compatibility.
- d) Producer's responsibilities: The producer of Ready-mix concrete has the following responsibilities
 - i) i)The concrete producer is responsible for the concrete slump as specified for a period of 30 minutes after the requested time or the time truck arrives at the placement site, whichever is later.
 - ii) ii) The concrete producer is required to deliver concrete at the requested slump and air content, within the accepted tolerances, as measured at the point of discharge from the transportation unit.

Note: The purchaser shall not alter the quality of concrete by any addition or modification at the job site. These include addition of water, admixture, fiber or special products into the ready-mix concrete supplied by the producer; in case the purchaser does this, then producer is not responsible.

24.1.6. The contractor shall provide self-compacting concrete/ concrete of specified grades from dedicated RMC plant. For smooth & uninterrupted flow of concrete, RMC plant of minimum 60cum/hr. capacity may be installed within the site. Contractor should ensure that RMC plant shall follow all the prevailing guidelines of MPCB, MCGM, NGT and any other concerned controlling authority in addition to the instructions issued by PMC/ SBI. No extra charges will be paid on this account.

25. GENERAL INFORMATION ABOUT RMC FACILITY

25.1. Location of RMC Plant

The RMC plant from where the concrete is being procured by the purchaser/user can be a commercial plant owned/operated by a third party or a captive plant owned and operated by the contractor.

In the case of commercial plants, the location is already decided as they are operational plants, and the user of RMC has no control on its location. The nearness to the site and availability of good roads can be the deciding factors in such cases.

However, when the RMC plants are captive plants and are erected on their site (other than project site) the constructor has to see the location of plant is suitable from all the considerations, the factors to be considered while deciding the location of plant can be.

- 1) Availability of land
- 2) Availability of Raw materials such as sand, aggregates, cements, fly-ash etc. and their leads.
- 3) Availability of Electric power
- 4) Availability of water
- 5) Nearness to site
- 6) Nearness to village
- 7) Environmental concerns; waste management, dust and noise control, safety etc. Hence, it is necessary that the owner of captive batching plant takes a judicious decision by considering all the above factors with respect to their technical and financial viability.

25.2. Components of RMC plant RMC plant/facility has in general the following components

25.2.1 PROPERTIES OF FRESH CONCRETE

25.2.1.1 Workability of concrete

Workability is a broad term which encompasses a range of properties of fresh concrete such as consistency (fluidity) , mobility (ability of concrete to move around the reinforcement and in restricted areas), compatibility, finishability and pumpability (for pumped concrete). The degree of workability varies depending upon the type of construction and method of placing, compacting and finishing. Workability is measured in terms of slump of concrete using the standard procedure laid down in IS 1199:1959. The IS 456 provides guidance on the range of workability requirements for different placing conditions and applications. Consistency of fresh concrete is considered to be a close indication of its workability and slump test has been the most widely used test for ascertaining consistency and hence workability. For applications requiring very high slumps (higher than 150mm) the IS 9103 recommends use of flow table test. For a majority of concrete supplied by RMC producers, slump test is the most commonly used test. The IS 4926 specifies the following tolerance limits of workability as criteria for acceptance.

- i) Slump: + 25 mm or + 1/3rd of the specified value whichever is less.
- ii) Compacting factor: + 0.03 For specified value > 0.9 + 0.04 For specified value < 0.9 > 0.8 + 0.05 For specified value > 0.8
- iii) Flow Table Test: Acceptance criteria to be established between the producer and the purchaser.
- iv) The test for workability needs to be performed upon discharge from producer's delivery vehicle on site or upon discharge into the purchaser's vehicle. On some occasions, lack of preparedness on the part of purchaser at construction site may result in delay of placement. RMC producer will be responsible for maintaining the slump within the permissible range for a

period of 30 minutes starting from arrival of transit mixers at job site. However, after 30 minutes, the IS 4926 clearly states that the responsibility for delay passes on the purchaser. Slump of concrete is quite sensitive to a variety of environmental and other factors such as concrete temperature, ambient temperature, surface rate of evaporation, changes in grading, batch mass differences, admixture dosage, presence of mineral admixtures or otherwise, variation in air content, variation in testing, etc.

25.2.1.2. Density of concrete

The plastic density (unit weight) of conventional normal-weight concrete varies depending upon the variation in the density of different ingredients, the amount of entrapped air and entrained air (if air – entraining agents are used), the maximum size of aggregate and water and cement contents in the mix. Increasing the aggregate volume and reducing the cement paste would increase the density of concrete. Ready Mixed concrete is measured on the basis of volume. The volume of fresh concrete can be determined by dividing the total weight of all batched materials by the unit weight or plastic density of concrete determined in accordance with IS 1199. Sometimes there is likelihood a discrepancy in the concrete ordered and that actually supplied. Also, it should be understood that the volume of hardened concrete may be or appear to be less than expected. There could be variety of reasons for this discrepancy. These include wastage and spillage of concrete, over excavation, miscalculation in form volume, deflection or distortions of forms, settlement of wet mixes, loss of entrained air, etc. Such difference can be reconciled if plastic density of concrete is monitored regularly. While carrying out mix Proportioning, the plastic density of designed (Proportioned) mix is measured and tallied with the theoretical density. It would be a good practice to measure the plastic density at regular interval so as that the quantities supplied match orders. The plastic density measurement can be done by filling a container of known volume with fully compacted concrete and taking the mass of concrete in that volume by following procedures detailed in IS 1199.

25.2.1.3 Air content of fresh concrete

In most parts of India, tropical weather prevails, necessitating adoption of adequate precautionary measures associated with hot weather concreting practices. Absence of adequate measures may lead to rapid loss of workability, accelerated stiffening of concrete, poor compatibility and finishability, and cracking of concrete owing to plastic and/or thermal shrinkage. To avoid adverse effect of hot weather, both RMC producer and the purchaser need to take adequate precautionary measures. It shall also be noted that generally retarding effect of retarder is smaller at higher temperatures and sometimes few retarders seem to be in-effective at extremely high temperatures. Thus, it is desirable to keep the temperature of concrete as low as possible. Although in the IS 4926:2003 the requirements of temperature of concrete has been deleted it is advisable that the temperature of concrete produced shall not be less than 50°C and shall not exceed 350°C.

As far as RMC is concerned, design the concrete mix using a combination of OPC and supplementary cementitious materials or blended cement for reducing the heat of hydration as approved by SBI/APMCF. The aggregate stockpiles in the plant should be covered to avoid direct exposure to sun and water should be sprinkled on the stockpile to bring down the temperature. Some RMC producers use chilled water or ice flakes to bring down the temperature of mixing water during hot summer months. Covering the drum of transit mixer by Hassien cloth helps in maintaining the temperature of concrete during transit. The requirements of extreme weather (hot weather conditions) concreting are given in IS 7861(part 1):1975 and shall be referred to.

25.3. PROPERTIES OF HARDENED CONCRETE

25.3.1. Strength of concrete

- i. Concrete Cubes: - When strength of concrete is used as a basis for acceptance of concrete, which is generally adopted parameter, the standard specimen shall be made, cured and tested at 28 days in accordance with IS 516. The compliance shall be assessed against the requirements of IS 456. The testing frequencies and sampling shall be as per sampling and testing of concrete. While the strength at 28 days has emerged as a basis for contract specification; in order to get relatively quicker idea of the quality of concrete, compressive strength at 7 days may be carried out; however, it is important to establish a relationship between early age and 28 days strength for a particular concrete. But in all cases 28 days compressive strength shall alone be the criteria for acceptance or rejection of concrete.
- ii. Concrete Cores: - The most widely accepted method of determining the in-place compressive

strength of concrete in existing structures, pavements and linings is the testing of core specimens obtained by drilling with a diamond core bit. While core strength tests are more reliable than the less expensive & less tedious non-destructive test methods now in use, the results can be affected by many structure and testing variables which must be controlled or taken into consideration while evaluating the concrete strength. Where possible, a length to diameter ratio L/D of 2 should be used but the diameter of core should be at least three times the nominal maximum size of coarse aggregate (MSA) and in no case shall the diameter of specimen be lower than twice the maximum nominal size of aggregate (MSA). Any specimen intended for strength testing shall not contain embedded reinforcing steel. Testing variables include considerations such as method of end preparation. Often sawing is necessary to thin cores so that ends are perpendicular to the axis of the core, to eliminate reinforcing steel or honeycombed areas or to eliminate surface irregularities. Usually, cores shall be capped to produce the required plainness for testing and it shall be ensured that good practices of capping are followed as per IS 516. The capping shall be thin with a strong material. The use of thick caps or ones that are not properly bonded to the specimen or are made with a weak material may cause markedly reduced core strengths specially in short cores of L/D less than 2.0.

- iii. Shorter cores with L/D less than 2.0 give a higher indicated strength which increases as L/d decreases; therefore, these higher strengths must be corrected by a factor (correction factor) given in IS 516 for each ratio which, on the average, will produce a corrected strength on a parity with the standard L/D = 2.0 specimen. The equivalent cube strength of the concrete shall then be determined by multiplying the corrected cylinder strength by 5/4.

25.3.2 Acceptance criteria for concrete

a. Cubes the IS 456:2000 provides guidance on the acceptance criteria of concrete based on compressive strength and shall be adhered to.

Accordingly,

- The test results of the sample shall be average of the strength of the three specimens. The individual variation should not be more than + 15% of the average. If more, the test results of the sample are invalid.
- The concrete shall be deemed to comply with the strength requirements when both the following conditions are met the mean strength determined from any group of four consecutive non overlapping test results complies with appropriate limits in column 2 of Table 1.
- Any individual test result complies with the appropriate limits in column 3 of Table 1.

Table1: Characteristic compressive strength compliance Requirement

Specified Grade	Mean of the group of 4 Non over lapping consecutive test results in N/sq.mm	Individual test results in N/sq. mm
M15	> or equal to $f_{ck} + 0.825 \times$ established standard deviation (rounded off to nearest 0.5 N/sq mm) or $f_{ck} + 3$ N/sq mm whichever is greater.	> or equal to $f_{ck} - 3$ N/sq mm
M20 And Above	> or equal to $f_{ck} + 0.825 \times$ established standard deviation (rounded off to nearest 0.5 N/sq mm) or $f_{ck} + 3$ N/sq mm whichever is greater.	> or equal to $f_{ck} - 3$ N/sq mm

b. Cores:

The IS 456:2000 gives the acceptance criteria for the core test on concrete. Accordingly, Concrete in the member represented by a core test shall be considered acceptable if the average equivalent cube strength of the cores is equal to at least 85% of the cube strength of the grade of concrete specified for the corresponding age and no individual core has strength less than 75%.

26. ENVIRONMENTAL CONCERNS AND SITE SAFETY

26.1. Site Safety

As in all civil engineering constructions, at the RMC plant safety shall be given a paramount importance and "Safety first" rule shall be followed. Ready Mixed Concrete plants are industrial operations relying on heavy equipment and vehicles with potential for accidents. So, safety of workers should be a critical

objective. All guardrails and machinery guards shall be fixed securely in position and walkways kept clean and with clear access. The facility shall provide suitable communication system between batching plant operator, transit mixer and delivery site. The RMC producer shall provide working conditions which have regard to the health and safety of employees. Plant should adopt a written safety program that includes formal safety training and provide incentives for workers who maintain safe practices. The following arrangements shall be ensured from safety point of view.

- i. Earthing arrangement: RMC equipment essentially needs to be earthed in view of abundant use of metal. Necessary earthing facility needs to be created by the owner/producer.
- ii. Air conditioning: Control unit of the RMC plant needs to be kept air-conditioned for trouble free running of computer systems and to provide good environment to the operators and staff.
- iii. Plumbing and drainage work: Water supply network needs to be laid at site for availability of water at different locations. Similarly, site drainage for rainwater or spillages need to be provided to keep it workable.

26.2. Environmental Considerations

Due regard shall be given to the environment in any RMC production facility. The technologies used shall be such that to reduce the environmental impact to the lowest realistic level at the same time the technologies shall be proven, economic and reasonable. The RMC facility shall endeavor that plant operations are well landscaped and screened from the surrounding residential or rural community such that the impact is minimal. The producer shall ensure that the traffic routes chosen are such as to avoid congested and sensitive areas wherever practicable and to minimize the fuel consumption. Concrete spillage on the public highway roads and pathways shall not be there. On the rural roads the dust menace shall be reduced by watering the pathways/roads near the rural community/site. As concrete producers in RMC plant the producer shall be aware and know the details of responsibilities regarding the environmental regulations such as Air Quality Permits; Discharge permits; Storm water management, clean water permits, Solids management, Hazardous waste regulations, Dust control, Recycling, reuse and sustainability.

26.3. Air and Noise Pollution and Vibration

Particulate matter emissions to air, also known as dust emissions, are the major air quality concerns at the ready mixed plant site. These very small particles can pose a health and safety risk to persons who may inhale those particles. The dust emissions can be process (point source) emissions and fugitive emissions. Process or point source emissions occur at discrete and definable locations during various activities such as silo filling; material handling and stacking; truck batching etc. Fugitive dust emissions are difficult to pinpoint and may arise from onsite vehicle movement, loading/transfer activities. The dust emissions can be reduced by plant enclosures and dust suppression wherein water is sprayed at the source of dust to prevent it from becoming airborne. There are many techniques and strategies available, and the producer shall utilize appropriate technology to prevent or minimize dust emissions in line with local and national regulations. Noise is defined as "unwanted sound" and is primarily a concern of surrounding community and plant employees. While sound is inherent to RMC facility there are many areas where noise can be minimized, and the producer should take steps to ensure that plant and vehicle noise are minimized through plant designs, landscaping, berms and sound walls, and through the use of appropriate technology and strategies.

26.4. Fuel, Oil and chemical spillage the risk of leaks and spills can be minimized by proper design of storage facilities. The producer shall take appropriate measures and employ best management strategies to prevent leaks and spills and prevent pollution of surrounding areas and ground water by accidental effluent discharges and fuel, oil and chemical spillage.

26.5. Waste management Waste is defined as materials disposed of in an unproductive manner for example being land filled or discarded in a quarry or back lot. Comprehensive waste management and programs will reduce environmental burden of waste disposal. Re use of the waste material alleviates the burden of raw materials extraction. Excess concrete and returned concrete mainly forms the solid waste in RMC industry and forms the major waste concern. The producer shall introduce processes, strategies and practices that minimize the production of waste.

27. LIST OF REFERRED STANDARDS & LITERATURE

INDIAN STANDARDS	Title
IS 383 : 1999	Coarse and fine aggregates from natural sources for concrete— Specification (third revision)
IS 456:2000	Plain and reinforced concrete— Code of practice (fourth revision) (Reaffirmed 2005)
IS 457:1957	Code of practice for Plain and reinforced concrete for Dams & other Massive structures.
IS 516:1959	Method of test for Strength of concrete.
IS 1199: 1959	Methods of sampling and analysis of concrete.
IS 1343:2012	Code of practice for prestressed concrete
IS 1791:1985	General requirements for batch type concrete
IS 2386(Part I):1963	Methods of test for Aggregates for concrete, Part I: particle size and shape, (Reaffirmed 2007)
IS 2386(Part III):1963	Methods for test for Aggregates for concrete, Part III: specific gravity, Density, Voids, Absorption and Bulking (Reaffirmed 2007)
IS 2430:1986	Methods for sampling of Aggregates for concrete(First Revision),(Reaffirmed 2005)
IS 3812(Part 1):2003	Pulverized fuel Ash For use on Pozzolana in Cement, Cement Materials and concrete
IS 3812(Part 2):2003	Pulverized fuel Ash For use on Admixture in Cement, Cement Materials and Concrete
IS 4082:1996	Recommendations on stacking and storage of construction materials and components at site (second revision)
IS 4634:1991	Methods for testing performance of batch-type concrete mixers (first revision)
IS 4925:1968	Specification for concrete batching and Mixing
IS 4926:2003	Ready Mixed Concrete – Code Of Practice
IS 5816:1999	Splitting tensile strength of concrete Method of
IS 5892:2004	Specification for concrete transit mixers and
IS 7861 (Part I):1975	Guidelines for concreting in Extreme weather conditions (Hot weather conditions)
IS 8142: 1976	“Method of test for determining setting time of concrete by penetration resistance.
IS 9103: 1999	Concrete admixtures—Specification (first revision) (Reaffirmed 2004)

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WORKMANSHIP

CLEARING OF SITE, EXCAVATION AND EARTH FILLING

Note: Workmanship for all items related to the construction work should be as per relevant I.S. Code.

1. General:

Trenches for wall foundations, column footings, raft foundations, pile caps, plinth beams, water tanks, cesspits, etc., shall be excavated to the exact length, width and depth shown in the figure on the drawing or as may be directed by the Architect. If taken out to greater length, width or depth than shown or required, the extra work occasioned thereby shall be done at the Contractors own expenses. Extra depth shall be brought up by plain cement concrete filling 1:4:8 proportion and extra length and width filled in by rammed earth or murum or if the Architect thinks it necessary for the stability of the work by 1:4:8 concrete, as may be directed by the Contractors costs.

Excavated material shall be used for filling in plinth, or each side of the foundation blocks or trenches or it shall be spread elsewhere on or near the site of work including watering, ramming and consolidating or carted away from site free of charge, as may be ordered. The Contractor shall at his own expenses and without any extra charge, make provision for supporting all utility services, lighting the trenches, separating and stacking, serviceable materials neatly, shoring, timbering, stuttering, bailing out of water either sub-soil or rain water including pumping at any stage of the work. Trenches shall be kept free of water while masonry or any concrete works are in progress and until the Architects consider that concrete is sufficiently set.

2. Excavation excluding in Hard Rock:

Excavation shall be carried out in any type of soil, murum (soft or hard), soft rock, boulders, old foundation, concrete asphalt or stone paved surfaces, old masonry or concrete (plain or reinforced).

3. Excavation in Hard Rock:

Rock which is in solid beds, which can only remove either by wedging or chiseling shall be treated as hard rock. A boulder or detached rock measuring one cubic meter or more, shall wedging or chiseling. Where hard rock is met with the blasting operations is considered necessary, the Contractor shall intimate about the same to the Architect.

The Blasting shall not be permitted in any case and contractor have to carry out hard rock excavation in hard rock by means of Wedging and chiseling only.

Excavation shall be done by wedging or chiseling and it shall be restricted to the quantity required to enable the necessary foundation etc. to be put in. In case, the dimension of trenches exceeds those shown in drawings or as directed by the Architect, the excess quantity shall not be paid for, the item also covers bailing out subsoil or rain water including pumping at any stage of work, shoring strutting, etc.

4. Earth Filling:

General: Filling shall be done with good earth, murrum, stone chips, or disintegrated building debris. It shall be free from salts, organic matter, black cotton or slushy earth and combustible material. All clods shall be broken.

a) Filling in Plinth :

Filling shall be done in layers not exceeding 25 cm., amply watered and consolidated by ramming with iron or wooden rammers weighing 7 to 8 kgs. and having base 20 cm. Square or 20 cm. diameter. When the filling reaches the finished level, surface shall be flooded with water for at least 24 hours, allowed to dry and then rammed and consolidated, after making good any settlement in order to avoid settlement at a later stage. Special care shall be taken to pack earth under plinth beams and column corners. Finished level of filling shall be kept to a slope intended to be given to the floor.

b) Filling in Outdoor portions and for Site Development:

Shall be done in layer of 30 cm. Each layer shall be adequately watered. When filling reaches the required level the top most layer shall be dressed to proper section, grade and camber and rolled by 8 to 10 ton's power roller and adequately watered to aid compaction.

5. DRY RUBBLE PACKING & LEVELING COURSE.

Dry Rubble Packing: Ground shall first be leveled up and thoroughly consolidated by means of heavy log hammer or frog rams. Rubbles of specified thickness shall then be laid and set with hand. It shall be consolidated by either hand roller or wooden log hammer; free use of water being made during consolidation. All hollows and interstices after consolidation shall be filled up with quarry spalls, stone chips etc., and the packing blinded with stone grit and watered and consolidated by log hammer.

Rubble packing in Road work shall be thoroughly consolidated by means of power rollers of 8 ton's capacity instead of log hammers and the surface shall be brought to proper grade and camber. After checking the level, grade and camber the surface will again be watered and rolled to receive road structure.

Leveling Course:

It shall be either plain cement concrete of leaner mix which shall be proportioned as stipulated in the relevant item and mixed and placed in position confirming to line and level show on the drawing and compacted by approved means and cured adequately.

6. PLAIN & REINFORCED CEMENT CONCRETE

A) VOLUMETRIC BASIS: -

General : In any case volumetric batching shall not be allowed.

Proportioning the Mix :

Design Mix shall be carried out and same shall be got approved from PMC/SBI. If aggregates are moist allowance shall be made for bulking in accordance with IS:2386/-. Allowance shall also be made for surface water present in aggregate when computing water contents. Surface water present shall be determined by one of the field methods described in IS:2386/- (Part III).

Mixing :

All Concrete shall be mixed in an approved mechanical mixer. The mixer and mixing platform shall be suitably protected from wind and rain. Aggregates shall be accurately measured as per design mix and mixed dry along with cement, water shall be then added in measured quantity and mixing shall be continued until there is a uniform distribution of the materials and the mass is uniform in colour and in consistency but in no case shall the mixing be done for less than 2 minutes.

Hand mixing shall not be permitted in any case.

Consistency:

Quantity of water for making reinforced concrete shall be sufficient so as to ensure that concrete shall surround and properly grip all the reinforcement. The best consistency shall be that which will flow sluggishly without flattening out and without separation of coarse aggregates from the mortar. The degree of plasticity shall depend on the nature of work and atmospheric temperature and whether the concrete is vibrated or hand compacted. The slumps shall be determined by standard slump test carried out in accordance with the procedure laid down in IS:119-1959 shall be adopted for different types of work and same shall be validated as per approved Design Mix Reports.

Admixtures:

The usage of admixtures is allowed only if approved by the structural consultant and his decision in this regard shall be final.

Transportation:

Concrete shall be conveyed from the place of mixing to the place of final deposit as rapidly as practicable by methods which will prevent segregation or loss of any of the ingredients. If segregation does occur during transport, the concrete shall remix before being placed. In no case, more than 30 minutes shall elapse between mixing the consolidation in its position.

Placing and Compacting:

Concrete shall be placed in layers of suitable thickness or in strips and compacted before initial setting commences and should not be subsequently disturbed. Method of placing shall be such as to preclude segregation and as far as practicable the placing shall be continuous. Special care shall be taken in accordance with IS:456 while laying concrete under extreme weather.

Concrete shall be thoroughly compacted during the operation of placing and thoroughly working around the reinforcement, embedded fixtures and spaded against corners of the form work and by punning, rodding, mechanically vibrating or by any other approved means. In addition, form work shall be tapped lightly by using wooden mallet at the pouring head. The number and type of vibrator to be used shall be subject to the approval of the Architects and in general immersion type vibrators shall be used. External vibrators shall also be used whenever directed.

The intensity and duration (of vibration shall be sufficient to cause complete settlement and compaction without any stratification of successive layers or separation of ingredients or formation of laitance. Vibrator shall be inserted vertically in the concrete at points not more than 45 cm. apart and withdrawn very slowly when air bubbles no longer come on the surface. Over vibration or vibration of very wet mixes is harmful and should be avoided. Care shall be taken to utilize the vibrator only to compact the concrete and not to spread it, sufficient number of reserve vibrator in good working condition shall be kept on hand at all times, so as to ensure that there is no slackening or interruption in compacting.

Construction Joints:

Concreting shall be carried out end to end continuously as far as possible and when construction joints are totally unavoidable; it shall be located in a predetermined position approved by the Architect. The joints shall be kept at places where the shear force is the minimum and these shall be straight and at right angles to the direction of main reinforcement. When the work has to be resumed, on a surface which has hardened, such surface shall be roughened. It shall be swept clean, thoroughly wetted and covered with a 13 mm. layer of mortar composed of cement and sand in the same ratio as the cement concrete mix. This 13 mm. layer of mortar shall be freshly mixed and placed immediately before the placing of the concrete.

Where the concrete has not fully hardened, all laitance shall be removed by scrubbing the Wet surface with wire or bristle brushes, care being taken to avoid dislodgment of particles of aggregate. The surface shall then be coated with neat cement grout. In horizontal joints the first layer of concrete to be placed on this surface shall not exceed 15 cm. thickness and shall be well rammed against old work, particular attention being paid to corners.

Expansion Joint:

Expansion joint shall be provided where required as shown on the drawings or as directed by the Architect / Consultant. The joints shall be filled by the approved quality filler.

Curing:

Concrete shall be carefully protected during first stage of hardening from harmful effects of excessive heat, drying winds, rain or running water. It shall be covered with a layer of sacking, sand canvas, hessian, or similar absorbent materials and kept constantly, wet for ten days from the date of placing of concrete. Alternatively, the concrete being thoroughly wetted and covered by layer of approved water-proof material which should be kept in contact with it for seven days.

Form Work:

The form work shall conform to the shape, lines and dimensions as shown on the plans and be so constructed as to remain sufficiently rigid during the placing and compacting of the concrete and shall be sufficiently watertight to prevent loss of cement slurry from the concrete. Form work or centering shall be constructed of steel or timber and adequately designed to support the full weight of wet concrete without deflection and retain its form during laying, ramming and setting of concrete. Timber used shall be properly seasoned so as to prevent deformation when wet.

All props shall be straight and of full height and no joints shall be allowed. Props shall be braced with MS supports and where additional staging is necessary, extra care shall be taken to use bigger diameters props with bracing at 4 or 5 levels. All props shall be supported on sole plates and double wedges. At the time of removing props these wedges shall be gently eased and not knocked out.

All rubbish, chippings, shavings and saw dust shall be removed from the interior of the forms before the concrete is placed and the form work in contact with the concrete shall be cleaned and thoroughly wetter or treated with non-staining mineral oil or any other approved materials is kept out of contact with the reinforcement.

All form work shall be removed without shock or vibration and shall be eased off carefully in order to allow the structure to take up its load gradually. Forms shall not be disturbed until concrete has adequately hardened to take up superimposed load coming on it and in no circumstances shall forms be struck until the concrete may be subjected at the time of striking.

In the normal circumstances (generally where temperatures are above 21 degrees centigrade) and where ordinary cement is used, forms may be struck after expiry of following periods:

a)	Walls, Columns and Vertical sides of beam}	48 hours
b)	Bottom of slab up to 4.5 m. span.	7 days.
c)	Bottom of slab up to 4.5 m. span.	14 days.
d)	Bottom of beams and arch rib over 6 m. span.	21 days.

However, this period may be increased or decreased at the discretion of Architects. Special care shall be taken while striking the centering of cantilevered slab canopies, portal frames, folded plate construction and period of striking centering shall be as determined by the Architect. If directed, form shall be given an upward camber to ensure that the beams do not have any sag. Surface that becomes exposed on removal of forms shall be carefully examined and any fins, burrs, projections etc., that are detected shall be removed. Any honeycombing of minor nature shall be finished neatly with cement mortar 1:2.

Any work showing signs of damage through premature or careless removal of centering or shuttering, shall be reconstructed by the contractor at his own cost.

Strength:

Concrete mixed in the proportion desired shall have compressive strength after placing, not less than the following:

Concrete mixed in the proportion desired shall have compressive strength after placing, not less than the targeted strengths as per requirement.

Tests:

Tests on concrete shall be carried out in accordance with IS-456/- and any other is applicable. The frequency of work test shall be at such intervals as ordered by the Architect and subject to that every 150 cu.m. of concrete placed or part thereof and for a day's concrete exceeding 30 cu.m. a batch of 6 cubes shall be made for every sample and 3 of them tested after 7 days and the remaining 3 cubes shall be tested after 28 days. The criteria for acceptance of a concrete as confirming to a specified proportion / grade of concrete shall be in accordance with IS:456 and the Contractor shall entirely re-do the rejected work at his own cost. Strength of 28 days shall alone be considered for acceptance.

The Contractor shall arrange to carry out the tests in accordance with the relevant Indian Standards Specifications in an approved laboratory and the test reports in original be submitted to Architect. The entire cost of testing shall be borne by the Contractor.

7. Steel Reinforcement:

Reinforcement shall be accurately fabricated, placed and adequately maintained in position as shown on the drawings or as directed by the Architect. All finished bars shall be free from cracks, surface flaws, laminations, jagged and imperfect edges. Cement mortar blocks shall be used to give requisite cover as shown be firmly tied with binding wire of 16 to 18 gauge. Reinforcement shall be bent in accordance with the procedure stipulated in IS:2502-1963 and will not be straightened in a manner which will injure the material.

All reinforcement shall immediately before placing in concrete be thoroughly cleaned of loose mill scale, loose rust, oil and grease or other deleterious matter that would destroy or reduce bond. Reinforcement

in reinforced concrete members shall not be connected by welding or coupling except in accordance with relevant ISS and with the previous approval of the Architect. Overlaps and joints shall be staggered and located at points, along the spans where neither shear nor bending moment is maximum.

Cover:

Reinforcement shall have cover as shown on the R.C.C. drawings and where not specified the thickness of cover shall be as follows.

- a) At each end of reinforcing bar not less than 25 mm. not less than twice the diameter of such rod or bar.
- b) For a longitudinal reinforcing bar in a column not less than the diameter of such rod or bar. In the case of columns of minimum of 20 mm. or under whose reinforcing bars do not exceed 13 mm. the cover of 25 mm. may be used.
- c) For longitudinal reinforcing bar in a column not less than 25 mm. not less than diameter of such rod or bar.
- d) For tensile, compressive, shear or other reinforcement in a slab not less than 13 mm. nor less than diameter of such reinforcement, and
- e) For any other reinforcement not less than 13 mm. not less than the diameter of such reinforcement.

8. BRICK AND STONE MASONRY

General:

All brick work should be carried out as shown on the drawings with setbacks, projections, cuttings things, etc. Wherever the proportion of cement mortar has not been specifically mentioned, cement mortar in the proportion of 1:6 shall be used. Flat bricks arches shall be provided wherever required without any extra cost. Brick work shall be kept wet while in progress, till mortar has properly set. On holidays or when work is topped, top of all unfinished masonry shall be kept wet. Should the mortar become dry, white or powdery, for want of curing work should be pulled down and rebuilt at the Contractor's expenses.

Brick Work 1stClass:

Bricks shall be thoroughly cleaned, well wetted and soaked for at least twelve hours in fresh water before being used on the work. Bricks shall be of locally, available best quality. English bond shall be used throughout in walling. A good bond shall be maintained throughout the work, both laterally and transversely. In walling, the courses shall be kept perfectly horizontal and in plumb with the frogs facing upwards. Vertical joints shall not exceed 10 mm. thickness and shall be full of mortar. No broken bricks shall be used except as closers. After day's work all joints shall be raked to 12 mm. depth to provide for proper key to plastering.

Mortar used shall be as specified in respective items and every third course of brick work shall be flushed with mortar grout.

Whole of the masonry work shall be brought up at one uniform level throughout the structure; but where breaks are unavoidable, joints shall be made in good long steps. All junctions of walls and cross walls shall be carefully bounded into the main walls. The rate of laying masonry may be up to a height of 60 cm. per day if cement mortar is used and 45 cm. Per day if lime mortar is used. Greater heights may be built only if permitted by the Architect. During rains, the work shall be carefully covered to prevent mortar from being washed away. Should any mortar or cement be washed away, the works shall be removed and rebuilt at the Contractor's expenses.

Bricks Work 2ndClass:

Shall be similar to 1st class brick work except that 2nd class bricks shall be used and joints shall be 10 mm. to 12 mm. thick.

Half Brick Masonry:

Shall be set in cement mortar as specified. Hoop iron bands of 2.5 cm. x 0.16 (1" x 1/16") shall be embedded in every fourth course with thick mortar band or 2 Nos. 6 mm. (1/4") dia. bars shall be used in every sixth course otherwise as specified under item.

9. RUBBLE MASONRY

General:

Stones shall be of the kind specified in the item and shall be from an approved quarry. Stones shall be well wetted before laying in position. The mortar shall be as specified in the item. Face stone shall not be less than in breadth than in height, it shall also tail into the work more than its height. Jambs of doors, windows and openings shall be formed with quoins. In case of battered walls, the courses on battered surface shall be at right angle to the batter.

Through stones or headers shall be laid in every course at a distance not exceeding 2 meters part and shall be staggered. They shall be in one piece for walls up to 1.5-meter width and shall be lap jointed in case of wall having thickness more than half meter. The face area of each header shall not be less than 0.50 sqm. 1:2:4 cement concrete may also be allowed where good length headers are not available. Headers shall be marked with oil paint for ready identification.

Height of quoins shall be same as that of the course. Length of quoins shall be 0.50 m. and shall be laid header and stretcher alternatively. Faces of quoins shall be fair dressed. No quoins stones shall be less than 0.30 cum. In content. Joints of masonry shall be raked out and unless otherwise stated, shall be raised cement pointed by using cement mortar 1:1 to all exposed surfaces. All masonry work shall be well watered for a period of seven days.

a) Coursed Rubble Masonry – First Sort:

Height of course shall not be less than 15 cm. and all courses shall be of uniform height. All stones in the course shall be of same height. In no case height of course shall be more than any of the course below it. Bed and sides shall be hammer or chisel dressed back from the face 75 mm. and 35 mm. respectively.

Faces of stones shall be hammer dressed and bushing shall not be more than 35 mm. Thickness of joints shall not be more than 10 mm. Stones shall break joints at least half the height of the course. Work on interior face shall be precisely the same, as on exterior face. Quoins shall be at least 0.5 m. long laid square on their beds and shall be fair dressed to a depth of at least 10 cm.

b) Uncoursed Rubble Masonry:

Stones shall be hammer dressed. Nearly fifty per cent of stones shall not be less than 0.30 cum. in content each, and twenty-five per cent of stone shall tail back in masonry by 40 cm. or more. Stones shall be so arranged as to break joints as much as possible.

Long vertical joints shall be carefully avoided. Thickness of joints shall in no case exceed 12 mm.

Pillar offsets shall be properly dressed with hammer or chisel to form proper angle. Stones used for the backing shall be of fairly large size.

c) Random Rubble Masonry – First Sort:

Stones shall be roughly chisel dressed. They shall be solidly bedded in mortar. Height of stone shall not be more than width of face or length of tail. Stones shall be of equal size and so arranged as to break joints as much as possible, avoiding long lines of horizontal or vertical joints. Quoins shall be same as described in Coursed Rubble Masonry – 1st Sort. All stones shall be carefully fitted. Thickness of face joint shall be not exceeded 25 mm. Edges of stones shall be chisel dressed for fitting in position properly.

10. Timber

Timber used shall conform to specifications described under Materials, Doors, Windows, Ventilators, walls, Paneling, False Ceiling, etc., shall be in accordance with Architect's drawing in every detail and all joiner's work shall be accurately set out, framed and finished in a proper workman-like manner, frames of doors, windows and ventilators etc. and shutter styles and rails shall be best solid teak of quality specified in the schedule of quantities. The scantlings shall be accurately planed smooth, rebates, rounding and mouldings shall be made as shown on the drawings, patching or plugging of any kind shall not be allowed. Joints shall be simple, neat and strong. Framed joints shall be coated with suitable

adhesive like glue or synthetic resin before the frames are put together. All mortise and tenon joints shall be fit and fully and accurately without wedging or filling. The joints shall be pinned with hard wood or bamboo pins of 10 mm. to 12 mm. dia. or rust resisting star shaped metal pins 8 mm. after the frames are put together and pressed in position by means of press. The frames are put together and pressed in progress of work by suitable boxing. All portions of timber abutting against or embedded in masonry or concrete shall be treated against termites by giving a coat of any approved wood preservative.

Unless otherwise specified all doors, frames shall have six M.S. flat hold fasts and window frames shall have four hold fasts shall be provided to the ventilators, if directed. Size of hold fasts shall be 30 mm. x 40 mm. x 6 mm. M.S. flat bent to shape worth fish tail end and it shall be fixed to frame with sufficient number of screws as directed. When door / window frames are to be fixed to R.C.C. column or R.C.C. wall, hold fasts shall be substituted by suitable arrangements such as coach crews, rawl bolts etc., to secure frames to R.C.C. column or R.C.C. wall as directed by the Architect.

Frames and shutter shall not be painted or erected before being approved by Architect.

11. Paneled Shutter:

Panels shall be of pattern and size as shown on the drawings or as directed by Architect. Solid teak wood panels shall be in one piece wherever possible. Where two or more pieces are permitted, they shall be of equal width. Panels shall be framed into grooves made in styles and rails to the full depth of groove and faces shall be closely fitted to sides of groove.

Where panels specified are block board, it shall be solid core with teak internal lipping and of approved make.

Partly paneled and partly glazed shutter shall be similar to paneled shutters except that such parts as are directed shall be glazed with plain or ground glass as specified. Styles and rails shall be rebated 12 mm. to receive glass. Sash bars shall be moulded and rebated and mitred on sides to receive the glass which shall be fixed with putty and beads.

12. Hardware Fittings:

Supply and Installation of All hardware, fittings and fixtures shall be the scope of Contractor the cost of same shall be included in the rate quoted. The fixing shall be done in the best workman-like manner in accordance with the manufacture's specifications. The Contractor shall be held responsible for working of all moving parts dependent on proper fixing. He will also be responsible for any breakage due to negligence during fixing or lack of protection before the building is handed over. The Contractor shall also take delivery of all hardware fittings etc., as and when supplied and arrange for safe storage etc.

Hardware required for fixing false ceiling, wall paneling etc., shall be arranged by the Contractor at his cost. Apart from the hardware fittings required for the joinery items, the Contractor shall have to fix all other items of hardware fittings to be supplied by the employer viz. coat / picture hooks, numerical, letters to denote buildings, hanging rods etc., as directed by the Architects.

Painting and polishing of wood work shall be as per specifications under respective heads.

13. Flush Doors:

All flush doors shall be solid core unless otherwise specified. It shall conform to the relevant specifications of I.S. 2202 and shall be obtained from approved manufactures. The finished thickness of the shutter shall be mentioned in the items. Face veneers shall be of the pattern and colour approved by the Architect and an approved sample shall be deposited with the Architect for reference.

The solid core shall be wood laminae prepared from battens of well-seasoned and treated good quality wood having straight grains. The battens shall be of uniform size of about 2.5 cm. width. These shall be properly glued and machine pressed together, with grains of each piece reversed from that of adjoining one. The longitudinal joints of the battens shall be staggered and no piece shall be less than 50 cm. in length. Alternatively, the core shall be of solid teak particle board. Edges of the core shall be lipped internally with 1st Class teak wood battens of 4 cm. (1.5") minimum depth, glued and machine pressed along with the core.

The core surface shall then have two or three veneers firmly glued on each face. The first veneer (called cross band) shall be laid with its grains at right angles to those of the core and the second and the third

veneers with their grains parallel to those of the core. The under veneers shall be of good quality, durable and well-seasoned wood. The face veneers shall be of minimum 1 mm. thickness and of well-matched and seasoned 1st class teak, laid along with grains of the core battens. The combined thickness of all the veneers on each face shall not be less than 4 mm. Thermosetting synthetic resin conforming to I.S. 303 or moisture-proof plywood grade MPF.I. shall be used in manufacture. In addition to internal lipping all doors shall have external lipping all round.

14. STEEL DOORS, WINDOWS, VENTILATORS, ROLLING SHUTTER, M.S. GRILLES ETC.

Steel used in the manufacture of rolled steel sections shall not have more than 0.060 per cent of Sulphur and 0.065 per cent of phosphorus. The carbon content shall not exceed 0.30 per cent and shall be of weldable quality. In all other respects, the rolled steel sections shall conform to I.S. 226-1955 and I.S. 1977-1962.

Frames shall be square and flat. Both the fixed and openable frames shall be constructed of sections which have been cut to length, mitred and electrically welded at corners. Subdividing bar units shall be tenoned and rivetted into the frames. All frames shall have the corners welded to a true right angle and welds shall be neatly cleaned off. Couplings, mouldings and weather bar shall be provided as directed by the Architects.

Outer frames shall be provided with fixing holes centrally in the web of the sections and fixing screws and lugs shall be used for fixing the frame to masonry. Mastic cement shall be used for making the joints watertight.

Hinges shall be strong projecting type. If directed friction type hinges shall be used in which case windows shall not be fitted with peg stays.

Projecting type hinged shutter shall be fitted with bronze or brass peg stays, 30 cm. Long with peg and brackets welded / riveted to the frame or as sated under item.

All windows shall be provided with handles of brass or bronze or otherwise as stated under them. Top hung ventilators shall be fixed with plain hinges riveted / welded to the fixed frame. A brass or bronze peg stay 30 cm. long as in windows shall be provided or as stated under item.

Center hung ventilators shall be hung on two pairs of brass or leaded tin bronze cup pivots riveted to the inner and outer frames of the ventilators to permit the ventilators to swing through an angle of approximately 85. The opening position of the ventilator shall be so balanced to keep it open at any desired angle under normal weather conditions. A bronze spring catch shall be fitted in the center of the top bar of the ventilator for the operation of the ventilator. This spring catch shall be secured to the frame with brass screws and shall close into a mild steel malleable iron catch plate riveted or welded to outside of the outer ventilator frame bar. A brass cord pulley wheel in mild steel or malleable iron brackets shall be provided along with card eye.

The windows and ventilators shall be painted/Powder coated if applicable. All the steel surfaces shall be thoroughly cleaned free of rust, scale or dirt and mill scale by picking or phosphating and before erection painted with one coat of approved primer and after erection painted with two finishing coats of synthetic enamel paint of approved shade and quality.

Glazing of specified thickness shall be provided on the outside of frames and unless otherwise specified, metal beading of approved shape, and section shall be used for fixing glasses. Special metal sash putty of approved make shall be used, if directed.

Rolling Shutters:

Shall be of approved manufacture suitable for fixing in the position ordered i.e. outside, inside, on or below lintel or between jambs. Shutters up to 12 sqm. (130 Sq.ft.) in area shall be manually operated or Push Up type while bigger sizes shall be of reduction gear type mechanically operated chain or handles.

These shall be consisting of 8 gauges or as specified with 75 mm. (3") M.S. laths of best quality mild steel strips machine rolled and straightened with an effective bridge depth of 16 mm. (5/8") and shall have convex corrugation. These shall be interlocked together throughout their entire length with end locks. These shall be mounted on specially designed pipe shaft. The spring shall be of approved make coiled type. These shall be manufacture from tested high tensile spring steel wire or strip of adequate

strength to balance the shutters in positions. The spring pipe, shaft etc., shall be supported on strong M.S. or malleable cast iron brackets.

Both the side guides and bottom rail shall be jointless and of single piece of pressed steel.

Top cover of shaft, spring etc., shall be of the same material as that of lath.

For rolling shutter with wicket-gate, night latch shall be provided free of cost.

The shutter and cover etc., shall be painted with one coat of anti-corrosive paint and two coats of synthetic enamel paint of approved quality and shade.

Collapsible Steel Gate:

It shall consist of vertical double channels at 10 cm. centers. The sizes of channels T-Section for top and bottom shall be as approved by the Architects. The gate shall be provided with necessary bolts, nuts, locking arrangements, stoppers and brass handles on both sides. The gate shall be painted with one coat of anti-corrosive paint before erection and two coats of synthetic enamel paint of approved quality and shade.

MS Grilles:

Grille shall be manufactured as per drawings and the welded joints shall be smooth. The grilles shall be painted with one coat of anti-corrosive paint before fixing and two coats of synthetic enamel paint of approved quality and shade.

15. Aluminum Doors, Windows, Ventilators & Partitions etc.:

These shall be obtained from approved and established manufactures and shall be of Aluminum alloy conforming to I.S. 733 and sections shall generally conform to I.S. 1948.

Theses shall be fabricated as per the detailed drawings,

Frames for windows, ventilators etc., shall be square and flat. Both fixed and openable frames shall be constructed of section which have been cut to length, mitered and welded at corners. Sub-dividing bars shall be tenoned and rivetted into the frames. All frames shall have corners welded to a true right angle. For side hung shutters, hinges shall normally be of projecting type made of Aluminum alloy and rivetted / welded to frames. Handles, peg stays etc., or approved quality Aluminum or its alloy conforming to IS Specifications.

All types of shutters shall be fabricated, supplied and fixed as specified in the IS:1948. The rate shall include supplying and fixing all fittings and fixtures required for proper and safe operation.

The doors shall be fabricated by using standard aluminum alloy extruded sections as specified in IS:1948. The rate shall include supplying and fixing all fittings and fixtures including approved locking arrangement as directed.

All aluminum fabricated work shall be anodized to the British Standard 1616:1961 to give an anodized film of 25 micron.

The Contractor shall take to stack the fabricated frames etc., on site under cover. They shall be handled with care, stacked on edge on level bearers and supported evenly. Before erecting, the frames coming in contact with concrete, masonry, plaster of dissimilar metals shall be coated with a coat of Zinc Chromate conforming to IS:104-1950. The Contractor shall cover all anodized finish work with a thick layer of clear transparent lacquer based on methacrylate or cellulose but rate to protect the surface from wet cement during installation. This coating shall remove on completion. Before handing over, the aluminum work shall be washed with mild solution of non-alkali soap and water.

Glazing: Glazing shall be approved especially quality glass of specified thickness and unless otherwise directed it shall be provided the exterior with metal beading.

16. FLOORING, SKIRTING, DADO AND STONE VENEERING

All flooring, skirting, dado and stone veneering etc., shall be executed strictly as per relevant IS Specification and in workman-like manner.

Indian Patent Stone:

Selection of materials, method of mixing, placing and compacting shall generally conform to the specifications under plain and reinforced cement concrete described earlier. A stiff mix consistent with workability shall be used.

Preparation of Surface:

Before the operation for laying topping is started the surface of base concrete shall be thoroughly cleaned of all dirt, loose particles coked mortar droppings and laitance if any, by scrubbing with coir or steel wire brush. Where the concrete has hardened so much that roughening of surface by wire brush is not possible, the surface shall have roughened by chipping or hacking at close intervals. The surface shall then be cleaned with water and kept wet for 12 hours and surplus water shall be removed by mopping before the topping is laid.

Laying:

The screed strips shall be fixed over the base concrete dividing it into suitable panels. Before placing the concrete for topping, neat cement slurry shall be thoroughly brushed into the prepared surface of the base concrete just ahead of the finish. Concrete of specified proportion and thickness shall be laid in alternate panels to required level and slope and thoroughly tamped.

Finishing the Surface:

After the concrete has been fully compacted it shall be finished by troweling or floating with neat cement rendering. Finishing operations shall start shortly after the compaction of concrete and the surface shall be troweled three times at intervals so as to produce a uniform and hard surface. The satisfactory resistance of floor to wear depends largely upon the care with troweling is carried out. The time intervals allowed between successive troweling is very important. Immediately after placing cement rendering, only just sufficient troweling shall be done to give a level surface. Excessive troweling in the earlier stages shall be avoided as this tends to bring a layer rich in cement to the surface. Sometime, after the first troweling, the duration depending upon the temperature, atmospheric conditions and the rate of the set of cement used, the surface shall be re-troweled to close any pores in the surface and to bring to surface and to scrape off any excess water in concrete or laitance. No dry cement shall be used directly on the surface to absorb moistures or to stiffen the mix. The final troweling shall be done well before the concrete has become too hard but at such time that considerable pressure is required to make any impression on the surface.

If directed by the Architect, approved mineral pigment shall be added to the rendering to give desired colour and shade to the flooring at no extra cost.

When instead of 1:2:3 or 1:2.5:3.5 mix, 1:2:4 is specified the topping shall be rendered with 1:1 cement mortar with a suitable mineral pigment, if directed, instead of cement only. If specified in the Schedule of Quantities, the flooring shall be machine polished as per the Architect's instructions. Wherever the patent stone flooring is used as finishing on roof the joints shall be filled with an approved bitumastic filler in workman like manner.

Ironite Topping:

Instead of finishing the top with rendering coat of 1:1 cement mortar, the top shall be finished with 12 mm. thick ironite topping. Unless otherwise specified, one part of ironite and four parts of ordinary cement by weight shall be mixed dry thoroughly. This dry mixture shall be mixed with stone grit 6 mm. (1/4") and down size or as otherwise directed in the ratio of 1:2 by volume and well turned over. Just enough water shall be added to this dry mix and mixed thoroughly well and laid to uniform thickness of 12 mm. and compacted. After initial set has started the surface shall be finished as directed.

Plain and Colored Cement Tiles, Marble Mosaic and Terrazzo Tiles Flooring:

The tiles shall conform to IS : 1237 having the colour approved the Architect and the rate shall include provision of border tiles and tiles of different colours in pattern if directed. The mosaic topping of lighter shade tiles shall be made of White Cement with an approved shade pigment and neutral shade shall be of Grey cement with an approved shade pigment. The type of tiles shall be as specified in respective items.

The sub-grade shall be thoroughly wet after cleaning of all dirt, laitance, and loose material. A bed of lime mortar consisting of one part of lime and two parts of sand shall be laid and properly leveled to an average thickness of 25 mm. and the surface shall be kept slightly rough to form a satisfactory key for

tiles. Neat cement paste of honey like consistency shall be spread over mortar bed, over such area at a time as would accommodate about 20 tiles. Tiles shall be soaked in water for 15 minutes and allowed to dry for the same duration. Tiles then be fixed with a thin coat of cement paste on back of each tile and then each tile being gently tapped with a wooden mallet till it is properly bedded and in level with adjoining tiles. Joints shall be fine and as imperceptible as possible.

After tiles have been laid in a room or a day's fixing work is completed, surplus cement grout that may have come out of the joints may be wiped off gently and joints cleaned. A thin slurry of coloured cement matching to the colour of tiles shall be spread over it and rubbed so as to seal even a thinnest joint between the tiles and make it impervious and the flooring cured for 7 days. The tiles shall be polished and finished according to IS:1443.

Dado, Skirting and Risers:

Tiles shall conform to IS:1237 and shall be of approved design. The tiles shall be fixed near cement grout on a blacking coat consisting of 1:4 cement sand plaster of 15 mm. thick. The top and bottom junctions of tiles shall be rounded off neatly as directed. The joints shall be filled with matching shade-coloured cement slurry. The surface shall be kept wet for 7 days and then polished with carborundum stone to obtain smooth surface and fine polish.

Shahabad / Tandur / Kotah / Cuddappa Stone Flooring:

The flooring shall be either with rough stone or machine cut and machine polished as specified in respective items and shall be of specified thickness (within tolerance limit of +/- 2mm) and of approved quality and size, free from cracks and flakes and shall be uniform in colour with straight edges. The sides of machine cut and machine polished stone shall have perfect right angles and surface smooth. The stone slabs shall be laid and finished as described under plain cement or colour cement tiles on a bedding of cement mortar of required average thickness. The finished stone surface thus laid shall then be polished to the required degree as approved by the Architect.

In Dado, Skirting, Risers etc.:

Stone slabs shall be laid on backing plaster of cement mortar as per BOQ specifications and finished as described under plain and coloured cement tile dado.

Marble mosaic / Terrazzo in situ work in flooring, dado, skirting etc.:

The terrazzo / mosaic finish shall be laid on an under layer of thickness as specified in the respective items. The topping shall consist of a layer of marble chips of selected sizes, colour and design approved by Architect, mixed with cement with desire shade of pigment. For lighter shade mosaic, terrazzo white cement shall be used and for neutral shade, grey cement shall be used. The proportion of terrazzo mix shall be three parts of cement one part of marble powder by weight. For every part of cement marble powder mix, the proportion of marble aggregate by volume shall be 1.5 parts unless otherwise specified. The topping shall be mixed and laid in panels as described in IS:2114 and as per decorative designs prepared by Architects. The dividing strips of panels shall be Aluminum or as specified in the Schedule of Quantities. It shall be polished as specified in IS: 2114.

Broken Mosaic Flooring:

Broken mosaic finish shall be laid on an underlayer of thickness as specified in the item. Pieces of mosaic tiles shall be obtained from broken marble mosaic tiles of approved shade conforming to IS:1257. The sizes of pieces shall be suitable to obtain the desired pattern of flooring as shown on the drawings or as approved by Architect.

Broken pieces shall be thoroughly wetted before fixing them. Ordinary or coloured cement grout shall be spread on the bedding. Mosaic tile pieces shall be fixed piece by piece to the desired pattern. The flooring shall be laid to correct level and slopes and compacted by straight screed tamper. The grout shall cream up to the surface. The junctions of the flooring and the wall shall be rounded and the flooring shall be extended along the wall to about 15 cm. (6"). After the day's work, the surplus cement grout that may have come out of the joints shall be cleaned off. The flooring shall be cured for seven days and then polished with a machine as stipulated in IS:1443.

Broken China Mosaic:

Broken China Mosaic flooring shall be exactly as per broken mosaic tile flooring except that the broken pieces shall be of China of approved colour and manufacturer and the floor shall not be polished.

Marble Flooring:

Marble slabs shall be of the best Indian marble of White or other approved colour as specified in the item. They shall be hard, dense, uniform and homogeneous in texture. They shall have even crystalline grain and free from defects and cracks. The surface shall be machine polished to an even and perfectly plane surface and edges machine cut true to square. The rear face shall be rough enough to provide a key for the mortar. No slab thinner than the specified thickness at its thinnest part. The sizes of the slabs shall be as specified in the respective items. The slabs shall be paid as described under mosaic tile flooring in every respect.

White Glazed / Ceramic Tiles / Vitrified Tiles in Flooring and Dado:

White Glazed Tiles from an approved manufacturer conforming to IS:777 shall be used. They shall be of specified size and thickness. All specials viz. coves, internal and external angles, corners, beads etc., shall be used wherever directed. Under layer of specified thickness and mortar of stipulated proportion shall be laid as described in marble mosaic flooring. Tiles shall be washed clean and set in cement grout and each tile being gently tapped with a wooden mallet till it is properly bedded and in level with the adjoining tiles. The joints shall be kept as thin as possible and in straight lines or to suit the required pattern. After the tiles have been laid, surplus cement grout shall be cleaned off.

The joints shall be cleaned off the grey cement grout with a wire brush or trowel to a depth of 5 mm. (3/16") and all dust and loose mortar removed. Joints shall then be flush pointed with white cement (with/ without colour pigments as required). The floor shall then be kept wet for seven days. After curing, the surface shall be washed with mild hydrochloric acid and clean water. The finished floor shall not

17. PLASTERING**Scaffolding:**

Scaffolding for carrying out plastering work shall be double steel scaffolding having two sets of vertical supports so that the scaffolding is independent of the walls.

Preparation of surface:

All put log holes in brick work and junction between concrete and brick work shall be properly filled in advance. Joints in brick work shall be raked about 10 mm. if not raked out while constructing brick masonry work and concrete surface hacked to provide the grip to the plaster, if not hacked earlier projecting burns of mortar formed due to gaps at joints in shuttering shall be removed. The surface shall be scrubbed clean with wire brush / coir brush to removed dirt, dust etc., and the surface thoroughly washed with clean water to remove efflorescence, grease and oil etc., and shall be kept wet for a minimum of six hours before application of plaster.

Neeru Plaster:

Cement mortar of specified proportion and thickness shall be prepared in small batches and applied to the wall surface / ceiling. To ensure proper thickness, gauged patches shall be made at 1.5 to 2 m. apart and the surface plastered true to line, level and plumb taking special care to finish jambs of windows, doors, wall returns, corners, junctions etc. A thin layer of neeru shall then be applied and rubbed into surface and finished by means of trowel until the surface is even and smooth. The surface shall be kept moist for seven days and then given a coat of white wash.

Sand-faced Plaster:

The surface shall be prepared as above.

The coat of cement mortar in proportion of 1:4 or as specified, shall be applied uniformly all over the surface to a thickness of 12 mm. and finished true to level and line and keys shall be formed on the surface. The surface shall be kept moist till the finishing coat is applied.

The finishing coat shall be applied a day or two after. The proportion of mortar for finishing coat shall be one part of cement and three parts of selected, well graded and washed sand, or as specified under item and it shall be applied in a uniform thickness of 6 mm. (1/4").

The surface shall be tapped to uniform grained texture by using sponge pads as directed. Curing shall start after 24 hours and the surface kept wet for seven days.

Rough Cast Plaster:

Except for the finishing coat the surface shall be prepared and base coat of plaster applied as under sand-faced plaster.

Finishing coat mortar shall be in proportion of one part of cement and one part of specially selected and graded sand and one part of gravel of 3 to 6 mm. size. It shall be flung upon the first coat with large trowel to form an even and decorative coat. The work shall generally conform to clause 16.5 of IS:1661-1960. The thickness of the coat shall be about 12 mm. (1/2"). It shall be cured for seven days.

Rough coat plaster with colour finish:

This finish shall be similar to Rough cast plaster above except a high-grade mineral pigment of approved shade shall be mixed with white cement instead of ordinary grey cement while preparing the mortar.

18. Water-proofing Treatment :

Unless otherwise specified, the Contractor shall carry out waterproofing treatment of basements, terrace and water retaining structures through reputed firms having specialization in the line and approved by the Architects. The Contractor shall also furnish full details of such treatment to the Architects and provide all information / proof etc., regarding the effectiveness of the treatment when called upon to do so. All such treatment shall have to be guaranteed in the form approved by the Employer for a minimum period of ten years. Any defects / leakages noticed during the guarantee period shall have to be rectified free of cost by the Contractor including reinstating the surface to its original condition and finish.

Waterproofing of sunk portions of floor slabs for baths, W.C. and kitchen moories etc., in residential buildings, unless otherwise specified, shall be done as specified in the schedule and shall generally comprise of :

a) A coat of chemical waterproofing as specified in the BOQ specifications.

b) Min. 20 mm. thick cement plaster in cement mortar 1:4 with approved water-proofing cement compound as per manufactures specifications. The plaster shall be cured by pounding for seven days.

The rate for the above treatment shall include drying and cleaning surfaces free of dust etc. and wiping with kerosene before application of bitumen. The vertical faces and returns shall also be treated similarly. The actual area treated including vertical faces and returns shall be measured and paid for. The work should be done in such a way that the finished flooring in bath has a minimum slope of 20 to 25 mm.

19. PAINTING

General:

Wherever scaffolding is necessary, it shall be double scaffolding.

The surface shall be thoroughly brushed free from mortar droppings and foreign matter. All steel work shall be cleaned of loose rust, mill scales etc. so as to expose the original surface. All broken edges, cracks, loose plaster and wavy surface shall be brought up either by patch plaster work or by plaster of paris.

All materials viz., dry distemper, oil bound distemper, oil paint, flat oil paint, synthetic enamel paint, plastic emulsion paint, cement primer, red lead and other primers and metallic paints shall conform to respective I.S. specifications and shall be obtained from approved manufactures. All paints shall be brought on site in sealed thins in ready mixed form and shall be applied directly with the addition of thinner, if recommended by the manufacturers.

White Washing:

White was shall be prepared from lime slaked on spot, mixed and stirred with sufficient water to make a thin cream. This shall be allowed to stand for 24 hours and shall be screened through clean cloth. Four kg. gum dissolved in hot water shall be added to each cubic meter of the cream (115 gm. per cft.).

Blue shall be added to give required whiteness. The approximate quantity of water to be added in making cream shall be five liters per kg. of lime.

White wash shall be applied in specified coats by using flat brushes or spray pumps. Each coat shall be allowed to dry before next coat is applied. If additional coats than what have been specified, are necessary to obtain uniform and smooth finish, it shall be given at no extra cost. The finished dry surface

shall not show any signs of cracking and peeling nor shall it come off readily on the hand when rubbed.

If directed by the Architects one coat of chalk and glue shall be applied before application of white / colour wash at no extra cost.

Colour Wash:

Colour wash shall be prepared by adding mineral colours not affected by lime to white wash. No colour wash shall be done until a sample of the colour wash to the required tint or shade has been got approved from the Architects.

Colour wash shall be applied as specified under white wash.

Dry Distemper:

Shade shall be got approved from the Architects before application of distemper.

The surface shall be prepared as specified earlier. A primer coat using approved primer or sizing shall be applied. Distemper prepared as per manufacturer's directions shall be applied and each coat shall be allowed to dry before subsequent coat is applied. The finished surface shall be free from chalking when rubbed, even uniform and shall show not brush marks. If additional coats are necessary, they shall be given at no extra cost.

Oil Bound Distemper:

The surface shall be prepared as specified above. A primer coat of either cement primer or any approved distemper primer shall be applied.

After the primer coat has dried, the surface shall be lightly sand papered and dusted to make to smooth to receive distemper.

Distemper shall be prepared as per the directions of the manufacturer and conforming to shade approved. It shall be applied in specified coats, taking care to allow for drying of each coat before subsequent coats are applied.

Water-proof Cement Paint / Sand-tex matt Paint:

The surface shall be prepared as specified above and thoroughly wetted with clean water before water-proof cement paint is applied.

The paint shall be prepared strictly as per manufacturers specifications and in such quantities as can be used up in an hour of its mixing, as otherwise the mixture will set and thicken, affecting flow and finish.

The paint thus prepared shall be applied on clean and wetted surface with brush or spraying machine. The solution shall be kept stirred during the period of application. It shall be applied on the surface which is on the shady side of the building so that the direct heat of the sun on the surface is avoided. The completed surface shall be watered after the day's work. Number of coats shall be as specified in the item.

Painting – Oil / Enamel / Plastic Emulsion/ Elastomeric Paint etc.:

Ready mixed oil paint, flat oil paint, plastic emulsion paint, ready mixed synthetic enamel paint, etc., shall be brought in original containers and in sealed tins. If for any reason thinner is necessary, the brand and quantity of thinner recommended by the manufacturer or as instructed by the Architect shall be used. The surface shall be prepared as specified above and a coat of approved primer shall be applied. After 24 hours drying approved or specified quality paint shall be applied evenly and smoothly. A filler putty coating may be given to give a smooth finish. Each coat shall be allowed to dry out thoroughly and then lightly rubbed down with sand paper and cleaned of dust before the next coat is applied. Number of coats shall be as specified in the item and if the finish of the surface is not uniform, additional coats as required shall be applied to get good and uniform finish at no extra cost. After completion no hair marks from the brush or clogging of paint puddles in the corners of panels, angles or mouldings etc., shall be left on the work. The glass panes, floor etc. shall be cleaned of stains.

When the final coat is applied, if directed, the surface shall be rolled with a roller or if directed, it shall be stippled with a stippling brush.

Note:- All paints shall be of LOW VOC content as per Green Building Norms

POLISHING AND VARNISHING

French Polishing: (LOW VOC AS PER GREEN BUILDING NORMS)

French spirit polish shall be of an approved make conforming to IS:348. If it has to be prepared on site, the polish shall be made by dissolving 0.7 kg. of best shellac in 4.5 liters of methylated spirit without heating. To obtain required shade pigment may be added and mixed.

Surface shall be cleaned. All unevenness shall be rubbed down smooth with sand paper and well dusted. Knots, if visible, shall be covered with a preparation of red lead and glue. Resinous or loose knots and gaps shall be filled with season timber pieces and make level with rest of the surface. Holes and indentations on surface shall be filled with putty made of whiting and linseed oil. Surface shall be give a coat of filler made of 2.25 kg. of whiting in 1.5 liter of methylated spirit. When it dries, surface shall again be rubbed down perfectly smooth with sand paper and wiped clean.

Piece of clean fine cotton cloth and cotton wool made into shape of pad shall be used to apply polish. The pad shall be moistened with polish and rubbed hard on the surface applying the polish sparingly but uniformly and completely over the entire surface. It shall have allowed to dry and another coat applied in the same way. To give finishing coat, the pad shall be covered with a fresh piece of clean fine cotton cloth, slightly damped with methylated spirit and fubbed lightly and quickly with a circular motion, till the finish surface attains uniform texture and high gloss.

Wax Polishing: (LOW VOC AS PER GREEN BUILDING NORMS)

Wax polish shall either be prepared on site or obtained ready made from market. Polish made on the site shall be prepared from a mixture of pure bee's wax, linseed oil, turpentine oil and varnish in the ratio of 2:1.5:1:½ by weight. The bees wax and the boiled linseed oil shall be heated over a slow fire. When the wax is completely dissolved the mixture shall be cooled till it is just warm, and turpentine oil and varnish added to it in the required proportions and the entire mixture is well stirred.

Surface shall be prepared as described under French polishing except that the final rubbing shall be done with sand paper which has been slightly moistened with linseed oil.

Mixture or polish shall be applied evenly, with a clean cloth pad in such a way that no blank patches are left and rubbed continuously for half an hour. When the surface is quite dry a second coat shall be applied in the same manner and rubbed continuously for an hour or until the surface is dry. Final coat shall then be applied and rubbed for two hours or more if necessary, until the surface has assumed a uniform gloss and is quite dry showing no sign of sickness when touched. Gloss of the polish depends on the amount of rubbing, therefore rubbing must be continuous and with uniform pressure and frequent change in direction.

Varnishing : (LOW VOC AS PER GREEN BUILDING NORMS)

Surface shall be prepared as described above. After preparation of surface, two coats of clean boiled linseed oil shall be applied at sufficient interval of time. After the linseed oil has dried two coats of varnish obtained from approved manufacturer shall be applied at sufficient interval of time. If the surface fails to produce the required gloss an additional coat shall be applied without any extra cost.

20. Road Concrete:

Specification for aggregate cement and concreting shall be as specified in the section under "Materials".

Before concreting, the surface shall be checked for the given profile. Wooden forms equal to the depth road slab thickness shall be erected to correct line and level and held by stakes driven into the ground along the outside edge at suitable intervals and two stakes being placed at each joint. Forms should be supported, strengthened or braced, whenever necessary so that they are able to prevent deformation and resist deformation under pressure of concrete or impact of tamping or vibrating. Working faces of all forms shall be thoroughly cleaned and oiled before use and forms which are used more than once, shall be carefully examined and trued, if necessary, before re-use.

Sub-grade shall be properly moistened before any concrete is deposited on it; care being taken to see

that there are no standing pools of water. It may be advisable to have the sub grade watered 12 to 24 hours in advance of placing concrete. Concrete shall be laid in alternate bays not exceeding 30 sqm. Concrete shall be deposited on sub-grade for the entire width of the slab and shall be kept sufficiently above the level of forms so that when tamped, it becomes a dense mass.

I.R.C. fabric reinforcement, if specified, shall be placed in correct position before commencing concrete.

The concrete shall be brought to the specified contour by means of heavy screed or tamper handles weighing not less than 10 kgs. / Meter and not less than 7.5 cm. wide or surface vibrator if directed by the Architects. This screed or tamper may be steel. It shall be drawn with a saw in motion in combination with a series of lefts and drops. At transverse joint tamper shall be drawn not closer than one meter towards the joint and shall then be lifted and set down at the joint and drawn backwards away therefrom. Surplus concrete shall then be taken up with shovels and thrown ahead of the joint. Immediately after the screeding or tamping has been completed the surface shall be inspected for high and low spots and any needed correction made by adding or removing concrete. The entire surface shall then be floated with hand floats one meter long and 7.5 cm. wide and this operation must be performed from bridge provided across the slab. The surface shall be roughened by brooming.

The longitudinal and transverse edges of the slab shall be properly formed with suitable tools and the same should be rounded to 10 mm. radius.

The finished surface of the slab must conform to the grade, alignment and contours as directed and cured for fourteen days.

After curing period is over the joints shall be filled up with approved bitumastic filler. Unless otherwise specified, the rate shall include filling of joints as specified.

21. STORM WATER DRAINAGE

The work shall be carried out in accordance with rules and regulations of local Drainage Authority. Necessary provision for sight rails, boning staves etc. shall be made.

Tests regarding watertightness of joint and cleanliness of pipes shall be performed before the trenches are covered.

Work of laying pipe lines and provided Manholes, Chambers, etc., shall include necessary excavation in any strata including old foundations of any description, refilling the trenches in layers of 20 cm. watering and consolidation.

Pipes:

All Hume pipes (Reinforced) shall conform to the relevant I.S.S. and shall be new, perfectly sound, free from cracks, cylindrical, straight and of specified nominal diameter. They shall be made of reinforced cement concrete manufactured by centrifugal or spun process and shall have even texture.

Trenches:

The trenches for laying shall be excavated to lines and levels as directed. The bed of the trench shall be truly and evenly dressed throughout from one change of grade to the next.

The gradient is to be set out by means of boning rods and should the required depth be exceeded at any point; the trench shall be brought to proper grade by means of cement or lime concrete of the specification of the bed concrete without any extra cost.

The bed of the trench, if in soft or made-up earth, shall be well watered and rammed and depressions thus formed filled with sand or other suitable materials as directed by the Architects.

If rock is met with, it will be removed to 15 cm. below the level of the pipe and the trench will be refilled with bed concrete, sand or other suitable material approved by the Architects.

The trench shall be kept free from water. Shoring and timbering shall be provided wherever required.

The width of trench shall be nominal diameter of the pipe plus 38 cm. but it shall not be less than 52 cm.

Laying of Pipes:

No concreting is ordinarily necessary. In cases where the soil is made up is very soft, concreting may be resorted to form the bed of the trench below the pipe, if directed by the Architects at no extra cost.

The pipes shall be carefully laid to levels and gradients shown in the plans and sections. Great care shall be taken to prevent sand etc., from entering the pipes. The pipes between two manholes shall be laid truly in straight lines without vertical or horizontal undulations. The body of the pipe shall for its entire length on an even bed in the trench and places shall be excavated to receive the collar for the purpose of jointing.

Jointing:

A few skeins of spun soaked in neat cement wash shall be inserted in the groove at the end of the pipe and the two adjoining pipes butted against each other. The collar shall then be slipped over the joint, covering equally both the pipes. Spun yarn soaked in neat cement wash shall be passed round the pipes and inserted in the joint by means of caulking tools from ends of the collar. More skeins of yarn shall be added and well rammed above.

The object of the yarn is to center the two ends of the pipes within the collar and to prevent the cement mortar of the joint penetrating into the pipes.

Cement mortar with one part of cement and one part of sand shall be slightly moistened and must on no account be soft or sloppy and shall be carefully inserted by hand in to the joint and more cement mortar added until the space of the joint has been filled completely with tightly caulked mortar. The joint shall be finished off neatly outside the collar on both sides at an angle of 45°.

Any surplus mortar projecting inside the joint is to be removed and to guard against any such projections sack or gunny bag shall be drawn past each joint after completion. Cement mortar joint shall be cured at least for seven days.

Testing:

All joints shall be tested to a head of 60 cm. of water above the top of the highest pipe between two manholes.

The lowest end of the pipe shall be plugged watertight. Water shall then be filled in manhole at the upper end of the line.

The depth of water in the manhole shall be 60 cm. plus the diameter of the pipe. The joint shall then be examined. Any joint found leaking or sweating shall be remade and embedded into 15 cm. layer of cement concrete (1:2:4) 30 cm. in length and the joint retested without any extra cost.

Manholes:

Size of manholes shall be as specified in the item and the sizes specified shall be internal size of the manhole. The work shall be done strictly as per standard drawing and specifications.

Bed Concrete:

Shall be in 1:4:8 cement concrete 23 cm. (9") thick or as specified in the BOQ specifications.

Brick Work:

Shall be with best quality local bricks and proportion of mortar shall be 1:4 unless otherwise specified.

Plaster:

Inside of the walls shall be plastered with 12 mm. thick cement plaster 1:4 or as specified in the BOQ. The external face shall be pointed with 1:3- 1:5 cement mortar.

Footrests:

M.S. square rods of 22 mm. (7/8") diameter or C.I. rungs shall be embedded in masonry where the depth of manhole exceeds one meter and they shall be fixed 35 cm. apart and projecting 11 cm. from the wall. Footrests shall be painted with bitumen as directed.

Manhole Covers:

Covers for manhole in the road proper shall be as per BOQ specifications or shall not be less than 200 kgs. on footpaths and backyards. Lightweight covers shall be used whose weight for 45 cm. dia. shall

not be less than 58 kgs. and that of 90 cm. x 45 cm. or 61 cm. x 45 cm. 90 kgs.

Drop Connection:

The case of drop connection C.I. pipes shall be provided with heel rest bend at the bottom and bend with access door at the top for cleaning purposes. The pipe shall be encased in 1:3:6 plain concrete.

22. Miscellaneous Items of Work:

The rates quoted by the Contractor for all miscellaneous items of work viz. Cooking platforms, moiries, built-in cupboards, counters, partitions, railings, electrical meter, switchboard cupboards, etc., shall be for the work as described in the schedule of quantities and as show in detailed drawings and shall be to the entire satisfaction of the Architects.

23. Doors and Windows

Functional need of window:

1. Windows should be fabricated with "Fusion welded corners". The Mullion /Transom can be either Fusion welded or mechanically joined with desired sealing.
2. Windows / doors must conform to the strength requirements based on wind load as per IS 875-3.
3. Appropriate thickness of steel reinforcement should be selected to meet the desired strength. The reinforcement must be installed within 6 to 50mm distance from the face of the weld.
4. For window size $\leq 1500\text{mm}$ tolerance is $\pm 3.0\text{mm}$ and sizes above $\geq 1500\text{ mm}$ tolerances $\pm 5.0\text{mm}$ on both height and width.
5. The window diagonal should be less than equal to 5mm for window up to 1500mm, above 1500mm, the diagonal difference should not be more than 10mm
6. The minimum overlap of sashes on Frame/ mullion should be 5mm, higher overlap is desirable.
7. Water drainage / ventilation slot should be provided in sash / frames as described in section 9 of relevant IS standard
8. Min Gap of 3mm should be maintained per face between aperture and window to allow expansion / contraction of uPVC windows
9. The gap between window and its aperture should be filled with weatherable & elastic material to allow expansion / contraction of PVC and performance over period of years.
10. Material Description: The Windows shall be manufactured using Rigid uPVC profile sections, using GI reinforcement.
11. Fabrication Description: uPVC Window, size as per drawing of wall thickness $1.2\text{mm} \pm 0.25\text{mm}$ made out of extruded section plain uPVC sections.
12. The window frame made out of rigid uPVC sections meter cut at 4 corners & joint with plastic fusion welding with internal reinforce required.
13. The shutter section made out of extruded rigid uPVC with internal reinforced with provision of 5mm glaze/wire mesh as per requirement.
14. The window shall be fixed to the wall using 65x100mm long M.S. fasteners at suitable place provided for each vertical and horizontal member etc. complete as per manufacture's specification and direction of Engineer-in-charge.
15. All the surface of Window should be uniform smooth finish and free from crack and seam. Passing joints shall be finished ensuring that no gaps are visible.
16. Tolerance: All Dimensions (except thickness of profile sections) shall be as per drawing 1.2mm

thickness should be in $\pm 0.25\text{mm}$.

17. Measurement: The unit of measurement shall be in square meter or square feet.

18. Payment: Price shall be covering the fabrication, handling and installation including labor, material and equipment. Manufacture should provide maintenance manual and basic material test report for certification.

24. General Requirements:

1. All teak wood shall be of medium seasoned teak, free from sap, knots, cracks and shall be of lighter grains and shall be of approved range. If required, teak wood shall be tested in laboratories at Contractors expenditure.

2. All teak wood, plywood's and any other wood used in the work shall be coated with approved quality anti-termite i.e. Touch wood or approved equivalent type. At site with a test certificate at contractors expenditure.

3. All door shutters shall be fitted with approved fittings.

4. All screws shall be machine made and of approved equivalent make.

5. Even if it is not mentioned in the specifications / drawings, it should be clearly understood that all open edges of fitted plywood (i.e. door, cupboard, shutter, drawers etc.) shall be provided with teak wood beadings. All teak wood beadings shall match laminate shade or in approved finish.

6. The rate quoted in all the items shall include the cost of all fixtures, materials, manufacturing, transportation, installation, sales tax and other taxes if any etc. All partitions / paneling shall have provisions for conducts of electricity/ telephone/ computer / audio- visual / Fire alarm / Intercom etc.

7. No work shall be executed at the site without prior approval of samples of all parts.

Note : Regarding the availability of the material and that only materials called for under preference shall be supplied and installed. In the event of the materials of make called for are not available and alternative makes are approved for incorporation in the work, the rates quoted shall be suitable amended bases on the price variation between the specified marks and alternative makes on the day the alternative makes are accepted.

Material :

The material shall conform to IS:2095.

☐ Plywood /Plywood Boards : Plywood boards are formed by gluing and pressing three or more layers of veneers with the grains of adjacent veneers running at right angles to each other. The veneers shall be either rotary cut or sliced and shall be sufficiently smooth to permit an even spread of glue. Face veneers may be either decorative on both sides or one side commercial and the other decorative. Plywood shall be of BWP grade.

☐ Adhesive: Adhesive used for bonding BWP grade of plywood boards shall be BWP type synthetic resins conforming to IS 848.

☐ Thickness: Plywood boards shall be as mentioned in the specified item / detailed drawings. Tolerance in thickness shall be $\pm 10\%$ for boards up to and including 5 mm; $\pm 7\%$ for boards from 6 to 9 mm and $\pm 5\%$ for boards above 9 mm thickness. The boards shall be of uniform thickness, and the surfaces of the boards shall be sanded to a smooth finish. Number of plies in plywood boards shall be as per the drawings.

☐ Moisture content of the plywood boards when tested in accordance with IS 1734 (Part 1) shall not be less than 5 per cent and not more than 15 per cent.

☐ Testing: One sample for every 100 sqm or part thereof shall be taken and testing done as per IS 303. However, testing may not be done if the total requirement of plywood boards is less than 30 sqm. All the samples tested shall meet the requirements of physical and mechanical properties of plywood

boards

□ **Particle Boards:** Particle boards shall be of medium density and manufactured from particles of agro waste, wood or lignocelluloses i.e. material blended with adhesive and formed into solid panels under the influence of heat, moisture, pressure etc. The particle boards shall be flat pressed three layered or graded and of Grade-I as per Table 1 of IS 3087. Both surfaces of the boards shall be sanded to obtain a smooth finish and shall conform to IS 3087.

□ **Thickness and Tolerance:** Thickness of particle boards shall be as specified. Tolerance in thickness shall be $\pm 5\%$ for boards up to and including 25 mm thick and ± 2.5 per cent for boards above 25 mm thickness. Each board shall be of uniform thickness.

Testing: One sample for every 100 sqm or part thereof shall be taken and testing done as per IS 3087. However, testing may not be done if the total requirement of particle boards in a work is less than 30 sqm. All the samples tested shall meet the requirement of physical and mechanical properties of particle boards specified.

Sanding: This application is to make the surface level without undulations. To make the gypsum wall surface level (in particular at joints, where there is excess bonding plaster), do sanding with sand paper at joints and other places, wherever you find uneven surface, otherwise joints will be visible after painting. It is important to sand all joints uniformly.

Primer Application: The purpose of the primer is to give a better adhesion to the paint and also to reduce consumption of paint on the wall. Water thinable (Low VOC) primers shall be used only.

Limits of Manufacturing Defects -- Defective bonds, pleats and overlaps, and gaps in faces are not permitted. Occasional gaps may be repaired using veneer inserts bonded with the proper adhesive.

Moisture Content -- BS 1088 plywood must have moisture content between 6% and 14% when it leaves the factory.

Length & Width -- The length or width of a board produced as a standard size shall not be less than the specified size nor more than 6.3 mm (0.25") greater than the specified size. The lengths of the diagonals of a board shall not differ by more than 0.25% of the length of the diagonal.

MATERIAL TESTING & FREQUENCY CHART

The Contractors will have to take necessary material test as per I.S. code, which is applicable, at their own cost for the following materials or any other material using in construction work periodically or as and when required by the Architects / Consulting Engineer.

The materials should be got tested in an approved Laboratory as per IS standard and test reports in duplicate should be submitted to the Architect's Office.

A chart showing the recommended time and quantity scheduled for conducting test on various building materials is given. Please ensure that tests are carried out according to the above guidelines. Contractor's rate should include for necessary expenditure for testing including transport of samples of following tests.

Sr. No.	Material	Test.	Frequency of Testing	Remarks.
1.	Sand	i) Fineness Modulus. ii) Silt Content.	At the beginning & if there is change in source.	
2.	Masonry Stone.	i) Comp. strength. ii) Specific Gravity. iii) Water Absorption.	A set of 5 stones for each quarry & for doubtful quality.	
3.	Metal.	i) Crushing Value. ii) Impact Value. iii) Abrasion value. iv) Water Absorption. v) Flakiness Index. vi) Stripping value. vii) Gradation.	One test per 200 cum or part thereof.	I.S. 2386 Part-II.
4.	Bricks.	i) Crushing Strength. ii) Water Absorption iii) Efflorescence	A set of 15 Bricks for each consignments of 50,000 or part thereof.	I.S. 1077
5.	Flooring Tiles.	i) Flexural strength. ii) Water Absorption.	A set of 12 Tiles for each 2,000 Tiles or part thereof.	I.S. 1237-1989.
6.	Glazed Tiles	Water Absorption.	A set of 16 Tiles for each 2000 Tiles or part thereof.	IS 13630 (Parts 1 to 15) : 2006
7.	Design Mix	For Every Grade of Concrete	Initially and on every source change of materials	IS-456 & MORTH 1700 Structural Concrete
8.	Cement Concrete.	i) Slump ii) Cube/Compressive Strength	For Each Batch Up to 5 cum. 1 set. 6-15 cum- 2 sets. 16-30 cum- 3 sets. 31-50 cum- 4 sets. 51 & above-4 sets + 1 additional set for each additional 50 cum or part thereof.	IS 456
9.	Cement.	i) Comp. strength. ii) Initial setting time. iii) Final setting time. iv) Specific Gravity. v) Soundness. vi) Fineness.	One test for each consignment of 50 M.T. (1000 bags) or part thereof.	I.S. 269-12269.
10.	Steel.	i) Wt. per meter. ii) Ultimate Tensile stress. iii) Yield stress. iv) Elongation.	As Specified in Part D- Technical specifications.	I.S. 432.

		v) Chemical Test		
11.	AAC Block	i) Density Test ii) Compressive Strength. iii) Thermal Conductivity iv) Drying Shrinkage	Every lot of 10000 Nos	IS 6441 (Part-I) & IS : 2185 (Part 3 j - 1984
12.	Paving Block	i)Compressive Strength ii)Water Absorption	Every lot Every lot	IS 15658 : 2021
13.	Aluminum Section	i) Thickness, Mass Per Running meter ii) Test on Powder Coating/ anodizing.	Test to be carried as per IS codes & Instructions of Engineer In-charge.	IS: 1081-1960 IS 1949 (1961) IS 1948:1961

Note : The Contractor will have to take necessary material test other than above test as per I.S. code for above material or other than above material, if required and as directed by the APMCF / SBI.

The contractor shall install testing equipment as per requirement on site. The contractor shall ensure and certify the calibration of the equipment so installed and shall maintain the same in working order throughout the period of construction.

The contractor should also provide the necessary trained staff for carrying out such tests for using such equipment.

The tests shall be carried out under the supervision of the Engineer-in-charge. 70% of the total tests to be done are to be carried out on site laboratory if the facilities are available as per tender terms and conditions, remaining 15% tests is to be carried out at govt./ Semi Govt. laboratory and 15% tests are to be carried out at Govt. recognized/NABL accredited laboratory.

THEORETICAL CEMENT CONSUMPTION STATEMENT (BASE CPWD)

No	Description of item of work.	Quantity of cement to be used per Unit Quantity of work.	Unit.
1	Cement Concrete (Cast in Situ) Plain or Reinforced.		
a.	1:1:2 (1 Cement: 1 Sand :2 Graded Aggregate).	12.20 Bags.	Cubic Meter
b.	1:1.5:3 (1 Cement: 1.5 sand: 3 Graded Aggregate).	8.00 Bags.	Cubic Meter
c.	1:2:4 (1 Cement: 2 Sand :4 Graded Aggregate).	6.40 Bags.	Cubic Meter
d.	1:3:6 (1 Cement: 3 Sand :6 Graded Aggregate).	4.40 Bags.	Cubic Meter
e.	1:4:8 (1 Cement: 4 Sand :8 Graded Aggregate).	3.40 Bags.	Cubic Meter
f.	1 Cement: 5 Sand :10 Graded Aggregate).	2.60 Bags.	Cubic Meter
g.	Providing and laying cement concrete 1:2:4 (1 Cement: 2 Coarse Sand: 4 Graded Aggregate of 20 mm. nominal size) including finishing exposed surface with 6 mm. thick cement mortar 1:3 (1 Cement: 3 Fine Sand). Krebs, Steps, and the like.	7.02 Bags.	Cubic Meter
h.	String or lacing courses, parapets, coping, bed blocks, anchor blocks, plain window sills and the like moldings in cornices, window sills etc.	7.62 Bags.	Cubic Meter
1.1	Ready Mix/Design mix Concrete		
	M20 & Above Grades	As per Approved Design Mix /RMC Batch Report	As per Approved Design Mix /RMC Batch Report
2.	Cement Mortar		
a.	1:1 (1Cement: 1 Sand)	20.40 Bags.	Cubic Meter
b.	1:2 (1Cement: 2 Sand)	13.60 Bags.	Cubic Meter
c.	1:3 (1Cement: 3 Sand)	10.20 Bags.	Cubic Meter
d.	1:4 (1Cement: 4 Sand)	7.60 Bags.	Cubic Meter
e.	1:5 (1Cement: 5 Sand)	6.20 Bags.	Cubic Meter
f.	1:6 (1Cement: 6 Sand)	5.00 Bags.	Cubic Meter
g.	1:2 (1Cement: 2 Stone Dust)	13.60 Bags.	Cubic Meter
h.	1:2 (1Cement: 2 Marble Dust)	13.60 Bags.	Cubic Meter
i.	1:5 (1Cement: 5 Marble Dust)	6.20 Bags.	Cubic Meter
j.	1:1:3 (1Cement: 1 Marble Dust: 3 Stone Dust)	7.60 Bags.	Cubic Meter
k.	White Cement Mortar 1:2 (1 White Cement: 2 Marble Dust)	13.60 Bags.	Cubic Meter
l.	White Cement Mortar 1:3 (1 White Cement: 3 Marble Dust)	10.20 Bags.	Cubic Meter
m.	White Cement Mortar 1:5 (1 White Cement: 5 Marble Dust)	6.20 Bags.	Cubic Meter
3.	Cement Lime Mortar		
a.	1:1:3 (1 Cement:1 Lime putty:3 Sand)	8.20 Bags.	Cubic Meter
b.	1:1:6 (1 Cement:1 Lime putty:6 Sand)	5.00 Bags.	Cubic Meter
4.	Brick Work in All Classes		
a.	In Cement Mortar 1:3 (1 Cement:3 Sand)	2.56 Bags.	Cubic Meter

b.	In Cement Mortar 1:4 (1 Cement:4 Sand)	1.90 Bags.	Cubic Meter
c.	In Cement Mortar 1:5 (1 Cement:5 Sand)	1.56 Bags.	Cubic Meter
d.	In Cement Mortar 1:6 (1 Cement:6 Sand)	1.24 Bags.	Cubic Meter
5.	Half Brick Work in All Classes		
a.	In Cement Mortar 1:3 (1 Cement:3 Sand) With or without hoop iron.	28.56 Bags per 100 Square Meter	
b.	In Cement Mortar 1:4 (1 Cement:4 Sand)	21.28 Bags per 100 Square Meter	
c.	In Cement Mortar 1:5 (1 Cement:5 Sand)	14.50 Bags per 100 Square Meter	
d.	Molding and cornices in brick masonry in cement mortar 1:4 Cement:4 Sand) Joining old brick work with new brick work.	0.18 Bags per 100 Square Meter per cm. Girth	
	a) Old Brick in metric or FPS. System with new brick work in metric system in cement mortar 1:4 (1 Cement: 4 Sand).	4.20 Bags per 100 Square Meter	
	b) Old Brick work in FPS. System with new brick work in cement mortar 1:4 (1 Cement: 4 Sand).	5.44 Bags per 100 Square Meter	
6.	Random Rubble Masonry		
a.	Cement Mortar 1:6 (1 Cement: 6 Sand)	1.70 Bags.	Cubic Meter
b.	Cement Lime Mortar 1:1:8 (1 Cement: 1 Lime Putty: 8 Sand)	1.32 Bags.	Cubic Meter
7.	Coursed Rubble Masonry		
a.	Cement Mortar 1:6 (1 Cement: 6 Sand)	1.50 Bags.	Cubic Meter
8.	Ashlar Masonry In plain ashlar punched (ordinary) in superstructure in cement mortar 1:6 (1 Cement: 6 Sand) including pointing with cement mortar 1:2 (1 Cement:6 Stone dust) with an admixture of pigment matching the stone shade.	1.08 Bags.	Cubic Meter
9.	Stone Veneering Work For wall lining etc., average thickness 40 mm. to 170 mm. in cement lime mortar 1:1:6 (1 Cement:1 Lime Putty:6 Sand) including pointing in White cement mortar 1:2 (1 White Cement: 2 Stone Dust) with an admixture of pigment matching the stone shade.	17.50 Bags per 100 Square Meter	
10.	Marble work in steps jambs, walls, pillars and other plain work in cement mortar 1:4 (1 Cement: 4 Sand) including pointing in White cement mortar 1:2 (1 Cement: 2 Marble dust).	0.136 Bags per 1.52 Bags per	Cubic Meter (Grey Cement) Cubic Meter (White Cement)
11.	Marble work in steps jambs, walls, pillars and other plain work in cement mortar 1:4 (1 Cement: 4 Sand) including pointing in cement mortar (1 Cement : 2 Marble dust).	1.66 Bags per	Cubic Meter

12.	Marble work for wall lining (Veneer) work) 2.5 cm. thick in cement mortar 1:3 (1 Cement : 3 Sand) including pointing in White cement mortar 1:2 (1 Cement : 2 Marble dust).	1.1. Bags per 100 Square Meter (Grey Cement) a. Bags per 100 Square Meter (White Cement)	
13.	Marble work for wall lining (Veneer) work) 2.5 cm. thick in cement mortar 1:3 (1 Cement : 3 Sand) including pointing in cement mortar 1:2 (1 Cement : 2 Marble dust).	17.68 Bags per	Square Meter
14.	Marble work for wall lining (Veneer) work) 4 cm. thick in cement mortar 1:3 (1 Cement : 3 Sand) including pointing in White cement mortar 1:2 (1 Cement : 2 Marble dust).	20.40 Bags per 100 Square Meter. (Grey Cement) 1.1. Bags per 100 Square Meter (White Cement)	
15.	Marble work for wall lining (Veneer) work) 4 cm. thick in cement mortar 1:3 (1 Cement : 3 Sand) including pointing in cement mortar 1:2 (1 Cement : 2 Marble dust).	23.80 Bags per 100 Square Meter.	
16.	Cement Concrete Flooring Flooring 1:2:4 (1 Cement: 2 Sand : 4 Graded Stone Aggregate) finished with a floating coat of neat cement including cement slurry rounding of edges and strips etc., but excluding cost of nosing of steps etc., complete.		
a.	25 mm. thick with 20 mm. nominal size stone aggregate.	0.244 Bags	Square Meter
b.	40 mm. thick with 20 mm. nominal size stone aggregate.	0.34 Bags	Square Meter
c.	50 mm. thick with 20 mm. nominal size stone aggregate.	0.404 Bags	Square Meter
d.	75 mm. thick with 20 mm. nominal size stone aggregate.	0.564 Bags	Square Meter
17.	Cement Plaster Skirting (up to 30 cm. height) with cement mortar 1:3 (1 Cement: 3 Coarse Sand) finished with a floating coat of neat cement including rounding of junctions with floor, including slurry complete.		
a.	18 mm. thick.	0.32 Bags	Square Meter
b.	21 mm. thick.	0.35 Bags	Square Meter
18.	Pavement (25 to 50 mm. thick) with 1:2:4 (1 Cement: 2 Coarse Sand : 4 Graded Stone Aggregate 20 mm. nominal size) including finishing complete.	6.80 Bags	Cubic Meter
19.	Terrazzo Flooring 40 mm. thick marble chips flooring rubbed and polished to granolithic finish, under layer 34 mm. thick cement concrete 1:2:4 (1 Cement: 2 Coarse Sand: 4 Graded Stone Aggregate 12.5 mm. nominal size) and top layer 6 mm. thick with white, black or white and black marble chips of size 1 mm. to 4 mm. nominal size laid in cement marble powder 3:1 mix. (3 Cement: 1 Marble Powder) by weight in proportion of 4:7 (4 Cement marble powder) by weight in marble powder		

	mix:7 Marble chips) by volume including cement slurry etc., complete.		
a.	Dark shade / Light shade pigment with ordinary cement.	0.339 Bags per	Square Meter
b.	Light shade pigment with white cement.	0.258 Bags per 0.081 Bags per	Square Meter (Grey Cement) (White Cement)
c.	Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	0.298 Bags Per 0.0440 Bags per	Square Meter (Grey Cement) (White Cement)
20	40 mm. thick marble chips flooring rubbed and polished to granolithic finish, under layer 31 mm. thick cement concrete 1:2:4 (1 Cement: 2 Coarse Sand: 4 Graded Stone Aggregate 12.5 mm. nominal size) and top layer 9 mm. thick marble chips, chips, size 4 to 7 mm. size, laid in cement marble powder mix. 3:1) (3 Cement : 1 Marble Powder) by volume in proportion of 4:7 (4 Cement marble powder mix. 7 Marble chips) by volume including cement slurry etc., complete.		
a.	Dark shade / Light shade pigment with ordinary cement.	0.357 Bags	Square Meter
b.	Light shade pigment with white cement.	0.241 Bags 0.116 Bags	Square Meter (Grey Cement) Square Meter (White Cement)
c.	Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	0.299 Bags 0.058 Bags	Square Meter (Grey Cement) Square Meter (White Cement)
21	40 mm. thick marble chips flooring rubbed and polished to granolithic finish, under layer 28 mm. thick cement concrete 1:2:4 (1 Cement: 2 Coarse Sand: 4 Graded Stone Aggregate 12.5 mm. nominal size) and top layer 9 mm. thick marble chips, chips, sizes 7 mm to 10 mm. nominal size, laid in cement marble powder mix. 3:1) by weight in proportion of 2:3 (2 Cement Marble Powder mix. 3 Marble Chips) by volume including cement slurry etc., complete.		
a.	Dark or Light shade pigments with grey cement.	0.381 Bags	Square Meter
b.	Light shade pigment or without any pigment with white cement.	0.219 Bags 0.162 Bags	Square Meter (Grey Cement) Square Meter (White Cement)
c.	Medium shade pigment with approximately 50% grey cement and 50% white cement.	0.300 Bags 0.081 Bags	S.M. (Grey Cement) S.M. (White Cement)
22	Marble chips skirting (up to 300 mm high) rubbed and polished to granolithic finish top layer 6 mm. thick marble chips of sizes from smallest to 4 mm. nominal size laid to cement marble powder mix. 3:1 (3 Cement: 1		

	Marble Powder mix. By weight in proportion of 4:7 (4 Cement Marble Powder mix: 7 marble chips) by volume including cement slurry complete.		
a.	18 mm. thick with under layer 12 mm. thick cement plaster 1:3 (1 Cement: 3 Coarse Sand) dark or light shade pigment with grey cement.	0.298 Bags	Square Meter
b.	Light shade pigment or no pigment with cement.	0.217 Bags Per Square Meter (Grey Cement) 0.081 Bags Per Square Meter (White Cement)	
c.	Medium shade colour pigment with 50% grey cement and 50% white cement.	0.258 Bags Per Square Meter (Grey Cement) 0.0406 Bags Per Square Meter (White Cement)	
d.	21 mm. thick with under layer 15 mm. thick cement plaster 1:3 (1 Cement: 3 Course Sand) dark or light shade pigment with grey cement.	0.327 Bags	Square Meter
e.	Light shade pigment or no pigment with white cement.	0.246 Bags Per Square Meter (Grey Cement) 0.081 Bags Per Square Meter (White Cement)	
f.	Medium shade pigment with 50% grey cement and 50% white cement.	0.286 Bags Per Square Meter (Grey Cement) 0.04 Bags Per Square Meter (White Cement)	
23.	Tile Flooring:		
a.	Precast terrazzo tiles 20 mm. thick white, black or white and black marble chips of size up to 6 mm. laid in floors treads of steps and landings jointed with neat cement slurry mixed with pigment to match the shade of the tile including rubbing polishing with precast tiles of 30 mm. thick bed of lime mortar 1:1.2 or 1:3 light shade using white cement.	0.088 Bags Per Square Meter (Grey Cement) 0.088 Bags Per Square Meter (White Cement)	
b.	Medium shade colour pigment with 50% white cement and 50% grey cement.	0.132 Bags Per Square Meter (Grey Cement) 0.044 Bags Per Square Meter (White Cement)	
c.	Dark shades using ordinary cement precast terrazzo tiles 20 mm. thick with marble chips of size 6 mm. in skirting and risers of steps not exceeding 30 cm. in height on wall, laid on 12 mm. thick cement plaster 1:3 mix. (1 Cement: 3 Sand) joint with neat cement slurry, light shades using white cement.	0.235 Bags Per Square Meter (Grey Cement) 0.044 Bags Per Square Meter (White Cement)	
d.	Medium shade colour pigment with 50% white cement and 50% ordinary cement.	0.257 Bags Per Square Meter (Grey Cement) 0.022 Bags Per Square Meter (White Cement)	
e.	Dark shades using ordinary cement.	0.279 Bags	Square Meter
24.	Checkered Terrazzo Tile Flooring		
a.	Checkered Terrazzo Tile 22 mm. thick with marble chips of sizes up to 6 mm. in floors, jointed with neat cement slurry mixed with pigment to match the shade of the tiles including robbing, polishing complete on 28 mm. thick bed of lime mortar 1:1.2 or 1:3.		
a.	Light shade using white cement.	0.088 Bags Per Square Meter (Grey Cement) 0.096 Bags Per Square Meter (White Cement)	
b.	Medium shades using 50% grey cement and 50% white cement.	0.136 Bags Per Square Meter (Grey Cement) 0.048 Bags Per Square Meter (White Cement)	
c.	Dark shade using grey cement.	0.184 Bags Per Square Meter (Grey Cement)	

d.	Checkerered Terrazzo Tile 30 mm. thick with marble chips of sizes up to 6 mm. in stairs, treads, jointed with neat cement slurry mixed with pigment to match the shade of the tiles including rubbing polishing rounding of nosing etc., complete on 20 mm. bed of : Lime mortar 1:1:1 (1 Lime putty:1 Surkhi:1 Coarse Sand) :		
i.	Light shade using white cement.	0.088 Bags Per Square Meter (Grey Cement) 0.136 Bags Per Square Meter (White Cement)	
ii.	Medium shades using 50% grey cement and 50% white cement.	0.154 Bags Per Square Meter (Grey Cement) 0.066 Bags Per Square Meter (White Cement)	
iii.	Dark shade using grey cement.	0.220 Bags Per Square Meter (Grey Cement)	
e.	Cement mortar 1 : 4 (1 Cement : 4 Coarse Sand)		
i.	Light shade using white cement.	0.258 Bags Per Square Meter (Grey Cement) 0.132 Bags Per Square Meter (White Cement)	
ii.	Medium shades using 50% grey cement and 50% white cement.	0.324 Bags Per Square Meter (Grey Cement) 0.066 Bags Per Square Meter (White Cement)	
iii.	Dark shade using grey cement.	0.39 Bags Square Meter (Grey Cement)	
25.	White Glazed Tiles.		
	White Glazed Tiles 5,6 or 7 mm. thick in flooring treads risers of steps skirting and dado on 12 mm. thick cement plaster 1:3 (1 Cement: 3 sand) in base and cement jointed with white cement slurry etc. complete.	0.188 Bags Per Square Meter (Grey Cement) 0.050 Bags Per Square Meter (White Cement)	
26.	Marble Stone Flooring		
	Marble Stone slab flooring over 20 mm. thick base of lime mortar 1:1:1 (1 Lime putty:1 Surkhi:1 Sand) and jointed with grey cement slurry etc. (all marble slabs).		
a.	20 mm. thick	0.098 Bags Per Square Meter	
b.	30 mm. thick	0.102 Bags Per Square Meter	
c.	40 mm. thick	0.107 Bags Per Square Meter	
	Marble stone slab flooring over 20 mm. thick base of cement mortar 1:4 (1 Cement:4 Sand) and jointed with grey cement slurry etc., (all marble slabs).		
d.	20 mm. thick	0.268 Bags Per Square Meter	
e.	30 mm. thick	0.273 Bags Per Square Meter	
f.	40 mm. thick	0.277 Bags Per Square Meter	
g.	Extra if white cement slurry is used instead of grey cement slurry in joints of marble stone flooring.	0.015 Bags Per Square Meter (White Cement)	
h.	Marble slabs 30 mm. thick in risers of steps, skirting dado, wall and pillars, laid on 12 mm. thick cement mortar 1:3 (1 Cement : 3 Sand) and jointed with grey cement slurry.	0.246 Bags Per Square Meter (White Cement)	
27.	Kotah Stone Flooring		
	Kotah stone slab flooring over 20 mm. thick base of lime mortar 1:1:1 (1 Lime putty:1 Surkhi:1 Sand) and jointed with neat cement slurry etc.		
a.	25 mm. thick	0.128 Bags Per Square Meter	
b.	30 mm. thick	0.136 Bags Per Square Meter	
c.	40 mm. thick	0.152 Bags Per Square Meter	

	Kotah Stone slab flooring over 20 mm. thick base of cement mortar 1:4 (1 Cement:4 Sand) and jointed with neat cement slurry etc.		
d.	25 mm. thick	0.298 Bags Per Square Meter	
e.	30 mm. thick	0.306 Bags Per Square Meter	
f.	40 mm. thick	0.322 Bags Per Square Meter	
g.	Kotah stone slab 25 mm. thick risers of steps, skirting, dado and pillar laid on 12 mm. thick cement mortar 1:3 (1 Cement:3 Sand) and jointed with neat cement slurry etc.	0.275 Bags Per Square Meter	
28	Sandstone Flooring		
a.	40 mm. thick sandstone flooring over 20 mm. thick base of cement mortar 1:5 (1 Cement :5 Sand) with joints finish flush.	0.155 Bags Per Square Meter	
b.	40 mm. thick sand stone flooring over 20 mm. thick base of cement mortar 1:5 (1 Cement :5 Sand) including pointing with cement mortar 1:2 (1 Cement: 2 Stone Dust).	0.186 Bags Per Square Meter	
c.	40 mm. thick sandstone flooring over 20 mm. thick base of lime mortar 1:1:1 (1 Lime :1 Surkhi:1 Sand) including pointing with cement plaster 1:2 (1 Cement :2 Stone Dust).	0.031 Bags Per Square Meter	
d.	40 mm. thick fine dressed and rubbed stone flooring over 20 mm. thick base of cement mortar 1:5 (1 Cement :5 Sand) with joints 5 mm. thick finished flush.	0.166 Bags Per Square Meter	
e.	40 mm. thick fine dressed and rubbed stone flooring over 20 mm. thick base of lime mortar 1:5 (1 Cement : 5 Sand) with joints 5 mm. thick including pointing with cement mortar 1:2 (1 Cement : 2 Stone Dust).	0.196 Bags Per Square Meter	
f.	25 mm. thick cast iron grid flooring using grid tiles of required size weighing 47 kg. per square meter on bed of 12 mm. thick cement concrete 1:2 (1 Cement : 2 Stone Aggregate 6 mm. nominal size) including filling the hollows with cement concrete same mix and tamping with 10 mm. dia. iron bars and grouting the joints with neat cement slurry complete.	0.025 Bags Per Square Meter	
g.	Filling cement concrete 1:2:4 (1 Cement :2 Coarse Sand : 4 Graded Stone Aggregate 12.5 mm. nominal size) in gaps of A.C.Sheet corrugations and wings of ridges.	3.82 Bags Per Square Meter	
29.	Cement Plaster		
a.	12 mm. 1:3 (1 Cement: 3 Sand).	14.68 Per 100 Square Meter	
b.	12 mm. 1:4 (1 Cement: 4 Sand).	10.94 Per 100 Square Meter	
c.	12 mm. 1:5 (1 Cement: 5 Sand).	8.92 Per 100 Square Meter	
d.	12 mm. 1:6 (1 Cement: 5 Sand).	7.20 Per 100 Square Meter	
e.	15 mm. 1:3 (1 Cement: 3 Sand).	17.54 Per 100 Square Meter	
f.	15 mm. 1:4 (1 Cement: 4 Sand).	12.08 Per 100 Square Meter	
g.	15 mm. 1:5 (1 Cement: 5 Sand).	10.66 Per 100 Square Meter	

h.	12 mm. 1:6 (1 Cement: 6 Sand).	8.60	Per 100 Square Meter
i.	20 mm. 1:3 (1 Cement: 3 Sand).	22.84	Per 100 Square Meter
j.	20mm. 1:4 (1 Cement: 4 Sand).	17.02	Per 100 Square Meter
k.	20 mm. 1:5 (1 Cement: 5 Sand).	13.88	Per 100 Square Meter
l.	20 mm. 1:6 (1 Cement: 6 Sand).	11.20	Per 100 Square Meter
30.	Cement Plaster with a Floating Coat of neat cement		
a.	12 mm. 1:3 (1 Cement: 3 Sand).	19.08	Per 100 Square Meter
b.	12 mm. 1:4 (1 Cement: 4 Sand).	15.34	100 Per Square Meter
c.	12 mm. 1:3 (1 Cement: 3 Sand).	21.94	Per 100 Square Meter
d.	12 mm. 1:4 (1 Cement: 4 Sand).	17.48	Per 100 Square Meter
e.	15 mm. 1:3 (1 Cement: 3 Sand).	27.24	Per 100 Square Meter
f.	15 mm. 1:4 (1 Cement: 4 Sand).	21.42	Per 100 Square Meter
31.	Cement Plaster in two coats		
a.	20 mm. Cement Plaster in two coats under layer 12 mm. cement plaster 1:4 (1 Cement :4 Sand) finished with a top layer 8 mm. thick cement plaster 1:3 (1 Cement: 3 Sand)	20.00	Bags per 100 Square Meter
b.	18 mm. thick Cement Plaster in two coats under layer 12 mm. thick cement plaster 1:5 (1 Cement :5 Sand) finished with a top layer 6 mm. thick cement plaster 1:3 (1 Cement: 3 Sand)	16.26	Bags per 100 Square Meter
32.	6 mm. Cement Plaster		
a.	6 mm. Cement Plaster to ceiling 1:3 (1 Cement :3 Sand)	7.34	Bags per 100 Square Meter
b.	6 mm. Cement Plaster to ceiling 1:4 (1 Cement :4 Sand)	5.48	Bags per 100 Square Meter
c.	6 mm. Cement Plaster to ceiling 1:3 (1 Cement :3 Sand) finished with a floating coat of neat cement.	11.74	Bags per 100 Square Meter
d.	Neat Cement Punning.	4.40	Bags per 100 Square Meter
33.	Sand Cement Neeru Finished Plaster		
a.	Sand cement smooth neeru finished plaster for ceiling in cement mortar mix 1:4 (1 Cement :4 Sand), 10 to 15 mm. thick average, finished top smooth with neeru.	13.00	Bags per 100 Square Meter
b.	Sand cement smooth neeru finished plaster for walls in cement mortar mix 1:4 (1 Cement :4 Sand), 18 to 20 mm. thick average, finished top smooth with neeru.	19.00	Bags per 100 Square Meter
34.	Rough Cast Plaster		
	Rough Cast Plaster with a mixture of sand and gravel or crushed stone from 2.36 mm. to 12.5 mm. nominal size dashed over and including the fresh plaster in two layers, top layer 10 mm. cement plaster 1:3 (1 Cement: 3 Sand) mixed with 10% finely grounded hydrated lime by volume of cement and under layer 12 mm. cement plaster : 1:4 (1 Cement: 4 Sand)		
a.	With ordinary cement finish or cement pigment finish.	23.18	Bags per 100 Square Meter
b.	With white cement and pigment finish.	10.94	Bags Per 100 Sqm. (Grey Cement)
		12.24	Bags Per 100 Sqm. (White Cement)
	1:5 Cement Sand (1 Cement:5 Sand)		

c.	With ordinary cement finish or cement and pigment finish.	21.16 Bags Per 100 Sqm. (Grey Cement)
d.	With white cement and pigment finish.	8.92 Bags Per 100 Sqm.(Grey Cement) 12.24 Bags Per 100 Sqm. (White Cement)
35.	Pointing on Stonework	
a.	Flush or ruled pointing on stone work with cement mortar 1:3 (1 Cement: 3 Sand)	2.34 Bags per 100 Square Meter
b.	Raised and cut pointing in stone work with cement mortar 1:3 (1 Cement: 3 Sand)	3.88 Bags per 100 Square Meter
36.	Waterproofing	
a.	Proprietary waterproofing treatment to the terrace with brickbat coba, cement base.	55.00 Bags per 100 Square Meter
b.	Proprietary waterproofing treatment to the canopy with brickbat coba, cement base.	45.00 Bags per 100 Square Meter
c.	Waterproofing chajja with sand cement plaster average 25 mm. thick in cement mortar 1:3 (1 Cement :3 Sand)	25.00 Bags per 100 Square Meter
d.	Proprietary waterproofing treatment to the sunk portion of toilet, cement base.	30.00 Bags per 100 Square Meter

Plumbing & Sanitary

No	Description of item of work.	Quantity of cement to be used per Unit Quantity of work.	Unit.
1.	Cast Iron Pipes		
	Providing and fixing on wall face C.I. rain water pipes including filling the joints with spun yarn soaked in neat cement slurry and cement mortar 1:2 (1 Cement: 2 Sand)		
a.	75 mm. dia pipe	0.132 Bags per 100 Running Meter	
b.	105 mm. dia pipe	0.176 Bags per 100 Running Meter	
c.	150 mm. dia pipe	0.264 Bags per 100 Running Meter	
2.	Cast Iron Accessories		
	Providing and fixing on wall face C.I. Accessories for rain water pipes including filling the joints with spun yarn soaked in neat cement slurry and cement mortar 1:2 (1 Cement : 2 Fine Sand)		
a.	75 mm. dia pipe C.I. Plain bend.	0.0052	Each
b.	100 mm. dia pipe C.I. Plain bend.	0.0062	Each
c.	150 mm. dia pipe C.I. Plain bend.	0.010	Each
d.	75 mm. dia C.I. head flat or corner type.	0.003	Each
e.	100 mm. dia C.I. head flat or corner type.	0.003	Each
f.	150 mm. dia C.I. head flat or corner type.	0.0052	Each
g.	75 mm. dia C.I. plain shoe.	0.003	Each
h.	100 mm. dia C.I. plain shoe.	0.003	Each
i.	150 mm. dia C.I. plain shoe.	0.0052	Each
j.	75 mm.dia C.I. single branch (plain)	0.0052	Each
k.	100 mm. dia C.I. single branch (plain)	0.0062	Each
l.	150 mm. dia C.I. single branch (plain)	0.0010	Each
m.	75 mm.dia C.I. double branch (plain)	0.008	Each
n.	100 mm. dia C.I. double branch (plain)	0.009	Each

o.	150 mm. dia C.I. double branch (plain)	0.0052		Each
p.	C.I. off-sets (plain) 75 mm. dia. 55 mm. projection.	0.0052		Each
q.	C.I. off-sets (plain) 75 mm. dia. 150 mm. projection.	0.0052		Each
r.	C.I. off-sets (plain) 100 mm. dia. 55 mm. projection.	0.0052		Each
s.	C.I. off-sets (plain) 100 mm. dia. 55 mm. projection.	0.0062		Each
t.	C.I. off-sets (plain) 100 mm. dia. 75 mm. projection.	0.0062		Each
3.	A.C. Fittings & Pipes			
	Providing and fixing on wall face asbestos cement rainwater pipes including jointing with spun yarn soaked in bitumen and cement mortar 1:2 (1 Cement 2 Coarse Sand) complete.			
a.	50 mm. dia.	0.150		Per 100 Running Meter
b.	80 mm. dia.	0.250		Per 100 Running Meter
c.	100 mm. dia.	0.300		Per 100 Running Meter
d.	150 mm. dia.	0.320		Per 100 Running Meter
e.	Providing and fixing A.C. Pipe (or any diameter) wall plugs and standard holder bat clamps comprising of two semi-circular halves of flat and cast-iron base screwed on wooden plugs.	0.0004		Per 100 Running Meter
f.	Providing and fixing on wall face asbestos cement rainwater pipes including jointing with spun yarn soaked in bitumen and cement mortar 1:2 (1 Cement 2 Coarse Sand) complete.			
		50 mm. (2")	80 mm. (3")	100 mm. (4")
g.	Bend of required degree with door or without door.	0.0072	0.012	0.015
h.	Off-set 52.2 mm. projection.	0.0058	0.0090	0.0116
i.	Off-set 76.2 mm. projection.	0.0058	0.0090	0.011
j.	Off-set 114.3 mm. projection.	0.0058	0.0090	0.0116
k.	Off-set 152.4 mm. projection.	0.0058	0.0090	0.0116
l.	Off-set 228.6 mm. projection.	0.0058	0.0090	0.0116
m.	Off-set 304.8 mm. projection.	--	0.0090	0.0116
n.	Off-set 457.2 mm. projection.	--	0.0090	0.0116
o.	Off-set 609.6 mm. projection.	--	--	0.0116
p.	Junction equal single of required degree with or without door.	0.0072	0.0116	0.0146
q.	Junction equal double with or without door or required degree.	0.0108	0.0174	0.0220
r.	Standard shoe.	0.00400	0.0058	0.0058
4.	Sanitary Fittings			
a.	Fixing long pan pattern or Orissa pattern squatting pan or pedestal type water closet 12.5 liters or 15 liters flushing cistern and brackets, telescopic flush pipe or bend with fittings and clamps, overflow pipe with specials and mosquito proof coupling complete including cutting and	0.10		Each

	making good the walls and floors.		
	Fixing flat back or wall corner type, lipped front, urinal basin of 430 x 260 x 350 mm. and 340 x 430 x 265 mm. size respectively, white glazed earthenware with automatic C.I. flushing cistern with fittings, brackets, standard size flush pipe and spreaders with brass union and G.I. clamps complete including painting of cistern and fittings, cutting and making good the walls and floors.	0.050	Each
b.	One urinal basin with 5 liters C.I. automatic flushing cistern.	0.050	Each
c.	Range of two urinal basins with 10 liters C.I. automatic flushing cistern.	0.08	Each
d.	Range of three urinal basins with 10 liters C.I. automatic flushing cistern.	0.134	Each
e.	Range of four urinal basins with 15 liters C.I. automatic flushing cistern.	0.190	Each
	Fixing white glazed fire clay stall urinal with automatic C.I. flushing cistern with fittings R.S. or C.I. brackets standard size C.P. brass flush pipe and spreaders with unions and clamps, C.I. trap with outlet grating and other coupling in C.P. brass including painting of cistern and fittings, cutting and making good the walls and floors.		
f.	Single stall urinal with 5 liters C.I. automatic flushing cistern.	0.102	Each
g.	Range of two urinal basins with 10 liters C.I. automatic flushing cistern.	0.204	Each
h.	Range of three urinal basins with 10 liters C.I. automatic flushing cistern.	0.306 Bags	Each
i.	Range of four urinal basins with 15 liters C.I. automatic flushing cistern.	0.406 Bags	Each
	Fixing one-piece construction white squatting plate urinal with an integral longitudinal flushing pipe 100 mm. dia. half round channel automatic C.I. flushing cistern with fittings R.S. or C.I. brackets, standard size. G.I. flush pipe for back and front flush with standard spreader pipes with fittings G.I. clamps, white vitreous tiling 1200 mm. high to the front and side walls with white vitreous china corners and angles set in neat cement, standard urinals C.I. trap 65 mm. diameter with vent arm and outlet grating and coupling in C.P. brass complete, including painting the cistern and fittings and making good the walls and floors.		
j.	Single squatting plate with 5 liters C.I. automatic flushing cistern.	0.102 Bags	Each
k.	Range of two squatting plates with 10 liters C.I. automatic flushing cistern.	0.204 Bags	Each
l.	Range of three squatting plates with 10 liters C.I. automatic flushing cistern.	0.306 Bags	Each
m.	Range of four squatting plates with 15 liters C.I. automatic flushing cistern.	0.406 Bags	Each
n.	Fixing lavatory basin with brackets, pillar taps, rubber plug, waste of standard	0.050 Bags	Each

	pattern, trap and unions complete including cutting and making good the walls.		
o.	Fixing white pedestal for wash basin completely recessed at the back for reception of pipes and fittings.	0.032 Bags	Each
p.	Fixing sink with brackets, 40 mm. rubber plus, brass chain, waste, trap with necessary unions complete including cutting and making good the walls.	0.050 Bags	Each
q.	Fixing teal-wood draining board with skirting and beading, wax polished with brackets painted white complete including making good the walls.	0.028 Bags	Each
5.	Sanitary Fittings (Items separately ordered)		
a.	Fixing long pan pattern or Orissa pattern squatting, or pedestal type W.C. pan.	0.050 Bags	Each
b.	Fixing a pair of white glazed earthenware or vitreous china footrests of standard pattern for Indian type W.C. pan.	0.010 Bags	Each
c.	Fixing flat back or wall corner type lipped front urinal basin of 430 x 260 x 350 mm. and 340 x 430 x 265 mm.	0.020 Bags	Each
d.	Fixing white glazed fire clay stall urinal of standard size.	0.04 Bags	Each
e.	Fixing white squatting plate urinal with integral longitudinal flush pipe. .	0.040 Bags	Each
f.	Fixing wash basin including making all connections excluding cost of fittings.	0.030 Bags	Each
g.	Fixing kitchen sink including making all connections complete.	0.030 Bags	Each
h.	Fixing in position 32 mm. diameter galvanised steel telescopic flush pipe complete including cutting and making good the walls and floor.	0.020 Bags	Each
6.	Sand Cast Iron Pipe and Fittings		
a.	Fixing M.S. holder bat clamp to 100 mm. dia. sand cast iron pipe embedded in cement concrete blocks 10 x 10 x 10 cm. of cement concrete 1:2:4 (1 Cement: 2 Sand : 4 Stone Aggregate) including cost of cutting holes and making good the walls etc.	0.010 Bags	Each
b.	Fixing M.S. stays and clamps for 100 mm. diameter sand cast iron pipe.	0.010 Bags	Each
c.	Fixing M.S. holder bat clamps for 50 mm. diameter sand cast iron pipe embedded in cement concrete block 10 x 10 x 10 cm. of 1:2:4 (1 Cement: 2 Sand : 4 Stone Aggregate) including cost of cutting holes and making good the walls etc.	0.010 Bags	Each
d.	Fixing M.S. stays and clamps for 50 mm. diameter sand cast iron pipe.	0.010 Bags	Each
e.	Fixing sand cast iron trap 100 mm. inlet 100 mm. outlet of self-cleaning design with sand cast iron screwed down or hinged grating with or without vent arm complete including cost of cutting without and making good the walls and floor.	0.050 Bags	Each

f.	Fixing 100 mm. inlet and 50 mm. outlet sand cast iron floor trap of self-cleaning design with sand cast iron screwed down or hinged grating with or without vent arm complete including cost of cutting and making good the walls and floors.	0.050 Bags	Each
7.	Asbestos Cement Soil, Waste and Vent Pipes and Fittings		
	Providing and fixing on wall face asbestos cement soil waste and vent pipe including jointing with spun yarn soaked in bitumen and cement mortar 1:2 (1 Cement: 2 Sand) complete.		
a.	For 100 mm. diameter.	0.300 Bags	100 Meter
b.	For 50 mm. diameter.	0.150 Bags	100 Meter
	Fixing wooden plugs and standards holder bat clamps comprising of two semicircular halves of flat iron and cast-iron base screwed on wooden plugs.		
c.	For 100 mm. diameter.	0.0004 Bags	Each
d.	For 50 mm. diameter.	0.0004 Bags	Each
	Providing and fixing A.C. bends of required degree with access door insertion rubber washer 3 mm. thick, bolts and nuts or plain bend of heel rest unitary bend including jointing with spun yarn soaked in bitumen and cement mortar 1:2 (1 Cement: 2 Sand)		
e.	For 100 mm. diameter.	0.0020 Bags	Each
f.	For 50 mm. diameter.	0.0010 Bags	Each
	Providing and fixing double equal or unequal A.C. junctions of required degree plain or with access door, insertion, rubber washer 3 mm. thick bolts and nuts, including jointing with spun yarn cement mortar 1:2 (1 Cement: 2 Sand) complete.		
g.	100 x 100 x 100 x 100 mm. double equal junctions or 100 x 100 x 50 x 50 mm. double unequal junctions.	0.004 Bags	Each
h.	50 x 50 x 50 50 mm. double equal junctions.	0.002 Bags	Each
	Providing and fixing single equal or unequal A.C. junctions of required degree plain or with access door, insertion, rubber washer 3 mm. thick bolts and nuts, including jointing with spun yarn cement mortar 1:2 (1 Cement: 2 Sand) complete.		
i.	100 x 100 x 100 x 100 mm. single equal junctions or 100 x 100 x 50 x 50 mm. single unequal junctions.	0.0030 Bags	Each
j.	50 x 50 x 50 50 mm. single equal junctions.	0.0016 Bags	Each
	Providing and fixing plain A.C. invert branch of required degree including jointing with spun yarn soaked in bitumen and cement mortar 1:2 (1 Cement: 2 sand).		
l.	50 x 50 x 50 x 50 mm.	0.002 Bags	Each
m.	50 x 50 x 50 x 50 mm.	0.0016 Bags	Each
	Providing and fixing A.C. offset including jointing with spun yarn soaked in bitumen		

	and cement mortar 1:2 (1 Cement: 2 Sand)		
n.	100 mm. dia. A.C. offset with any projection.	0.002 Bags	Each
o.	50 mm. dia. A.C. offset with any projection.	0.0010 Bags	Each
	Providing and fixing A.C. loose socket including jointing with spun yarn soaked in bitumen and cement mortar 1:2 (1 Cement: 2 Sand) complete.		
p.	100 mm.	0.002 Bags	Each
q.	50 mm.	0.0010 Bags	Each
	Providing and fixing A.C. Terminal guard including jointing with spun yarn soaked in bitumen and cement mortar 1:2 (1 Cement: 2 Sand).		
r.	100 mm.	0.002 Bags	Each
s.	50 mm.	0.0010 Bags	Each
t.	Cutting chase in brick masonry walls for fixing 100 mm diameter sand cast iron pipes and making good the same with brick work in cement mortar 1:3 (1 Cement: 3 Sand)	10.00 Bags	100 Meter
u	Cutting chase in brick masonry walls for fixing 50 mm. diameter sand cast iron pipes and making good the same with the brick work in cement mortar 1:3 (1 Cement: 3 Sand).	6.66 Bags	100 Meter
8.	Drainage		
	Jointing glazed stone ware pipes grade "A" with stiff mixture of cement mortar in the proportion of 1:1 (1 Cement: 1 Sand)		
a.	100 mm. dia.	4.34 Bags	100 Meter
b.	150 mm. dia.	6.46 Bags	100 Meter
c.	200 mm. dia.	8.66 Bags	100 Meter
d.	230 mm. dia.	9.74 Bags	100 Meter
e.	250 mm. dia.	10.80 Bags	100 Meter
f.	300 mm. dia.	12.94 Bags	100 Meter
g.	450 mm. dia.	19.54 Bags	100 Meter
	Laying cement concrete 1:5:10 (1 Cement: 5 Sand : 10 Graded Stone Aggregate 40 mm. nominal size) around S.W. pipe including bed concrete 15 cm. thick.:		
h.	100 mm. dia. S.W. Pipe.	47.32 Bags	100 Meter
i.	150 mm. dia. S.W. Pipe.	50.70 Bags	100 Meter
j.	200 mm. dia. S.W. Pipe.	58.24 Bags	100 Meter
k.	230 mm. dia. S.W. Pipe.	62.92 Bags	100 Meter
l.	250 mm. dia. S.W. Pipe.	66.04 Bags	100 Meter
m.	300 mm. dia. S.W. Pipe.	73.58 Bags	100 Meter
n.	350 mm. dia. S.W. Pipe.	81.12 Bags	100 Meter
o.	400 mm. dia. S.W. Pipe.	88.40 Bags	100 Meter
p.	450 mm. dia. S.W. Pipe.	96.20 Bags	100 Meter
	Laying cement concrete 1:5:10 (1 Cement: 5 Sand : 10 Graded Stone Aggregate 40 mm. nominal size) up to haunches of S.W. pipe including bed concrete 15 cm. thick.:		
q.	100 mm. dia. S.W. Pipe.	31.72 Bags	100 Meter
r.	150 mm. dia. S.W. Pipe.	34.84 Bags	100 Meter
s.	200 mm. dia. S.W. Pipe.	40.56 Bags	100 Meter

t.	230 mm. dia. S.W. Pipe.	44.20 Bags	100 Meter
u.	250 mm. dia. S.W. Pipe.	46.54 Bags	100 Meter
v.	300 mm. dia. S.W. Pipe.	52.26 Bags	100 Meter
w.	350 mm. dia. S.W. Pipe.	58.24 Bags	100 Meter
x.	400 mm. dia. S.W. Pipe.	62.96 Bags	100 Meter
y.	450 mm. dia. S.W. Pipe.	69.94 Bags	100 Meter
z.	Laying light duty non-pressure NP2 or P1 class R.C.C. pipes with collars jointed with stiff mixture of cement mixture of cement mortar in the proportion of 1:2 (1 Cement: 2 Sand) including joints etc.		
Z1.	100 mm. dia. R.C.C. pipe (NP2) or (P1)	1.00 Bags	100 Meter
Z2.	150 mm. dia. R.C.C. pipe (NP2) or (P1)	1.20 Bags	100 Meter
Z3.	250 mm. dia. R.C.C. pipe (NP2) or (P1)	1.80 Bags	100 Meter
Z4.	300 mm. dia. R.C.C. pipe (NP2) or (P1)	2.20 Bags	100 Meter
Z5.	450 mm. dia. R.C.C. pipe (NP2) or (P1)	4.80 Bags	100 Meter
Z6.	500 mm. dia. R.C.C. pipe (NP2) or (P1)	5.20 Bags	100 Meter
Z7.	600 mm. dia. R.C.C. pipe (NP2) or (P1)	6.40 Bags	100 Meter
Z8.	700 mm. dia. R.C.C. pipe (NP2) or (P1)	7.40 Bags	100 Meter
Z9.	800 mm. dia. R.C.C. pipe (NP2) or (P1)	8.40 Bags	100 Meter
Z10.	900 mm. dia. R.C.C. pipe (NP2) or (P1)	9.80 Bags	100 Meter
Z11.	1000 mm. dia. R.C.C. pipe (NP2) or (P1)	11.00 Bags	100 Meter

TECHNICAL SPECIFICATION FOR PLUMBING WORKS

1. BASIC MATERIALS AND METHODS

1.1 GENERAL

All materials shall be new and of the best of their kind and shall conform to the latest Indian Standard Specification.

All materials shall be approved quality as per samples and origins approved by the Architects/Consultants.

As and when required by the Architects/Consultants, the contractor shall arrange to test the materials and/or portions of the works at his own cost to prove their soundness and efficiency. If after tests any materials, work or portions of work are found defective or unsound by the Architects/Consultants, the contractor, shall remove the defective materials from the site, pull down and re-execute the works at his own cost to the satisfaction of the Architects/Consultants. To prove that the materials used are as specified, the contractor shall furnish to Architects/Consultants with original vouchers on demand.

All works executed shall be as directed and to the satisfaction of the Architects/Consultants.

Pipes and fittings shall be fixed truly vertical, horizontal or in slopes as required in a neat workmanlike manner.

Short or long bends shall be used on all main pipelines as far as possible.
Use of elbows shall be restricted for short connections.

Pipes shall be fixed in a manner as to provide easy accessibility for repair and maintenance and shall not cause obstruction in shafts, passages etc.

Pipes shall be securely fixed to walls and ceilings by suitable clamps at interval specified.

Clamps, hangers and supports on RCC walls, columns & slabs shall be fixed only by means of approved made of expandable metal fasteners inserted by use of power drills.

All pipes clamps, supports, nuts, bolts, washers shall be galvanized MS steel throughout the building. Painted MS clamps & nuts, bolts & washers shall not be accepted.

Valves and other appurtenances shall be so located as to provide easy accessibility for operations, maintenance and repairs.

Access, doors for fittings and clean outs shall be so provided that they are easily accessible for repair and maintenance.

2. PLUMBING FIXTURES AND TRIM

2.1 SCOPE OF WORK

Work under this section shall consist of furnishing all materials & labor necessary and required to completely install all sanitary fixtures, chromium plated fittings and accessories as required by the drawings specified hereinafter and given in the Schedule of Quantities.

Without restricting to the generality of the foregoing the sanitary fixtures shall include the following:

Sanitary fixtures

Chromium plated fittings

Porcelain or stainless steel sinks

Accessories e.g. towel rods, toilet paper holders, soap dish, towel rack, coat hooks etc.

Connections to all kitchen, pump headers and other equipment requiring water and drainage connections.

Whether specifically mentioned or not all fixtures and appliances shall be provided with all fixing devices, nuts, bolts, screws, hangers as required.

All exposed pipes within toilets and near fixtures shall be chromium plated brass or copper unless otherwise specified.

2.2 GENERAL REQUIREMENTS

Sanitary fixtures shall be of the best quality approved by the Architects/Consultants. Wherever particular makes are mentioned, the choice of selection shall remain with the Architects/Consultants.

All fixtures and fittings shall be provided with all such accessories as are required to complete the item in working condition whether specifically mentioned or not in the Schedule of Quantities, specifications and drawings. Accessories shall also include proper fixing arrangement, brackets, nuts, bolts, screws and required connection pieces.

Fixing screws shall be half round head chromium plated brass screws with C.P. washers where necessary.

Contractor shall furnish without cost all such accessories and fixing devices that are necessary and required but not supplied along with the Plumbing Fixtures & CP fittings by the manufacturers as a part of the original and standard supply.

All fittings and fixtures shall be fixed in a neat workmanlike manner true to level and heights shown on the drawings and in accordance with the manufacturer's recommendations. Care shall be taken to fix all inlet and outlet pipes at correct positions. Faulty locations shall be made good and any damage to the finished floor, tiling or terrace shall be made good at Contractor's cost.

Contractor should seal all fixtures fixed near wall, marble and edges with an approved type of polysulphide sealant appropriate for its application.

2.3 VITREOUS CHINA SANITARYWARE

All glazed Vitreous China Sanitary ware fixtures shall conform to Indian Standard IS: 2556. The details, make and type to be provided are given in the Schedule of Quantities. The Vitreous China Sanitary ware shall be of first quality only. They shall be non-porous and fully vitreous, with all the visible portions perfectly glazed and should absolutely be free from hairline cracks pin- holes and local depressions. It shall be perfectly symmetrical, uniform and smooth and curves. All sanitary fixtures and fittings shall be stores under covered roof and handled carefully to prevent any damage.

2.4 CHROMIUM PLATED FITTINGS

All Chromium plated fittings shall be of brass/copper, heavy chromium plated, of the make and design approved by the Architects/Consultants. The fittings shall be cast fittings of screw type, machined and threaded properly for fixing to the supply pipes.

The plating shall conform to Indian Standard IS:482 (Electroplated coating of nickel and chromium of copper and copper alloys).

The fittings shall be supplied complete with chromium plated matching flanges, wall cover plates, nuts and extension pieces of required lengths. Metallic washers where required shall also be of chromium plated brass. All bib cocks and stop cocks shall conform to Indian Standard IS:781. Brass screw down pillar taps shall conform to IS:1701 and all other fittings shall match the supply fitting in construction and appearance. All fixing accessories and screws shall be similar to fittings. All washers shall conform to Indian Standard IS:4346.

All waste fittings (Waste, Chain, Overflow, Spreaders Caps etc.) shall be of brass/copper heavy chromium plated of the make and design specified and match the supply fittings. They shall conform to Indian Standard IS:2963.

Bottle traps (for wash basins, sinks, urinals etc) shall be deep seal (Min. 6cm seal) cast brass bottle traps, heavy chromium plated. All bottle traps shall be provided with suitable cleaning eye, extension piece, flare nuts of all chromium plated.

Wall flanges shall be provided on all walls, floors, columns etc. wherever supply and disposal pipes pierce through them. These wall caps shall be of chromium plated brass snugly fittings and the receiving pipes and shall be large enough to cover the punctures properly.

2.5 INSTALLATION OF SANITARY FIXTURES AND FITTINGS

General Requirement

The fixtures and fittings shall be provided with all such accessories as are required to complete the item in satisfactory working conditions, whether specifically mentioned or not in the schedule of Quantities, Specifications and drawings.

The sanitary fixtures and fittings shall be installed at the correct assigned position as shown on the drawings and as directly by the Architects / Consultants and shall fully meet with the aesthetic and symmetrical requirements as demanded by the Architects/Consultants.

All fixtures and accessories shall be fixed in accordance with a set pattern matching the tiles or interior finish as per Architects/Consultants requirements. Wherever necessary, the fittings shall be centered to dimensions and pattern as called for.

Fixtures shall be installed by skilled workman with appropriate tools according to the best trade practice. Manufacturer's instructions shall be followed for the installation of fixtures. Fixtures in all toilets shall be standard height mounting as called for on the drawings. Fixtures shall be mounted rigid, plumb, and true to alignment.

Mock up and Trial Assembly

The installation of the sanitary fixtures and fittings shall be as per the shop drawings approved by the Consultants.

The Contractor shall have to assemble at least one set of each type of sanitary fixtures and fittings in order to determine precisely the required supply and disposal connections. Relevant instructions from manufacturers shall be followed as applicable. This trial assembly shall be developed to determine the location of puncture holes, holding devices etc. which will be required for final installation of all sanitary fixtures and fittings. The above assembly shall be subject to final approval by the Architects/Consultants.

The fixtures in the trial assembly can be re-used for final installation without any additional payments for fixing or dismantling of the fixtures.

Supporting and Fixing Devices

The contractor shall provide all the necessary supporting and fixing devices to install the sanitary fixtures and fittings securely in position. The fixing devices shall be rigidly anchored into the building structure. The devices shall be rust resistant and shall be so fixed that they do not present an unsightly appearance in the final assembly. Where the location demands, the Architects/Consultants may instruct the contractor to provide chromium plated or other similarly finished fixing devices. In such circumstances the contractor shall arrange to supply the fixing devices and shall be installed complete with appropriate vibration isolating pads, washers and gaskets.

Final Installation

The contractor shall install all sanitary fixtures and fittings in their final position in accordance with approved trial assemblies and as shown on drawings. The installation shall be complete with all supply and waste connections. The connection between building and piping system and the sanitary fixtures shall be through proper unions and flanges to facilitate removal/replacement of sanitary fixtures without disturbing the built in piping system. All unions and flanges shall match in appearance with other exposed fittings.

Fixtures shall be mounted rigid, plumb and to alignment. The outlets of water closet pans and similar appliances shall be examined to ensure that outlet ends are butting on the receiving pipes before making the joints. It shall be ensured that the receiving pipes are clear of obstruction. When fixtures are being mounted, attention shall be paid to the possibility of movement and settlement by other causes.

Overflow shall be made to ensure that necessary anchoring devices have been provided for supporting water closets, wash basins, sinks and other appliances.

Protection against damage

The contractor shall take every precaution to protect all sanitary fixtures against damage, misuse, crazing, staining, breakage and pilferage by providing proper wrapping and locking arrangement till the completion of the installation. At the time of handing over, the contractor shall clean, disinfect and polish all the fixtures and fittings. Any fixtures and fittings found damaged, cracked chipped stained or scratched shall be removed and new fixtures and fittings free from defects shall be installed at his own cost to complete the work.

2.6 MEASUREMENT

Sanitary fixtures and accessories shall be measured by numbers in the unit given in the Schedule of Quantities.

Rates for all items shall be inclusive of cutting holes and chases and making good the same, C.P Brass screws, nuts, bolts and any fixing arrangements required and recommended by manufacturers, testing and commissioning.

3. WATER STORAGE TANKS

3.1 GENERAL

The following provisions should be made by the plumbing contractor in the Water Storage Tank

- a) 600 dia opening with sealed manhole cover.
- b) Manhole with double sealed manhole cover.
- c) Inlets, outlets over flows, drains and insect proof vent cowls as required on Drawings.

Necessary perforations (including puddle flanges where required) for the inlet and outlet pipes shall be provided. Each storage tank shall be equipped with overflow, warning pipes, and vent pipes, and drain sumps (located directly below the manhole).

With watertight hinged cover and locking device.

3.2 TANK FITTINGS

Overflow pipes shall be:

- a) Dipped down inside tanks so as to terminate in a position maximum 150mm above bottom of pump room floor.
- b) Provided with metal hinged flaps at outlets.
- c) Discharge in conspicuous position.

Strainers with Anti-vortex Plate shall be:

- a) 3mm thick brass strainer
- b) With 3mm thick galvanized mild steel anti-vortex plates
- c) With total perforation area of 3mm maximum diameter holes not less than twice of the cross section area of pump suction pipe.

3.3 LEVEL CONTROLLERS

All level controllers, level sensors, level switches and alarms shall be provided and installed by the contractor.

Each probe shall be of the correct length for its particular application and tank location. Electrodes shall be polished stainless steel 20mm outside diameter. Electrode holders shall be weather proof in all respect.

The earthing probes shall be connected and wired to the building earth system.

Each set of electrodes shall be installed inside a 200mm-diameter PVC pipe acting as a wave barrier.

The level switch set shall be operated with a stepped down voltage at 24V maximum. Stepped down transformer shall be provided for each set of control probes and shall be installed inside centralized control cubicles inside pump room.

Provide interface unit at the pump control panel for the levels to be monitored by BMS.

The BMS control panel will monitor the following outputs from the location for each Tank.

BMS Tank High Level (O/P): Whenever the liquid level in the tank reaches a high level, a high level dry contact will be provided by the contractor at the level controller and monitored by the BMS.

BMS Tank Low Level (O/P): Whenever the liquid level in the tank reaches a low level, a low level dry contact will be provided by the contractor at the level controller and monitored by the BMS.

BMS Level Sensing (O/P): The BMS will monitor the level of liquid in specified tanks via a level sensor provided by the M&E contractor. All level controllers will provide a 4-20mA signal to the ELV contractor.

3.4 EXECUTION

Tank Disinfection:

Disinfect all potable water tanks including concrete water tanks according to the following method:

A solution of 200-mg/L available chlorine shall be applied directly to the surfaces of all parts of the storage facility that would be in contact with water when the storage facility is full to the overflow elevation.

The chlorine solution may be applied with suitable brushes or spray equipment. The solution shall thoroughly coat all surfaces to be treated, including the inlet and outlet piping, and shall be applied to any separate drain piping such that it will have available chlorine of not less than 10mg/L when filled with water. Overflow piping need not be disinfected.

The disinfected surfaces shall remain in contact with the strong chlorine solution for at least 30-min. after that entire water should be flushed out. After which potable water shall be admitted, the drain piping shall be purged of the 10-mg/L chlorinated water, and the storage facility shall then be filled to its overflow level. Following this procedure, and subject to satisfactory bacteriological testing and acceptable aesthetic quality, water may be delivered to the distribution system.

4. DOMESTIC WATER PUMPS

4.1 GENERAL

The pump heads in the Schedules or Drawings are given for tendering guidance only and the Contractor shall be responsible for checking the total final resistance of each system, based on the actual pipe runs and equipment offered, prior to ordering. Any modifications to the system or any of the components, i.e. pumps, pipe work, motors etc., which may be required to meet the scheduled duties and space limitations shall be carried out at the Contractor's expenses.

All pumps shall be constructed to meet the required working conditions and test pressure of the system concerned.

The water supply pumps shall conform to IS 2065 and installation and maintenance of fire-fighting pumps shall conform to IS 15301 or the updated versions.

All pumps shall be self primed and mounted on a common bed-plate together with the motor and directly driven through a flexible coupling and shall be aligned in the manufacturer's factory. Locally fabricated bed-plates will not be accepted. Base plate shall incorporate a drain pan and is to be provided with a screwed socket outlet for drain connection.

Unless otherwise specified, pumps shall be selected for an impeller speed of 1450/1500 rpm. All pumps shall be as commercially silent in operation. A unit which is considered to be noisy shall be removed from the site and replaced by silent unit at the Contractor's expense.

The Contractor shall submit for the approval the characteristic curves of the pump offered. The operating conditions shall be indicated and curves of pumps having excessive shut-off head will not be accepted.

All pumps shall be fitted with an air-cock and drain plug. Each pump shall be provided with connection for pressure gauges at the suction and delivery connections, the gauges for all pumps shall be mounted on common hardwood gauge board on the wall inside the pump room. Gauges shall have 100mm diameter dial. Pressure gauge piping shall be copper comprising a loop siphon at the junction to the gauge and each gauge shall be supplied through 12mm diameter brass pig tail and pet cock. Drains shall be piped to floor drains/waste pit.

The Contractor shall supply and install flexible pipe connectors at the pump suction and discharge sides for each pump.

All flexible connectors shall have flanges joints to the table specified for the connecting pipe work and must be suitable for the working conditions and test pressure of the system concerned.

Gland drains shall be provided in each pump packing gland with suitable nipple/outlet for copper/PVC pipes to be connected. All drains shall have a minimum internal bore of 15mm diameter and arranged in a proper pattern for piping the same to the channels or floor outlets provided inside the pump room.

Where necessary, an automatic air relief valve shall be fitted in the pump to purge any air likely to be trapped inside the pump casing. Suitable fittings shall be use for such connection and the relief valve shall not discharge at a location directly over or near to the pump motor or any other electrical equipment.

Each pump shall be fitted with valves to enable it to be isolated, and in addition, where shown on the Drawings, a check/non-return valve shall be fitted in each pump discharge.

Transfer pump set shall be mounted on an inertia block. Holding down bolts shall be set in position during the manufacture of the structural base. Filtered pump set shall be mounted on base plate. Generally all pump sets shall be mounted on the reinforced concrete plinth 150mm high or as instructed by the Engineer.

Inertia block shall be concreted. Anchor bolts of base shall be provided.

Pump inertia bases shall be provided and shall be twice the operating weight of the pumps and the associated pipe work as mounted thereon.

The base plates shall be of approved make with pads machined for correct alignment of the pump and motor. At least two locating dowels shall be fitted to each component after alignment.

All exposed shafts, couplings and moving parts of pumps shall be provided with suitable galvanized angle iron wire mesh guards which shall be stoutly constructed and easily removable; and shall be provided with lifting handles. Care shall be taken that these guards of not cause "Ringing" sound and/or vibrate so causing noise.

An identification plate of metal shall be fixed to each pump unit. This plate shall include full details and diameter of the impeller installed, pump size, model and serial number, r.p.m., amps, etc. pump head and delivery for the duty specified, and lubricant required.

All pumps shall bear the manufacturer's designation plate which shall indicate the type of services and serial number of the unit.

4.2 PUMP ARRANGEMENT

Pumps for plumbing services are single stage or multi-stage centrifugal vertical, horizontal split case or end suction type as shown in Schedule of quantity/ Drawing.

De watering / Sewage pumps shall be submersible type, or centrifugal self priming.

Each pump shall be directly coupled with the electric motor and aligned in the manufacturer's factory.

4.3 PUMP SPEED AND NOISE

Pumps shall be selected for an impeller speed as indicated in the Schedule/Drawing. All pumps shall be as commercially silent in operation.

The noise level shall be not more than NC 70 in pump room, NC 50 in the same floor and NC 35 for other floors.

4.4 PUMP BASES

Except for submersible sump pumps, each pump set shall be mounted on an anti-vibration pump base. Details shall be submitted by the Contractor well before the manufacture of the structural base. All holding down bolts shall be provided by the Contractor.

4.5 SAFETY GUARDS

All exposed shafts, couplings and moving parts of pumps shall be provided with suitable galvanized angle iron wire mesh guards which shall be stoutly constructed and easily removable; and shall be provided with lifting handles. Care shall be taken that these guards do not cause "ringing" sound and/or vibrate so causing noise.

4.6 IDENTIFICATION

An identification plate of metal shall be fixed to each pump unit. This plate shall include full details and diameter of the impeller installed, pump size, model and serial number, r.p.m., sumps, etc pump head and delivery for the duty specified, and lubricant required.

All pumps shall bear the manufacturer's designation plate which shall indicate the type of services and serial number of the unit.

4.7 ELECTRICAL WORKS FOR PUMPS

All electrical works associated with the pumps shall comply with the NBC regulations (the latest edition) code of practice, supply rules and regulations and described in the relevant section of this Specification.

Anti-vibration mountings shall be inertia base, springs, neoprene pads or rubber-in-shear isolators, with the specified static deflection and selected to provide isolating efficiency of not less than 95%.

The control panel shall be of the pressure switch/microprocessor. The complete control panel assembly and all the internal devices shall be UL508. The panel shall be complete with IP54/NEMA1 enclosure and include door interlocked main disconnect and magnetic motor starters with fused motor protectors, adjustable time delays, Hand-off auto selector switch and for each pump, power on light, minimum run timers, low suction pressure switch and pilot light. The control circuit shall include fault relay circuit to turn on the next pump should the lead pump fail.

Should the duty pump fail to start and the pressure of the system shall continue to fall, the standby pump shall be initiated to cut into operation.

Either pump shall be able to be selected as duty/standby/jockey. The operating pump shall also be able to select as automatic or manual and interlock devices shall be provided to isolated the system for maintenance

Automatic changeover shall be provided to alternate between the duty/standby pumps on each cycle.

A time delay (0 to 10 seconds) switch shall be installed in the control of each pump set to provide a time lag between the initiation of signals by the pressure switches (both pump on and off) and the actual start and stop of the pump set to avoid excessively frequent on/off cycles of the pumps due to surges in the long pipe runs.

The pump set together with the required control panel and fitted with hoot shall be mounted on a common bedplate, factory assembled in the manufacturer's workshop prior to delivery. Control panel for all pump sets shall be as per Indian standard.

4.8 RAW WATER PUMPS

Where required transfer water pumps shall conform to the following Specification:

Pump efficiency shall not be less than 70%,

Materials shall be as follows:

Casing	Cast iron and minimum working pressure not less than 250 PSI or 150% of maximum discharge pressure.
Impellers	Bronze and hydraulic balancing.
Wearing	Bronze
Shaft Sleeve	Bronze
Shaft	Stainless steel
Seal	Mechanical seal/gland packing (asbestos-free)
Couplings	Flexible metallic coupling, complete with coupling guards
Bearings	Ball thrust type, grease lubricated, rated bearing life not less than 100,000 Hr.

4.9 SUBMERSIBLE SEWAGE AND DRAINAGE PUMPS

Submersible sewage and drainage pumps shall be installed in sump pits consisting of discharge connection, galvanized steel guide rail, galvanized chain and lifting handle as per detail. Pump efficiency shall not be less than 40%. Can be handle solid particle upto 50mm.

Materials shall be as follows:

Casing	High grade grey cast iron with internal sea water corrosion resistant coating
Impellers	Cutter type for sewage pumps or non-clog type for drainage pumps, chromium-alloyed cast iron material of abrasion resistant.
Shaft	Stainless steel.
Seal	Double mechanical seal.
Bearings	Ball thrust type, grease lubricated
Maximum	As shown on the Equipment Schedule/Drawing speed
Bolt & Nut	Stainless Steel

Level switches (high/low levels alarm and indications) shall be supplied and installed in the sump pits for the proper operation of the pumps as indicated on Drawings. The switches shall be of mercury float switch type. The lead-lag selection of pumps shall be changed automatically after each cycle.

Attention shall be paid to the internal dimensions of the sump pits and covers when selecting pumps.

Motor shall be class F insulation and inside the casing to prevent water ingress.

The Contractor shall supply and fix pump discharge pipe work, vents, control and power wiring and associated conduit. Three spare packing shall be provided for each pump.

The installations shall be complete with the following accessories:

- Seal monitor in oil chamber for leakage checking.
- Thermal switches to protect over heated motor.
- Automatic control box.
- Cable of sufficient length for connection of the control box to the pump.
- A.C. alarm bells.

- Cable holders.

4.10 EXECUTION

Completely align and level pumps, motors and bases. Where pumps and motor are shipped as a unit, realign them in the field.

Grout base plates completely to provide a non-deflecting support.

Install and align mechanical seals in accordance with the manufacturer's recommendation.

Pump manufacturer or his represented to set packing, adjust impellers and check alignment prior to start-up.

Pump shall be mounted according to vibration isolation section.

Repair all surface damage during shipping and installation or prior to client's acceptance of the building.

5. INTERNAL & EXTERNAL WATER SYSTEM GENERAL

Water supply system to cater for drinking, washing, flushing & other requirements shall be installed as called for on the drawings.

The piping system consists of medium class galvanized iron pipes and fittings, CPVC/PPR Pipes or other material as called for. The size and makes are specified in the Schedule of Quantities.

5.1 REQUIREMENTS

- A. Supply and installation of the cold water system including incoming water mains, all distribution pipe work, transfer pumps, multistage pressure booster pump sets, valves, fittings, hangers, equipment supporting frameworks, etc. as per the Drawings.

Water connections shall be provided at the plot boundary by Municipal Authority.

The incoming water mains shall be routed to water storage tanks in the vicinity. Water supply distribution system to various usage points will be through hydro-pneumatic system.

Water hose points shall be provided to the basement plant rooms, roof and other areas as required.

5.2 FERRULES

The ferrules for connection with C.I main shall generally conform to IS: 2692. It shall be of non-ferrous materials with a C.I bell mouth cover and shall be of nominal bore as specified. The ferrule shall be fitted with a screw and plug or valve capable of completely shuttering of the water supply to the communication pipe, if and when required.

5.3 WATER METERS

Water meters of approved make and design shall be supplied for installation at location as shown. The water meters shall meet with the requirement and approval of local water supply authorities. Suitable valves and chambers or wall meter box to house the meters shall also be provided along with the meters.

The water meters shall be provided with straight reading, dry dial, hermetically sealed register, and magnetic drive. The rotation of the turbine is directly transmitted to be register, and magnetic drive. The meters shall have totalization of flow volume in cubic meter (cum). The meter shall conform to local municipal authority, and shall be suitable for connection to the BMS for remote monitoring/reading.

The meters shall conform to Indian Standard IS: 779 and IS: 2373. Where called for, the water meters shall be located in masonry chambers of appropriate size.

Provision shall also be made to lock the water meter. The provision shall be such that the lock is conveniently operated from the top. Where the provision is designed for use in conjunction with padlocks, the hole provided for padlocks shall be a diameter not less than 4mm.

5.4 MAKING WATER CONNECTION

A pit of suitable dimension shall be dug at the point where the connection is to be made with ring main and earth removed up to 150mm below the main. The flow of water in main shall be disconnected by operating the nearest sluice valve on the main. The main shall be drilled and slopped at 45 to the vertical and the ferrule of required size shall be screwed in. the ferrule shall be fitted in a manner so that no portion of projection of the shank shall be left projecting within the main into which it is fitted. Ferrule shall be non-ferrous material with a CI bell mouth cover and shall be of nominal bore as required.

5.5 INSTALLATION OF WATER METER AND STOP COCK

The GI lines shall be cut to the required lengths at the position where the meter and stop cock are required to be fixed. Then end of the pipe shall be threaded. The meter and stop cock shall be fixed in a position by means of connecting pipes, GI jam nut and socket etc. The stop cock shall be fixed near the inlet of the water meter. The paper disc inserted in the ripples of the meter shall be removed. And the meter is installed exactly horizontal or vertical in the flow line in the direction shown by the arrow on the body of the meter. Care shall be taken that the factory seal of the meter is not disturbed. Wherever the meter shall be fixed to a newly fitted pipe line, the pipe line shall have to be completely washed before fitting the meter.

5.6 DISINFECTION OF PIPING SYSTEM AND STORAGE TANKS

Before commissioning the water supply system, the contractor shall arrange to disinfect the entire system as described in the succeeding paragraph.

The water storage tanks and pipes shall first be filled with water thoroughly flushed out. The storage tanks shall be filled with water again and disinfecting chemical containing chlorine shall be added gradually while tanks are being filled to ensure thorough mixing. Sufficient chemical shall be used to give water a dose of 50parts of chlorine to one million parts of water. If ordinary bleaching powder is used, the proportions will be 150gms of powder to 1000 liters of water. The powder shall be mixed with water in the storage tank. If a proprietary brand of chemical is used, the proportions shall be specified by the makers. When the storage tank is full, the supply shall be stopped and all the taps on the distributing pipes are opened successively. Each tap shall be closed when the water discharged begins to smell chlorine. The storage tank shall then be filled up with water from supply pipe and added with more disinfecting chemical in the recommended proportions. The storage tank and pipe shall then remain charged at least for three hours. Finally the tank and pipes shall be thoroughly flushed out before any water is used for domestic purposes.

5.7 TESTING

All water supply system shall be tested to hydrostatic pressure test of at-least two and half time the maximum pressure but not less than 7kg/sq.cm. for a period of not less than 24hours. All leaks and defects in joints revealed during the testing shall be rectified and got approved at site.

Piping required subsequent to the above pressure test shall be resettled in the same manner.

System may be tested in sections and such sections shall be entirely resettled on completion.

5.8 SHIFTING OF EXCAVATED SURPLUS MATERIAL

Contractor shall make his own arrangement to shift the surplus excavated material within the site limits as directed by Engineer-in-charge.

5.9 PROTECTION AGAINST CORROSION

All embedded piping material and accessories shall be suitably protected against corrosion. All embedded GI pipes shall be wrapped throughout with 1mm thick fiberglass RP tissue laid in bitumen

with overlaps in joints and final coat of bitumen paint. Where GI pipes are laid under floors, trenches etc shall be encased with 100mm thick fine sand all-round in additions to the protective coating as stated above.

5.10 LAWN HYDRANT

Lawn hydrants shall be of 25mm size unless otherwise indicated. All hydrants shall be provided with ball valves and nipple to receive hose pipes. Lawn hydrant valves shall be of approved make and design. Where called for lawn hydrants shall be located in masonry chambers of appropriate size.

5.11 MEASUREMENT

All pipes above ground shall be measured per linear meter (to the nearest cm) and shall be inclusive of all fittings e.g. coupling, tees, bends, elbows, unions, flanges and U clamps with nuts, bolts & washers fixed to tall or other standard supports.

Jointing as per procedure specified above.

Cutting holes, and chases in walls, floors, any pipe support required for pipes below ground & making good the same.

6. PIPING AND FITTING MATERIAL

6.1 PIPING SCHEDULE:

Notes on Pipe work

All PVC pipes where installed in direct contact with concrete shall be wrapped with polythene tape/fibreglass tissue for protection.

All fittings for pipe work shall conform to the relevant IS Specification.

Where purpose made fittings are required, they shall also be generally in accordance with the relevant IS Standard.

All pipe work shall be free from burst, rust and scale and shall be thoroughly cleaned before erection. Open ends during the progress of work shall be blanked-off with purpose-made metal or plastic caps and the use of wooden plugs is forbidden. Should any stoppage in the circulation occur after the various systems have been put into operation, owing to non-compliance with this requirement, the Contractor shall rectify the matter at no extra cost to the Contract.

Pipes shall be installed with correct falls for venting and draining and attention shall be paid to neatness of installation, i.e. groups of pipes shall be accurately spaced and valves, joints, etc., symmetrically arranged. Where two or more pipes are visible and change direction together bends shall be struck from a common radius point. Adequate clearances shall be maintained in all cases, to allow for the application of the insulation materials and finishing.

The Contractor shall note the restricted space available in certain service ducts and the absolute necessary to ensure that all pipe work is installed in the correct sequence, manner and position to ensure that operation of all valves and maintenance is possible.

All reductions in pipe sizes shall be made using approved standard reducing fittings. Reducing bushes shall not be used without approval. Connections between lengths of pipes shall be made using standard fittings. Long threads must not be used.

All pipe work in pump rooms, plant rooms, and on roof level shall be installed with flanges or unions at intervals not exceeding 6m for ease of dismantling.

Changes in direction of pipes shall be made with long radius bends or elbows, as far as practicable. Under no circumstances shall pipes be bent without the use of fittings.

All pipe runs when not installed underground or in duct shall be concealed as far as possible by careful positioning or shall be chased into walls, laid in screed etc, or as directed by the Architect/Consultant.

All pipe work within the fire protected areas must be enclosed to satisfy the local municipal/fire authority requirements and such enclosures shall be supplied and installed by the Contractor.

Where pipes and rainwater outlets pass through roof slabs, the Contractor shall leave suitable openings during casting. The Contractor shall install the pipe, rainwater outlets, etc., taking into account the water proofing installation method as detailed by the Engineer.

Floor drains shall be installed in the same manner as rain water outlets. Openings will be formed in the slabs. Detailed dimensions and locations shall be supplied by the Engineer.

The Contractor shall locate all valves, traps, cleaning eyes, and similar items where they are easily accessible for operation, inspection, and maintenance.

Water Supply System

S. No.	Description	Type of Material
1.	Potable water tank drains, overflows and warning pipes up to and including 65mm diameter.	CPVC/ PPR.
2.	Incoming potable water pipe from Municipal authority.	CPVC/Galvanized Iron medium class
3.	External Potable cold water main distribution pipe	CPVC/ PPR
4.	Potable cold water branch pipe	CPVC/ PPR

Soil/Waste and Rain Water System

S. No.	Description	Type of Material
1.	Rain water pipe above ground 50mm and above	Cast Iron Pipe to IS 3989 / UPVC SWR SYSTEM.
2.	Rain water pipe below ground 80mm and above.	RCC pipes NP2 class
3.	Waste water pipe above ground 50mm and above	Cast Iron Pipe to IS 3989/PVC SWR SYSTEM/UPVC Pipes.
4.	Waste water pipe below ground 50mm and above	Stoneware Pipes / UPVC Pipes/RCC Pipes NP2 class

6.2 PIPING SPECIFICATIONS

6.2.1 Galvanized Iron Pipes

The pipes shall be galvanized mild steel welded (ERW) or (HFW) screwed and socketed conforming to the requirements of IS: 1239. The Galvanizing shall conform to IS: 4736. The Zinc coating shall be uniform, adherent, reasonably smooth and free from imperfections such as flux, ash and drops inclusions, bare patches, black spots, pimples, lumpiness, runs, rust stains, bulky white deposits and blisters. The pipes and sockets shall be cleanly finished, well galvanized in and out and free from cracks, surface flaws, laminations and other defects. All screw threads shall be clean and well cut. The ends shall be cut cleanly and square with the axis of the pipe.

The fittings shall be malleable cast iron as per IS: 1879 and comply with all the requirements that of pipes. The sizes of pipes and fittings are specified in the schedule of quantities.

6.2.2 UPVC PIPES

UPVC Pipes and fittings shall be heavy class manufactured to Indian Standard IS: 4985 (Specification for UPVC pipes & fittings). The Pipes shall be of desired working pressure as specified in the Schedule of Quantities. On delivery to site, the pipes and fittings shall be inspected for thickness, cracks etc. The pipes shall be stored in a sheltered roof as protection against direct sunlight.

6.2.3 CPVC Pipes

Chlorinate Polyvinyl Chloride (CPVC) Pipes and fittings shall be heavy class manufactured to international Standard with the CPVC compound shall conform to (NSF) certification for use in drinking /portable water. Pipes shall be of desired working pressure as specified in the Schedule of Quantities. On delivery to site, the pipes and fittings shall be inspected for thickness, cracks etc. The pipes shall be stored in a sheltered roof as protection against direct sunlight.

6.2.4 Copper Pipes

Copper pipes shall be of 99.9 percent copper. Pipes shall be either hard or soft or temper drawn straight lengths. The pipes shall be rigid enough to hold its shape without support and yet is flexible enough to permit flaring and bending by hand bending machine. Pipes in nominal sizes from 15 to 54mm (inclusive) shall be connected with capillary fittings with integral tin-copper or silver solid ring to B.S. 864:1972 Part 2.

Pipes in nominal sizes from 67mm to 219mm (inclusive) shall be connected with gunmetal silver brazing fittings by welding to the relevant welding code B.S. 1723:1963 (brazing). All brazing fittings shall be cast in one piece. Local welded lobster fittings shall not be accepted. All welders shall be fully qualified. Welding rods shall be of not less than 50% silver content and suitable for the piping materials and atmospheric conditions to B.S. 1845:1977.

6.2.5 CAST IRON PIPES & FITTINGS

Soil, Waste, Vent, Anti-Siphonage and Rainwater Pipes shall be cast Iron Pipes. All pipes shall be straight and smooth and inside free from irregular bore, blow holes, cracks and other manufacturing defects. Pipes shall be either of sand cast to IS: 1729 or centrifugal cast to IS: 3989.

6.2.6 PPR PIPES

Polypropylene Random (PPR) pipes and fittings shall be heavy class as per DIN 8077 and fittings as per DIN: 16962. The Pipes shall be of desired working pressure as specified in the Schedule of Quantities. On delivery to site, the pipes and fittings shall be inspected for thickness, cracks etc. The pipes shall be stored in a sheltered roof as protection against direct sunlight.

6.2.7 Composite Pipes.

Composite pipes consists of aluminium and polyethylene composite pipes is combination of two materials i.e, aluminium core tube bonded by adhesive to the interior and exterior tube layers of polyethylene. The pipes and fittings as manufactured as per internal standards and meet the requirements of BS: 7291, the specifications for thermoplastic pipes and associated fittings for hot and cold water for domestic purposes.

The pipes are available in coils and easily formed into curves set by hand and only requires bending spring when forming light bends down to radius equivalent to 5 times the diameter of pipe. Inside & outside layers comprise High Density

6.2.8 STAINLESS STEEL PIPES

The stainless steel pipes shall conform to the following standards and codes.

ASTM – A 312- Seamless and Welded Austenitic Stainless Steel Pipes

ASTM – A530- Testing of Welded Austenitic Stainless Steel Pipes

The stainless steel pipes shall be seamless or welded annealed tubing with plain ends conforming to ASTM 312 – TP 304/316 as indicated in the bill of materials. The pipes shall be internally electro polished with 300 grade finish.

The fittings shall be butt /socket welded conforming to ASTM A – 182.

The pipes thickness shall be as per Schedule 10 for pipes up to 50 mm dia and schedules for pipes above 50 mm dia.

6.2.9 Fittings

Fittings shall confirm to the same Indian Standard as for pipes. Contractor shall use pipes and fittings of matching specifications.

Fittings shall be of the required degree of curvature with or without access door. Access door shall be made up with 3mm thick insertion rubber washer and white lead. The bolts shall be lubricated with grease or white lead for easy removal. Fixing shall be air and water tight.

6.2.10 PVC SWR SYSTEM

PVC SWR systems for soil, waste & rain water shall be of manufactured to Indian Standard 13592. The pipes and fittings shall be of the rubber ring joints type with in-built groove for insertion of rubber seal lubricant. The rubber ring shall conform to IS: 5382. The Pipes shall be of desired working pressure as specified in the Schedule of Quantities. On delivery to site, the pipes and fittings shall be inspected for thickness, cracks etc. The pipes shall be stored in a sheltered roof as protection against direct sunlight.

6.3 PRESSURE RATING

The pressure rating of all pipes from cistern water tanks to the elevated water tanks shall be PN12.5. For all other pipe works, pressure rating shall be PN10 unless otherwise specified.

The pressure rating of valves and fittings shall be not less than that of the pipe work.

6.4 EXECUTION

The execution of works and materials used shall be as per the latest relevant I.S. specification. Whether reference has been made to Indian Standard or any other specifications, the same shall mean to refer to the latest specification irrespective of any particular edition of such specification being mentioned in the specifications below or Schedule of Quantities.

All pipes shall be anchored as required to control the movement of pipes due to thermal changes. Generally anchors shall also be fitted at the lower end of vertical pipes. Details of all anchor points shall be submitted to the PMC for approval. The anchors shall securely hold the pipe in a rigid position to resist the attempted movement due to expansion and the weight of pipe and contents.

The Contractor shall submit full details to the PMC of all expansion and anchor joints proposed for use throughout the building prior to installation.

General

In all pipe system, long radius bends and pitcher tees shall be used wherever possible and practicable. All fittings shall be suitable for a working and test pressure not lower than that specified for the pipe work system concerned elsewhere in this Specification.

Reduction in bore of pipe shall be generally made eccentric for horizontal pipes (to prevent air locking) and concentric taper fittings shall be fitted for vertical pipes unless specifically required to achieve the required space and invert levels.

The workmanship shall be best of its kind and shall conform to the specifications, as below or Indian Standard Specifications in every respect or latest trade practices and shall be subject to approval of the Architects/Consultants. All materials and/or Workmanship which in the opinion of the Engineer is defective or unsuitable shall be removed immediately from the site and shall be substituted with proper materials and/or workmanship forthwith.

Pipes shall be fixed in a manner as to provide easy accessibility for repair and maintenance and shall not cause obstruction in shafts, passages etc.

Pipes shall be securely fixed to walls and ceilings by suitable clamps at interval specified.

Clamps, hangers and supports on RCC walls, columns & slabs shall be fixed only by means of approved made of expandable metal fasteners inserted by use of power drills.

All pipes clamps, supports, nuts, bolts, washers shall be galvanized MS steel throughout the building. Painted MS clamps & nuts, bolts & washers shall not be accepted.

Valves and other appurtenances shall be so located as to provide easy accessibility for operations, maintenance and repairs.

Tender drawings indicated schematically the size and location of pipes. The Contractor, on the award of the work, shall prepare detailed working drawings, showing the cross-section, longitudinal sections, details of fittings, locations of isolation and control valves, drain and air valves, and all pipe supports. He must keep in view the specific openings in buildings and other structures through which pipes are designed to pass.

Piping shall be properly supported on, or suspended from, on stands, clamps, hangers as specified and as required. The contractor shall adequately design all the brackets, saddles, anchor, clamps and hangers, and be responsible for their structural stability. All pipes shall be secured at every joint.

Piping supports shall be steel, adjustable for the height and primer coated with rust prevent paint and finish coated black. Where pipe and clamps are of dissimilar materials, spacing b/w the supports shall not exceed the following:

Pipe size	Spacing between supports
Up to 12mm	1.5 meters
15mm to 25mm	2.0 meters
30mm to 50mm	2.0 meters
Above 50mm	2.5 meter

Vertical risers shall be parallel to walls and columns lines and shall be straight and plumb. Risers passing from floor to floor shall be supported at each floor by clamps or collars attached to pipe and with a 15mm thick rubber pad or any resilient materials. Where pipes pass through the terrace floor, suitable flashing shall be provided to prevent water leakage. Riser shall have a suitable drain out in the valve at the lowest point and air vent at the highest point.

Pipe sleeves, 50mm larger diameter than pipes, shall be provided wherever pipes pass through walls and slabs, and annular space filled with fiberglass and finished with retainer rings.

Insulated piping shall be supported in such a manner as not to put undue pressure on the insulation 14gauge metal sheet shall be provided between the insulation and the clamp, saddle or roller, extending at-least 15cm. on both sides of the clamps saddles or roller.

All pipe work shall be carried out in a workman like manner, causing minimum disturbance to the existing services, buildings, roads and structure. The entire piping work shall be organized in consultation with other agencies work, so that area shall be carried out in one stretch.

Cut-outs in the floor slab for installing the various pipes area are indicated in the drawings. Contractor shall carefully examine the cut-outs provided and clearly point out wherever the cut-outs shown in the drawings do not meet with the requirements.

The contractor shall make sure that the clamps, brackets, saddles and hangers provided for pipe supports are adequate. Piping layout shall take due care for expansion and contraction in pipes and include expansion joints where required.

All pipes shall be accurately cut to the required sizes in accordance with relevant BIS codes and burrs removed before lying. Open ends of the piping shall be closed as the pipe is installed to avoid entrance of foreign matter. Where reduces are to be made in horizontal runs, eccentric reduces shall be used for the piping to drain freely in other locations, concentric reduces may be used.

Flanged inspection pieces 1.5meters long, with bolted flanges on both ends, shall be provided not more than 30meter centers wherever shown in approved for construction shop drawings to facilitate future cleaning of all welded pipes.

Insulation

Hot water pipe lines in chase shall be insulated by wrapping 6mm thick insulation rope around the pipe and finished with a coat of magnesia cement containing 85% magnesia.

Fixing Ferrules

For fixing ferrule, the empty main shall be drilled and tapped at 4-5 to the vertical and the ferrule screwed in. The ferrule must be so fitted that no portion of the shank shall be left projecting within the main into which it is fitted.

Cutting Chases in Masonry Walls

The chases up to 7.5x7.5cm shall be made in the walls for housing GI pipes etc. These shall be provided in correct positions as shown in the drawings or directed by the Architects/Consultants. Chases shall be made by chiseling out the masonry to proper line and depth. After GI pipes etc are fixed in chases, the chases shall be filled with cement mortar 1:2:4 or as specified may be made flush with the masonry surface. The concrete surface shall be roughened with wire brush to provide a key for plastering.

Water Fittings

Unless otherwise specified, all gunmetal fittings such as gate, globe, check & safety valves shall be fitted in pipe line in workman like manner. Necessary unions shall be provided on both ends of the valves for easy replacement. The joints between fittings and pipes shall be leak-proof when tested to a pressure. The defective fittings and joints shall be replaced or redone.

6.5 PIPE INSTALLATION

Laying and Jointing of GI Pipes

All pipes and fittings shall be fixed truly vertical and horizontal unless unavoidable. The pipes shall be fixed to walls with standard slotted angles 'U' shape threaded bolts, nuts for clamping pipes to angles. Slotted angles shall be grouted to RCC work with dash fasteners of size so as to fit tightly on the pipes when tightened with screwed bolts. These slotted angles shall be spaced at regular intervals in straight lengths and heights.

The galvanized pipes and fittings shall run in wall chase or ceiling or as specified. The fixing shall be done by means of standard pattern holder bat clamps keeping the pipes about 1.5cm clear of the wall where to be laid on surface. Where it is specified to conceal the pipes, chasing may be adopted or pipes fixed in the shafts, ducts etc. provided there is a sufficient space to work on the pipes with usual tools. As far as possible, pipes may be buried for short distances provided adequate protection is given against damage and where so required special care to be taken at joints. Where directed by the Architects/Consultants, pipe sleeves shall be fixed at a place where the pipe is passing through a wall or floor reception of the pipe and allow freedom for expansion and contraction and other movements. In case of pipe is embedded in walls or floors, it should be painted with anticorrosive bitumastic paints of approved quality. Under the floors, the pipes shall be laid in layer of sand filling.

Galvanized iron pipes shall be jointed with threaded and socket joints, using threaded fittings. Care shall be taken to remove any burr from the end of the pipes after threading. White lead or an equivalent jointing compound of proprietary make shall be used, according to the manufacturer's instructions, with a grummet of a few strands of fine yarn while tightening compounds containing red lead shall not be used because of the danger of contamination of water. Any threads exposed after jointing shall be painted with bituminous paint to prevent corrosion.

Laying and Jointing of UPVC Pipes

UPVC pipes and fittings shall be jointed using solvent cement. It is recommended to use formulated heavy duty solvent cement supplied by the company for trouble free performance of the system.

The jointing procedures are as follows:

- Cut the pipe square to the required length by cutter.
- Chamfer the cutting edges.
- Insert the pipe into the fitting and check to that the interface occur about 1/3rd to 2/3rd of the socket depth.
- Remove any dirt, moisture or grease from pipe end and fitting sockets with clean dry rag.
- Roughen the contacting surfaces.
- Application of solvent cement – Apply cement lightly but uniformly to inside of socket and outside of pipe end with suitable applicator.
- Immediately after applying the last coat of cement to the pipe and while the cement is still fluid or wet, forcefully bottom the male end of the pipe in the socket, giving pipe or fitting ¼ turn (but not after pipe is bottomed) to distribute the cement evenly. Remove excess cement from the end of the fitting socket. The joint must not be disturbed immediately after cementing so that the joint can be properly cured.

Laying and Jointing of CPVC Pipes

CPVC pipes and fittings shall be jointed using solvent cement. It is recommended to use formulated heavy duty solvent cement supplied by the company for trouble free performance of the system.

The jointing procedures are as follows:

- Cut the pipe square to the required length by cutter.

- Chamfer the cutting edges.
- Insert the pipe into the fitting to check that the interface occur about 1/3rd to 2/3rd of the socket depth. Remove any dirt, moisture or grease from pipe end and fitting sockets with clean dry rag. Roughen the contacting surfaces.
- Application of solvent cement. – It is recommended that to use Only CPVC cement or all-purpose cement conforming to ASTM F-493 for pipe jointing. Apply cement lightly but uniformly to inside of socket and outside of pipe end with suitable applicator.
- Immediately insert the tubing into fitting socket, rotating the tubing 1/4" to 1/2" turn while inserting. This motion ensures an even distribution of cement within the joint. Properly align the fitting. Hold the assembly for approximately 10 seconds, allowing the joint to set-up. An even bead of cement should be evident around the joint. Wipe excess cement from the tubing and fitting surfaces.

Laying and jointing of Copper Pipes

Pipes in nominal sizes from 15 to 54mm (inclusive) shall be connected with capillary fittings with integral tin-copper or silver solid ring to B.S. 864:1972 Part 2.

After the tubing is cleaned, the ring is slipped into the fittings and heat is applied to the fittings by high-temperature gas flame. When a sufficient temperature is reached, the solder melts and a bead appear at the edge of the fittings around the entire circumference. As this type of fittings is usually of considerable thickness, it requires a hot flame to melt with the solder. The contractor should maintain every care in order to avoid damage to other fittings or valves that may be near the fittings being heated.

Pipes in nominal sizes from 67mm to 219mm (inclusive) shall be connected with gunmetal silver brazing fittings by welding to the relevant welding code B.S. 1723:1963 (brazing). All brazing fittings shall be cast in one piece. Local welded lobster fittings shall not be accepted. All welders shall be fully qualified. Welding rods shall be of not less than 50% silver content and suitable for the piping materials and atmospheric conditions to B.S. 1845:1977.

Pipes shall be supported by cast brass pipe brackets or hangers of approved pattern. End or end butt-welded joints shall not be permitted. Vertical pipes shall be fixed with copper or copper alloy holder bats or pipe brackets of approved pattern. Horizontal pipes shall be fixed with heavy duty copper or copper alloy pipe brackets or hangers of approved pattern.

Pipe Cleaning

Upon completion of the domestic water system, disinfect this system, including water service connections, with chlorine in accordance with local practice.

The water storage tanks and pipes shall first be filled with water thoroughly flushed out. The storage tanks shall be filled with water again and disinfecting chemical containing chlorine shall be added gradually while tanks are being filled to ensure thorough mixing. Sufficient chemical shall be used to give water a dose of 50parts of chlorine to one million parts of water. If ordinary bleaching powder is used, the proportions will be 150gms of powder to 1000liters of water. The powder shall be mixed with water in the storage tank. If a proprietary brand of chemical is used, the proportions shall be specified by the makers. When the storage tank is full, the supply shall be stopped and all the taps on the distributing pipes are opened successively. Each tap shall be closed when the water discharged begins to smell chlorine. The storage tank shall then be filled up with water from supply pipe and added with more disinfecting chemical in the recommended proportions. The storage tank and pipe shall then remain charged at least for three hours. Finally the tank and pipes shall be thoroughly flushed out before any water is used for domestic purposes.

All valves, after they have been disinfected, shall be opened and closed several times during the eight-hour period. Disinfection shall be repeated until piping meets the local bacteriological test criteria.

All other piping systems shall be flushed free of foreign matter with clean water and left with strainers clean upon completion of the work.

6.6 HANGERS, SUPPORTS, ANCHORS AND GUIDES

Provide hangers of heavy construction suitable for the size of pipe to be supported. All materials to be of hot dipped galvanized steel (touch up all cut ends with galvanized paint) except rollers which are to be of wrought or malleable iron. Hangers for pipes up to and including 125mm to be adjustable swivel ring, split ring, wrought pipe clamp, or adjustable wrought clevis type.

Support vertical piping with double bolt riser clamps attached to the pipe, resting on pre-engineered spring hanged attached to the floor slab. Use one clamp at every floor. Where pipes are in open shafts, provide forged steel bar brackets fixed to wall.

The following tables will establish a minimum level of acceptance for pipe hangers, supports and attachments.

Hangers and Supports

Service	Hanger Type	Maximum Pipe Size
Un-insulated PVC	Clevis	300mm
Un-insulated	Clevis	300mm
Copper		
Copper Pipe	Riser Clamp	150mm
All Insulated	Full Clamp & Insulation	600mm
	Shield	
All	Trapeze	600mm
All	Wall Bracket	125mm
All	Wall Bracket	300mm

Structure Attachments

Type	Maximum Rod Size	Maximum Pipe Size
Beam Clamp	20mm	200mm
Beam Clamp	40mm	600mm
Side Mount	20mm	200mm
Clamp		
Channel Clamp	20mm	200mm
Expansion Shield	20mm	200mm

Job or site fabricated products will not be allowed.

Execution

Provide hanger rod sizes in accordance with the following schedule:

Pipe Size	Rod Size
25mm and smaller	10mm
32mm to 50mm	10mm
65mm to 80mm	15mm
100mm to 125 mm	15mm
150 mm to 200 mm	20 mm
200 mm to 300 mm	25 mm
350 mm to 400 mm	25 mm
450 mm	30 mm
500 mm	35 mm
600 mm	40mm

A. Provide hangers in accordance with the following schedule:

Hanger Spacing Schedule		
Piping Material	Pipe Size	Maximum hanger Spacing
Copper	35mm & less	1800mm
Copper	42mm & larger	3000mm
UPVC	50mm & larger	1350mm
GI	50mm & less	2000mm
GI	65mm & larger	2500mm

Provide hangers at each change in direction on both sides of each valve and on both sides of cast iron pipe fittings. For flat slab construction only, support hangers from concrete inserts. Furnish, locate and set such inserts and make sure that such inserts are in place when the concrete is poured. Construct inserts of malleable iron or pressed steel with space for rods of all sizes. Install all inserts for pipes 80mm and larger in size with a reinforcing rod 15mm in diameter run through a slot in the insert specifically provided for this purpose.

For flat slab construction only, if any pipe is to be hung in a space where no inserts have been provided, drill holes in the slab (subject to the Engineer prior approval) and provide rods and hanger attached to an approved fishplate or install double expansion shields connected by a 50mm x 50mm angle, from which the hanger rod is to be suspended. For pipe size 50mm and under, use single shields but the hanger spacing defined herein before to be reduced to 1.5m. The carrying capacity and size of each shield to be calculated on the basis of the spacing indicated above but the minimum size to be 10mm. Install additional shields of the same size so that the number of hangers are of adequate size to support the loads which they carry. Shields may be used in flat concrete slabs only. Regardless of the type of construction (i.e., concrete, concrete-deck-steel or other variations) take particular care to support all main lines and all large and heavy pipes in an approved manner, including the furnishing and installation of supplementary steel, if required. Supplementary steel sections are to be mill-rolled. Submit shop drawings, indicating support methods, point loadings to the building structure and hanger locations for review sufficiently in advance of concrete pouring schedules to permit evaluation, critique and any necessary changes to handling and support methods.

Set all inserts for all pipes in ample time to allow concrete work to be performed on scheduled time. Hangers may be directly attached to steel beams of building construction, where they occur, if approved by Engineer. Smaller pipes may be suspended from crosspieces of pipe or steel angles, which in turn, are to be securely fastened to building beams. The intention is to provide supports which, in each case, will be amply strong and rigid for the load, but which will not weaken or unduly stress the building construction.

Provide approved roller support, floor stands, wall brackets, etc., for all lines running near the floor or near walls, which can be properly supported or suspended by the floors or walls. Pipelines near walls may also be hung by hangers carried from approved wall brackets at a level higher than the pipe.

Do not hang piping from other piping. Support of hangers by means of vertical expansion bolts is not permitted.

Anchor piping as required to localize expansion or to prevent undue strain on piping and branches. Anchors to be entirely separate from hangers. All anchor designs to be submitted flex bellows flanged or screw type for approval and to include piping reactions which respective anchors are capable of supporting. Provide all indicated or required expansion loops.

Support all lines of copper tubing individually by approved type hangers not more than 2m apart, or as shown on the drawings. Use hangers especially designed for copper tubing and of exact outside diameter of tubing. On hangers for covered tubing, use broad straps fitting outside of covering.

Hangers used for cold piping will support the pipe without piercing the insulation. Use insulation shields to protect the insulation on cold pipes. Weld insulation protection saddles to insulated hot pipes, or any piping subject to axial movement, at roller supports. Space between pipe and saddle to be filled with insulation.

For piping 100mm and larger, support the elbows of the piping adjacent to the pumps with steel base elbow supports from the inertia base which pump is on to prevent loading heavy weights of piping on pump casing. Where inertia base is not provided, base elbows to be supported on floor with 25mm neoprene pad.

All pipes supports shall be of such design and type to allow for the removal of any pipe section without the necessity of disconnecting other adjacent pipes.

In all plant and pump rooms discharge pipe work from pumps, and all pipe work where vibration could be transmitted to the building structure, shall be supported with brackets having a tough rubber lining in contact between the bracket and the pipe shall have anti-vibration hangers/supports.

Special attention shall be paid to that adequate pipe anchors shall be provided at the ends or turning points of UPVC/PPR pipes to prevent fracture of the pipe resulted from water hammer. Such anchor points shall be fixed in a manner that the inertia in the flow will be directly transmitted onto the anchors thus releasing the UPVC/PPR pipes from being under stress.

Where pipes pass through walls, floors, etc., the Contractor shall provide pipe sleeves. Sleeves for UPVC/PPR pipes & copper pipes shall be galvanized steel. For pipes, where sleeves are fitted through floors/wall they shall extend to at least 20mm clear stud out from the finished floor level.

The space between pipe sleeves and the pipe or insulation shall be completely backfilled with materials having the same fire resistant rating as the walls and floors by the Contractor. Should multi-services or more than one single pipe be laid through box-out, the space between the individual pipe sleeves shall be sealed off. The space between the pipe sleeves and the service pipes shall be back-filled.

Sleeves shall be of sufficient size to allow free movement of pipes and furthermore where pipes are insulated the sleeves shall be oversized to allow the insulation to be carried through the sleeves. The Contractor's attention is drawn to the vertical pipe ducts at all floor levels. Where pipes pass through floors, sleeves shall be provided. The Contractor shall supply sleeves for setting in walls, floors, etc., and in the shuttering before concrete is poured.

Details of pipes through walls and floors shall be approval by the Engineer before installation.

Where pipes pass through flat roofs covered with roofing material: -

- a) Cast or build in galvanized mild steel sleeve with 2 to 12mm clearance projecting 150mm above roof finish.
- b) Caulk space and point both ends with approved sealant.
- c) Cover tops of sleeves with lead collars as per roofing Specification.

7. VALVES

General

- The Contractor shall supply and install all valves as indicated on the drawings and as required for the proper and efficient operation and maintenance of the entire systems.
- All valves supplied shall be suitable for the working pressure and test pressure of the various water supply systems.
- All valves other than automatic control valves shall be full line size.
- Each valve shall have a purpose made reference number plate or label engraved or stamped indicating the manufacturer's catalogue number, pressure and temperature ratings. Valves shall be arranged so that clockwise rotation of the spindle will close the valve. Demo labels are not acceptable.
- Furnish all valves and accessory material necessary in the piping whether or not shown on drawings as follows.
- All valves shall be packed with an approved packing and threads shall be coated with oil and graphite. Packing should be replaced when found deteriorated on site.

- Where possible locate all valves at convenient positions of operation from the floor with valve stems upright.
- Valves that are flanged shall have flanges comply with BS 4504 as for pipe work.
- Provide valves of the same manufacturer throughout, where possible.
- All valves shall be with screwed or flanged ends as required by the piping system in which they are installed.
- Valves shall be selected for the working pressure required.
- Plastic or metal plates (rust-less) shall be provided to indicate the open/close status as well as the use of each valve in the pump and tank rooms, and in the town main.

Valves & Controls - Specifications

All valves (gate, globe, ball, check, and safety) shall be of gun metal non rising spindle valves suitable for the particular service as called for. All valves shall be of the particular duty and design as called for. Valves shall either be of screwed type or flanged type, with suitable flanges and non-corrosive bolts and gaskets. Tail pieces as required shall be supplied along with valves. Gate, globe and check valves shall conform to Indian Standard IS:778 and non-return valves to swing check type reflux to IS:5312.

Sluice valves, where called for shall be flanged sluice valves of cast iron body. The spindle, wall seat and wedge nuts shall be gunmetal. They shall generally have non-rising spindle and shall be of the particular duty and design called for. The valves shall be supplied with suitable flanges, non-corrosive bolts and asbestos fiber gasket. Sluice valve shall conform to Indian standard IS:780 and IS:2906.

Ball valves with floats to be fixed in storage tanks shall consist of cast brass lever arm having copper balls (26 SWG) screwed to the arm integrally. The copper ball shall have bronze welded seams. The closing/opening mechanism incorporating the position and cylinder shall be non-corrosive metal and include washers. The size and construction of ball valves and float shall be suitable for desired working pressure operating the supply system. Where called for brass valves shall be supplied with brass hexagonal back-nuts to secure them to the tanks and a socket to connect to supply pipe.

Type of Valve	Size	Contraction	Ends
a) G.M valve	15mm to 50mm 50mm and above	Gun metal Gun metal	Screwed Flanged
b) Sluice valve & butterfly valve	65mm and above	Cast iron/ iron	Flanged
c) G.M Non return valve	15mm to 50mm 50mm and above	Gun metal Gun metal	Flanged Flanged
d) Flap type Non return	above 65mm and above	Cast iron	Flanged

Execution

All valves shall be installed only in the upright vertical or horizontal positions unless specifically otherwise required by the drawings.

All valves shall be installed in accessible locations to facilitate easy removal for repair or replacement.

Install with operating clearance for handle and stem.

Install isolation valves on equipment so that valve and piping do not interfere with equipment removal or maintenance. Install unions or flanges on equipment side of valves.

Provide 25mm drain valves with threaded ends for hose connections at drain points, at main shutoff valves, low points of piping systems, bases of vertical risers, and at equipment.

Provide required manual or automatic vent valves at high points of piping systems to facilitate venting of air and to ensure quiet operation.

Provide renewable bronze seat rings and bronze spindles for cast iron body valves. Provide chain operated sheaves and chains for all valves which are more than 2.5m above the floor in Mechanical Equipment Rooms.

Furnish and install other valves, check valves, cocks, etc., as required for the complete and proper valving of the entire installation.

Install butterfly valves in horizontal piping with stem in the horizontal position so that bottom of disk lifts in the direction of flow.

Install butterfly valves in vertical piping at pumps with stem perpendicular to pump shaft.

8. STRAINERS

Strainers shall be installed in all pump suctions and tanks.

Strainer under 65mm dia shall be pipeline Y-type strainer. Strainer of 65mm or above shall be of vertical (bucket) type, cast iron body with flanged connectors.

Strainer mesh shall be stainless steel 316 with filtering area of at least three times the sectional area of the pipe inlet.

9. FLEXIBLE CONNECTOR

Flexible connectors shall be of synthetic fiber reinforced rubber, twin sphere type for flush water and 316 stainless steel bellow type for type fresh water system. Details shall be submitted to the Architect at the time of tender.

Pressure rating of the flexible connectors shall be the same as the system pipe work unless otherwise noted on drawings.

Flexible connectors shall be installed at where shown on drawings and at all pump suctions and discharges. The materials of the flexible connectors shall be suitable for the substances to be conveyed.

10. EXTERNAL DRAINAGE (SOIL, WASTE & RAINWATER)

10.1 SCOPE OF WORK

Work under this section shall consist of furnishing all labor, materials, equipments and appliances necessary and required to completely install the drainage system as required by the drawings and specified hereinafter or given in the schedule of quantities.

Without restricting to the generality of the foregoing, the drainage system shall include:-

Sewer Lines including excavations, pipe lines, manholes, drop connection and connections to the municipal or existing sewer, underground storm water drains, including pipes, manholes, catch basins and drop connections.

10.2 GENERAL REQUIREMENTS

All materials shall be new of the best quality conforming to specifications and subject to the approval of Consultants.

Drainage lines shall be laid to the required gradients and profiles.

All drainage work shall be done in accordance with the local municipal bye-laws.

Contractor shall obtain necessary approval and permission for the drainage system from the Municipal or any other competent authority.

Location of all manholes, catch basins etc. shall be got confirmed by the Architect/Consultants before the actual execution of work at site.

All works shall be executed as directed by Consultants.

All pipe lines shall be subjected to testing as per specification.

10.3 ALIGNMENT AND GRADE

The Sewer and Storm water drainage pipes shall be laid to alignment and gradient shown on the drawings but subject to such modifications as shall be ordered by Architect/Consultants from time to time to meet the requirements of the works. No deviations from the lines, depths of cutting or gradients of sewers shown on the plans and sections shall be permitted except by the express direction in writing of Architect/Consultants.

10.4 EXCAVATION

The Excavation for Sewers works shall be open cutting unless the permission of Architect/Consultants for the ground to be tunnelled is obtained in writing. Where sewers have to be constructed along narrow passages, Architect/Consultants may permit the excavation to be made partly in tunnel and in such cases the excavated soil shall be brought back later on for refilling the trenches or tunnel.

10.5 OPENING OUT TRENCHES

In excavating the trenches, etc. the solid road metalling, pavement, curbing, etc. and turf is to be placed on one side and preserved for reinstatement when the trenches or other excavation shall be filled up. Before any road metal is replaced, it shall be shifted the surface of all trenches and holes shall be restored and maintained to the satisfaction of Architect/Consultants and of the owner of the roads or other property traversed and the contractor shall not cut out or Break Down any live fence of trees in the line of the proposed works but shall tunnel under them, unless Architect/Consultants shall order to the contrary.

The contractor shall grub up the clear the surface over the trenches and other excavations of all trees, stump roots and all other encumbrances affecting executing of the work and shall remove them from the site to the approval of Architect/Consultants.

10.6 REMOVAL OF FILTH

All night soil, filth or any other offensive matter met with during the execution of the works, immediately after it is taken out of any trench, sewer or cesspool, shall not be deposited on to the surface of any street or where it is likely to be a nuisance or passed into any sewer or drain but shall be at once put into the carts and removed to a suitable place to be provided by the contractor.

10.7 EXCAVATION TO BE TAKEN TO PROPER DEPTHS

The Trenches shall be excavated to such a depth that the sewer shall rest on concrete as described in the several clauses relating there to and so that the inverts may be at the levels given in the sections. In bad ground, the Architect/Consultants may order the contractor to excavate to a greater depth than that shown on the drawings and to fill up the excavation to the level of the sewers with concrete, broken stone, gravel or other materials. For such extra excavation and concrete, Broken Stone, Gravel or other

materials, the contractor shall be paid extra at rates laid down for such works in the schedule. If the extra work was operated by the Architect/Consultants in writing, but if the contractor should excavate the trench to a greater depth than is required without a specific order to that effect in writing of the Architect/Consultants the extra depth shall have to be filled up with concrete at the contractor's own costs and charges to the requirements and satisfactions of the Architect/Consultants.

10.8 REFILLING

After the Sewer or other work has been laid and proved to be water tight, the trench or other excavations shall be refilled. Utmost care shall be taken in doing this, so that no damage shall be caused to the sewer and other permanent work. The filling in the haunches and up to 75cms above the crown of the sewer shall consist of the finest selected materials placed carefully in 15cms layers and flooded and consolidated. After this has been laid, the trench and other excavation shall be refilled carefully in 15cms layers with materials taken from the excavation. Each layer being watered to assist in the consolidation unless the Architect/Consultants shall instruct otherwise.

10.9 CONTRACTOR TO RESTORE SETTLEMENT AND DAMAGES

The Contractor shall, at his own costs and charges, take good promptly during the whole period the works surfaces of roads, beams, footpaths, gardens, open spaces etc. Whether public or private caused by his trenches or by his other excavations and he shall be liable for any accidents caused thereby. He shall also, at his own expense and charges, repair and make good any damage done to buildings and other property. If in the opinion of the Architect/Consultants, he fails to make good such works with all practicable dispatch. The Architect/Consultants shall be at Liberty to get the work done by other means and the expense thereof shall be paid by the contractor or deducted from any money that may be or become due to him or recovered from him in any other manner according to the law of the land.

10.10 DISPOSAL OF SURPLUS SOIL

The contractor shall at his own costs and charges provide places for disposal of all surplus materials not required to be used on the works. As each trench is refilled the surplus soil shall be immediately removed, the surface properly restored and roadways and sides left clear.

10.11 TIMBERING OF SEWER AND TRENCHES

The contractor shall at all times support efficiently and effectively the sides of the sewer trenches and other excavations by suitable timbering, piling and sheeting and they shall be closed. Timbered in loose or sandy strata and below the surface timbering, piling and sheeting and they shall be closed.

All Timbering, sheeting and piling with their walling and supports shall be of adequate dimensions and strength and fully braced and strutted so that no risk of collapse or subsidence of the walls of the trench shall take place.

The contractor shall be held responsible and will be accountable for the sufficiency of all timbering, bracing, sheeting and piling used as also for all damage to persons and property resulting from improper quality, strength, placing, maintaining or removing of the same.

10.12 REMOVAL OF WATER FROM SEWER, TRENCH ETC.

The contractor shall at all times during the progress of the work keep the trenches and excavations free from water which shall be disposed of by him in a manner as will neither cause injury to the public health nor to the public or private property nor to the work completed or in progress nor to the surface of any roads of streets, nor cause any interference with the use of the same by the public.

If any excavation is carried out at any point or points to a greater width than the specified cross section of the sewer with its envelope, the full width of the trench shall be filled with concrete by the contractor at his own expenses and charges to the requirements of the Architect/Consultants.

10.13 WIDTH OF TRENCH

The Architect/Consultants shall have power by giving by order in writing to the contractor to increase the maximum width in respect of which payment will be allowed for excavation in trenches for various classes of sewer, manholes, and other works in certain lengths to be specifically laid down by him, where on account of bad ground or other unusual conditions, he considers that such increased widths are necessary in view of the site conditions. Maximum width of the bed concrete shall also be as below. No additional payment is admissible for widths greater than specified.

Recommended width of trenches at the bottom of the trench are as follows:-

100 mm dia Pipe	55 cms
150 mm dia Pipe	55 cms
225-250 mm dia Pipe	60 cms
300 mm dia Pipe	75 cms

10.14 STONEWARE PIPES

Stoneware pipes shall be of first class quality salt glazed and free from rough texture inside and outside and straight. All pipes shall have the manufacturer's name marked on it and shall comply to I.S. 651 of makes approved by the Architect/Consultants

10.15 LAYING AND JOINTING OF STONEWARE PIPES

Pipes are liable to be damaged in transit and notwithstanding tests that may have been made before dispatch each pipe shall be examined carefully on arrival at site. Each pipe shall be rung with a wooden hammer or mallet and those that do not ring true and clear shall be rejected. Sound pipes shall be carefully stacked to prevent damage. All defective pipes should be segregated, marked in a conspicuous manner and their use in the works prevented.

The pipes shall be laid with sockets leading uphill and should rest on solid and even foundations for the full length of the barrel. Socket holes shall be formed in the foundation sufficiently deep to allow the pipe jointer room to work right round the pipe and as short as practicable to admit the socket and allow the joint to be made.

Where pipes are not bedded on concrete the trench bottom shall be left slightly high and carefully bottomed up as pipe laying proceeds so that the pipe barrels rest on firm bound. If excavation has been carried too low it shall be made up with cement concrete at the contractor's cost and charges.

If the bottom of the trench consists of Rock or very hard ground that cannot be easily excavated to a smooth surface, the pipes shall be laid on cement concrete bed to ensure even bearing.

Jointing of Pipes

Tarred gaskin shall first be wrapped round the spigot of each pipe and the spigot shall then be placed into the socket of the pipe previously laid, the pipe shall then be adjusted and fixed in its correct position and the gaskin caulked tightly home so as to fill not more than one quarter of the total length of the socket.

The remainder of the socket shall be filled with stiff mix of cement mortar (1 cement : 1 clear sharp washed sand). When the socket is filled, a fillet should be formed round the joint with a trowel forming an angle of 45 degrees with the barrel of the pipe the mortar shall be mixed for immediate use and no mortar shall be used after it has begun to set. After the joint has been made any extraneous materials shall be removed from inside of the joint with a suitable scraper of "Badger". The newly made joints shall be protected until set from the sun, drying winds, rain or dust. Sacking or other materials which can be kept damp shall be used. The joints shall be exposed and space left all round the pipes for inspection by the Architect/Consultants. The inside of the sewer must be left absolutely clear in bore and free from cement mortar or other obstructions throughout its entire length, and shall efficiently drain and discharge.

Testing

All lengths of the sewer and drain shall be fully tested for water tightness by means of water pressure maintained for not less than 30 minutes. Testing shall be carried out from manhole to manhole. All pipes shall be subjected to a test pressure of at least 1.5meter head of water. The test pressure shall, however, not exceed a meter head at any point. The pipes shall be plugged preferably with standard design plugs with rubber plugs on both ends. The upper end shall, however, be connected to a pipe for filling with water and getting the required head.

Sewer Lines shall be tested for a straightness by:

Inserting a smooth ball 12mm less than the internal diameter of the pipe. In the absence of obstructions such as yarn or mortar projecting at the joints the ball should roll down the invert of the pipe and emerge at the lower end.

Means of a mirror at one and a lamp at the other end. If the pipeline is straight the full circle of light will be seen otherwise obstruction of deviation will be apparent.

The contractor shall give a smoke test to the drains and sewer at his own expense and charges, if directed by the Architect/Consultants.

A test register shall be maintained which shall be signed and dated by contractor, and representative of Architect/Consultants.

10.16 GULLY TRAPS

The gully traps shall be of the first quality and free from rough texture.

Gully traps shall be fixed in cement concrete 1:5:10mix and a brick masonry chamber 30x30cms inside in cement mortar 1:5 with 15x15cms grating inside and 30x30cms C.I. sealed cover and frame weighing not less than 7.3kg to be constructed as per standard drawing. Where necessary, sealed cover shall be replaced with C.I. grating of the same size (1 cement :5 coarse sand :10 stone aggregate 40mm nominal size).

10.17 REINFORCED CEMENT CONCRETE PIPES

All underground storm water drainage pipes and sewer lines where specified (other than those specified cast iron shall be centrifugally spun RCC pipes of specified class. Pipes shall be straight with uniform bore, throughout. Cracked, warped pipes shall not be used on the work. All pipes shall be tested by the manufacturer and the contractor shall produce, when directed a certificate is that effect from the manufacturer.

Laying

R.C.C. Spun Pipes shall be laid on cement concrete bed or cradles as specified and shown on the detailed drawings. The cradles may be pre-cast and sufficiently cured to prevent cracks and breakage in handling. The invert of the cradles shall be left 12mm below the invert level of the pipe properly placed on the soil to prevent any disturbance. The pipe shall then be placed on the bed concrete or cradles and set for the line and gradient by means of sight rails and boning rods etc. cradles or concrete bed by be omitted, if directed by the Architects/Consultants.

Jointing

After setting out the pipes the collar shall be centered over the joint and filled in with tarred gaskin, so that sufficient space is left on either side of the collar to receive the mortar. The space shall then be filled with cement mortar 1:2 (1cement: 2fine sand) and caulked by means of proper tools. All joints shall be finished at an angle of 45degrees to the longitudinal axis of the pipe on both sides of the collars neatly.

Testing

All pipes shall be tested to a hydraulic test of 1.5M head for at least 30minutes at the highest point in the section under test. Test shall also be carried out similar to those for stoneware pipes given above. The smoke test shall be carried out by the contractor, if directed by the Architect/Consultants, at the expense and charges of the contractor. A test register shall be maintained which shall be signed and dated by contractor, and representative of Architect/Consultants.

10.18 CAST IRON PIPES FOR DRAINAGE

All drainage lines passing under Building, floors and roads with heavy traffic, in exposed position above ground E.G. service floor and basement ceiling shall be cast iron pipes. Position of such pipes shall generally be shown on the drawings.

Cast Iron pipes shall be centrifugally spun iron pipes conforming to I.S.1536-1967. Quality certificates shall be furnished.

Fittings and Inspection Chambers

Fittings used for C.I. Drainage pipe shall conform to I.S :1538-1967. Wherever possible junction from branch pipes shall be made by a Y Tee.

Contractor shall provide as indicated on drawings or directed by Architect/Consultants cast Iron inspection chamber at all Junction. Inspection chambers shall be specially cast with inlet, outlet and branches of appropriate and required sizes. Branches shall be Y type wherever possible.

Cleanout plugs shall be provided on head of each drain and at location indicated on plans or directed by Architect/Consultants. Cleanout plugs shall be of size matching the full bore of the pipe. Plugs shall be made out with G.I coupling caulked into the socket of the pipe or fittings. The end shall be provided with a brass screwed plug with suitable key for opening.

Laying

All Cast Iron Pipes and fittings shall be jointed with best quality soft pig lead which shall be free from impurities. In wet trenches joints shall be made from lead wool. Nothing extra will be paid for lead wool joints. Depth of pig lead and weight for joints shall be as given in the sub section "B" of Section II.

The Spigot of pipe or fittings shall be centered in the adjoining socket by caulking. Sufficient turns of tarred gaskin will be given to leave unfilled the required depth of socket for depth of 45mm when the gaskin has been caulked tightly home. Joining ring shall be placed round the barrel and against the face of the socket. Molten pig lead shall then be poured to fill the remainder of the socket. This shall then be done in one pouring. The lead shall then be solidly caulked with suitable tools and hammers weighing not less than 2 kg.

For lead wool joints the socket shall be caulked with tarred gaskin. As explained above. The lead wool shall be inserted into the sockets and tightly caulked home skein by skein with suitable tools and hammers of not less than 2 kg. Weight until joint is filled.

Alternatively, cast iron pipes and fittings shall also be jointed with drip seal joints as per manufacturer's specifications where called for.

Testing

All Cast Iron Pipes for drainage shall be tested to a hydraulic test of 3 meter head. Test for straightens shall be same as for stoneware pipe given above. A test register shall be maintained which shall be signed and dated by contractor, and representative of Architect/Consultants.

10.19 CEMENT CONCRETE AND MASONRY WORKS (FOR MANHOLES AND CHAMBERS ETC.)

Materials

Water: Water used for all the construction purposes shall be clear and free from oil, acid, alkali, organic and other harmful matters, which shall deteriorate the strength and/or durability of the structure. In general, the water suitable for drinking purposes shall be considered good enough for construction purpose.

Aggregate for concrete: The aggregate for concrete shall be in accordance with I.S. 383 and I.S. 515. In general, these shall be free from all impurities that may cause corrosion of the reinforcement. Before actual use these shall be washed in water, if required as per the direction of Architect/Consultants. The size of the coarse aggregate shall be done as per I.S. 383.

Sand: Sand for various construction purposes shall comply in all respects with I.S. 650 and I.S. 2116. It shall be clean, coarse hard and strong, sharp, durable, uncoated, free from any mixture of clay, dust, vegetable matters, mica, iron impurities soft or flaky and elongated particles, alkali, organic matters,

salt, loam and other impurities which may be considered by the Architect/Consultants as harmful for the construction.

Cement: The cement used for all the constructional purposes shall be ordinary portland cement or rapid hardening portland cement conforming to I.S. 269.

Mild Steel reinforcement: The Mild Steel for the reinforcement bars shall be in the form of round bars confirming to all requirements of I.S. 432 Grade I.

Bricks: Bricks shall have uniform color, thoroughly burnt but not over burnt, shall have plain rectangular faces with parallel sides and sharp right angled edges. They should give ringing sound when struck. Brick shall not absorb more than 20% to 22% of water, when immersed in water for 24 hours. Bricks to be used shall be approved by the Architect/Consultants.

Others Materials: Other materials not fully specified in these specifications and which may be required in the work shall conform to the latest I.S. All such materials shall be approved by the Architect/Consultants.

Cement concrete (Plain or Reinforced): Cement concrete pipes bedding, cradles, foundations and R.C.C. Slabs for all works shall be mixed by a mechanical mixer where quantities of the concrete poured at one time permit. Hand mixing on properly constructed platforms may be allowed for small quantities by the Architect/Consultants. Rate for cement concrete shall be inclusive of all shuttering and centering at all depth and heights.

Concrete work shall be of such thickness and mix as given in the schedule of quantities.

All concrete work shall be cured for a period of at least 7 days. Such work shall be kept moist by means of gunny bags at all times. All pipes trenches and foundations shall be kept dry during the curing period.

10.20 MASONRY WORK

Masonry work for manholes, chambers, and such other works as required shall be constructed from 1st class bricks as specified in the schedule of quantities in cement mortar 1:5 mix (1 cement : 5 coarse sand). All joints shall be properly raked to receive plaster.

Cement Concrete for pipe Support

Wherever specified or shown on the drawings, all pipes shall be supported in bed all round or in haunches. The thickness and mix of the concrete shall be given in the schedule of quantities. Width of the bedding shall be as per para 4.14.

Unless otherwise directed by the Architect/Consultants cement concrete for bed, all round or in haunches shall be laid as follows: -

	Up to 1.5m	Up to 3m	Beyond 3m
	Depth	Depth	Depth
RCC or S.W pipes in all round		In Haunches	all round
open ground (No sub soil water)			
	(1:4:8:)	(1:4:8:)	(1:4:8)
RCC or S.W. in sub All Round		In Haunches	In Haunches
soil water			
	(1:3:6)	(1:3:6)	(1:3:6)
C.I. Pipes (In all All Round		In Haunches	In Haunches
conditions)	(1:3:6)	(1:3:6)	(1:3:6)

RCC Pipes or CI Pipes All round In Haunches In Haunches under of Building (1 cement: 3coars: 6 stone aggregate 40mm nominal size)

R.C.C Pipe or C.I. Pipes may be supported on brick masonry or pre-cast R.C.C. or in situ cables. Cradles shall be as shown on the drawings.

Pipes in loose soil or above ground shall be supported on brick or stone masonry pillars as shown on the drawings.

10.21 MANHOLES AND CHAMBERS

All Manholes, Chambers, and other such works as specified shall be constructed in brick masonry in cement mortar 1:5 (1cement 5 coarse sand) or as specified in the schedule of quantities.

All manholes, chambers, etc. shall be supported on base of cement concrete of such thickness and mix as given in the schedule of quantities or shown on the drawings.

Where not specified, manholes shall be constructed as follows :-(All Dimensions Internal clear in cms)

Size of Manhole	900x800	20x90	90dia	140dia
	Rect.	Rect.	Conical	Conical
Maximum Depth	200	200	250	250
Average Thickness of R.C.C Slab	15	15	-	-
Size of cover and frame	900x450	50dia	50dia	50dia

All Manholes shall be provided with cement concrete benching in 1:2:4 mix. The benching shall have a slope of 10cms towards the channel. The depth of the channel shall be full diameter of the pipe. Benching shall be finished with a floating coat of neat cement. (1 cement: 2coarse sand: 4 stone aggregate 20 mm nom. size)

All Manhole shall be plastered with 12/15 mm thick cement mortar 1:3 (1 cement: 3 coarse sand) and finished with a floating coat of neat cement inside. Manhole shall be plastered outside as above but with rough plaster.

All Manholes with depths greater than 1m. shall be provided with 20mm square or 25mm round rods catch rings set in cement concrete blocks 25x10x10cms in 1:2:4 mix 30cms vertically and staggered. Foot rests shall be coated with coal tar before embedding.

All Manholes shall be provided with cast iron covers and frames and embedded in reinforced cement concrete slab. Weight of cover, frame and thickness of slab shall be as specified in the schedule of quantities or given above.

10.22 DROP CONNECTION

Drop connections shall be provided between branch sewer and main sewer itself in steep ground when the difference in invert level of the two exceeds 45 cms of the required sizes.

Drop connections from gully traps to main sewer on rectangular manholes shall be made inside the manholes and shall have H.C.I. special type door bend on top and heel rest bend at bottom connected by a H.C.I. pipe. This pipe shall be supported by holder bat clamps at 180cms intervals with at least one clamp for each drop connection. All joints shall be lead caulked joints 25mm deep.

Drop connections from branch sewer to main sewer shall be made outside the manhole wall with glazed stoneware pipe tee connections, vertical pipe and bend at the bottoms. The top of the tee shall be finished up to the surface level and provided with a C.I. Hinged type frame and cover 30cms x 30cms. The connection shall be embedded in cement concrete 1:2:4 mix 5cms all round the pipe and tee up to the surface chamber of the tee.

Drop connection made for vertical stacks directly into manholes shall not be considered as drop connections. They shall be paid for under the relevant soil and waste pipes.

10.23 MAKING CONNECTIONS

Contractor shall connect the new sewer line to the existing manhole by cutting the walls, benching and restoring them to the original condition. A new channel shall be cut in the benching of the existing manhole for the new connection. Contractor shall remove all sewage and water if encountered in making the connection without additional cost.

10.24 ROUTE MARKERS

Markers indicating the particular service installed shall be provided along the routes of pipe trenches. Markers shall be of mild steel indicating the type of service installed and the direction of flow painted on it. The markers shall be set firmly in a concrete base and installed at all corner and turning points. Over straight runs, markers shall be spaced at 50 meter centers generally.

10.25 MEASUREMENTS

Excavation

Measurement for excavation of pipe trenches shall be made per linear meter under the respective category of soil classification encountered at site.

- Ordinary soil
- Hard soil (Hard Moorum & Soft Rock)
- Hard Rock requiring chiseling
- Hard Rock required blasting.

Trenches shall be measured between outside walls of manhole at top and depth shall be the average depth between the two ends to the nearest cm. The rate quoted shall be for a depth up to 1.5 m or as given in the schedule of quantities.

Payment for trenches more than 1.5 m in depth shall be made for extra depth as given in the schedule of quantities and above the rate for depth up to 1.5 m.

Timbering and Shoring

Timbering and shoring as described above shall be measured per Sqm and paid for as per the type of timbering or shoring done at site and as per the relevant item in the schedule of quantities. Rate for timbering and shoring shall be for all depths and types of soil classifications including saturated.

Saturated Soil

No extra payment for pumping and bailing out water shall be made for excavation with an average depth of 1.5m in saturated soil, surface water from rain falls or broken pipes lines, or sieves and other similar sources. An extra rate as quoted in the schedule of quantities shall be paid for excavation in saturated soil for pipe trenches above average depth of 1.5 m. No payment is admissible for water collected from surface sources and broken pipe lines or sewers.

Refilling, Consolidation and Disposal of surplus earth

Rate quoted for excavation of trenches shall be inclusive of refilling, consolidation and disposal of surplus earth within a lead of 50 m.

Stoneware Pipes/R.C.C. Pipes/C.I. Pipes

Stoneware Pipes/R.C.C. Pipes/C.I. Pipes shall be measured for the finished length of the pipe line per linear meter i.e., (A) lengths between manholes shall be recorded from inside of one manhole to inside of other manhole, (B) Length between gully trap and manhole shall be recorded between socket of pipe near gully trap and inside of manhole. Rate shall include all items given in the schedule of quantities and specification.

Gully Trap

Gully traps shall be measured by the number and rate shall include all excavation, foundation, concrete brick masonry, cement plaster inside and outside, C.I. grating and sealed cover and frame.

Cement Concrete for Pipes

Cement concrete in bed and all-round or in Haunches shall be paid per running meter between the outside wall of manholes at bottom of the trench. No additional payment is admissible in respect of concreting done for widths greater than specified, for shuttering or centering and concreting in sub soil water conditions.

Manholes

All manholes shall be measured by numbers and shall include all items specified above and necessary excavation, refilling and disposal of surplus earth.

Manholes with depth greater than specified under the main item shall be paid for under "Extra Depth" and shall include all items as given for manholes. Measurement shall be done to the nearest cm. depth of the manholes shall be measured from top of the manhole cover to bottom of channel.

Drop Connections

Drop connections shall be measured by number for a depth of 60cms or part thereof between invert levels. Additional depth shall be paid for as extra per meter depth as per the actual length of the drop connection, measured to the nearest cm.

Making Connections

Item for making connection to municipal sewer shall be paid for by number and shall include all items given in the schedule of quantities and specifications.

Masonry Drains

Payment for masonry drains shall be made under individual items of masonry, cement concrete and plaster by volume or area as given in the schedule of quantities. Except for such drawings which are specifically provided in schedule of quantities.

11. SOIL, WASTE & RAINWATER SYSTEM- GENERAL

Water supply system to cater for drinking, washing, flushing & other requirements shall be installed as called for on the drawings.

The piping system consists of PVC SWR System, UPVC Pipes or other material as called for. The size and makes are specified in the Schedule of Quantities.

11.1 SCOPE OF WORK

Work under the section shall consist of furnishing all labor, materials, equipment and appliances necessary and required to completely install all soil, waste, vent and rainwater pipes as required by the drawings, specified hereinafter and given in the schedule of quantities.

Without restricting to the generality of the foregoing the soil, waste, vent, and rainwater pipes system shall include the following:-

Vertical and Horizontal soil, waste and vent pipes, rainwater pipes and fittings, joints, clamps and connections to fixtures.

Connection of all pipes to sewer and storm water lines as show on the drawings.

Waste pipe connections from all fixtures W.C. Wash Basin, Sinks, Urinals, Kitchen equipment and Plant room equipment.

11.2 TRAPS

External Traps

All traps for drains shall be socket-less tar free cast iron with water seal. Traps shall be provided with an expansion plug and a with a watertight plate at the base for traps located above ground.

Nahani Trap or Floor Traps.

Nahani traps or floor traps shall be cast iron/PVC, deep seal with an effective seal of 50mm. The trap and waste pipes shall be set in cement concrete blocks firmly supported on the structural floor. The blocks shall be in 1:2:4 mix (1cement: 2 coarse sand: 4 stone aggregate 20mm nominal size) and extended to 40mm below finished floor level. Contractor shall provide all necessary shuttering and centering for the blocks. Size of the block shall be 30x30cms. of the required depth.

11.3 FLOOR DRAINS AND ROOF DRAINS

All floor drain shall have 80mm trap.

Floor drains for mechanical room shall have cast iron grating and body. These floor drains shall be heavy duty.

Roof Drains shall have cast iron body, flushing clamp and gravel stop with cast iron dome (JR Smith or equivalent).

11.4 FLOOR TRAP INLET

Bathroom traps and connection shall ensure free a silent flow of discharging water. Where specified, contractor shall provide a special type fabricated GI pipe inlet Hopper without or with one, two or three inlet sockets to receive the waste pipes. Joint between waste hopper inlet socket shall be lead caulked joint. Hopper shall be connected to a CI P or S trap with at least 50mm seal (Hopper and traps shall be paid for separately) floor trap inlet hoppers and the traps shall be set in cement concrete blocks as specified in para above without extra charge.

11.5 C.P./STAINLESS STEEL GRATINGS

Floor and Urinal traps shall be provided with square or round C.P./ Stainless steel grating, with rim of approved design and sharp. Minimum thickness shall be 4 to 5 mm or as specified in the schedule of quantities.

11.6 JOINTING

Cast Iron Soil, Waste vent and rainwater pipes shall be jointed with refined Pig lead confirming to IS: 89-1977 sufficient skein of jute row shall be caulked to live a minimum space for the pig lead as given in 3.1.2 to be poured in. After the pouring the lead shall be caulked into the joint with caulking tool and hammer. All surplus lead should be cut and joint left flush with the edge of socket neatly. Alternative to lead joint, cast iron pipes may also jointed with Drip Seal Joint as per manufacturer recommendations. All PVC SWR systems for drainage shall be jointed with rubber lubricant ring joints.

11.7 CLEANOUTS

A. Provide cleanouts at the base of all soil, waste and leader stacks, all changes in direction of horizontal piping, every floor of sanitary, vent and rain water stacks, all branch connections to stacks and every 15m of horizontal pipe runs. Cleanout plugs shall be threaded and provided with key holes for opening. Cleanout plugs shall be fixed to the pipe by a GI socket and lead caulked joint. UPVC Pipe Cleanouts. Heavy plastic cleanout screw plug in UPVC fitting with sound square nut.

Extend cleanouts to walls and floor with long sweep ells or "wye" and 1/8 bends with plugs and face or deck plates to conform to the architectural finish in the room. Where no definite finish is indicated on drawings provided by other Divisions of this Specification, use stainless steel wall plates and floor plates of nickel bronze.

11.8 WASTE PIPE FROM APPLIANCES

Waste Pipe from appliances e.g. washbasins, sinks, urinals, water coolers, shall be of galvanized steel, as given in the schedule of quantities.

All pipes shall be fixed in gradient towards the out falls of drains. Pipes shall be in chase unless otherwise shown on drawings.

11.9 CONDENSATE INSULATION

Provide 25mm THK expanded rubber insulation on branch waste pipe works for condensate drain from air conditioning system.

11.10 INSTALLATION

General

All materials shall be new and installed in a first class manner.

All drainage piping, unless otherwise indicated, shall be pitched at a minimum rate in direction of flow, in accordance with Indian Standard.

Branch connections to stacks or main drain shall not be made in a manner which will permit backflow. The Contractor shall handle fittings into and out of store to job site and they shall be well protected from damage. After the installation, the fixtures shall be covered by protective crating with carton cardboard and plastic sheets until handover to the employer.

The Contractor shall supply all jointing and fixing materials and seal to structure with polysulphide sealant and make all connections to water supply services, overflows and waste.

All pipe work and fitting connected to pit shall be cast in prior the pouring of concrete.

Rainwater drain shall be set in position and make good. The Contractor shall ensure that the works are coordinated.

Testing

All testing to be carried out in accordance with the local authority requirements and Indian Standards.

During the progress of the work, test the waste and storm drainage systems to permit general construction and building in of rough work to proceed.

Provide all apparatus and temporary work for tests. Take all due precautions to prevent damage to any part of the building.

No caulking of pipe joints to remedy leaks will be permitted except where lead and oakum joints are used.

Each sections of drainage and roughing piping tested shall have all openings tightly closed with screw plugs, or equal device, and shall stand without loss of level for a period of four hours when filled with water to produce at least a 3meter head at the highest point of the section tested.

The Contractor shall maintain system until handover and ensure that at handover the systems are clean and free of blockage.

Test Procedure

Before use at site, all CI soil pipes shall be tested by filling up with water for at least 10 minutes. After filling, pipes shall be struck with a hammer and inspected for blow holes and cracks. All defective pipes shall be rejected and removed from the site.

Pipes shall be tested after installation, by filling up the stack with water. All openings in connection shall be suitably plugged. The total head in the stack shall be, however, not exceed 3 m.

Alternatively, contractor may test all soil and waste stacks by smoke testing machine. Smoke shall be pumped into the stack after plugging all inlets and connections. The stack then be observed for leakages and all defective pipes and fittings removed or repaired as directed by the owner/consultant.

A test register shall be maintained and all entries shall be signed and dated by contractors and got approved from owner/consultant.

11.11 CEMENT CONCRETE

Cast Iron soil and Waste pipes under floor in sunken slabs and in wall chases (vane cut specially for the pipe) shall be encased in cement, concrete in 1:2:4 mix. The encased pipe shall be supported with suitable cement, concrete, pillars of required height and size at intervals as directed by owner/consultant.

11.12 PAINTING

Soil, waste vent and rainwater pipes in exposed location, in shafts and pipe spaced shall be painted with two or more coats of synthetic enamel paint to give an ever shade.

Paint shall be of approved quality and shade, where directed pipe shall be painted in accordance with approved pipe color code.

Waste pipe in chase shall be painted with two coats of bitumen paint covered with polythene tape/fiberglass tissue wrapping and final coat of bitumen paint. Exposed pipes shall be painted with two or more coats of synthetic enamel paint.

Pipe content	Basic colour (150mm)	Colour code indication (100mm)	Basic colour (150mm)
Water			
Drinking	Green	Blue	Green
Grey Water	Green	Black	Green
Reclaimed Water	Green	Black – White – Black	Green
Cooling (primary)	Green	White	Green
Boiler feed	Green	Crimson – White – Crimson	Green
Condensate	Green	Crimson – Em Green – Crimson	Green
Chilled	Green	White – Em Green – White	Green
Heating < 100°C	Green	Blue – Crimson – Blue	Green
Heating > 100°C	Green	Crimson – Blue – Crimson	Green
Cold distribution	Green	White – Blue – White	Green
Hot distribution	Green	White – Crimson – White	Green
Fire distribution	Green	Safety Red	Green
Sea, river-untreated		Green	
Gas			
Natural	Yellow Ochre	Yellow	Yellow Ochre
Manufactured	Yellow Ochre	Em Green	Yellow Ochre
Oil			
Diesel fuel	Brown	White	Brown
Compressed air		Light Blue	
Vacuum	Light Blue	White	Light Blue
Steam		Silver Grey	

11.13 CUTTING AND MAKING GOOD

Pipes shall be fixed and tested as work proceeds. Contractor shall provide on necessary holed out, cuts and chase in structure member as required. Wherever, holes are cut or left originally, they shall be made good with cement concrete 1:2:4 (1 cement :2 coarse sand :4 stone aggregate 20 mm nominal size) or cement mortar 1:2 (1cement : 2 coarse sand) and the surface restored as in original condition.

11.14 MEASUREMENT**General**

Rates for all items quoted shall be inclusive of all work and items given in the above mentioned specification and schedule of quantities and applicable for the work under floor, in shafts or at ceiling level at all heights and depths.

All rates are inclusive of cutting holes and chases in RCC and masonry work and making good the same.

All rates are inclusive of pre-testing and on site testing of the installations, materials and commissioning.

Pipes (Unit of Measurement). Linear meter to the nearest centimeters.

CI (L.A.) pipes shall be measured net when fixed correct to a centimeter including all fittings along its length.

CI soil, waste vent and rain water pipes shall be measured overall along the centre line correct to a centimeter including all fittings along its length. The rate for these pipes shall be inclusive of all fittings, holder bat clamps, lead caulked joints painting and all other items described in the schedule of quantities. Traps structural clamps and cement concrete shall, however, be paid separately under the relevant item.

GI pipes shall be measured per running meter correct to a centimeter for the finished work, which shall includes fittings e.g. bends, tees, elbows, reducers, crosses, sockets, nipples, nuts and painting. The length shall be taken along the centre line of pipes and fittings. All pipes and fittings shall be classified according to their diameter, method of jointing and fixing substance, quality and finish diameter shall be nominal diameter of internal bore. Pipe shall be described as including all cutting and waste. Incase of fitting of an equal bore, the large bore shall be measured.

Cement concrete around pipe shall be measure per along centre of the pipe line measured per linear meter.

Slotted angels/channels shall be measure per linear meter of finished length and shall include support bolts and nuts embedded in machinery which cement concrete blocks nothing extra will be paid for making good the same.

Painting

Painting of pipes shall be measured in meter along the centre line of pipes installed. The rate shall include the painting of clamps, suspenders and supports.

Structural Clamps

Structural Clamps and U clamps shall be paid for by weight per Kg. Rate shall be inclusive of all anchor fasteners, nuts, bolts, drilling, cutting, welding and painting. Weight of clamps shall be calculated from the actual length used in structural member multiplied by its theoretical weight given in manufactures catalogue. Weight of nuts, bolts shall not be taken into account.

Excavation for soil pipes.

No extra payment shall be admissible with respect to excavation / refilling and disposal of surplus earth for cast iron pipes.

11.15 Grease Trap

Grease trap separates grease waste-water, before being discharged. Grease Traps should be easy to clean and the covers should be fitted with stainless steel (SS) bolts. The grease traps should include a SS tray which collects the grease.

Grease Traps should be air-tight with zero chances of leakage or foul odour suitable to install as required.

Should be made of C I casting IS: 210 FG: 220 grade & SS-316 as specified in BOQ & final design.
The standard design parameters should be as under.

Flow - ltr/min	Inlet / Outlet	Holding capacity - ltrs	Bottom to centre of inlet	Bottom to centre of outlet	Width	Length	Height	Grease capacity - kgs
18	50mm	15.9	187.5	187.5	262.5	355	280	3kgs
28	50mm	25.5	203.125	203.125	337.5	404.5	303.125	7kgs
38	60mm	34.3	203.125	203.125	362.5	503.125	303.125	9kgs
57	75mm	47.5	259.375	259.375	400	512.5	343.75	13kgs
75.5	87.5mm	57.5	287.5	287.5	400	562.5	375	18kgs
95	87.5mm	72	300	300	437.5	612.5	412.5	23kgs
132	87.5mm	104	353	353	450	712.5	481.25	32kgs
189	112mm	156	406.25	406.25	556.25	762.5	537.5	45kgs

Grease Trap - Cast Iron (IS:210 FG:220 Grade)	Length	Width	Height	Flow rate	Grease capacity
4" inlet outlet	1305mm	615mm	490mm	100 ltrs/min	25 kgs
6" inlet outlet	1305mm	615mm	490mm	145 ltrs/min	42 kgs
Grease Trap - MS/SS	Length	Width	Height	Flow rate	Grease capacity
2" inlet outlet	510mm	380mm	445mm	50 ltrs/min	11 kgs
2.5" inlet outlet	510mm	380mm	445mm	70 ltrs/min	21 kgs

11.16 Patel Pattern or Collection Pipe/inspection chamber

Patel Pattern Inspection Chamber Box will be used for draining soil & waste pipes from multiple locations into one single outlet typically installed at service floor.

- Chambers should be air-tight with zero chances of leakage or foul odour . Possibility to hung on ceilings at Service floor or basements.
- Should be able to act as hub for numerous pipes (LA class and/or PVC) to converge, for directing flow
- Should be able to customize chamber boxes offering any angles required by design.
- Chamber should have a cover fitted with 10 stainless steel bolts. Easy to clean if chamber clogs.
- Basic material of construction should be C I.

E. DECLARATION

(To be typed and given on the letterhead of the Vendor)

I/We have inspected all the sites of the **Proposed Composite Construction Of A Multi-Storey Building at DN Nagar, Andheri West, Mumbai including civil, plumbing, Electrical and Other Services (Green Building With Gold Rating in LEED /GRIHA) For 122 residential flats for State Bank of India** and I/We have made me/ us fully acquainted with the local conditions in and around the sites of works and proposed layout drawings of works, drawings of each items, TENDER, Technical bid & Price bid, etc. complete.

I/We hereby declare that I/ We have carefully gone through the conditions laid down in the Notice Inviting Tender, General notes, General Conditions of Contract, Special conditions, Schedule of approximate quantities and rates , Form of Agreement, General Specification, Approved manufacturers/ natural source of materials (i.e. all parts of Technical bid), Technical Specifications of schedule of quantities (i.e. all parts of Price bid), and clearly understood all the same and on the basis of the same I/ We have quoted our rates in the Schedule of Quantities/ Price Bid attached with the tender documents.

We accept all the terms and conditions of tender documents. We will abide by the technical specification mentioned in the tender. We here by undertake to use only specified material/ make as per the tender schedule.

I/ We hereby declare that, in particular during execution of all works at site; it will be my/ our sole responsibility to strictly adhere to/ meticulously follow the General Specification, Approved manufacturers/ natural source of materials; Safety, Health and Environmental (SHE) guidelines; Labour Laws; Technical Specifications of schedule of quantities, all drawings of layout and items.

For any type of deviation (to any of above or subsequent instructions), it will be my/our responsibility to obtain the written instruction of the PMC/SBI, appropriate Government Authorities, local bodies for the same failing which it shall be deemed that I have carried out any such deviations at my own and I shall be duty bound to replace all the deviated material/ works from the site at my/ our cost as well as I shall be liable to penalized by the employer as deemed fit and for all such loses made thereof, I/ we shall not have any right to arbitrate in any manner.

I/ We hereby declare that I/ We shall obtain necessary clarifications from PMC/SBI in time and shall uniformly maintain such progress as may be directed by the APMCF/SBI to ensure completion of same within the target date/ time as mentioned in the tender document.

Date:

Signature and seal of Contractor/Tenderer

TECHNICAL SPECIFICATION

ELECTRICAL

INDEX

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I **TECHNICAL SPECIFICATION FOR HT SWITCHGEAR**

1.1 **SCOPE**

This specification covers the technical requirements of design, manufacture, test at works, Supply of HT Switchgear panels complete with all accessories for efficient and trouble-free operation.

1.2 **STANDARDS**

The equipment covered by this specification shall unless otherwise stated, be designed, constructed and tested in accordance with the latest revisions /Amendments of relevant Indian Standards.

IS: 13118 & IEC: 62271	:	Circuit breaker
IS: 3231 & IEC: 60255	:	Electrical relays for power system protection.
IS: 2705	:	Current transformer
IS: 3156	:	Voltage transformer
IS: 375	:	Marking and arrangement for switchgear busbar main connections and auxiliary wiring.
IS: 1248	:	Direct acting electrical indicating instruments.
IS: 6875	:	Control Switches for voltage upto and including 1000V AC and 1200V DC.
IS: 722	:	AC Electricity Meters.
IS: 3247	:	Insulator
IS: 5082	:	Bus-bar and Connections
IEC: 62271	:	A.C. metal-enclosed switchgear and control gear for rated voltages above 1KV and up to and including 52KV.
IEC: 62271	:	Alternating current disconnectors (Isolators) and earthing switches.
IEC: 60529	:	Classification of degrees of protection provided by Enclosures

The panel supplied shall satisfy all the requirements of local statutory authorities and modification if any required at site shall be carried out by the vendor without extra cost.

1.3 GENERAL REQUIREMENTS

The circuit breaker shall be Vaccum.

The Switchgear Panel shall be outdoor, floor mounting, and metal-clad, Horizontal Isolation & Horizontal drawout type, incorporating enclosures for the circuit breaker unit, busbar chamber, current transformers, potential transformers, with necessary meters, relays, switches and auxiliary wiring etc.

The Switchgear shall be of robust construction designed for service in the specified tropical climate.

Each circuit breaker panel shall be extendable in both directions so that individual units may be joined together in any order to form a composite flush fronted switchboard. For this purpose, the busbar chamber shall have removable end covers.

Enclosures shall be manufactured from folded and welded cold rolled sheet steel of minimum 2.0 mm thick. Cable sealing boxes shall have a removable gland plate. They shall be amply sized for the cable and shall be positioned to give adequate ground clearance to facilitate cable termination. The distance between the terminal and the gland plate shall be 600mm minimum.

All retaining catches, screws and bolts for doors and covers shall be Nickel-plated / Zinc Passivated. Gasketing shall be done at suitable locations to achieve IP52 / IP55 degree of protection for dust proof and waterproof.

Equipment's to be mounted on the face of the cubicle shall be flush mounted.
Panel shall be fabricated in suitable transport sections.

All hardware for the complete equipment including foundation bolts, lifting lugs etc., shall be supplied along with the panel.

1.4 LABELS

Labels should be provided to describe the duty of or otherwise identify every instrument, or other item of equipment mounted internally and externally. Switch positions shall be fully identified, wording shall be clear, concise and unambiguous. Each label shall be permanently secured to the panel surface below the item to which it refers.

The labels shall be engraved / anodised aluminium with white letters on black background or non-removable stickers. The nameplates shall not be screwed on.

Each component shall be labeled with a unique tag number corresponding with the tag number

given in the respective wiring diagram.

In additions to component labels, each cubicle door shall bear a large identification label and the board shall include a large, prominent overall identification Label.

Danger labels in local language shall be affixed at prominent places.

1.5 PAINTING

Care shall be taken in workmanship and selection of materials to prevent the occurrence of any form of damage or corrosion due to damp or humid conditions.

The board shall be prepared primed, filed and painted to the highest standards.

All items shall be cleaned and deburred, after fabrication and welding is completed. External surfaces shall be filed and rubbed down as necessary to obtain a perfectly flat smooth surface, free from blemishes and imperfections.

The painting process shall be using a seven-tank phosphating system with primer coating within 4 hours of phosphating and two coats Epoxy Powder Coating.

The paint shall not rust especially in corners and holes.

1.6 CLEARANCE

Minimum insulator lengths and clearances in air shall be not less than those specified for class A switchgear in IS: 3427.

1.7 TEMPERATURE RISE

The design ambient shall be as given in Cl. 1.30 (Particular Requirement). The limits of temperature rise determined in accordance with the relevant standard, shall be reduced such that the final temperature obtained at site does not exceed the final temperature computed by the addition of the design ambient temperature to the permissible temperature.

1.8 CURRENT RATINGS

Every current carrying part of the equipment, including circuit breakers, current transformers, isolating switches, busbars, connectors and joints, shall be capable of carrying its rated current continuously and in no part shall the permissible temperature rise be exceeded.

All Current Ratings indicated are for 'In Panel' Rating.

In addition, all parts of the switchgear, including current transformers, shall be capable of withstanding, without thermal or mechanical damage, the instantaneous peak and the three second short time currents corresponding to the rated making and breaking capacity of the

circuit breaker.

1.9 MAKING AND BREAKING CAPACITIES

Each circuit breaker shall be capable of making and breaking short circuit at 3 phase symmetrical faults and asymmetrical faults.

The supplier shall submit for approval certified test certificates for making and breaking tests carried out on similar equipment by approved independent testing authorities.

1.10 CIRCUIT BREAKER TYPE

All circuit breakers shall be of same make and type and shall be interchangeable.

The breaker shall be able to fulfill the operation duty cycle as specified in Cl. 1.30.

1.11 OPERATING MECHANISM

Circuit breaker closing mechanism shall be 230V single phase 50 Hz Ac motor operated spring charged type.

Tripping / Closing shall be effected by means of a D.C. coils.

Each breaker shall be provided with a visual, mechanised, indicating device for 'Open' and 'Closed' position of the breaker. It shall be operative when the circuit breaker is in the 'Service', 'Test' and 'Isolated/Earthed' position. Operation counters shall be provided on each mechanism to count the number of trips and cannot be reset.

Means shall be provided for coupling the secondary circuits on the fixed portion to those on the movable portion when the circuit breaker is in 'Test' position in order to permit closing, tripping and interlock circuits to be checked for operation test purposes.

It shall not be possible, without the use of tools, to gain access to the tripping toggle or any part of the mechanism which would permit defeat of the locking of the mechanical tripping feature or closing feature, casually or carelessly. The door shall not be openable unless the breaker is tripped.

Operating mechanism shall not hunt if a closing command and an opening command are simultaneously given. It shall first "Close" and then "Open" or first "Open" and then "Close". It shall not be possible to render the electrical tripping feature in-operative by any mechanical locking device.

Tripping shall be by working with current principle.

1.12 CIRCUIT BREAKER ISOLATING FEATURES

The circuit breaker shall have the positions, 'Service', 'Test' and 'Isolated', with the cubicle door closed.

Each circuit breaker shall be connected to the busbars and feeder circuit through plug in type isolating devices.

Plug and socket arrangement shall be provided for connection of secondary circuit.

1.13 CONTACTS

All contact parts shall be readily and quickly replaceable. Where contact parts are, by design, not intended to be dismantled, the complete contact shall be easily replaceable with the minimum of special tools.

1.14 INTERLOCKS

All mechanical interlocks shall be of the preventive type and shall be arranged to prevent mal operation as close as possible to the point at which mechanical force is applied, in order to prevent defeat of the interlocks by distortion of linkages. Electrical interlocks shall also function so as to prevent the closing of the circuit breaker.

Undetachable mechanical interlocks shall be provided which are designed to prevent:

- a. A closed-circuit breaker from withdrawn or inserted into the isolating contacts.
- b. Tripping by attempted isolation.
- c. The closing of a circuit breaker except when correctly located in service, and test Position.
- d. Defeat the inter locks manually by the operating personnel.
- e. Moving the Circuit Breaker from Test to Service Position when the secondary circuit between fixed and moving positions are not completed.
- f. Opening the Door of the Circuit Breaker compartment unless circuit breaker is in Test Position.
- g. Closing of Circuit Breaker when the Earthing Switch in on and vice versa.

1.15 SAFETY SHUTTER DEVICES

A set of metallic shutters shall be provided to cover stationery isolating contacts.

The shutters shall open automatically by a positive drive initiated by the movement of the circuit breaker.

The closing operation shall also be automatic.

When padlocked/closed, the shutters shall completely shroud the stationary contacts, and it shall not be possible to force the shutters or part of the shutters, to gain access to the stationary contacts.

To facilitate testing, means other than locking shall be provided for securing the shutters in the open position. However, such means shall be automatically cancelled and the automatic operation of the shutters restored upon reconnection of the circuit breaker to the fixed isolating contacts.

Busbar shutters shall be painted signal red, and shall be clearly and indelibly labelled 'BUSBARS' in large white letters. Circuit shutters shall be painted yellow, but shall not be lettered, except that on incoming feeders the circuit shutters shall be clearly and indelibly labelled 'DANGER LIVE CABLES' in large red letters.

Durable phase colour identification shall be provided in a prominent position. Provision or access shall be made to lubricate the mechanical linkages.

All shutters shall be effectively earthed.
Shutters shall not operate towards the fixed isolating contacts.

Shutters shall not have less than 20 mm gap from the moving isolating contacts of the circuit breaker throughout the movement of the truck.

1.16 LOAD BREAK SWITCH

Load Break Switch shall be of three pole, manually operated with operation counter and 2 NO + 2 NC Spare Auxiliary Contact in addition to contacts necessary for functional requirement. It shall be provided with spring assistance mechanism for fast making and breaking operation. The LBS shall conform to IS : 9920.

The Load Break Switch shall be provided with tripping facility to trip the switch on fusing of HRC fuses, by strike pin arrangement.

1.16.1 HRC FUSE

The HRC Fuse links shall conform to IS – 12534. All fuse links shall be fixed inside panels so as to face the boards. Sufficient space shall be provided to facilitate easy withdrawal of fuse link.

The Fuse shall be provided with striker pin to trip the Load Break Switch in the event of Fuse Blow Out.

1.17 BUSBARS AND CONNECTIONS

Busbars and connections shall comply with applicable clauses of IS:5082.

The busbars and connections shall be made of high conductivity copper. The busbars shall be amply sized to carry the rated continuous current under the specified ambient temperature without exceeding the total temperature as per Standards.

The busbars and their connections shall be capable of withstanding, without damage, the thermal and mechanical effects of a through fault current equivalent to the short time rating of the switchgear.

Busbars shall be of uniform cross-sectional area throughout their length with connections as short and straight as possible. The resistance of any length shall not be greater than an equal length of similar section conductor, without a joint.

Busbars shall be contained in a separate compartment within the general casing of the switchgear.

Access to busbars and the connections directly thereto shall be gained only by the removal of covers secured by bolts or screws.

Busbars shall be extendable at both ends; such extension shall entail the minimum possible disturbance to the busbar chambers. Compound-filled busbar chambers are not acceptable.

Busbars shall be fully sleeved using heat shrink sleeves of suitable voltage rating.

Phase colours shall be red-yellow-blue and colour sleeves shall be provided at regular intervals.

1.18 EARTHING

All metal parts, other than those forming part of an electrical circuit, shall be connected to a hard-drawn, high conductivity, 25 x 6 mm copper earth conductor on each unit.

It shall be bolted to the main frame and shall be located so as to provide convenient facilities for earthing cable sheaths and for use with earthing devices.

Earth bars of adjacent units shall form a continuous earth bus.

Particular care should be taken to ensure that the truck of draw-out circuit breakers shall be connected to the earth bar through a plug type contact and the plug shall be long enough to allow the busbar and feeder shutters to close before breaking contact.

Current transformer and voltage transformer secondary circuits shall be earthed at one point

only.

The panel shall be provided with two earthing stud terminals with suitable nuts, washers etc., for connection to ground bus.

1.19 INSULATORS

Insulators of moulded or resin bonded material shall have a durable, non-hygroscopic surface finish having a high anti-tracking index.

Provisions shall be made to accommodate expansion and contraction of the connection, to take care the temperature rise likely to be attained during fault conditions.

1.20 CONTROL SWITCHES

Control switches shall be of 'Spring return to neutral' type and selector switches shall be of stay put type. The contacts of all switches shall be shrouded to minimise ingress of dust and accidental contact.

1.21 AUXILIARY SWITCHES

Each circuit breaker shall be provided with auxiliary switches to interrupt the supply to the closing mechanism and to complete the trip circuit, when the circuit breaker is in the 'Closed' position, and to cover all the necessary indication, interlocking and control facilities.

All secondary connections between the fixed and moving portions of circuit breaker equipment shall be by means of plug and socket connections, arranged so as to eliminate, positively any false indication when the moving portion is racked into the service location.

Each circuit breaker shall be provided with 4 NO + 4 NC auxiliary contacts as spare in addition to the other functional requirements.

1.22 CURRENT TRANSFORMERS

The current transformers shall have synthetic resin insulation and be of the single-phase type, with separate core for metering and separate core for protection. The primary and secondary connections shall be clearly labelled. The burdens and accuracy Class of all cores shall be as specified.

All current transformers shall be dimensioned to carry continuously a current of 120% of the rated current. The rated secondary winding current shall be 1A.

1.23 VOLTAGE TRANSFORMERS

The voltage transformers shall be of synthetic resin insulation and be of the single phase type. Rated secondary voltage shall be $110\text{ V}/\sqrt{3}$.

All primary and secondary connections shall be clearly marked and shall be protected by HRC fuses.

The burden and accuracy class of all windings shall be as specified.

1.24 INSTRUMENTS

The instruments shall be digital type with provision of RS 485 communication port.

1.25 RELAYS

All Protection Relays shall be Numerical Type and other Relays can be of Electro Mechanical Type.

1.26 INDICATION LAMPS

Indicating lamps shall be of LED type (22.5mm).

1.27 PANEL WIRING

The switchboard should be completely factory wired and ready for external connections.

The wiring shall be carried out with flexible stranded 1.1 KV grade PVC insulated copper conductor cable (FRLS) of not less than 2.5 sq.mm. size for C.T / current carrying connections.

For other control circuits 1.5 Sq. mm wires shall be used.

All wiring shall be marked by numbered ferrules for easy identification.

The supplier shall provide single line diagram and control wiring diagrams for the switch boards inside the compartment/ panel.

1.28 TESTS**1.28.1 Factory Tests**

Complete tests shall be made at the manufacturer's works to determine the performance and operating characteristics of the assembled circuit breakers, and their respective accessories and to determine whether or not the guarantees have been met. The bidder shall give a complete description of the proposed test methods.

Unless otherwise specified, all tests shall be carried out in accordance with IS : 13118 / IEC: 62271 and shall include the following.

1.28.2 Type Tests

The Bidder shall furnish the (2) sets of type test certificates for all the tests, conducted on similar panel.

1. Short time current test.
2. Temperature rise test
3. One minute power frequency voltage withstand test.

1.28.3 Routine Tests (All Units)

Mechanical operation tests

Power frequency voltage test

Tests on auxiliary & control circuits

Measurement of resistance of the main circuit

All routine and acceptance tests shall be witnessed by the Client.

1.29 DRAWINGS AND MANUALS

All relevant Drawings / Documents / Manuals shall be submitted in six copies for Approval by Client / Consultant.

1.30 PARTICULAR REQUIREMENTS

1.30.1 11KV RMG Panel (Outdoor)

The Ring main Gear panel shall be metal clad, totally enclosed, dust & vermin proof, floor mounted, outdoor type and shall comprise of the following:

Incomer Panel: 2Nos.

Panel shall comprise of the following

- 1No. 11KV 26.3KA, 630A Load Break Switch with Earth Switch
- 1 Set of Copper Bus Bar rated for 630A
- 1 Set of LBS ON / OFF Indication Lamps
- 1No. Cable Termination Box suitable for Terminating 3C x 240 Sq.mm 11KV XLPE AL. Ar. cable
- 1 Set of Cu. Bus bars for earthing.

Outgoing VCB Panel - 1No.

Panel shall comprise of the following.

- 1No. 11KV, 26.3KA, 630A Horizontal Draw out Type Vacuum Circuit Breaker with Shunt Trip / Closing Coil, spring Charged closing mechanism, operation counter & 4NO + 4NC spare auxiliary contacts in addition to contacts necessary for functional requirements.
- 1 Set of Copper Bus Bar rated for 630A
- 1No. Flush Mounting non- lockable breaker control switch with pistol grip handle and spring return to neutral.

- 1No. Master trip Relay
- 1No. Anti Pumping Relay
- 1No. Panel illumination Lamps
- 1Set breaker ON / OFF indication lamps
- 1Set of Phase Indication Lamps
- 1No. Auto trip Indication lamp
- 1No. spring charged indication Lamp
- 1No. Trip Circuit Healthy Indication lamp with test push button
- 3Nos. Resin Cast CTs of 60-30A /5- 5A, 15VA, Cl. 0.5 accuracy
- 3Nos. Resin Cast CTs of 60-30A /5- 5A, 15VA, Cl. 5P 10 accuracy
- 1No. Three Phase Resin Cast Drawout type PT 11000V / 110V, 100VA per phase burden, Cl. 0.5 accuracy, neutral earthed complete with HT & LT fuses.
- 1No. T.P. Non directional IDMT, O/C and E/F Relay
- 1No. Cutout for mounting EB supplied electronic trivector meter
- 1 Set of Space Heater with switch and thermostat
- 1No. Cable termination box suitable for 3C x 240 Sq.mm 11KV XLPE Al. Ar. Cable.
- 1Set of copper busbar for earthing
- 1No. Self powered power pack with 24V DC output for closing / shunt trip Coil and IDMT Relay along with 1KVA Control Transformer.

Note : The panel shall be as per EB standards. The responsibility of obtaining drawings & panel approval from EB shall be in the scope of supplier.

1.30.2 11KV HT SINGLE Panel (Outdoor)

1 No. 11KV, 26.3KA, 630A fully draw out type Vacuum circuit breaker with shunt trip/closing coil, spring charged closing mechanism, operation counter and 4 N/O + 4 N/C spare auxiliary contacts in addition to contacts necessary for functional requirements & interlocks.

- 1 set of copper busbars rated for 630A
- 1 No. panel space heater with switch and thermostat.
- 1 No. flush mounting non lockable breaker control switch with pistol grip handle and spring return to neutral.
- 1 No. anti pumping relay.
- 1 No. panel illumination lamp.
- 1 set Breaker ON/OFF indication lamps.

- 1 No. auto trip Indication lamp.
- 1 No. spring charged Indication lamp
- 1 No. 'Trip circuit healthy' indication lamp with Test PB
- 3 Nos. resin cast CTs of 60/30 / 1-1A, 15VA, Cl 1.0 / Cl 5P 20 accuracy.
- 3Nos. Single phase Resin Cast PTs – 11KV / $\sqrt{3}$ / 110 / $\sqrt{3}$, 100VA per phase and Class.0.5 accuracy.
- The following protections by combined Numerical Transformer Protection Relay shall be provided.
 - IDMT over current relay
 - IDMT Earth Fault relay
 - Instantaneous Over Current relay
 - Instantaneous Earth Fault relay
- Monitoring / Control**
 - 1No, Trivector Meter with MDI & Controller.
 - 1No. Power quality monitoring
 - Cable Termination box suitable for 3C x 240 sq.mm XLPE Al. Ar. Cable
 - 1 Set of copper busbar for earthing.
 - 1No. Self powered power pack with 24V DC/110V AC output.

1.30.3 11KV, Three Panel Board (Indoor Type)

Incomer Panel - 1No.

The panel shall comprise of the following:

- 1 No. 11KV, 26.3KA, 630A fully drawout type circuit breaker with shunt trip/closing coil, spring charged closing mechanism, operation counter and 4 N/O + 4 N/C spare auxiliary contacts in addition to contacts necessary for functional requirements & interlocks.
- 1 Set of Copper busbars rated for 630A
- 1 No. Panel space heater with switch and thermostat.
- 1 No. Flush mounting non lockable breaker control switch with pistol grip handle and spring return to neutral.
- 1 No. Anti pumping relay.
- 1 No. Panel illumination lamp.
- 1 Set Breaker ON/OFF indication lamps.
- 1 No. Auto trip Indication lamp.
- 1 No. Spring charged Indication lamp

- 1 No. Trip Circuit Health Indication Lamp.
- 3 Nos. Resin cast CTs of 60-30A / 1-1A, 15VA Cl. 1.0 / 5P 20 accuracy.
- 3Nos. Single Phase Resin Cast PTs - 11KV / $\sqrt{3}$ / 110 / $\sqrt{3}$, 100VA per phase Cl.0.5 accuracy.
- The following protections by combined Numerical Relay shall be provided.

Protection / Control

IDMT Over current relay

IDMT Earth Fault relay

- Combined Digital Trivector Meter with Maximum Demand Indicator with RS485 Event

Recorder, Harmonic Distortion Analyser

- 1No. Master Trip Relay
- 1No. Cable termination box suitable for 3C x 240 Sq.mm. XLPE Al. Ar. Cable.
- 1No. Trip Circuit Supervision Relay
- 1 Set of Voltage Surge Diverter with separate earthing arrangement
- 1 Set of Copper busbar for earthing.
- 1No. DC Bell for non trip alarm
- 1No. Hooter for trip alarm
- 1No. 4 way static annunciator with communication port and accept / Test / Reset PB with suitable potential free contacts for remote annunciation.

Outgoing Panel - 2Nos.

The panel shall comprise of the following:

- 1 No. 11KV, 26.3KA, 630A fully draw out type Vacuum circuit breaker with shunt trip/closing coil, spring charged closing mechanism, operation counter and 4 N/O + 4 N/C spare auxiliary contacts in addition to contacts necessary for functional requirements & interlocks.
- 1 set of copper busbars rated for 630A
- 1 No. panel space heater with switch and thermostat.
- 1 No. flush mounting non lockable breaker control switch with pistol grip handle and spring return to neutral.
- 1 No. anti pumping relay.
- 1 No. panel illumination lamp.
- 1 set Breaker ON/OFF indication lamps.
- 1 No. auto trip Indication lamp.

- 1 No. spring charged Indication lamp
- 1 No. 'Trip circuit healthy' indication lamp with Test PB
- 3 Nos. resin cast CTs of 60/30 / 1-1A, 15VA, Cl 1.0 / Cl 5P 20 accuracy.
- 3Nos. Single phase Resin Cast PTs – 11KV / $\sqrt{3}$ / 110 / $\sqrt{3}$, 100VA per phase and Class.0.5 accuracy.

The following protections by combined Numerical Transformer Protection Relay shall be provided.

IDMT over current relay

IDMT Earth Fault relay

Instantaneous Over Current relay

Instantaneous Earth Fault relay

Monitoring / Control

- 1No, Multifunction Meter with MDI & Controller.
- 1No. Power quality monitoring
- Cable Termination box suitable for 3C x 240 sq.mm XLPE Al. Ar. Cable
- 1 Set of copper busbar for earthing.
- 1No. Self powered power pack with 24V DC output.
- 1No. 16 way static annunciator with communication port and accept / Test / Reset PB with suitable potential free contacts.

1.31 TECHNICAL REQUIREMENTS OF HT SWITCH GEAR (11KV)

1.	Service	-	Indoor/Outdoor
2.	Type	-	VCB
3.	Nominal System Voltage (KV rms)	-	11
4.	Highest System Voltage (KV rms)	-	12
5.	Rated Continuous Current (Amps)	-	As specified
6.	No. of Poles	-	3
7.	Rated Frequency Hz	-	50
8.	System Earthing	-	Earthed
9.	Rated Short Circuit Breaking	-	26.3 KA
10.	Current (3 phase) KA	-	26.3KA
11.	Duty Cycle t-0.3min	-	O-t-CO
12.	Insulation Levels		
	a. PF withstand voltage dry Kv rms	-	28KV

	b. Impulse withstand Voltage Kv peak	-	70KV
13.	Operating Voltage		
	a. Trip coil	-	24V DC
	b. Closing Coil	-	24V DC
	c. Spring charging motor	-	230V AC
14.	Number and type of spare auxiliary contacts	-	4NO+4NC rotary.
15.	Busbars		
	a. Nominal current rating	-	As Specified
	b. Material	-	Copper
16.	Reference Standards	-	IS: 3427
17.	Painting	-	Epoxy Powder Coating Shade RAL: 7032
18.	Ambient Temperature	-	45° C
19.	Ambient RH %	-	85%
20.	Degree of Protection	-	IP55 (Outdoor) IP52 (INDOOR)
21.	Degree of Protection between the Compartments	-	Metal clad

1.32 GUARANTEED TECHNICAL PARTICULARS FOR 11KV SWITCH GEAR (to be filled by the Electrical Contractor before supply)

1.0	Make	:
2.0	Service	:
3.0	Type	:
4.0	Nominal System voltage	:
5.0	Highest system voltage	:
6.0	Rated continuous current	:
7.0	No. of poles	:
8.0	Rated frequency	:
9.0	System earthing	:
10.0	Rated short circuit breaking current (3 phase)	:
	a. Symmetrical breaking current	:
11.0	Short time making current	:

12.0	Short time withstand current	
	for 3 seconds	:
13.0	Duty cycle	:
14.0	Insulation levels	
a.	Power frequency withstand voltage - dry	:
b.	Lightning impulse withstand voltage	:
15.0	Operating voltage	
a.	Trip coils	:
b.	Closing coils	:
c.	Spring charging motor	:
16.0	Number and type of spare aux. switches	:
17.0	Busbars	
a.	Normal current rating	:
b.	Material and size	:
18.0	Reference standards	:
19.0	Relays Meters & Accessories	:
a.	Type & make	:
b.	Rated current	:
c.	Characteristics	:
d.	Range	:
e.	Drawout / flush mounting	:
f.	Test Facility	:
g.	Reference standard	:
20.0	Auxiliary Relay	
a.	Type & make	:
b.	Rated voltage	:
c.	Operating voltage range	:
d.	No. of contacts	:
21.0	Control switches	
a.	Type & make	:
b.	Type of handle	:

- c. Contact development :
- 22.0 Test terminal block
 - a. Type & make :
 - b. System :
- 23.0 Wiring & Accessories
 - a. Auxiliary wiring minimum cross section of wire for control circuits :
 - b. VT Circuits :

II. TECHNICAL SPECIFICATION FOR DISTRIBUTION TRANSFORMER

2.1 SCOPE

This specification covers the minimum technical requirements for Design, Manufacture, Test at Works and Supply, Installation, Testing and Commissioning of Low Loss, Resin Cast Distribution Transformer.

2.2 STANDARDS

The equipment covered by this specification shall, unless otherwise stated, be designed, constructed and tested in accordance with latest revision / Amendments of relevant Indian Standards / IEC Specification and shall conform to the Regulations of local Statutory Authorities.

IS: 11171	:	Dry Type Transformer
IS 12063	:	Degree Of Protection Provided by Enclosures
IS 3347	:	Porcelain Transformer Bushing
IS 5	:	Colours for ready mixed paint and enamels
IEE C57.12.01-1988	:	Dry Type Transformers
IEC 60726	:	Dry Type Transformers Power Transformers – Insulation levels, dielectric tests and external clearances in air
IEC 60905	:	Loading Guide for dry type transformers
BS 171	:	Power transformers Specification for insulation levels and dielectric tests

2.3 GENERAL REQUIREMENTS

The Transformer shall be of core type, 2 winding, 3 phase Resin Cast Insulated, having a rating by natural air cooling and shall be suitable for indoor installation. The transformer shall be provided with the enclosure with IP20 degree of protection.

2.3.1 Voltage and Frequency Variations

Transformer shall operate without injurious heating on any Tap under the following conditions provided.

- i. At rated MVA at any voltage within $\pm 10\%$ of the rated voltage.
- ii. At 110% rated current at a voltage equal to 105% of the rated voltage and also 85% rated Voltage.

2.3.2 Overloads

It shall be possible to operate the transformer satisfactorily upto overloads of 110% of the

rated value of power. There shall be no limitation imposed by bushings, tap changers, auxiliary equipments to meet this requirement.

2.3.3 Rated Power and Temperature Rise

The transformer shall be capable of operating at rated power specified in the technical schedule on a continuous basis and with the cooling classification given at the most onerous simultaneous climatic conditions specified.

The temperature rise specified shall not be exceeded when the transformer is operated continuously at the rated power at the design ambient temperature of 45° C.

2.3.4 Rated Voltage and Tapping

The principal tapping rated voltage shall be as specified. The tapplings shall provide a variation in the transformation ratio without producing phase displacement. All tapplings shall be full power tapplings.

2.3.5 Impedance Voltage and Short Circuit Impedance

The impedance voltage at principal tapplings shall be as specified.

2.3.6 Short Circuit Withstand Capability

The transformer shall be designed and constructed to withstand capability of 25KA without damage to the thermal and dynamic effects of external short circuits between phases or between phase and ground.

2.3.7 Insulation Levels

The insulation shall withstand the rated lightning impulse voltage and power frequency with stand voltages.

2.3.8 Radio Interference

The transformer shall be designed and constructed so as not to cause any undesirable interference in radio, communication and Networking (PCs) Circuits.

2.4 CONSTRUCTIONAL FEATURES

2.4.1 General

Screws, studs, nuts and bolts shall be hot dip galvanised or of stainless steel or brass as appropriate. The transformer shall be provided with 4Nos. bi-directional flat rollers.

2.4.2 Core

The core shall be built up, interleaved with high grade, non-ageing, low loss, high permeability, cold rolled grain oriented, silicon steel laminations properly treated for core material.

The final core assembly shall be rigidly clamped to ensure adequate mechanical strength and to prevent vibrations during operation.

The frame, clamps and core bolts shall be electrically insulated from the core. Such insulation

shall be capable of withstanding maximum temperature existing in the core. For this purpose, class B or better insulation shall be used.

The core and windings shall be suitably braced to prevent displacement or distortion during transportation or abnormal electrical conditions in service.

The core shall be provided with lugs suitable for lifting the complete core and coil assembly.

All metal parts of the transformer core assembly except individual core laminations, core bolts and side-clamping plates shall be earthed.

2.4.3 Windings

The coils shall be of electrical grade high conductivity copper conductor, properly insulated and stacked.

Coil assembly shall be suitably supported between adjacent sections by insulating spacers and barriers which shall be permanently secured in place.

Windings shall not contain sharp bends which might damage the insulation or produce high dielectric stresses.

Coils shall be so insulated that the effect of impulse and power frequency voltage stresses are minimum. The transformer winding shall be connected in group of vector symbols specified.

2.5 CONSTRUCTIONAL FEATURES

2.5.1 Winding Temperature Scanner

Winding temperature scanner shall be fitted locally in a box so as to be readable at a standing height from Floor level. The device shall be provided with two sets of electrical contacts, one for alarm and one for trip. The set points shall be field settable and can be accessed even when the transformer is in operation.

The scanner shall be provided with RS 485 Modbus RTU for Connection to SCADA.

2.5.2 HV Cable Box

The HV Terminals of the Transformer shall be brought out in to a self supporting sheet steel cable box. The cable box shall be provided with undrilled gland plate.

Adequate space shall be provided within the box so that the 3 core XLPE Aluminium cable may be formed into the box without undue bending or stress on the terminals.

Adequate electrical clearance shall also be provided.

2.5.3 Voltage Surge Protector

The Voltage Surge Protector shall be heat resistant, molded type, designed, manufactured and tested as per IS. The VSP shall be provided with integral thermal protection and with a signaling indicator.

The VSP shall be of Class B and suitably rated for 50KA Surge. The VSP shall be DIN rail mounted and shall form part of each distribution board.

The VSP shall provide suppression from voltage spikes and surges that can occur in all modes. This protection shall be achieved by using high energy absorbing elements.

The VSP shall feature high surge current handling capability to ensure continuity of transient suppression.

2.5.4 Busduct Termination

LV terminals of the transformer shall be brought out to terminate the busduct enclosure.

2.6 ORIENTATION OF TERMINALS

The orientation of the HV and LV terminals shall at 90° Deg.

2.7 TAP CHANGER

2.7.1 Off Circuit tap changing link

The off-circuit tap changing link shall be provided on the H.V. windings for a variation of no load voltage as specified.

2.8 NEUTRAL BUSHING CURRENT TRANSFORMER

Single core current transformer in secondary neutral bushing shall be provided as per the rating given in the technical particulars. The above Bushing Current Transformer shall be provided for restricted earth fault protection.

These Current Transformers shall confirm to IS : 2705.

2.9 PAINTING

Care shall be taken in workmanship and selection of materials to prevent the occurrence of any form of damage or corrosion due to damp or humid conditions.

The enclosure shall be prepared, primed, filled and painted to the highest standards. All items shall be cleaned and deburred, after fabrication and welding is completed. External surfaces shall be filled and rubbed down as necessary to obtain a perfectly flat smooth surface, free from blemishes and imperfections.

The painting process shall be using a seven tank phosphating system with primer coating within 4 hours of phosphating and two coats Epoxy Powder Coating. The paint shall not rust especially in corners and holes.

2.10 TESTS

2.10.1 Factory Tests

Complete test shall be made at the manufacturer's plant to determine the performance and operating characteristics of the assembled transformer (in enclosure), and accessories to determine whether or not the guarantees have been met. Unless otherwise specified, all tests shall be carried out in accordance with the standards and shall include the following.

2.10.2 Routine Tests

- a. Resistance measurement of all windings on the rated voltage connection and on all taps.
- b. Ratio tests on the rated voltage connection and on all taps.
- c. Polarity and phase rotation tests.
- d. Impedance and load loss at rated current on the rated voltage connection and on all the taps.
- e. No-load loss and current at rated voltage on the rated voltage connection and at 95,110 and 115% of rated voltage.
- f. Power frequency withstand test.
- g. Insulation resistance tests.

2.10.3 Type Tests

The Vendor shall furnish two (2) copies of the following type test certificates conducted on similar transformer.

- a. Temperature rise test.
- b. Zero phase sequence test at all taps.
- c. Audible sound level test.
- d. Dry and Wet Impulse voltage withstand test.

2.11 DRAWINGS AND MANUAL

The following drawings and manual in shall be submitted

- a. General arrangement and dimensional drawing with foundation details.
- b. General Arrangement Drawing of Marshalling Box.
- c. Wiring Diagram of Marshalling Box.
- d. Name plate drawing with Terminal Marking and connection Diagrams.
- e. Technical Particular Data-sheet duly completed and filled-in.
- f. Type Test Certificate of Similar rating transformer.
- g. Routine Test Certificates.
- h. Operation maintenance and instruction manuals for transformer and associated fittings.

2.12 TECHNICAL REQUIREMENT FOR DISTRIBUTION TRANSFORMER

SL.NO	DESCRIPTION	RATING
1.	Service	Indoor
2.	Type	Dry Type (RESIN CAST)
3.	Rating (ON) KVA	1250
4.	Rated frequency Hz	50
5.	Number of phase	3
6.	Rated Voltage	
	a. HV winding KV	11
	b. LV winding KV	0.433
	Transformation Ratio	11000V / 415V
	Short Circuit Current Rating	25KA
7.	Vector group	Dyn 11
8.	Type of cooling	AF/AN
9.	Max. winding temperature rise above ambient of 45° C	90° C
10.	Tapping	
	a. Range %	+5/-15%
	b. No. of step	14
	c. Tapping provided on HV side	YES
11.	Tap changer	ONLOAD
12.	Impedance voltage at 75° C	
	a. At principal tapping %	6.25
13.	Terminals	
	a. HV side	Cable Box
	b. LV side	Busduct
14.	Insulation level	
	a. Impulse	
	i) HV KV peak	75
	b. Power frequency	
	i) HV KV RMS	28
	ii) LV KV RMS	3

SL.NO.	DESCRIPTION	RATING
15.	Winding Material	
	a. HV	Copper
	b. LV	Copper
16.	Neutral Bushing CT	
	a. Ratio	2000 / 5A
	b. Accuracy Class	5P20
17.	Quantity	2 Nos.
18.	Vendor shall provide required insulators on the body of the enclosure for fixing the copper Neutral Bus)	

2.13 GUARANTEED TECHNICAL PARTICULARS FOR DRY TYPE
TRANSFORMER (TO BE FILLED IN BY THE TENDERER)

SL. NO.	DESCRIPTION	RATING
		1250KVA
1.	Make	
2.	Service (outdoor/indoor)	
3.	Rating :	
	a. AN	KVA
4.	Rated Frequency	HZ
5.	Number of phases	
6.	Rated voltage	
	a. HV winding	KV
	b. LV winding	KV
7.	Vector group symbol	
8.	Type of cooling	(AN)
9.	Tapping :	
	a. Range	
	b. Number of steps	
	c. Tapping provided on HV side	
10.	Losses (at 75° C and principle tapping)	
	a. No load loss at rated voltage and frequency (KW)	
	b. Load loss at rated current (AN) (KW) - Subject to IS Tol.	
	c. Total loss at maximum rated power (KW)	
11.	Impedance voltage at 75° C	
	a. At principle tapping	%
	b. At maximum tapping	%
	c. At minimum tapping	%
12.	Reactance at rated current and frequency	%

SL. NO.	DESCRIPTION	RATING
13.	Resistance at rated current and at 75° C	
a.	HV	Ohms/phase
b.	LV	Ohms/phase
14.	Zero sequence impedance	%
15.	Zero sequence capacitance of HV winding	micro F
16.	Hottest spot temperature in winding limit to - over an maximum yearly weighted average ambient temperature of 32°C	
17.	Efficiency at 75° C and 0.9 PF	
a.	At full load (AN)	%
b.	At 75% load (AN)	%
c.	At 50% load (AN)	%
18.	Load and PF at which maximum efficiency occurs % full load	
19.	Maximum efficiency	
20.	Regulation at full load and at 75° C	
a.	At unit PF	%
b.	At 0.8 PF lagging	%
21.	No-load current referred to HV and 50 Hz (approx)	
a.	At 90% rated voltage	A
b.	At 100% rated voltage	A
c.	At 110% rated voltage	A
22.	Maximum current density at rated power	
a.	HV winding	

b. LV winding

SL. NO.	DESCRIPTION	RATING
23.	Maximum flux density in iron at rated voltage ratio and at rated frequency	
a.	at 100% rated voltage	Tesla
b.	at 110% rated voltage	Tesla
24.	Maximum clearances in air	
a.	Between phases (HV/LV)	mm
b.	Between phases and ground	mm
25.	Insulation level	
a.	Impulse	
i)	HV	KV (peak)
ii)	LV	KV (peak)
26.	Power frequency	
i)	HV	KV (rms)
ii)	LV	KV (rms)
27.	Winding Type	
a.	HV winding	
b.	LV winding	
28.	Insulation material	
a.	HV winding	
b.	LV winding	
c.	Tapping leads	
d.	Core to LV	
e.	HV to LV	
29.	Details of core	
a.	Core lamination material	
b.	Thickness of lamination	mm

- c. Insulation of lamination
 - d. Type (shell or core)
 - e. Specific loss of core steel at 1.5 tesla
30. Approximate overall dimension
- a. Height mm
 - b. Breadth mm
 - c. Length mm
31. Details of Enclosure
- a. Sheet steel Thickness
 - b. Degree of Protection
 - c. Paint Shade
 - d. Noise Level
32. Weight
- a. Core and windings (approx.) Kg
 - b. Total weight Kgs
 - c. Shipping weight Kgs
33. Neutral Current Transformer details
- a. Type
 - b. Ratio
 - c. Accuracy class
 - d. Burden
34. Reference Standards

TECHNICAL SPECIFICATION- SECTION- III**LT PANEL****3.0 SCOPE:**

3.1 This specification covers the requirements of design, fabrication, testing, packing and supply and erection of Medium Voltage Switchgear (MV Switchgear) consisting of Main MV Panel, APFC and Sub Switch Boards.

3.2 CODES AND STANDARDS:

3.2.1 The Switchgear assembly shall conform to following Indian Standards

- | | | |
|-------|------------------|---|
| 3.2.2 | IS : 375
1963 | - Switchgear bus-bars, main connections and auxiliary wiring, marking and arrangement for (revised). |
| 3.2.3 | IS : 2147 | Degrees of protection provided by enclosures for low-voltage switch gear and control gear. |
| 3.2.4 | IS : 3914 | Selection of AC Induction Motor Starters (voltage not exceeding 1000 volts). |
| 3.2.5 | IS : 4237 | General requirements for switch gear and control gear for voltages not exceeding 1000 V. |
| 3.2.6 | IS : 8623 | Factory-built assemblies of switch gear and control gear for voltages up to and including 1000V AC and 1200 V DC. |
| | | Part II particular requirements for bus bar, trunking system (bus ways). |

3.3 The individual equipment mounted in the switch gear shall conform to following Indian Standards.

- | | | |
|-------|-------------------|--|
| 3.3.1 | IS : 772 Part - 1 | AC Electricity Meters : Part I General requirements and tests (Second Revision). |
| 3.3.2 | IS : 1248 | Direct acting electrical indication instrument(First Revision). |
| 3.3.3 | IS : 1822 | Motor Starters, AC, of voltage not exceeding 1000 Volts (First Revision). |
| 3.3.4 | IS : 2208 | HRC Cartridge fuse links up to 650 V. |
| 3.3.6 | IS : 2607 | Air break isolators for voltages not exceeding 1000 Volts (First Revision). |
| 3.3.7 | IS : 2705 | Current Transformer : |

Part I General requirements.

Part II Measuring Current Transformers.

Part III Protective Current Transformers.

3.3.8 IS : 2959 Contactors for Voltage not exceeding 1000 V AC or 1200 V DC (First Revision).

3.3.9 IS : 3156 Voltage Transformers :

Part I General requirements (First Revision).

3.3.10 IS : 3231 Electrical relays for power system protection.

3.3.11 IS : 4047 Heavy duty air-break switches and composite units of air-break switches and fuses for voltage not exceeding 1000 Volts.

3.3.12 IS : 4064 Air-break switches, air-break disconnectors, air-break switch disconnectors and fuse combination units for voltages not exceeding 1000 V AC or 1200 V DC.

Part I General requirements (First Revision).

3.3.13 IS : 6875 Control switches (switching devices for control and auxiliary circuits including contactor relays) for voltages up to and including 1000 V AC and 1200 V DC

TYPE TESTED PANEL

- IEC 61439-1/2 Low voltage switchgear & controlgear assemblies – Part 2 Power switchgear and controlgear assemblies
- IEC60529 Degrees of protection provided by enclosures
- IEC60947-2 Low voltage switchgear & controlgear – part 2 Circuit breakers
- IEC60947-3 Low voltage switchgear & controlgear – Part 3 Switches, disconnectors, switch-disconnectors & fuse combination units
- IEC60 947-4-1 Contactors and motor starters
- IEC60044-1 Current transformers
- IEC60186 Voltage transformers
- IEEE693 Environmental testing/ Seismic Withstand Level

- IEC61140 Protection against electric shock – Common aspects for installation and equipment – Basic safety publication
- IEC61641 TR Internal Arc Test report

3.4 GENERAL REQUIREMENTS:

- 3.4.1 Medium Voltage Switch Gear shall be metal enclosed, Compartmentalized, Modular type suitable for Indoor installation. The MV Switch gear shall be dust and vermin proof and the enclosure shall be provided with a degree of protection of not less than IP:52 as per IS:2147.
- 3.4.2 The Switch Gear shall be assembled out of vertical sections of uniform height not exceeding 2100 mm and operating handle of fuse switch unit shall not be more than 1800 mm not less than 300 mm.
- 3.4.3 The Switch Gear shall be easily extensible on either side by the addition of vertical sections. It shall be possible to extend the switch gear, irrespective of the type of end panel and the design shall be such as to permit addition of extension panels of a type other than the type of end panel.
- 3.4.4 The switch gear shall be designed to ensure maximum safety during operation, inspection, connection of cables, relocation of outgoing circuits and maintenance, with the bus bar system energized and without taking any special precautions. Means shall be provided to prevent shorting of power and / or control terminal due to accidental dropping of maintenance tools etc., inside the switch board. Checking and removal of components shall be possible without disturbing adjacent equipment. All auxiliary equipment should be easily accessible.
- 3.4.5 All identical equipment and corresponding parts shall be fully interchangeable without any need for resorting to structural modifications.

3.5 FABRICATION REQUIREMENTS:

- 3.5.1 The frame of vertical individual vertical panels shall be fabricated using pressed and shaped cold rolled sheet steel of adequate thickness or by using suitable mild steel structural sections. **The frames and doors of all floor mounted panels shall be fabricated by using minimum 14 SWG cold rolled sheet, 12SWG for Gland plate and 16 SWG sheet steel for wall mounted Sub Switch Boards unless specified otherwise in BOQ.** Wherever required, stiffeners for large size doors and covers shall be provided.
- 3.5.2 The switch gear shall be provided with integral base frame for each vertical panel. The switch gear integral base frame shall be suitable for directly bolting with the help of foundation bolts and shall also be suitable for tack welding at two diagonally opposite points to the embedded base frame.
- 3.5.3 All openings, covers and doors shall be provided with neoprene gaskets around the perimeters to make the switch gear dust and vermin proof.
- 3.5.4 All hardware shall be corrosion resistant. All joints and connections of the panel members shall be made by galvanized, zinc passivated or cadmium plated high quality steel bolts, nuts and washers, secured against loosening.
- 3.5.5 Suitable lifting hooks and / or jacks shall be provided on each panel or shipping section for ease

of lifting of switch board.

3.5.6 Each switch board section shall have removable gland plates with adequate number of holes drilled for fixing glands suitable for the incoming and outgoing cables as specified in box (Cost of glands and lugs are not to be included).

3.5.7 The opening for fixing ammeters, voltmeters, push button indicating lamps, etc., shall be machine punched.

3.6 **PANEL DESIGN REQUIREMENTS:**

3.6.1 The Switch Gear shall be formed using distinct vertical sections each comprising following compartments.

3.6.2 A Completely metal enclosed horizontal bus bar compartment running horizontally at top unless specified at bottom. In case of double bus bar arrangement the bus bars shall be at top and bottom.

3.6.3 Individual feeder modules organized in multitier mode.

3.6.4 Completely enclosed vertical bus bars serving all feeder modules in the vertical panel.

3.5.5 Cable termination compartment.

3.5.6 Perforated sheet steel / insulating material enclosed horizontal auxiliary bus way for control, interlock, indication and metering wiring running horizontally.

3.5.7 Metal sheets shall be provided between two adjacent vertical panels running to the full width of the wider panel and up to full useful height of the switch gear.

3.5.8 Panels shall be in single front execution.

3.5.9 The Switch Gear shall be of compartmentalized, modular construction. All equipment associated with a particular circuit shall be housed in a separate and independent compartment earmarked for that particular circuit in the fixed portion of the vertical panel.

3.5.10 All auxiliary devices for control, indication, measurement and protection such as push buttons, control and selector switches, indicating lamps, Ammeters, Voltmeters, KWH Meters, Protective Relays except motor thermal overload relay and single phasing preventer relay, shall be mounted on the front side of the respective compartment only.

3.6 **Bus bar sizing, connection and supports:**

3.6.1 Bus bars shall be of high conductivity copper or Aluminium as specified and supported on insulators made of SMC/DNC. The current density of copper bus bar shall not exceed 1.2 A /sqmm and that of Aluminium bus bar shall not exceed 0.8 A / sqmm. The main bus bar shall have uniform current rating throughout their length as indicated in the BOQ. The current rating of the neutral shall be half that of the phase bus bars throughout the length of the switch board for bus bar rating above 100 A. For bus bar rating of 100A & below the neutral bus bar shall be rated for phase current. Removable neutral links shall be provided on feeders to permit isolation of the neutral bus bar.

3.6.2 The bus bar and supports shall be capable of withstanding the short circuit currents stated in the BOQ. Only zinc passivated or cadmium plated high tensile strength steel bolts, nuts and washers shall be used for all bus bar joints and supports.

3.6.3 The thermal design of the bus bars shall be based on installation of the switch gear in poorly ventilated condition. The cooling air volume shall take into account only the bus bar

enclosure and not the air volume of the vertical panels. The maximum operating temperature of bus bars at maximum design temperature shall not exceed 85 Deg. C. for normal operating condition and 200 Deg. C for short circuit condition.

- 3.6.4 All bus bars shall be colour coded by means of identifying colour rings at every power tap off point. Red, Yellow and Blue colour shall be used for phase bus bars and Black Colour shall be used for neutral bus bars. They shall be provided with heat shrinkable PVC Insulation of appropriate colour.
- 3.6.5 Minimum clearance between live parts, between live parts / neutral to ground shall be as per CEIG regulations.

3.7 Auxiliary wiring and terminals:

- 3.7.1 Inside the switch boards the wiring for power, control, signaling, protection and instrument circuits shall be done with PVC Insulated copper conductors. The insulation grade for these wires shall be 660 / 1100 Volts. All control wiring shall preferably be enclosed in plastic channels or neatly bunched together.
- 3.7.2 1.5 sqmm size wires shall normally be used provided if the control fuse rating is 10 Amps or less. For 16 Amps control fuse circuit 2.5 sqmm size wires shall be used. Each wire shall be terminated at a separate terminal. Termination of two outgoing wires on a single terminal will not be acceptable. Wires shall not be joined or tied between terminal parts. Shorting links shall be provided for all C.T terminals.
- 3.7.3 Each wire shall be identified at both ends by self-sticking wire marker tapes or PVC Ferrules. Ferruling of wire shall be as per relevant IS.
- 3.7.4 Control Cables shall enter the switch gear from top or bottom. Supporting facilities shall be provided for clamping the control cables in the cabling compartment.
- Inter panel wiring within each shipping section shall be contractors responsibility. For wiring between shipping sections, Contractor shall provide terminal blocks on adjoining shipping sections and supply suitable jumpering wires, interpanel wiring shall be taken through PVC Sleeves or rubber grommets.
- 3.7.5 A minimum of 10% spare terminals shall be provided on each terminal block if requirements of terminals are not specified in the BOQ for starter / control modules. Conductors shall be terminated with adequately sized compression type lugs. "Elmex" (direct conductor termination) type terminals will be acceptable for wires up to 10 sq.mm size and bolted type terminals with crimping type copper lugs shall be provided for all outgoing cable connections larger than 10 sqmm.

3.8 Earthing connections:

All the metal parts of all equipment supplied within the panel (including doors and gland plates) other than those forming part of all electric circuit, shall be connected by means of two independent Earth conductors to continuous copper Earth bar of size 25 x 6 mm. for Main LT panel and 25 x 3mm for Sub panel running along the full length of the panel.

The panel shall be provided with two brass Earthing stud terminals, with suitable nuts, washers etc. for connection to ground bus if required.

3.9 Name plate:

A name plate with the switch gear designation shall be fixed at the top of the control panel. A separate name plate giving feeder details shall be provided for each compartment. A separate name plate giving details of bus section shall also be provided for switch gears having more than one bus sections.

Engraved name plates shall be provided for each equipment (lamps, push buttons, switches, relays, auxiliary contactors etc.) mounted on the switchboard. Special warning plates, one each on each front of a shipping section shall be provided on removable covers or doors giving access to cable terminals and bus bars. Special warning labels should be provided inside the switch board also, wherever considered necessary. Identification tags shall be provided inside the panels matching with those shown on the circuit diagram.

Engraved name plates shall preferably be of 3-ply (Black-White Black) lamicoid sheets or anodized aluminum. Hard paper name plates will not be acceptable. Name plates shall be fastened by screws and not by adhesives.

The Name Plates for feeder compartments shall be in two parts. One part shall have necessary details pertaining to the switch board. The other part shall be removable and shall contain all details regarding the drives / equipment controlled by the particular module.

3.10 Painting:

3.10.1 All metal surfaces shall be thoroughly cleaned and degreased to remove mill scale, rust, grease and dirt. Fabricated structures shall be pickled and then finished to remove any trace of acid. The under surface shall be prepared by applying a coat of phosphate paint and coat of yellow zinc chromate primer. The under surface shall be made free from all imperfections before undertaking the finishing coat.

3.10.2 After preparation of the under surface, the switch board shall be **powder coated** with two coats of final paint. The painting thickness shall not be less than 60 micron. Colour shade of final paint shall be as per manufacturer standard. The finished panels shall be dried in stoving ovens in dust free atmosphere. Panel finish shall be free from imperfections like pin holes, orange peels, runoff paint etc.

3.10.3 The inside surface of all switch boards shall be given a coat of fire retardant paint.

3.10.4 All unpainted steel parts shall be cadmium plated or suitably treated to prevent rust formation. If these parts are moving elements then they shall be greased.

3.11 SPECIFIC REQUIREMENTS OF CIRCUIT BREAKER PANELS:

3.11.1 The Circuit Breaker compartment enclosure shall be IP: 31 as per IS: 2147 with the circuit breaker in position or withdrawn from the compartment. Spring actuated and gravity operated safety shutters shall be provided for isolating the bus bars and other live parts when the breaker is completely withdrawn. The design of the compartment shall be such as to prevent movement of vermin from a particular circuit breaker compartment of any other portion of the panel when the breaker is withdrawn and the compartment door is closed. Suitable isolating barriers shall be provided between circuit breaker and all auxiliary devices for control, indication, metering and protection.

3.11.2 Separate compartment totally enclosed from all sides shall be provided for cable termination, preferably on the rear side of circuit breakers. Access to cables shall be from the rear side

after opening the cabling compartment door. The incoming / outgoing cable termination shall be staggered for each circuit and barriers of sheet steel or insulating material shall be provided between terminations of two circuits such that maintenance on one circuit could be carried out while the other circuit is live. Suitable clamping arrangements shall be provided for cables and cable termination / terminal blocks shall not be used for supporting the cables.

The incoming supply for PCC Panels shall be through top entry bus ducts or through bottom entry single core or multi core Aluminium conductor cables. Similarly the outgoing cabling shall have bottom or top entry as per site requirement. The cable terminations shall be suitably sized for receiving specified number of cables per termination and provision shall be made for terminating each outgoing cable with a separate bolted connection. In case the total number of cables entering a particular panel cannot be accommodated in the cabling compartment of PCC Panels an add on type of panel shall be added to the cabling compartment for accommodating extra cables. The enclosure design of cabling compartment shall be such as to prevent the movement of vermin from this compartment to any other part of the switchgear and shall be minimum IP-31 as per IS-2147 with the door closed. In case of busduct entry, the bus duct shall be terminated on the TOP of the panel only and the rear portion of panel shall not be used for this purpose.

3.11.3 Circuit breaker compartment:

3.11.3.1 The Circuit breaker compartment shall be fully draw out, that is, bus side & outgoing power and control connections shall be through draw out contacts (It shall be possible to draw out the withdraw able trolley without having to unbolt or unscrew any power and control connections). Suitable guides shall be provided to facilitate easy withdrawal of the trolley. All identical feeder compartments shall be interchangeable.

3.11.3.2 The Current Transformers for the Ammeter / Protection Circuits shall be mounted on the fixed portion of the compartment. However, current transformers associated with built-in releases may be mounted on the breaker trolley. All terminals except wiping / sliding type control terminals shall be shrouded with plastic covers to prevent accidental contact. For direct termination - clip on shrouded type (ELMEX Type) terminals shall be provided.

There shall be three positions for the draw out trolley viz:

- “Full in” or “service” position in this position both power and control circuits shall be connected. This shall be then normal operating position of the circuit breaker.
- “Test” position - The power contacts shall be disconnected in this position but the control connections shall not be disturbed, it shall be possible to close and trip the breakers in this positions.
- “Full out” position - Both power and control circuits shall be disconnected in this position. It shall be possible to close the cubicle door in this position.

The circuit breaker shall be lockable in “service”, “test” and “Full out” positions.

The earth connections must remain connected in “test” position, earthing of the unit shall be done with a “pin” or with scrapping earth connections. The earth connection shall be made before the main power / control make and break after the power / control contacts are disconnected. Earthing connection through a plug and socket

connection shall not be acceptable.

3.12 Interlocks:

Following interlocks shall be provided.

- 3.12.1 Compartment doors shall be interlocked against opening when breaker is in closed or ON condition. However, it shall be possible to bypass this interlock for inspection purposes.
- 3.12.2 It shall not be possible to push “in” a drawn out circuit breaker in closed condition or withdraw a circuit breaker in closed condition.
- 3.12.3 It shall be possible to operate a circuit breaker only in the defined “Full in” or “Service” and “test” position inside the panel. It shall not be possible to operate the breaker in intermediate positions while inserting or withdrawing a circuit breaker or while in “Full out”, position inside the panel.
- 3.12.4 Mechanical “Castell” interlocks shall be provided for ACB as shown in the BOQ’ / SLD.
- 3.12.5 In case of multiple ACBs of different rating in a panel, a common cutout shall be provided for the entire range of ACBs. These ACBs shall have rating error prevention device to avoid accidental interchange of ACBs.

3.13 SWITCH GEAR COMPONENTS:

3.13.1 Circuit breakers:

Air Circuit Breakers:

- 3.13.2 Circuit Breakers shall be air break, draw-out type. The ratings shall be as indicated on switch board data sheets and rated for 1000 V insulation rated impulse voltage of 12kV.
- 3.13.2 The circuit breaker shall be provided with microprocessor based for overload, short circuit and earth fault protection.
- 3.13.3 ACB shall have the breaking performance $I_{cs} = I_{cu} = I_{cw} (1 \text{ sec}) = 50 \text{ kA}$
- 3.13.4 ACB shall have minimum Mechanical life of 20000 operations
- 3.13.5 The circuit breakers shall be provided with mechanically operated emergency tripping device. This device shall be available on the front of the panel. If available as a manufacturer’s standard design, mechanically operated ‘closing’ device shall be provided for all breakers except circuit breakers controlling motor feeders
- 3.13.6 The Circuit Breakers shall be provided with minimum 6 Nos. normally open and 6 Nos. normally close spare auxiliary contacts, wires and available for Purchaser’s use, if spare auxiliary contacts are not available, an auxiliary relay of approved make shall be used to multiply the available spare auxiliary contact.

3.13.7 Circuit breaker positions shall be indicated electrically. The following indicating colours shall be used.

BREAKER ' CLOSE '	-	RED
BREAKER ' OPEN '	-	GREEN
BREAKER ' AUTO TRIP '	-	AMBER

In addition, if available as a manufacturer standard design, the following positions / conditions shall also be indicated electrically. The colour to be used shall be as indicated below:

BREAKER DRAW-OUT IN "TEST POSITION"	-	BLUE
BREAKER SPRING 'CHARGED'	-	WHITE

3.13.8 Circuit Breaker positions (OPEN AND CLOSE), location (SERVICE AND TEST) and spring charged condition shall also be indicated mechanically.

3.13.9 All ACBs shall preferably be of single frame size up to 3200A to optimize requirement for spares management.

3.13.10 ACB shall have a Ready to close mechanism preferably having a ready to close mechanical indication on front of ACB. All EDO ACBs ready to close indication contact which shall be used to give a single indication via indicating lamps on panel door if ACB is ready to be closed, after checking all the given conditions (UV release energized, Shunt release de-energized, spring charged, Breaker is not "ON", Breaker has not tripped on fault, Breaker is not mechanically interlocked with other breaker and ACB is not racked in completely in service position) ensuring safety for user and electrical distribution.

3.13.11 Operating Mechanism:

- a. Electrical Power operating mechanism shall be motor would spring charged energy type. However, manual operating mechanism may be of the spring charging stored energy type or the spring assisted type. For circuit breakers with electrical power operating mechanism, provision shall be made for manual spring charging. Closing time of circuit breakers with manual operating mechanism shall be independent of the speed of the operating handle.
- b. All stored energy operating mechanism shall be equipped with following features.
 - (i) Failure of springs, vibrations or shocks shall not cause unintended operating of breaker or prevent intended tripping operation.
 - (ii). Closing of circuit breaker shall be prevented unless the spring is fully charged.
- c. All Electrical power operating mechanism shall be suitable for remote operation and shall be equipped with following features.
 - (i) Provided with universal motor operating on 230 V A.C designed to operate satisfactorily from 85% to 110% of rated voltage.

- (ii) Provided with emergency manual charging facility. The motor shall be automatically, (mechanically) delinked once the manual charging handle is inserted.
- (iii) Closing operation of circuit breaker shall automatically initiate charging of the spring for the next closing operation.
- (iv) One complete closing operation shall be completed once the closing impulse is given and the first device in the control scheme has responded even though the control switch / P.B. is released provided no counter trip impulse is present.

3.13.12 Circuit breaker trip and closing coils in case of electrically operated breakers and trip coil in case of mechanically operated breakers and circuit breaker indications shall be suitable for satisfactory operation on 230 V A.C

3.13.13 Circuit breaker trip and closing coils shall be rated for satisfactory operation under the following conditions of supply voltage variation:

- a). Trip coil : 50% to 110% of rated voltage.
- b). Closing coil : 85% to 110% of rated voltage.

3.13.14 The Circuit Breaker closing time shall be approximately in the range of 100 to 150 Millisecs. and the tripping time shall not exceed 100 Millisecs.

3.13.15 The under voltage relay shall be suitable for operation on 415 V 50 Hz supply and shall be provided with a fixed time delay of 3 to 1 sec.

3.13.16 The breakers shall be trip free. If available as a manufacturer's standard feature, the breakers shall be provided with a release to trip the breakers instantaneously if closed on to a fault.

3.13.17 Shunt trip and closing coil (having common AC/DC supply upto 250V) shall be continuous rated. For Incomer ACBs delayed type under voltage release shall be used to avoid nuisance tripping during voltage surges

3.13.18 Breaker selection shall be done for 50 degree ambient temperature, suitable deration of ACB shall be considered for carrying full load current.

3.13.19 ACB shall have vertical connection for better heat dissipation to avoid over heating problem

3.13.20. Releases:

If provided, the releases shall have the following features:

- ◆ The overload release shall be provided with a setting of 50% to 100% of the breaker rated current.

- ◆ The short circuit release shall be provided with settings corresponding to 200% to 1000% of the rated current and shall be provided with adjustable time delay range of 100 MS to 350 MS.
- 3.13.21 In case of static / microprocessor based relays, separately powered, individual fault trip indication LEDs (For overload, short circuit, earth fault and trip unit failure) shall be available on the trip unit which shall function even if the display fails.
- ◆ All trip units provided shall have thermal memory as standard
- 3.13.22 In case of breakers controlling motor feeders the setting values shall correspond to the motor capacity.

ACB Communications:

Main MV Panel incomer ACBs shall be provided with individual Modbus TCP/Ethernet ports/modules for better speed of data transfer and the parameters available in the trip unit shall be monitor remotely monitor for faster indication of undesired events like circuit breaker tripping on electrical fault, circuit breaker tripping on overload/short circuit/earth fault/trip unit failure etc.

All other Sub panels incomer and Outgoing breakers shall have the capability of status monitoring (ON/OFF/TRIP) feature in required in future.

3.14 Molded Case Circuit Breaker (MCCB):

3.14.1 The circuit breaker 3 or 4 pole shall fully comply with IEC60947(1&2) & IS13947(1&2). Certificate for all the sequences (sequence 1 ,2& 3) should be available. The circuit-breakers shall have no line-load restriction.

3.14.2 Circuit Breakers shall be air break and shall be of current limiting type with positive isolation. They shall be fixed type unless specified otherwise. The ratings shall be as indicated on switch board boq.

3.14.3 MCCBs should have double break mechanism to limit the let through energy and increase the life of the breaker.

3.14.4 They shall be rated for not less than 650 V insulation voltage. The Service breaking capacity shall not be less than the value specified in the switch board boq. ICu of breaker shall be equal to ICs of the breaker.

3.14.5 The MCCBs shall have a rated service breaking capacity (Ics) equal to the ultimate breaking capacity (Icu) at 415V. (Ics = 100% Icu)

3.14.6 In case of motorized MCCB, the motor voltage will be 230 V.

3.14.7 Where accessories such as shunt release, UV Release etc are specified they shall be of drop in type and shall be common for all ranges.

3.14.8 The MCCBs of all ranges shall have CPRI certification or any recognized foreign body.

3.14.9 The breakers shall be operated by a toggle, which shall clearly indicate the three fundamental position ON, OFF & TRIPPED.

3.14.10 All MCCBs shall be provided with Phase Separator, Terminal Covers & Spreader link if the same is available in the Manufacturer's range.

3.14.11 The protection scheme of MCCB shall be as follows unless specified otherwise in the BOQ.

- a) Fixed Thermal over load & Fixed Magnetic protection up to 63 A rating(sub panels)
- b) Variable Thermal over load & Variable Magnetic protection up to 160 A rating(sub panels)
- c) Micro Processor based Variable Overload and Variable Short Circuit shall be provided in Main MV panel

3.15 Switch disconnect fuse units:

3.15.1 All switches shall be load break, heavy duty air break type provided with quite make / break manual operating mechanism suitable for AC23 A duty. The operating handle shall be mounted in the door of the compartment having the switch.

3.15.2 Switches shall be designed to carry the rated current continuously without overheating. The neutral rating shall be same as phase rating. The switches shall have current breaking feature at four points / phase and have self contact cleaning facility.

3.15.3 Barriers shall be provided to prevent interphase arcing and live terminals shall be shrouded to avoid accidental contact.

3.15.4 Rating of heavy duty switches for a particular starter module shall be as indicated in the BOQ. Motor duty switches of smaller ratings meeting the requirements of AC 23 duty as per IS:4064, may be accepted in place of specified heavy duty switch rating for particular starter module only if, motor duty switches have been tested and certified for use with the larger size of fuse link specified in the BOQ for the same Starter module.

3.16 Fuses:

3.16.1 Fuses shall be non-deteriorating HRC Cartridge link type. Diazed fuses are not acceptable.

3.16.2 The fuses shall be provided with operation indicator which shall be visible without removal of fuses from service.

3.16.3 Fuses shall be pressure fitted type and shall preferably have ribs on the contact blades to ensure good line contact.

3.16.4 It shall be possible to handle fuses during off load conditions with full voltage available on the terminals. Wherever required fuse pullers shall be provided. The fuse bases shall be so

located in the modules to permit insertion of fuse pullers and removal of fuse links without any problems.

3.17 **Contactors:**

- 3.17.1 The contactors shall be air break type, equipped with three main contacts and minimum 2 No + 2 NC auxiliary contacts. The main contacts of a particular contactor shall have AC 3 ratings as shown in the relevant switch board data sheet for that contactor.
- 3.17.2 The auxiliary contacts shall be rated for minimum 5 Amps at 240 V AC and 1.3 Amps at 110 V DC (Inductive Load).
- 3.17.3 Unless specified otherwise, the coil of the contactor shall be suitable for operation on 240 V, 1 Ph. AC supply and shall work satisfactorily between +10 and -15% of the rated voltage.
- 3.17.5 The contactor drop off voltage shall be between 15% to 65% of the rated coil voltage.

3.18 **Thermal overload relay:**

- 3.18.1 Thermal overload relays shall be three element, positive acting, ambient temperature compensated type with adjustable setting range.
- 3.18.2 Thermal overload relays shall be manually reset type with the reset push button brought out on the front of the panel. The reset push button shall be capable of being operated without opening the compartment door.
- 3.18.3 Thermal overload relays shall be provided with minimum 1 NO + 1 NC contacts with a rating of 5 Amps at 240 V, 1 Ph, 50 Hz AC and 1.3 Amps at 110 V DC (Inductive Load).

3.19 **Single phasing preventer (SPP):**

- 3.19.1 Unless specified otherwise, SPPs shall be provided in all motor starter modules. The SPP shall be of the current operated type and shall operate on the principle of sensing negative sequence component of current.
- 3.19.2 In case of single phasing, the SPP shall operate after a time delay of 2 to 3 secs. The relay shall be of the hand reset type and visual indication of the relay operation shall be available.
- 3.19.3 The SPP shall be suitable for protection of the non-reversible and reversible motors. The relay operation shall be independent of the loading and RPM of the motor prior to the occurrence of single phasing.
- 3.19.4 The SPP shall be fail safe type and shall operate without any external power supply. In case of failure of internal wiring the relay shall trip the motor.

3.20 **Instrument transformers:**

- 3.20.1 Current Transformers shall be tape wound or wire wound type as specified in data sheets. The CT Ratio and VA burden shall be as indicated in the BOQ.

3.20.2 Current Transformers shall generally conform to IS:2075. The C.T ratio and ratings shall be as indicated in the BOQ. For general guidance the vendor shall note that the protective current transformers shall have an accuracy class "5P" and an accuracy limit factor greater than "10". Low reactance CTs shall be used for protection.

3.20.3 Current Transformers for instruments shall have an accuracy class 1.0 and accuracy limit factor less than 5.0. However, accuracy class of 3.0 is acceptable for Ammeters only. If a metering load is fed from a protection C.T., suitable 1/1 or 5/5 ratio saturable interposing C.Ts shall be used. The current transformers shall be capable of withstanding the peak momentary short circuit and the symmetrical short circuit current for 1.0 second. The neutral side of the current transformers shall normally be earthed through a link.

3.21 **Measuring instruments:**

3.21.1 All measuring instruments shall of the type (Digital or Analogue) as specified in BOQ. They shall be of 96 x 96 mm square pattern, flush mounting type.

3.21.2 In case of Analogue Meters all A.C Ammeter, Voltmeters, KW Meters shall be of moving iron type for A.C and permanent magnet type for D.C. Accuracy class shall be 1.0 for KW / KWH Meters and 1.5 for Ammeters and Voltmeters as per IS : 1248. The range shall be as indicated in the BOQ. Ammeters for motor feeders shall have a non-linear compressed scale at the end to indicate motor starting current. Voltmeter shall be suitable for direct line connection.

3.22 **Control switches:**

3.22.1 All control switches shall be rotary, back connected type, having a cam operated contact mechanism. Preferably phosphor bronze contacts shall be used on the control switches. They shall have pistol grip handles for circuit breaker control and knob type handle for other applications. Ammeter, Selector Switches shall have make before break feature on its contacts. The selector switch shall generally have 4 positions for reading 3 phase currents and fourth position for OFF. The voltmeter selector switch shall also have 4 positions. These shall be used to measure phase to phase voltage and fourth shall be OFF Position.

3.23 **Push buttons:**

3.23.1 Push Buttons Colours shall be as follows:

Stop / open / emergency	-	Red
Start / Close	-	Green
Reset / Test	-	Yellow/Black/White

3.24 **Indicating lamps:**

3.24.1 Indicating Lamps shall be LED type or filament type as specified. In case of filament type the lamp shall be suitable for 6.0 volts A.C complete with built in 240 / 6 V Transformer. Lamps shall be provided with translucent covers to diffuse light.

3.25 **ERECTION:**

3.25.1 It will be the responsibility of Contractor to unload the panel from lorry, store in temporary stores and final shifting for installation.

3.25.2 Before grouting of panels, assembling, paneling and alignment of panels has to be completed.

The panel level shall be checked with spirit levels.

3.25.3 The panels and all components shall be cleaned and checked for tightness of all components. All contacts shall be cleaned with CTC. All relays shall be noted by secondary injection.

3.25.4 Provision of MS channels or angle iron for mounting the panels wherever required across the trench etc., shall be included in the cost of erection and nothing extra shall be paid against the same.

3.26 **MODE OF MEASUREMENTS:**

3.26.1 Panels shall be considered as one unit for purpose of measurement as per BOQ.

TECHNICAL SPECIFICATION- SECTION- IV**CAPACITOR PANEL****4.0 SCOPE:**

This specification covers the technical requirements for Design, Engineering, Manufacture, Testing at Works, Supply at Site, Install, Test & Commissioning of 415V **APFCR Panel**, Complete with all accessories for efficient and trouble – free operation.

4.1.1 STANDARDS:

The equipment covered by this specification shall, unless otherwise stated, be designed, constructed and tested in accordance with the latest revisions of relevant Indian Standards and shall conform to the regulations of local statutory Authorities.

IS: 722	A.C. electricity meters
IS:732	Code of practice for Electrical wiring installation.
IS:375	Marking and arrangement for switchgear Bus bar main connections and auxiliary wiring.
IS:1248	Direct acting electrical indicating instruments.
IS:13947	LV Switchgear and Control Gear.
IS:2705	Current Transformers.
IS :3231	Electricity relays for power system protection.
IS:8623	Specification for factory Built assemblies for Voltages. Up to 1000V A.C. and 1200V.D.C
IS:4237	General requirement of switchgear and Controlgear for Voltages not exceeding 1000V.
IS:2959	AC Contactors for voltage not exceeding 1000V.
IS:162	Specification for Electric Power Switchgear and Associated apparatus.
IS:2834	Capacitors.

The panel supplied shall satisfy all the requirement of Local Statutory authorities and modification if any required site shall be carried out by the Contractor without extra cost.

a) 4.2 CONSTRUCTION:**4.2.1 General:**

- The panel shall be metal-enclosed, free standing, modular type, suitable for indoor installation. The panel shall be dust and vermin proof and the enclosure shall provide a degree of protection of not less than IP-52.
- The fabrication shall be rigid, robust, flaw less and shall have a smooth finish.
- The board shall be fabricated out of 3mm thick mild steel structural sections. The frames shall be enclosed with steel of not less than 2.0mm thickness. The internal partition shall of 1.6mm thick.
- The panel shall be extensible type.
- Each capacitor shall be provided with a fast acting controller which shall be capable of a very fast response.
- Due consideration shall be given to the following during design of the panel and locating the various components viz. Circuit Breakers, Instruments & Relays, Bus bar, Capacitors and secondary wiring.
 - a. Facility for inspection, maintenance & repair
 - b. Minimum vibration and noise.
 - c. Risk of accidental short circuit, open circuit and damage to personnel due to accidental contact with live parts.
 - d. Inter chargeability of components.
 - e. Secure and vibration proof connection for power and control circuit.
 - f. Shrouding of all live parts in feeder component and cable chamber.
- All retaining catches, Screws and bolts for doors and covers shall be zinc passivated. Screws and bolts shall be captive. All covers, doors and joints shall be gasketed.
- Equipment to be mounted outside cubicles shall be flush mounted on cubicle door. No externally mounted equipment shall be mounted above 1.9m or below 0.4m from floor level.
- All components including ACB/MCCB/Contractor shall be of same make.
- Control Circuit shall be protected by MCBs and not by fuses.

BUS BARS:

The bus bars shall be air insulated and made up of high conductivity, electricity Aluminium. All bus bars shall be fully screened by means of PVC sleeves.

The bus bars shall be TP and N, the section of the neutral bus being half the size of the phase bus bar.

The bus bar shall be PVC sleeved with colour strips of red, yellow, blue and black and the same shall be arranged in accordance with IS-375.

The bus bar shall be properly segregated, suitable braced with insulated supports (SMC) placed at appropriate intervals to withstand the electromagnetic stresses during short circuit.

Minimum electrical clearances shall be maintained between phase, neutral and body as per standards.

The insulation used shall be non – hygroscopic and shall be treated for preventing fungus growth.

INTERCONNECTION:

For each and every tapping from the bus bars, separate connections shall be made. No direct tappings from the bus bar shall be made for any feeder without control and protection. All interconnections shall be by rigid bus bars only.

Wherever lugs are used for terminated of rigid bus bars, it shall be soldered and not crimped.

4.5 MOLDED CASE CIRCUIT BREAKER:

The MCCB shall confirm to IS:13947 / IEC947 in all respects. The MCCB shall comprise of switching mechanism, contact system, arc extinguishing device and tripping unit all mounted in a moulded case, made of high strength heat resistant and flame retardant thermosetting insulating material.

MCCB shall employ quick make and quick break switching mechanism independent of the speed of operation of the operating handle. The operating mechanism shall be trip free. The operating handle shall indicate the position of the MCCB in ON / OFF / TRIPPED.

The MCCB shall be provided with Field settable thermal magnetic relay suitable for short circuit and over load protection.

MCCB shall be suitable for horizontal and vertical mounting and with line load reversibility. All MCCB shall be of current limiting type.

b) 4.6 AUTOMATIC POWER FACTOR CORRECTION:

Due to high load fluctuation, automatic control of power factor by capacitors shall be provided to achieve, as far as possible, a power factor of 0.95 lagging.

To achieve the present desired power factor, suitable microprocessor based reactive power controller to switch ON and OFF the regulating stages shall be provided in the panel. The capacitor shall be sub divided into eight regulating stages.

In case of supply interruption, the zero voltage relay shall be provided to reset the control devices to their neutral position as that on restoration of supply, the capacitor stages are switched on again stage by stage, thus preventing any undesirable current / voltage peaks.

The panel shall be provided with auto and manual control of capacitors.

4.7 CAPACITOR:

The dielectric system shall consist of polypropylene film and zinc plating electrode. The dielectric shall be wound to form a winding. The winding shall then be coated under vacuum in self-extinguishing resin which forms the casing.

The Capacitor shall be rated for 550V. Each capacitor unit shall be provided with inrush current limiting devices within the container.

The capacitors shall be enclosed in heat proof, dust proof, water proof containers.

Permissible tolerance of capacitor :

a.	Operating Voltage	:	+/-10% Max.
b.	Output KVAR	:	-5% to +10%
c.	Measured loss angles	:	+10%
d.	Line Current	:	+30% (including transients)

- e. Losses : < 0.2Watts per KVAR.

Note :

The discharge resistor contained in the capacitor unit shall reduce the residual voltage to 50 Volts in less than within 5 minutes after the capacitor unit is disconnected from the source of supply.

4.8 CONTACTORS

Contactor shall be suitable for capacitor duty.

The contactor shall be triple pole, solenoid operated air break type with three main contacts and minimum 2 NO+2 NC spare auxiliary contacts. The main contactor shall be suitable for the duty class III category AC6b as per IS : 2959.

The auxiliary contacts shall be rated for minimum 5 Amps. at 230 V AC single phase.

The contactor drop off voltage shall be between 10% to 75% of the rated coil voltage.

All spare auxiliary contacts shall be wired to the terminal blocks for future purposes.

4.9 REACTORS

The reactors to be specially wound ensuring low hysteresis loss and high harmonic suppression. The reactors shall be provided with filters to take care of 3rd, 5th & 7th 14 th harmonics.

Permissible tolerance of reactors :

- a. Tolerance of Inductance : +/-3%
- b. Saturation Limit : 1.8In

4.10 PUSH BUTTON:

The push buttons shall be provided for ON 1 OFF control.

The push button contacts shall be rated for a minimum of 5A at 230V single phase.

Push button actuator shall be shrouded with dust tight cover, associated springs and contacts shall be of robust construction.

Push button colour shall be as follows:

Stop Open Emergency	-	Red
Start Close	-	Green
Reset Test	-	Yellow / black white

4.11 INDICATING LAMPS:

Indicating lamps shall be LED type.

4.12 MEASURING INSTRUMENTS:

Measuring instrument shall be of digital type with LED display. They shall be capable of carrying the normal full load current (via CTs) and shall not be damaged by effects of rated fault current. The instruments shall have an accuracy class of 1.0 as per IS - 1248.

4.13 CONTROL WIRING:

Panel shall be supplied with all internal wiring comprising of PVC insulated 1.1 KV grade, multi-stranded flexible copper conductor of 1.5 sq.mm cross section.

Wiring associated with a particular phase shall be the colour of that phase viz. Red / Yellow, or Blue, wiring associated with earthing shall be with green colour insulation and for neutral it shall be with black colour insulation.

Wiring shall be neatly laid and run on insulated cleats of limited compression type insulated straps.

All cables shall have crimped terminations and shall be identified by means of glossy plastic ferrules at both ends, showing the wire number as indicated in the schematic diagrams. The ferrules shall be indelibly marked.

Wiring to items mounted on hinged doors or wiring that is subject to movement, shall run in helical binding. The binding shall be securely anchored at both ends and sufficient slack provided to prevent any strain being imposed on wiring.

4.14 EARTHING:

All the metal parts of all equipment supplied within the panel (including doors and gland plates) other than those forming part of all electric circuit, shall be connected by means of two independent earth conductors to continuous copper earth bar of size 25 x 3 mm running along the full length of the panel. The panel shall be provided with two brass earthing stud terminals, with suitable nuts, washers etc. for connection to ground bus.

4.15 TERMINAL BLOCK:

Terminal blocks shall preferably be grouped according to circuit functions and each terminal block group shall have at least 20% spare terminals. Terminal blocks for control circuit shall be of 650V grade with contact ratings not less than 10A and stud type.

Not more than two wires shall be connected to any terminal block.

4.16 LABELS:

Labels shall be provided to describe the duty of or otherwise identify every Instrument, or other item of equipment mounted internally and externally. Switch positions shall be fully identified. Wording shall be clear, concise and unambiguous.

Each label shall be permanently secured to the panel surface below the item to which it refers.

The labels shall be engraved in plastic with white letters in black background.

In addition to component labels, each cubicle door shall bear a large identification label and the panel shall include large, prominent overall identification label.

4.17 PAINTING:

Care shall be taken in workmanship and selection of materials to prevent the occurrence of any form of damage or corrosion due to damp or humid conditions.

The Board shall be prepared, primed, filled and painted to the highest standards.

All items shall be cleaned and deburred, after fabrication and welding is completed. External surfaces shall be filled and rubbed down as necessary to obtain a perfectly flat smooth surface, free from blemishes and imperfections.

The painting process shall be using a seven tank phosphating system with primer coating within

4 hours of phosphating and two coats Epoxy Powder Coating.

The paint shall not rust especially in comers and holes.

4.18 Type Tests

The Bidder shall furnish the (2) sets of type test certificates for all the tests, conducted on similar panel.

1. Short time current test.
2. Temperature rise test

4.19 Routine Tests

1. Mechanical operation test
2. Dielectric tests
3. One minute power frequency voltage withstand test.
4. Any other Test as stipulated by the relevant standards.

4.20 Drawings and Documents

The following drawings and documents shall be furnished.

- a. General Arrangement drawing of the panel showing.
 - i. Overall Dimensions
 - ii Terminal locations
 - iii. Total weight
 - iv Foundation details
 - v. Sectional view
 - vi. Sill of materials
- b. Single line diagram and wiring diagram.
- c. Technical details for Switchgear, lamps, meters etc.
- d. Manufacturing schedule and test schedule,
- e. Calculation for sizing of bus bars.

4.21 MODE OF MEASUREMENTS:

Panels shall be considered as one unit for purpose of measurement as per BOQ.

TECHNICAL SPECIFICATION - SECTION-V**MCB – DB****5.1 SCOPE:**

- 5.1 This chapter covers the Supply, Installation, Testing and Commissioning of **MCB Distribution Boards for lighting and power.**

5.2 STANDARDS:

1. Indian Electricity Act 1910 and Indian Electricity Rule 1956 amended up to date.
2. Relevant Indian Standards Code of Practice for the type of work as per appendix.

5.3 MATERIALS AND CONSTRUCTION:**5.3.1 Enclosure:**

- 5.3.1.1 The Distribution Board shall be fabricated out of 16 SWG Sheet Steel and shall be factory fabricated and shall be duly prewired in the works ready for installation at site.

- 5.3.1.2 The board shall be wall mounted and shall be with stove enameled paint finish. The M.S Sheet shall have undergone degreasing, acid nickeling and phosphating before powder coated painting.

- 5.3.1.3 The board shall be provided with loose wire box as a separate compartment for the complete width and depth of the board and with maximum height of 125 mm in case of TPN DBs and 100 mm in case of Single Phase Distribution Boards.

- 5.3.1.4 The board shall be provided with a hinged cover of 1.6 mm thick sheet steel in the front. DB shall be suitable for indoor / outdoor installation as specified in the BOQ , wall mounting free standing type, in double door construction. The Distribution Boards shall be totally enclosed, completely dust and vermin proof and shall be with hinged doors, Neoprene gasket, padlocking arrangement. All removable/ hinged doors and covers shall be grounded by 4.0 sq mm tinned stranded copper connectors. Only the knobs of the MCB shall protrude out of the sheet covers, through openings neatly machine made for this purpose.

- 5.3.1.5 Knock out holes at the bottom and detachable plates with knock out holes at the top of the board shall be provided.

- 5.3.1.6 Three phase boards shall have phase barriers and wire channel on three sides.

- 5.3.1.7 The board shall be complete with the following.

- a. Solid tinned copper bus bars (Rated 200A)
- b. Solid tinned neutral bars with tapped holes and chase headed screws. For 3 Phase DBs 3 Nos. independent neutral bus shall be provided.
- c. Common Earth Bar.
- d. DIN bar for mounting MCBs.
- e. Elmex type terminal connector suitable for incoming.

- f. Earthing studs.
- g. A set of Neon Indication lamps with HRC Cartridge fuse for each phase of incoming supply where specified.

5.3.1.8 The board shall be fully prewired with single wire PVC Insulated copper conductor of size (as specified below) insulated solid copper links and terminated to extended type terminal connections suitable for connections to the size of respective conductors up to 15 A 2.5 sqmm 25 A 4.0 sqmm 32 A 6 sqmm 63 A 16 Sqmm.

5.3.1.9 All incoming wirings to the MCB DBs shall be terminated only on the elmex type extended connectors provided within the DB. The terminal connections shall therefore be so provided so as to facilitate easy cable connections and subsequent maintenance.

5.3.1.10 A common copper earth bar shall be provided within the loose wire box. The common neutral bar as well as terminal connections shall however be provided within the main compartment just below the loose wire box.

5.3.1.11 Circuit diagram indicating the load distribution shall be pasted on the inside of DB as instructed and covered with transparent plastic sheet. Each board shall have engraved plastic name plates fixed to it in the front.

5.3.1.12 The board shall be provided with two brass earthing stud terminals with suitable nuts & washer for connection to earth bus outside the boards.

5.3.2 **Miniature Circuit Breaker:**

5.3.2.1 The MCBs shall have quick make and break non welding self-wiping silver alloy contacts. They shall have short circuit rating of 10 KA. Each pole of MCB shall be provided with inverse time thermal overload and instantaneous over current magnetic tripping elements with trip free mechanism. In case of multipole breakers, the tripping must be on all poles and operating handle shall be common. Breaker shall conform to BS 3671 with facility for locking in / Off Position.

5.3.2.2 Pressure clamp terminals for conductors up to 2.5 sqmm copper is acceptable and for higher sizes the terminals shall be suitably shrouded.

5.3.2.3 'B' series MCBs shall be used for normal lighting circuits. 'C' series MCBs shall be used for motor loads, Halogen lamp fittings, sodium/ mercury discharge lamps and all power circuits.

5.3.2.4 The ELCB/RCCB's shall have sensitivity as specified in the BOQ and shall be of 10KA breaking capacity. In case they have only overload capability additional MCB for breaking under short circuit condition shall be provided at no extra cost.

5.4 **TERMINALS:**

5.4.1 The distribution board shall be recessed in wall unless specified otherwise with the outer surface of door flush with the wall.

5.4.2 The frame work shall be suitably grouted in the wall. Minor civil works such as cutting of walls and plastering after erection will be included.

- 5.4.3 Shop drawings showing the detailed dimensions and design including the disposition of mountings shall be submitted for approval of consultant before fabrication,
- 5.4.4 All wires to the boards shall be bushed at the entries to avoid damage to the insulation.
- 5.4.5 When it is required to terminate number of conduits a suitable M.S Adaptor box shall be provided as per direction of the Consultant.
- 4.6 5.All distribution shall be touched up for damaged painting before handing over.

5.4 RESIDUAL CURRENT CIRCUIT BREAKER CURRENT OPERATED TYPE (RCBO):

5.5.1 SYSTEM OPERATION

Residual Current Circuit Breaker Operator shall conform to IEC 61008. RCBO shall work on the principle of core balance transformer. The incoming shall pass through the toroidal core transformer. As long as the current in the phase and neutral shall be the same, no electro motive force shall be generated in the secondary winding of the transformer. In the event of a leakage to earth, an unbalance shall be created which shall cause a current to be generated in the secondary winding, this current shall be fed to a highly sensitive miniature relay, which shall trip the circuit if the earth leakage current exceeds a predetermined critical value. RCBO shall be current operated independent of the line voltage, current sensitivity shall be of 30 / 100 mA at 240/415 volts AC and shall have a minimum of 20,000 electrical operations. Earth leakage miniature circuit breakers current operated with a sensitivity of 100 mA. The RCBO shall have Trip free mechanism and shall operate even on neutral failure.

It should provide full protection as envisaged by IE rules – 61-A, 71 – ee, 73 – ee, 1985 and also rule 50 of IE rule 1956.

5.5.2 Mechanical Operation

The moving contacts of the phases shall be mounted on a common bridge, actuated by a rugged toggle mechanism. Hence, the closing /opening of all the three phases shall occur simultaneously. This also shall ensure simultaneous opening of all the contacts under tripping conditions.

5.5.3 Neutral Advance Feature

The neutral moving contact shall be so mounted on the common bridge that, at the time of closing, the neutral shall make contact First before the phases; and at the time of opening, the neutral shall breaks last after allowing the phases to open first. This is an important safety feature which is also required by regulations.

MCB should be having an integrated label holder with dual side din rail locking facility. Incoming & Outgoing should have facility for termination of Bus bar& Cable separately.

Cable termination facility should be up to 35 sq. mm.

5.5.4 Testing Provision

A test device shall be incorporated to check the integrity of the earth leakage detection system and the tripping mechanism. When the unit is connected to service, pressing the test knob shall trip the ELCB / RCCB and the operating handle shall move to the "OFF" position. The ELMCB shall be provided with a Test Push Button to stimulate leakage and test the ELMCB. The ELMCB shall operate and switch off the circuit within 300 milliseconds in case of a fault.

The enclosure of the ELMCB shall be moulded from High quality insulating materials, which shall be fire retardant, anti-tracking, non-hygroscopic, and impact resistant and shall withstand high temperatures.

5.6 MEASUREMENT:

The distribution board complete with the various components specified, indicating lamps supporting trays, internal wiring etc., shall be treated as one unit for purpose of measurement and payment.

TECHNICAL SPECIFICATION - SECTION- VI
LT BUS DUCT (SANDWICH TYPE)

6.1 SCOPE

This specification covers the Technical requirements of design, manufacture, test at works and Supply, Installation, Testing and Commissioning of LT Sandwich type Bus duct.

The Installation of the Bus duct shall be supervised and shall be certified by the manufacturer before charging.

6.2 STANDARDS

The equipment covered by this specification shall unless otherwise stated, be designed, constructed and tested in accordance with the latest revisions of relevant Indian / International Standards.

IS: 8623 (Part II) : Particular Requirement for Bus Trunking System.

IEC: 60529 : Degrees of protection provided by Enclosure.

IEC: 60439 - 1 / 2 : Low Voltage Switchgear and Control gear Assemblies
(Busway)

6.3 CONSTRUCTION

The Bus duct shall comprise of suitably rated bus bars in sandwich configuration, enclosed in a GS / Al. Enclosure.

The Enclosure shall be provided with necessary gaskets / sealant to provide a dust and vermin proof Enclosure.

The construction of the Enclosure shall be rigid and robust and shall be treated to prevent any corrosion. The Enclosure shall act also as an integral ground.

6.4 BUS BARS

The bus bars shall be made up of high conductivity (99.9% Purity) Aluminium in sandwich configuration. The bus bars shall have a continuous current rating as mentioned in the Bill of Quantities and shall have fault withstand capacity as specified. The bus bars shall be TP & N (as specified), the cross section of the Neutral Bus shall be half the size of Phase Bus.

Bus bars shall be identified with red, yellow, blue black and Green colour tape bands at appropriate places and at terminations / Joints.

Both ends of bus duct shall be provided with necessary flexible connector with appropriate termination chamber.

The insulation of bus bars shall be made with Class F insulation to withstand a temperature of 155°C.

6.5 JOINTS

All joints shall be maintenance free and shall employ single bolt operating system. It shall be possible to check the tightness of the joints without de-energizing the Bus duct system.

It shall also be possible to remove one section of the Bus duct in the system, without disturbing the adjacent sections.

6.6 EARTHING

The Enclosure of the Bus duct shall have earth continuity and shall act as integral Earthing.

To meet the local statutory regulations, 2Nos. 25 x 3mm Copper external earth continuity tapes shall be run along the full length of the Bus duct. Necessary cleats / clamps for fixing these Copper tapes to the Enclosure shall also be provided.

6.7 WALL FRAME ASSEMBLY/ WALL FLANGE

Whenever, the Bus duct crosses a wall, suitable wall flange shall be provided to have fire containment. Suitable wall frame assembly shall be supplied along with the Bus duct.

6.8 METHOD OF MEASUREMENT

The length of the Bus duct will be measured along the center line of Bus duct from the flanges with Bus bars. The Vertical and Horizontal Bends will be measured by numbers, excluding the Bus bars.

6.9 TESTS**6.9.1 Type Tests**

The Vendor shall furnish type test reports for the following tests conducted on similar equipment.

1. Short time current test.
2. Temperature rise test
3. One minute power frequency voltage withstand test.

6.9.2 Routine Test and Inspection at Manufacturer's Works

- a. Fully assembled Bus duct shall be offered for inspection at works with prior intimation.
- b. Following routine tests shall be conducted on the Bus duct.
 1. Physical verification check
 2. Megger Test
 3. Power frequency with stand test
 4. Millivolt drop test on Rigid and expansion joints
 5. Temperature Rise Test
 6. Any other tests as stipulated by the relevant standards
- c. The Vendor shall submit the following drawings / documents for Approval.
 1. GA Drawings for Bus duct.
 2. Routing Drawings.
 3. Support Arrangement.
 4. End Termination Box with Copper flexible.
 5. Design Calculation for Copper flexible.

6.10 TECHNICAL PARTICULARS- LT BUSDUCT (SANDWICH)

S. No.	Description		
1.	Service	Indoor / Outdoor	
2.	Type	Sandwich/Compact Air insulated	
3.	Normal System Voltage KV rms	415V	
4.	Rated continuous current	A	As specified in BOQ
5.	Frequency	Hz	50 Hz
6.	System Earthing	Earthed	
7.	No. of Phases	3 ϕ Neutral	
8.	Max. Temp. Rise over design ambient of 50°C	40°C	
9.	Insulation levels		
a.	Power frequency with-stand voltage dry	KV rms	2.5
10.	Short time current for a duration of 1 Second	KA rms	As specified in BOQ
11.	Busbar		
a.	Material	Copper	
12.	Enclosure		
a.	Material	GS / AL	
b.	Degree of Protection	IP : 52 (Indoor)/ IP : 65(Outdoor)	
13.	Bus Insulation	Class B Type with glass- mica & Polyester / Epoxy	

TECHNICAL SPECIFICATION - SECTION-VII**UNINTERRUPTED POWER SYSTEM (UPS)****7.1 SCOPE**

This specification covers the general design and requirements of On Line Uninterrupted Power Supply System for efficient and trouble free operation.

7.2 STANDARDS

The equipment covered by this specification shall unless otherwise stated, be designed, constructed and tested in accordance with the latest revisions of relevant standards.

7.3 OPERATION PRINCIPLE

The main elements of UPS shall consists of IGBT rectifier, inverter, static bypass switch, service bypass with suitably rated batteries for energy storage.

The rectifier output shall be converted by the inverter into 3 Phase AC which shall supply the connected load with constant voltage and frequency.

In the event of mains failure, the battery shall feed power to the loads through the inverter. The rectifier shall be designed such as to supply the inverter as well as charge the battery.

Service by pass shall permit easy maintenance to be carried out.

7.4 DESCRIPTION

The UPS shall be True On Line, Double Conversion Type, Capable of working in standalone, hot standby and parallel redundant configuration.

The Control Circuits shall be Digital Signal Processor based and shall be controlled by internal software

The UPS shall be able to accept an input voltage range of 400 - 20%, +15%at full Load, and up to with a frequency range of 45 - 55 Hz.

The constant output voltage and frequency of UPS shall be as specified in the Technical Particulars.

The Rectifier / Charger shall be fully Digital Signal Processor based, temperature compensated type. Input current harmonics shall be restricted to <3% and power factor should be maintained at > 0.99. The rectifier shall be provided with fuse protection.

The Ripple voltage shall be restricted to less than $\pm 1\%$.

The rating of the rectifier shall be such that it shall not take more than 8 hrs to charge batteries after full discharge.

The inverter shall be fully Digital Signal Processor based vector control technique with IGBTs.

The efficiency of the UPS shall be above 94% at optimum load and in any case shall not be less than 92% at any load.

The Charger shall not be affected by change in phase rotation of incoming supply.

The UPS shall be provided with static bypass switch and transfer to bypass and retransfer shall be automatic. Bypass transfer inhibit switch to be provided to disable bypass for each unit.

The total voltage harmonic distortion at 100% non linear load shall be less than 5% and the THD for 100% linear load shall be less than 1%.

The phase displacement for shall be 120 ± 1 deg.

The UPS shall be capable of withstanding an overloading of 125% for 10 minutes and 150% for 1 minute and 1000% for 20ms (Through Static by pass).

The UPS shall be provided with the following protection.

Over Load

Short Circuit

Input Low Voltage

Output Over Voltage

Battery Over Charging

Battery Over discharging

7.5 CONTROL AND INDICATION

The UPS shall be provided with a control panel for alarm and annunciation of the UPS faults. Also it should be able to store 200 history events

The following alarm and annunciation shall be provided.

Alarm

1. Mains Failure
2. UPS Fault
3. Battery Discharged
4. Inverter Off / Failed
5. Rectifier Off / Failed
6. Over Temperature
7. Over Load
8. Emergency Stop

Annunciation / Indication

1. Input Voltage
2. Input current per phase
3. Bypass input voltage
4. Bypass input frequency
5. Inverter output voltage
6. Inverter output current per phase
7. Input, output and bypass frequency
8. Percentage load at the inverter output
9. Inverter output power factor
10. Inverter output in KVA and KW
11. DC voltage
12. Load crest factor
13. Battery Current

UPS System shall be provided with Serial LAN Port and Serial RS 232 / USB Port and with RS485 for IBMS compatibility.

7.6 DEGREE OF PROTECTION

The degree of protection for the enclosure shall be IP 20 when the door is open.

7.7 COOLING METHOD

Cooling method shall be by forced air with integral fans, filtered with alarms.

7.8 CREST FACTOR

For meeting sensitive electronic loads drawing non linear currents, UPS shall be with a crest factor of 3:1

7.9 BATTERIES

The batteries shall be of sealed maintenance free, lead acid type and shall provide a backup for 15 Min.

7.10 TESTS

All routine tests as per Standards shall be conducted, for proper functioning of the equipment.

7.11 DRAWINGS & DOCUMENTS

The following drawings and documents shall be furnished.

- a. General arrangement drawing showing
 - i. Overall dimensions
 - ii. Total Weight
 - iii. Sectional View
 - iv. Bill of Materials
- b. Manufacturing Schedule & Test Schedule
- c. Calculation for Selection of Batteries for Backup
- d. SLD for Main AC Circuits, Main blocks and Connections
- e. Schematics for Battery Fuses / Switches and Numbered Battery Units
- f. Protocol of Functional Tests and Measurements

7.12 TECHNICAL PARTICULARS-UPS

1.	Rated O/P Power in KVA-	—	10KVA	=	1No.
2.	Input Voltage (V)	-	3 Phase, 415 - 20%, +15%		
3.	Input Frequency (Hz)	-	45 - 55 Hz		
4.	O/P Voltage (V)	-	415V, 3Ø ± 1%		
5.	O/P Frequency (Hz)	-	50 ± 1% (± 2Hz for tracking with mains)		
6.	Efficiency (%)	-	> 94% (Specify actual in GTP)		
7.	Degree of Protection	-	IP 20		
8.	Battery	-	Maintenance Free Lead Acid		
9.	Ambient Temperature	-	45 Deg. C.		
10.	Noise Level (Max)	-	65Db		

11.	Over Load	-	125% for 10 Min
		-	150% for 1 Min
12.	Total Harmonic Content		
	I/P current THDi	-	<3%
	O/P THDv	-	<1%
	With 3 :1 Crest Factor Load	-	<5%
13.	I/P power factor	-	> 0.99
14.	Backup time	-	15min

7.13 GUARANTEED TECHNICAL PARTICULARS FOR UPS

(TO BE FILLED IN BY THE VENDOR)

1.	Make	-	
2.	Model	-	
3.	Rated O/P Power in KVA	-	
4.	Input Voltage (V)	-	
5.	Input Frequency (Hz)	-	
6.	O/P Voltage (V)	-	
7.	O/P Frequency (Hz)	-	
8.	Efficiency (%)	-	
9.	Degree of Protection	-	
10.	Noise Level (<u>Max</u>)	-	
11.	Over Load Capacity	-	
12.	Total Harmonic Content		
	I/P current THDi	-	
	O/P THDv	-	
	With 3:1 Crest Factor Load	-	
13.	I/P power factor	-	-
14.	Backup time	-	
15.	Rectifier	-	
16.	Inverter	-	
17.	Static Bypass	-	
18.	Dimensions	-	
19.	Weight	-	
20.	Heat Dissipation	-	
21.	Warranty	-	

BATTERY

1.	Make	-	
2.	Type	-	
3.	Numbers	-	
4.	AH Capacity	-	
5.	Dimension	-	
6.	Weight	-	

TECHNICAL SPECIFICATION - SECTION-VIII**AVIATION OBSTRUCTION LIGHT****8.1 INTRODUCTION**

This specification covers the Technical Requirements of Design, Manufacture, Test and Supply & Installation of Aviation Obstruction Lighting for Building top.

8.2 STANDARDS

The equipment shall meet all Relevant Indian Standards and shall also meet the requirements of International Civil Aviation Organization.

8.3 CONSTRUCTION

The light shall be Cluster Type LED Type with Twin Lamp configuration suitable for medium intensity. The fitting shall be provided with an inbuilt photo sensor for controlling the lights. The twin lamps shall be independently controlled such that even if one fails, the other one shall function normally. The necessary converters / Adaptors shall be housed in a weatherproof enclosure with IP65 degree of Ingress protection. The fitting should be suitable for operating in hot, humid atmospheric conditions and shall be suitable for operating at 45°C Ambient Temperature without determination of lamp life and internal wiring. The fittings shall be shocking proof and vibration resistant. The fixing arrangement shall be robust and shall be suitable to withstand the wind pressure at the height where it is mounted. The fitting shall be provided with the necessary cable box with a connector for terminating the incoming cable. The fitting shall be provided with earth studs. The complete control gear shall be housed in an enclosure made of non-rust material. The lamp shall have a long life (1,00,000HRs) and shall be maintenance free.

TECHNICAL SPECIFICATION - SECTION- IX**CONDUIT & WIRE****9.1 SCOPE:**

- 9.1 This chapter covers the detailed requirement for **wiring work in Metallic Conduits, Rigid PVC Conduits, Rigid PVC Conduits in surface and recess.**

9.2 STANDARD APPLICABLE:

- 9.2.1. Indian Electricity Act 1910 and Indian Electricity Rules 1956 amended up to date.
- 9.2.2 Relevant Indian Standard codes of Practice for type of work as per appendix.

9.3 MATERIALS:**9.3.1 Conduits and accessories steel conduits:**

The Conduits shall be of steel; the wall/conduit thickness shall not be less than 1.6 mm (16 SWG) for conduits up to 32 mm dia and not less than 2 mm (14 SWG) for conduits above 32 mm dia. They shall be solid drawn or seamed by welding and finished with galvanized or black stove enameled surface as specified in Schedule of Works.

9.3.2 Accessories:

- (i) All conduits' accessories shall be threaded type and under no circumstances pin grip type or clamp grip type accessories shall be used.
- (ii) Bends, Couplers etc., shall be solid type in recessed type of works and may be solid or inspection type as required in surface type works.
- (iii) Minimum 60 mm depth junction boxes shall be used in roof slabs in recessed conduit system and in other places shall be as per IS 2667 - 1977.
- (iv) Saddles for surface conduit work on walls shall not be less than 0.55 mm (24 Gauge) for conduits up to 25 mm dia and not less than 0.9 mm (20 Gauge) for larger diameter. The corresponding width shall be 19 mm & 25 mm.

9.3.3 Rigid PVC conduits:

- (i) The conduit pipes and accessories shall be of suitable material complying with IS:2509 - 1973 and IS:3419 - 1976 for rigid conduits. The interior of the conduits shall be free from obstruction. The make shall be as specified.
- (ii) The conduits shall be circular in cross section. The conduits sizes are designated by their nominal outside diameter. The dimensional details shall be as follows:

S.No.	Nominal Outside Diameter	Maximum Outside Diameter	Minimum Inside Diameter	Maximum Permissible Eccentricity	Maximum Permissible
1	20	20 + 0.3	17.2	0.2	0.5
2	25	25 + 0.3	21.6	0.2	0.5
3	32	32 + 0.3	28.2	0.2	0.5
4	40	40 + 0.3	35.8	0.2	0.5
5	50	50 + 0.3	75	0.5	0.8

Note: All Dimensions are in MM.

9.3.4 Conduit accessories:

- (i) Rigid Conduit accessories shall be normally grip type.
- (ii) Bends, Couplers etc., shall be solid type in recessed type of work and may be solid or inspection type as required in surface type works.
- (iii) Saddles for fixing conduit shall be heavy gauge nonmetallic type with base.
- (iv) The minimum width and thickness of ordinary clips or guarder clips shall be as follows:

ORDINARY CLIPS OR GUARDER CLIPS:

SIZE OF CONDUIT	WIDTH	THICKNESS
20 MM & 25 MM	19 MM	20 SWG (0.9144 MM)
32 MM & ABOVE	25 MM	18 SWG (1219 MM)

- (v) For all sizes of conduits the size of clamping rod shall be 4.5 mm (7 SWG) diameter.

9.3.5 Wires:

- (i) All Wires shall be Multistrand Electric Grade Copper, Zero Halogen FRLS PVC Insulated as per IS 694 and 660 V Grade.
- (ii) The colour coding of wires shall be as follows :

PHASE	COLOUR OF WIRE
R	RED
Y	YELLOW
B	BLUE
N	NEUTRAL – BLACK
EARTH	GREEN (Insulator)
CONTROL	GREY

9.3.6 Outlets:

- (i) **Switch Box:** A rectangular box shall be made of metal on all sides except the front. The boxes shall be made from 16 SWG Galvanized Sheet Steel or MS Cadmium Plated.
- (ii) The depth of the boxes shall not be less than 60 mm and this shall be increased suitably to accommodate mounting of fan regulators in flush pattern.
- (iii) An earth terminal with stud and 2 metal washers shall be provided in each MS box for termination of protective conductors and for connection to socket outlet / metallic body of fan regulator.
- (iii) The size of switch boxes shall be suitable for fixing modular plates of switches of the makes specified.
- (v) The cover of the Switch Boxes shall be molded plates of modular design to fix switches and socket outlets.

9.3.7 Switches & sockets outlets:

- (i) Control Switches shall be of molded plate type of modular design with silver plated contacts.
- (ii) The socket outlets shall be 3 pins with switch and plate type cover. The sockets shall be of shuttered type where specified in the schedule of work.

9.4 INSTALLATION:

9.4.1 Sizing of Conduits:

- (i) Conduit size shall be so selected that the maximum number of PVC Insulated cables including earth conductor that are drawn in one conduit shall not exceed the numbers shown in the table below :

Maximum number of PVC insulated 650 V / 1100 V Grade Aluminium / Copper Conductor Cable as per IS: 694 - 1990.

Nominal Cross Sectional Area of Conductor in Sqmm	20 mm	25 mm	32 mm	38 mm	51 mm
	S B	S B	S B	S B	S B
1.50	5 4	10 8	18 12		
2.50	5 3	8 6	12 10		
4.00	3 2	6 5	10 8		
6.00	2	5 4	8 7		
10.00	2	4 3	6 5	8 6	

16.00	2	3	6	10	
	2	3	5	7	

Note:

- (i) Column headed 'S' apply to runs of conduits which have distance not exceeding 4.25 mm between draw in boxes and which do not deflect from straight by an angle of more than 15 degrees. The columns headed 'B' apply to runs of conduit which deflect from the straight by an angle of more than 15 degrees.
- (ii) The minimum size of conduit shall be 20 mm for lighting and 25 mm for power wiring.

9.4.2 Conduit Points:

- a. Conduit work of each circuit section shall be completed before cables are drawn.
- b. Conduit Pipes shall be joined by sleeved coupler and sleeved accessories only and threads on conduit pipes in all cases shall be 15 mm long, sufficient to accommodate full threaded portion of coupler and accessories in case of metallic conduits. In case of rigid PVC Conduit, all points shall be sealed / cemented with an approved cement.

Cut ends of conduit pipes shall have no sharp edges nor any burs left to avoid damage to the insulation of conductors while pulling than through the pipes.

9.4.2.1 Bends in Conduit:

All necessary bends in the system including diversion shall be done by either neatly bending the pipes without cracking with a bending pipe of not less than 7.5 cm. In respect of rigid PVC Pipes, the bending shall be achieved with an approved method of heating. Alternatively, the bends may be formed by using suitable accessories such as bends, elbows and Junction Boxes (cast iron / MS in case of rigid steel conduits and PVC in case of rigid PVC Conduits).

No length of conduit shall have more than equivalent of four quarter bends from outlet to outlet.

9.4.2.2 Making Chase:

The chase in the wall shall be neatly made and of ample dimension to permit the conduit to be fixed in the wall.

In case of buildings under construction, the conduit shall be buried in the wall before Plastering and shall be refilled and brought to the finish of the wall neatly after erection of the conduit.

In case of exposed brick / marble masonry work, special care shall be taken to fix the conduit and accessories like Switch Boxes and Junction Boxes along with building.

9.4.2.3 Fixing Conduits in Chase:

The Conduit pipes shall be fixed by means of staples, hooks or means of saddles not more than 60 cm apart or by any other approved means of fixing.

All threaded points of conduit pipes shall be treated with some approved preservative compound to ensure protection against rust.

9.4.2.4 Fixing Conduits in RCC Work:

The Conduit pipes shall be laid in position and fixed to the steel reinforcement bars by steel binding wires before the concreting is done. The conduit pipes shall be fixed to the steel reinforcement bars to avoid their dislocation while cement concreting and subsequent curing of the same.

Fixing of standard bends or elbows shall be avoided as far as possible, and all bends shall be maintained by bending the conduit pipe itself with a long radius which will facilitate easy drawing of conductors.

Where conduits pass through expansion joints in the building adequate expansion fittings shall be used to take care of relative movement.

9.4.2.5 Fixing Inspection Boxes:

Suitable inspection boxes to the minimum requirement shall be provided to permit inspection and to facilitate the replacement of wires if necessary.

These shall be mounted flush with the wall or ceiling concrete in case of recessed conduit system. They shall be spaced at not more than 12 meters apart or 90-degree solid bends or equal.

All junction boxes shall be covered by 6 mm clear perspex / 3 mm hylam sheet of approved colour truly cut and fixed with Cadmium Plated Screws & Washers.

9.4.2.6 Surface Conduits:

- a. Conduit Pipes shall be fixed by saddles, with sleeves in an approved manner at an interval of not more than one meter, in respect of rigid steel conduit and 60 cm in respect of rigid PVC Conduit but on either side of coupler or bends or similar fittings. Saddles shall be fixed at a distance of 30 cm from the centre of such fittings.
- b. In long distance straight run of conduits, inspection type couplers at reasonable intervals shall be provided.
- c. In respect of rigid PVC Conduits, if the pipes are susceptible to mechanical damage, they shall be adequately protected.

9.4.2.7 Fixing of Outlet Boxes:

Only a portion of the switch box shall be sunk in the wall, the other portion being projected out for suitable entry of conduit pipes into the box.

9.5 LIGHT AND POINT WIRING:

9.5.1 Wiring System:

- (i) Unless and otherwise specified in the tender documents, wiring shall be done only by the

- looping system and the phase conductor shall be looped at the switch boxes and neutral conductors at the point outlets.
- (ii) Where joint box system is specified in the Tender Documents, all joints in the conductors shall be made by means of approved mechanical connectors in suitable and approved junction boxes.
 - (iii) The wiring throughout the installation shall be such that there is no break in the neutral wire except in the form of linked switch gear.
 - (iv) while drawing insulated wires into the conduits, care shall be taken to avoid scratches and kinks which may cause breakage of conductors. There shall be no sharp bends.
 - (v) Insulation shall be shaved off for a length of 15 mm at the end of wire like sharpening of a pencil and it shall not be removed by cutting it square or ringing. PVC insulated copper conductor wire ends before connection shall be properly soldered (at least 15 mm length) with soldering flux / copper solder, for copper conductor. Strands of wires shall not be cut for connecting to the terminals. The connecting brass-screws shall have flat ends. All looped joints shall be soldered and connected through terminals block / connectors. The pressure applied to tighten terminal screws shall be just adequate, neither too much nor too less. Conductors having nominal cross section are exceeding 4 sq. mm shall always be provided with crimping type cable sockets. At all bolted terminals, brass flat washer of large area and approved steel spring washers shall be used. Brass nuts and bolts shall be used for all connections.
 - (vi) Only certified wire man and cable jointers shall be employed to do joining work.

9.5.2 Joints in Wiring:

- 9.5.2.1 No joints in wiring are permitted. If the length of final circuit to a submain is more than the length of a standard coil, thus necessitating a through joint, such joints shall be made by means of approved mechanical connectors in suitable junction boxes.
- 9.5.2.2 The termination of multi strand conductors shall be done using suitable crimping type lugs for size more than 4 sqmm.
- 9.5.2.3 All final branch circuits conductors shall be as shown in schedule of quantities and a drawing.
- 9.5.2.4 No other wiring shall be bunched in the same conduit except those belonging to the same phase and each conduit shall not have more than three branch circuits. In case of three phase loads, separate conduit shall be run from distribution boards to the load points or outlets as the case may be.

9.5.3 Earthing Requirements:

- (i) The entire system of metallic conduit work including the outlet boxes and other metallic accessories shall be mechanically and electrically continuous by properly screwing at points or by double check nuts at termination. The conduit shall be continuous when passing through walls or floors.

- (ii) Protective (loop earthing) conductor (s) shall be laid along runs of the conduit between the metallic switch boxes and the distribution boards / switch boards, terminated thereto. The protective earth conductors shall be drawn either inside conduits along with cables, or shall be laid external to the conduits. When laid external to the conduits, this shall be properly clamped with the conduits at regular intervals.
- (iii) The protective conductors shall be terminated properly using earth studs, earth terminal block etc., as the case may be.

9.5.4 Fixing of Wiring Accessories:

- (i) Control Switches shall be 'ON' when the knobs are down. Control Switches shall be placed only on the live conductors of the circuit. No single pole switch or fuse shall be provided in the protective earth conductor or earthed neutral conductor of the circuit.
- (ii) The Switch Box and socket outlets shall be installed based on interior drawing. The contractor shall submit for approval, the shop drawing of conduit layout indicating the route of the conduits, number and size of the conduits, location of junction / inspection / pull / outlet boxes, size and location of switch boxes, number and size of wires pulled through each conduit and all other necessary relevant details prior to laying of conduits. Only after the drawings are approved, the contractor shall proceed with the work of conduit laying. In respect of rooms the location shall follow approved mock-up room layout. As location of outlets are liable for change, only such changes which are done after approval of shop drawing in respect of public area and after approval of mock-up room in respect of rooms will be eligible for payment as actuals.

9.6 MEASUREMENTS:

9.6.1 Point Wiring:

The wiring for light points, fan points, exhaust fan points, bell points and 5A Socket outlets shall be measured on unit basis by counting.

The following shall be deemed to be included in the point wiring.

- ◆ Conduits and its accessories and wiring cables between the switch box and point outlets.
- ◆ All fixing accessories such as saddles, screws, rawl plugs etc., as required.
- ◆ Metal / PVC Switch Boxes for control switches, regulators sockets etc., and moulded plate cover / plastic laminated sheet cover for the same.
- ◆ Outlet boxes, junction boxes, pull through boxes etc., along with their covering sheets including metal boxes provided with switch boards for housing wire / conduit termination.
- ◆ Control Switch / MCB as specified.
- ◆ 3 Pin or 6 Pin Socket, ceiling rose connector as required.
- ◆ Connection to ceiling roses, connector socket outlet, lamp holder / switch etc.,
- ◆ Inter connecting wiring between points of the same circuit on the same switch box and from another.

- ◆ Protective (loop earthing) conductor from one metallic switch box to another in the distribution circuits and for socket outlets, fittings etc., (The length of protective conductor run along with the circuit / submain is excluded from scope of points).

9.6.2 Group Controlled Point Wiring:

In case of point with more than one point controlled by same switch / MCB, such points shall be measured as one point depending on the number of points controlled by one switch / MCB as specified in the BOQ. The wiring from the MCB the first point connected to the MCB shall also be considered as one point.

9.6.3 5/6 Amps Socket Outlet:

9.6.3.1 Dependent Socket Outlet:

Each socket outlet fixed in switch box and phase, neutral and earth looped from adjacent / same switch box shall be measured as one dependent socket outlet.

9.6.3.2 Independent Socket Outlet:

Socket outlet including boxes, socket, switch etc., for which conduit is laid and wires drawn from DB or another switch board shall be measured as Independent Socket outlet.

9.6.4 15/16 Amps Socket Outlet:

The wiring for these outlets shall be paid as linear basis if wiring is not specified in the BOQ and the outlets shall be measured as unit basis.

9.6.5 Submain and Circuit Wiring:

These shall be paid on a linear basis along the run of the conduit, including junction boxes, bends etc., from the top of the distribution board to the top of the switch board. **No extra rate shall be paid for light circuit mains for looping switch board to switch board.**

9.6.6 System of Wiring :

The system of wiring is with ZH - FRLS PVC insulated copper wire with ZH-FRLS PVC insulated copper wire as for earth protection wire.

TECHNICAL SPECIFICATION- SECTION-X**CABLING****10.1 SCOPE:**

This specification covers the requirements for supply, the installation, testing and commissioning of the **cabling system**.

10.2 STANDARDS:

The following standards shall apply :

IS : 1554 Part I

IS : 2098 Part II

IEE Wiring regulation.

10.3 CABLE SPECIFICATIONS:**10.3.1 Power Cables :**

Power cables for use on 415 V systems shall be of 1100 Volts grade, aluminium stranded conductor, PVC insulated, PVC sheathed steel wire / tape armoured and overall PVC sheathed. Power Cables for 11 KV systems shall be aluminium conductor, XLPE insulated, screened, PVC bedded galvanized steel flat armoured and PVC sheathed cable. Cable shall confirm the relevant IS Specification.

10.4 MISCELLANEOUS MATERIALS SPECIFICATIONS:**10.4.1 Connectors:**

Cable termination shall be made with aluminium / tinned copper crimped type solderless lugs of approved make for all aluminium conductors and stud type terminals.

10.4.2 Method of Construction:

Before fixing the lugs to the cable end, the cable end to the equivalent length of the lug shall be prepared by removing the outer PVC insulation along with the steel armouring and then, the inner PVC insulation. The paste shall be applied to the cable lead and inside the lug prior to the inserting of lug on the cable lead. The lug shall then be crimped with hydraulic / mechanical type heavy duty crimping tool. The crimping shall be done in such a manner that there shall be no air gap. Then the crimped portion shall be wrapped with the PVC insulation tape. (Colour of tape shall be of that of cable lead) The above method shall be carried out for all the cores. The cable end with lug shall then be terminated into the terminal and then be tightened with either brass nuts or Cadmium plated nuts as directed by Engineer in-charge.

10.4.3 Cable Identification:

Cable tags shall be of 2 mm thick, 20 mm wide aluminium strap or plastic identification tags of suitable length to contain cable number, equipment no. etc.,

10.4.4 Cable Glands:

Cable Glands to be supplied shall be double compression type. Made of high purity brass metal, nickel plated with brass washers, rubber rings, threaded stud with washers and nuts.

10.4.5 Method of Construction:

Before erection of gland, the cable end shall be prepared by removing the outer PVC insulation up to the point where gland to be fixed, by assessing the length of leads required. Bottom portion of gland shall be inserted over the steel armouring, and then armour strips shall be bent for the length of collar of gland, remaining length of armoring shall be cut. The cable end shall then be, inserted through the entry of plate where the cable is to be terminated. The top portion of gland with washer shall be then inserted in such a manner that the bent armour strip should be touching the surface of the entry. The nuts shall be tightened with spring washers over the projected stud portion. Fixing of gland shall be at right angle to the gland plate. Tightening shall assure continuity of earth. Hole to the gland plate shall be punched / knocked out, of correct diameter with respect to gland size.

10.4.6 Cable Route Marker

Cable route marker shall be cast iron Circular plate having 100 mm dia. and 6 mm thick with Iron rod for fixing of cable indicator plate: 700 mm long galvanized iron rod of 12mm dia., and 150 mm long cross bar welded at bottom or hook to be made with same continuous bar.

10.4.7 Method of Construction:

Cable indicator plate fixed/welded to the 700 mm long iron rod or angle, with 150 mm cross bars welded at bottom as fasteners or bent in 'J' shape to hook the cable in the bent portion, shall be buried along the route of cable in the trench made for laying the cable. For clear visibility, the Cable indicator plate shall be buried in such a manner that the plate should be minimum 200 mm above the ground level and shall be provided at not more than 30metre in straight run, at both ends of road crossing and immediate before and after turning point of cable.

10.5 CABLE LAYING:

10.5.1 Cables shall be laid in the routes marked in the drawing. Cable routes given on the layout drawings shall be checked in the field to avoid interference with structures, heat sources, drains, piping, air conditioning duct, etc., and minor adjustments shall be done to suit the field conditions wherever deemed necessary without any extra cost. Procurement of cables shall be on the basis of actual site measurements.

10.5.2 High voltage, medium voltage and other control cables shall be separated from each other by adequate spacing or running through independent pipes, trenches or cables trays, as applicable.

10.5.3 Bending radius of different type of the cable shall not be less than values shown the Table Below:

Table Minimum bending Radius for Cables

S.No.	Voltage level of cables	Single core		Multi core
		Unarmoured	Armoured	Multi core
1	Up to 11 kV	20 D	15 D	12 D
2	Up to 22 kV	25 D	20 D	15 D
3	Up to 33 kV	30 D	25 D	20 D

Note: D diameter of cable.

Wherever possible, 25 percent larger radius than the specified above should be used.

10.5.4 Cables shall be laid in complete, uncut lengths from one termination to the other whenever the length required is not more than Standard lengths manufactured.

10.5.5 Cables shall be neatly arranged in the trenches / trays in such a manner so that crisscrossing is avoided and final take off to the motor / switch gear is facilitated.

10.5.6 All cables will be identified close to their termination point by cable numbers as per cable schedule. Cable numbers will be punched on aluminium straps (2 mm thick) securely fastened to the cable and wrapped around it. Alternatively plastic cable tags may be used.

Even underground cables shall be provided with identity tags of lead securely fastened every 30 Mtr. of its underground length with at least one tag at each end before the cable enters the ground. In unpaved areas, cable trenches shall be identified by means of markers. These posts shall be placed at location of changes in the direction of cables and at intervals of not more than 30 Mtr. and at cable joint locations.

10.5.7 RCC Cable trenches with removable covers will be provided by the Owners. Cables shall be laid in 3 or 4 tiers in these trenches on steel supports. Removal of concrete covers for purposes of cable laying and reinstating them in their proper positions after the cables are laid shall be done by the electrical contractor at no extra cost.

10.5.8 TRENCHING

10.5.8.1 WIDTH OF TRENCH AS PER SITE CONDITION REQUIREMENT:

- The minimum width of the trench for laying a single cable shall be 35cm.
- Where more than one cable is to lay in the same trench in horizontal formation, the width of the trench shall be increased such that the inter-axial spacing between the cables, except where other specified, shall be at least 20cm or as per cable manufacturer recommendation.
- There shall be a clearance of at least 15cm between axis of the end cables and the sides of the trench.

10.5.8.2 DEPTH OF TRENCH

- Where the cables are laid in a single tier formation, the total depth of trench shall not be less than 75cm for cables up to 1.1 KV and 1.2 m for cables above 1.1 KV.
- When more than one tier of cables is unavoidable and vertical formation flying is adopted, the depth of the trench in above shall be increased by 30cm for each additional tier to be formed.
- Where no sand cushioning and protective covering are provided for the cables as per above, the depth of the trench as per above shall be increased by 25cm.

10.5.8.3 LAYING OF CABLE IN TRENCH

- a. The trench shall then be provided with a layer of clean, dry sand cushion of not less than 8cm in depth, before laying the cables therein.
- b. However, sand cushioning as per above need not be provided for MV cables, where there is no possibility of any mechanical damage to the cables due to heavy or shock loading on the soil above. Such stretches shall be clearly specified in the tender documents.

10.5.8.4 TESTING BEFORE LAYING

At the time of issue of cable for laying, the cables shall be tested for continuity and insulation resistance.

10.5.8.5 SAND COVERING

Before cables are placed, the trenches bottom shall be filled with a layer of sand. This sand shall be leveled and cables laid over it. These cables shall be covered with 150 mm of sand on top of the largest diameter cable and sand shall be tightly pressed. A protective covering of 75 mm thick second class red bricks shall then be laid flat. The remainder of the trench shall then be back-filled with soil, rammed and levelled.

10.5.8.6 EXTRA LOOP CABLE

- a. At the time of original installation, approximately 3m of surplus cable shall be left on each terminal end of the cable and on each side of the underground joints. The surplus cable shall be left in the form of a loop. Where there are long runs of cables such loose cable may be left in the form of a loop. Where there are long runs of cables such loose cable may be left at suitable intervals as specified by the Engineer-in-charge.
- b. Where it may not be practically possible to provide separation between cables when forming loops of a number of cables as in the case of cables emanating from a substation, measurement shall be made only to the extent of actual volume of excavation, sand filling etc. and paid for accordingly

10.5.9 Where cables pass through foundation walls or other underground structures, the necessary ducts or openings will be provided in advance for the same. However, should it become necessary to cut holes in existing foundations or structures, the electrical contractor shall determine their location and obtain approval of the Engineer-in-charge before cutting is done. The cable entries through pipes from outside to inside the buildings shall be sealed water tight with approved type of sealant to avoid water entering into the building.

10.5.10 At road crossing and other places where cables enter pipe sleeves adequate bed of sand shall be given to that the cables do not slack and get damaged by pipe ends.

10.5.11 Cables installed above ground shall be run in trays, exposed on walls, ceilings or structures and shall be run parallel or at right angles to beams, walls or columns.

10.5.12 Individual cables or small groups which run along structures / walls etc., will be clamped by means of 16 SWG G.I saddles on 25 x 6 mm saddle bars at intervals not exceeding 300 mm for cables up to 25 mm dia and at 500 mm for higher size cables. The cost of saddle and saddle bars shall be deemed to have been included in the installation of cables and no separate payments shall be made on this account.

10.5.13 Cable laid on supporting angle in cable trenches, structures, columns and cable trays shall be suitably clamped by means of G.I Saddles / Clamps.

10.5.14 Supporting steel shall be painted before laying of cables. The painting shall be done with one coat of red lead paint and two coats of approved bituminous aluminium paint unless otherwise specified.

10.6 TERMINATION:

10.6.1 All PVC Cables up to 1.1 KV grade shall be terminated at the equipment / switch boards by means of compression type cable glands.

10.6.2 The cable armouring shall be earthed at both the ends.

10.6.3 Cable termination up to 4 sqmm shall be insertion type and others solderless crimped type Aluminium / Copper Sockets. Conductor jelly shall be applied before crimping after cleaning the surface.

10.7 TESTING:

10.7.1 M.V Cables testing shall be done with 1000 Volts Meggar and that of H.V Cables with 5 KV Meggar. The following test shall be done

- a. Continuity in all phases.
- b. Insulation Resistance between Conductor and between Conductors to earth.

All tests and recorded done in the presence of Engineer - in - charge and shall form part of completion document.

10.8 MODE OF MEASUREMENT:

10.8.1 Cables will be measured on the basis of common rate per unit length irrespective of the type of laying. The cost shall include cable, clamps, excavation, sand filling, protective brick layer and refilling, cable markers and installation, testing and commissioning. **Cable length inside Switch Board/ Panels etc shall not be paid.**

10.8.2 Cable termination will be measured as one unit for payment. The cost shall include glands, lugs, nuts, joint material, earthing of glands, installation, testing and commissioning.

TECHNICAL SPECIFICATION - SECTION-XI**CABLE TRAY****11.1****GENERAL:**

The cable management system shall consist of well designed, field proven products based on popularly accepted international practices.

It shall be flexible, possible to use in various configurations. Accessories required to use in such configuration must be available as standard.

Manufacture and finish shall be complete at the works. No welding should be expected to be done at site for installation.

All products used indoors /outdoors shall be Hot Dip Galvanized.

The successful cable management products vendor should be in a position to offer various products to support and route, like power, data & voice cabling, on floors, under floors and above floor locations.

11.2**SUPPORT FOR CABLE TRAYS:**

Shall be Flexible system consisting of Slotted Verticals usable in single fashions with racks on one side only or in box configurations with racks on two opposing sides.

The system should be fixable to walls, columns, ceilings, floors and also from floor to ceiling.

Wall clamps, footplates, ceiling plates and telescopic coupler channels etc., which are required for fixing as above should be available as standard equipment.

Racks with Tapered section lowest at free end should be engaged easily with Vertical stand sections.

Racks should have slots to fix cables / conduits or trays.

All sections like stand, racks, clamps shall be of minimum 2.5mm thickness (12swg).

All mounting shall be of 3.5mm to 6mm thickness.

All products for the support system shall be Hot Dip Galvanized as per IS: 2629 to achieve uniform zinc coating not less than 60 microns at any spot.

11.3**LADDER TYPE CABLE TRAYS:**

Shall consist of a Bolted rung assembly of field proven design. It shall consist of side rails of 2.5 mtr standard lengths and Slotted rung spaced 250mm center to center.

Coupler, Coupler fasteners, tray assembly fasteners should form the part of the equipment.

The Ladders should be light sheet metal constructions yet robust enough to carry a cable load of 50 kg/m on a span of 1.25 mtrs.

The design shall be flexible enough to accommodate change of widths at site.

Sections of 1.6/2mm shall be standard for different widths of trays as detailed in BOQ.

However where locations demand lighter sections those too shall be detailed in BOQ.

Horizontal bends, Vertical internal / external, tees, crosses shall be standard products.

Hinged or flexible coupler assemblies for Vertical / Horizontal route changes shall be available as standard as also reducers (offset couplers).

Covers shall be of Pre-galvanized sheet 18swg (1.25mm up to 300w) & 16swg (1.6mm on higher widths) construction possible for easy mounting on to trays.

Provision to add on earth flat holding clamps, trays fixing clamps should be part of cable tray design.

Perforated cable trays of 1.6mm / 2mm thickness and standard lengths 2.5 mtrs as detailed in BOQ shall be offered by bidder if required. Finish shall be as specified earlier.

Wire ways of enclosed type with covers, cover screws, coupler, coupler screws of Pre-Galvanized sheet constructions or of powder coated finish with all accessories shall be offered by bidder as detailed in BOQ. Such Wire ways should be suitable for surface laying or on / under floors.

11.4 CONSTRUCTION:

The cable trays shall be either ladder or perforated type. The cable trays and accessories shall be fabricated out of hot rolled steel sheets, which shall be hot dipped galvanized. The complete assembled cable tray sections shall be corrosion resistant, high strength and with extreme smooth surface. Accessories Cable tray manufacturer shall supply suitable accessories for clamping the cable trays on cable tray supports from ceiling and beams of the building structure. Cable trays shall be supplied with GI coupler plates, hardware, nuts bolts and washers for joining the standard lengths of cable tray section.

11.5 LOAD DEFLECTION CRITERIA:

For tray system design in addition to self-load, following criteria shall be applied to determine section and thickness of cable tray.

Support span	3000	mm
Cable load for 300 mm	50	Kg / m
Cable load for 400 mm	75	Kg / m
Cable load for 500 mm	85	Kg / m
Cable load for 600 mm	100	Kg / m

In addition to this 70 Kg/m concentrated load at the center span shall be considered.

Allowable mid span deflection < 7 mm

11.6 SURFACE PREPARATION:

For treatment and preparation of surface of fabricated trays, seven tank process shall be employed comprising of the following cleaning all the members, plates shall be free from grease, paints or any foreign matter. A chemical solvent as trichloroethylene / carbon tetrachloride or a combination of solvent cleaning and heating shall be employed. Immediately after degreasing the material shall be rinsed with hot / cold water. Pickling Hydrochloric acid and sulphuric acid solutions may be used from pickling. After pickling the material shall be rinsed in running water. After surface treatment, cable trays shall be galvanized.

11.7 GALVANIZING:

Zinc conforming to grade Zn 98 of IS 209 shall be normally used for galvanizing. As far as practicable, the fabricated tray and accessories shall not be sunk to the bottom of the bath. Tray and accessories shall be galvanized at the lowest possible temperature, which will allow free drainage of zinc from the work piece during withdrawal. The thickness of the coating shall be minimum 100 microns. The zinc deposited shall not be less than approximately 720 gm/sq.m. The thickness shall be checked by a magnetic method as per IS 3203.

Small components handed in baskets shall be centrifuged to remove excess of Zinc immediately after galvanizing while the coating is still in molten condition. Hardware such as nuts and bolts shall be Electro galvanized / zinc passivated.

11.8 PAINTING:

If necessary at site - only, especially after cutting / jointing. The metal surface after cleaning shall be prepared by applying a coat of phosphate paint and a coat of yellow zinc chromate primer. After preparation the tray surface shall be spray painted with two coats of yellow paints.

11.9 EARTHING:

The cable tray /floor raceways shall be provided with brass earth clamps every 3000mm for connecting suitable copper wire earth

TECHNICAL SPECIFICATION - SECTION-XII
LOW CURRENT SYSTEM

12.1 Scope

To provide wiring for telephone on surface of wall or ceiling concealed in slab, wall, under flooring, etc., through existing metallic conduits, rigid PVC conduits, PVC trunking, with all necessary hardware, material, etc. as specified.

To provide, install, test & commission the instruments / equipment and accessories used in telephone system, such as; Main Distribution Frames (MDF), Krone Modules, Over Voltage Magazine, PBX / EPABX, CO-axial cable, Rosette box, Jumper wire, etc.

12.2 Conduits:

12.2.1

Conduits shall be as given below:

Indoor: medium gauge Rigid PVC conduit.

The conduit shall generally be as specified under section 'CONDUIT WIRING'.

12.3 Cables and Wires:

12.3.1

The type of cables and the services shall be as follows:

- i) Indoor Multi pair, PVC insulated sheathed armored and sheathed.
- ii) Inside Twin core PVC insulated with conduit twisted cores.

12.3.2

All multi core cables and wires shall be of tinned copper conductor of not less than 0.5 mm dia and shall be colour coded twisted pairs with rip cord.

12.3.3

The conductor resistance shall be less than 150 ohms per KM and the insulation resistance between the conductors not less than 50 mega ohms and the nominal capacitance of about 0.1 micro farad per kilometer.

12.3.4

Cables laid underground or locations subject to dampness and flooding shall be filled with polyethylene compound and shall have sufficient protection against moisture and water ingress.

12.3.5 All armouring shall be of galvanized steel wires and protected against corrosion by an outer sheath of PVC in the case of indoor cables and polyethylene in the case of outdoor cables. Outer sheathing must be fire retarding and anti-termite.

12.3.6

All un-armoured single core cables and inner sheath of armoured cables shall be provided with rip cord.

12.3.7

All single pair cables for final extension to the telephone outlet box shall be un-armoured tinned copper conductors of not less than 0.5 mm. diameter and shall be drawn in conduits. All

telephone outlets shall consist of 2 A 2 pair polythene connector in G.I box with modular plate and modular RJ 11 Telephone socket..

- 12.3.8 Saddles: Saddles fabricated from G I sheet of required gauge (16/18 gauge) either galvanized finish or painted with superior quality enamel black paint, with necessary shearing for mechanical strength, semicircular shaped with extended piece having suitable holes for fixing on spacer.

- 12.3.9 Junction box: Manufactured by reputed manufacturer of specified capacity, facility for wall mounting, with door & lock, aluminium frame for fixing of Krone, duly enclosed in cabinet made from 18 SWG CRCA sheet with powder coating of required colour. The depth of the box should consider the height of KRONE module plus protection magazine.

12.4 Tag blocks

12.4.1

The telephone tag blocks shall be suitable for the multi core telephone cables and shall have two terminal blocks, cross connect type. All incoming and outgoing cables shall be terminated on separate terminal blocks and termination shall be silver soldered. The cross connecting jumpers shall be insulated wires of same diameter and screw connected.

12.4.2

The tag blocks shall be mounted inside fabricated sheet steel boxes with removable hinged covers and shall be fully accessible. The enclosure shall be painted with 2 coats of red oxide and stove enameled.

- 12.4.3 KRONE Module: Disconnection type KRONE module having capacity to connect 10 pairs with silver-plated terminal contacts.

12.5 Installation:

12.5.1

The installation of conduits shall generally be as specified under section 'CONDUIT WIRING'.

12.5.2

All cables shall be on cable racks and neatly stitched together.

- 12.5.3 The connection at the tag blocks shall be silver soldered so as to achieve minimum contact resistance.

12.5.4

The final branch connections with single pair cables in conduits and the maximum number of cables in each conduit shall be as follows :

Conduit diameter	Max. No. of cables
------------------	--------------------

inch	mm.
------	-----

3/4"	20	2 Nos. single pair
------	----	--------------------

1"	25	6 Nos. single pair
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1¼" 32 12 Nos. single pair

1½" 40 18 Nos. single pair

12.5.5 Erection of Jelly filled armoured Telephone cable:

Erection shall be done as per the layout finalized, in perfect level and plum. Before fixing the cable shall be straightened as far as possible for good aesthetics look. Cable shall be fixed with saddles firmly clipped on cable. Saddles shall be fixed to wall with minimum 50x8 mm SM screws with plugs/wooden gutties (Distance between two saddles shall be minimum 600 mm). Wooden gutties shall be used wherever required (Especially for stone wall). The entries made in wall, floor slab, etc., for laying the cable shall be made good by filling and finishing with plastering the same.

12.5.6 Erection of MDF Junction box / Rosette box / PBX / EPABX, etc.,:

Specified equipment shall be fixed to wall with minimum 50x8 mm SM screws, with necessary plugs, wooden gutties, etc. or may be fixed on Table Top if required.

12.6 Mode of measurement:

12.6.1

The main telephone cables shall include supply and laying of multi pair cables on ceiling/wall/on cable trays/racks including all supports and shall be measured and paid on running length basis. Cable trays/racks shall be paid for separately.

12.6.2

The multi pair tag blocks shall consist of two telephone connectors strips, jumpered interconnections silver soldered enclosure etc. and shall be measured and paid as one unit.

12.6.3

The Telephone Socket outlet including GI Box, Modular Plate and modular RJ 11 Socket shall be paid as one unit.

TECHNICAL SPECIFICATION - SECTION - XIII**EARTHING****13.1 SCOPE:**

- 13.1.1 This Specification covers the essential requirement of **Earthing system** consisting of earthing stations, earth conductor laying and connection to Panel boards/ Distribution Boards etc.
- 13.1.2 For details not covered in these specifications, IS Code of Practice on Earthing (IS : 3043-1987) shall be referred to.

13.2 APPLICATION

- i) The electrical distribution system is with earthed neutral (i.e. neutral earthed at the transformer/ generator end). In addition to the neutral earthing, provision is made for earthing the metallic body of equipment and non-current carrying metallic components in the substation, as well as in the internal/ external electrical installations.
- ii) An Earthing system is also required for lightning protection, computer installations etc., for function reasons.
- iii) Earthing requirements are laid down in Indian Electricity Rules, 1956 as amended from time to time, and in the Regulations of the Electricity Supply Authority concerned. These shall be complied with. The basic system of earthing shall be TNS(Tetra Neutral Separate).

13.3 STANDARDS:

The earthing installation shall be carried out in the best workman like manner confirming to this specification and to the latest Indian Standards and other statutory provisions.

- a) Code for practice for earthing IS : - 3043
- b) Indian Electricity Act and Rules.
- c) Local Electrical Inspectorate Regulations. In case of conflict, the instructions of Consultant shall be binding.

13.4 EARTHING STATION:

- 13.4.1 The earthing station may be pipe earth electrode or plate earth as per Bill of Quantities and shall conform to the attached drawings.
- 13.4.2 The earth electrode shall be driven preferably to reach permanently moist soil. It is preferable that soil around the electrode has fine texture, which is packed by watering and ramming as tightly as possible. Wherever possible the soil shall be dug up and all stone and lumps removed from around the electrode.
- 13.4.3 Earthing electrode shall not interfere with roads, building foundation col. etc., They shall be located atleast 1.5 Mtr away from the building. The location and numbers shall be got approved from the Consultant before execution.
- 13.4.4 The resistance of each earth station shall not exceed 5 Ohms. In case of higher resistance, further improvement of soil resistivity shall be carried out as per instruction of Consultant.

13.4.5 The G.I Plate and pipe earth electrode shall be hot drip galvanized and the thickness of galvanizing shall not be less than 75 Micron.

13.5 EARTHING LEAD AND CONNECTION:

13.5.1 The G.I Flat and wire earth leads shall be hot drip galvanized with galvanizing thickness of not less than 75 Micron. Phosphate content of copper leads shall not exceed 0.15 percent.

13.5.2 The earth leads when laid in ground shall be at 500 mm depth below grade level. When laid in masonry cable trenches along walls beams etc., the strips shall be 6 mm thick galvanized steel saddles at 500 mm interval.

13.5.3 The earthing conductor from the electrode up to the building shall be protected from mechanical injury by a medium class, 15mm dia GI pipe in the case of wire, and by a minimum of 40mm dia, medium class GI pipe in the case of strip. The protection pipe in ground shall be buried at least 30 cm deep (to be increased to 60 cm in case of road crossing and pavements). The portion within the building shall be fixed on walls.

13.5.4 The earthing conductor shall be securely connected at the other end to the earth stud/ earth bar provided on the switch board by Soldered or preferably crimped lug, bolt, nut and washer in the case of wire, and, Bolt nut and washer in case of strip conductor.

13.5.5 In case of copper electrode jointing shall be by copper riveting with 80 mm fish plate and brazing. In case of G.I the same shall be by means of lap welding with minimum 50 mm lap.

13.5.6 All joints below grade shall be protected by giving two coats of bitumen and covered with Hessian tape / polythene based hessian.

13.5.7 All hardware used for earth lead connections shall be hot drip galvanized zinc passivated. Spring washers shall be used for equipment earth connection.

13.5.8 All metallic junction and switch boxes shall be provided with an earth stud to which earth lead shall be connected. The stud shall be painted distinct green or yellow.

13.5.9 Armoured Cables shall be earthed to two distinct earth connections to the armouring at both the ends.

The size of earth lead for equipment earthing shall be as follows:

EQUIPMENT EARTHING - SIZE OF EARTH LEAD (TRANSFORMERS, MOTORS, GENERATORS, SWITCHGEARS ETC.)

RATING OF 50 HZ EQUIPMENT IN KVA	SIZE OF EARTHING CONDUCTOR
--	----------------------------

	BARE COPPER	ALUMINIUM PVC INSULATED	GALVANIZED IRON CONDUCTOR
UP TO 5	10 SWG	10 Sq.mm	7/22
5 TO 15	10 SWG	16 Sq.mm	6 SWG
15 TO 50	8 SWG	16 Sq.mm	1" X 1/16" (25 X 1.5 mm)
51 TO 75	6 SWG	25 Sq.mm	1" X 1/16" (25 X 1.5 mm)
75 TO 100	6 SWG	35 Sq.mm	1" X 1/8" (25 X 3 mm)
101 TO 125	4 SWG	50 Sq.mm	1" X 1/4" (25 X 6 mm)
126 TO 150	2 SWG OR 1" X 1/15" (25 X 1.5 mm)	70 Sq.mm	1" X 1/4" (25 X 6 mm)
151 TO 200	1" X 1/15" (25 X 1.5 mm)	70 Sq.mm	1 1/2" X 1/4" (40 X 5 mm)
201 AND ABOVE	1" X 1/8" (25 X 3 mm)	185 Sq.mm	2" X 1/4" (50 X 6 mm)

NOTES:

1. CONDUCTOR TO BE PROTECTED SO THAT NO MECHANICAL DAMAGE COULD BE CAUSED.
EARTH CONNECTIONS FOR ALL H.V. MOTORS AND SWITCH GEARS SHOULD BE NOT LESS THAN 0.1 Sq.Inch COPPER TAPE OR 185 Sq.Mm PVC INSULATED STRANDED ALUMINIUM CONDUCTOR OR 2" X 1/4" GALVANIZED STRIP.
2. IS : 3043/1987 TO BE REFERRED FOR CONFIRMATION.

13.6 INSTALLATION – Installation shall be as per the specifications mentioned in the tender document**13.7 Electrodes****13.7.1 Various Types Of Electrodes**

- a) Pipe electrode shall be buried in the ground vertically with its top at not less than 3 mtr below the ground level. The installation shall be carried out as shown in the drawing.
- b) In locations where the full length of pipe electrode is not possible to be installed due to meeting a water table, hard soil or rock, the electrode may be reduced length, provided the required earth resistance result is achieved with or without additional electrodes, or any alternative method of earthing may be adopted, with the prior approval of the Engineer-In-Charge. Pipe electrodes may also be installed in horizontal formation in such exceptional cases.
- ii) Plate electrode shall be buried in ground with its faces vertical, and its top not less than 3 m below the ground level. The installation shall be carried out as shown in drawing.
- iii) When more than one electrode (plate/ pipe) is to be installed, a separate of not less than 3m shall be maintained between adjacent electrodes.

13.7.2 Artificial Treatment of Soil

When artificial treatment of soil is to be resorted the electrode shall be surrounded by

charcoal/ coke.

13.7.3 Watering Arrangement

- i) In the case of plate earth electrodes, a watering pipe of 20mm dia. medium class GI pipe shall be provided and attached to the electrodes as shown in the drawing and a funnel with mesh shall be provided on the top of this pipe for watering the earth.
- ii) In the case of pipe electrodes, a 40mm x 20mm reducer shall be used for fixing the funnel with mesh.
- iii) The watering funnel attachment shall be housed in a masonry enclosure of size not less than 30 cm x 30 cm x 30 cm.
- iv) A cast iron/MS frame with MS cover of 6mm thick, and having locking arrangement shall be suitably embedded in the masonry enclosure.

13.8 Earth Conductor

In the case of plate earth electrodes, the earthing conductor shall be securely terminated on to the plate with two bolts, nuts, check nuts and washers.

In the case of pipe earth electrodes, wire type earthing conductor shall be secured as indicated in drawing using a through bolts, nuts and washers and terminating socket.

The earthing conductor from the electrode up to the building shall be protected from mechanical injury by a medium class, 15mm dia GI pipe in the case of wire, and by a minimum of 40mm dia, medium class GI pipe in the case of strip. The protection pipe in ground shall be buried at least 30 cm deep (to be increased to 60 cm in case of road crossing and pavements). The portion within the building shall be fixed on walls.

The earthing conductor shall be securely connected at the other end to the earth stud/ earth bar provided on the switch board by Soldered or preferably crimped lug, bolt, nut and washer in the case of wire, and, Bolt nut and washer in case of strip conductor.

13.9 Earth Bus And Main Earthing Terminal

In all installations, main earthing terminal shall be provided at the main switchboard. This may be in the form of earth stud or single earth bar depending on the type of the switchboard.

Following conductors shall be terminated on to the main earthing terminal.

- a) Earth connection from electric supply company (where provided)
- b) Earthing conductor from electrode.
- c) Protective conductors
- d) Equipotential bonding conductors.

13.10 Protective (Loop Earthing/ Earth Continuity) Conductor

Earth terminal of every switchboard in the distribution system shall be bonded to the earth bar/ terminal of the upstream switchboard by protective conductors.

Two protective conductors shall be provided for a switchboard carrying a 3-phase switchgear thereon.

13.11 Earth Resistance

The earth resistance at each electrode shall be measured. No earth electrode shall have a

greater ohmic resistance than 5 ohms as measured by an approved earth testing apparatus. In rocky soil the resistance may be up to 8 ohms.

Where the above stated earth resistance is not achieved, necessary improvement shall be made by additional provisions, such as additional electrode (s), different type of electrode, or artificial chemical treatment of soil etc., as may be directed by the Engineer-In-Charge, at additional cost as per the provisions of the contract.

13.12 Marking

- i) Earth bars/ terminals at all switch board shall be marked permanently, as 'E'.
- ii) Main earthing terminal shall be marked 'SAFETY EARTH- DO NOT DISCONNECT'.
- III) The Cable armouring shall be connected to earth stud in case of street light poles.

13.13 TESTING:

The following tests shall be conducted and recorded as per the attached Earthing Installation Testing Report Proforma.

- i. Earthing resistance of individual electrodes (the electrode shall be disconnected from grid during test).
- ii. The Earthing Resistance of grid after connecting all electrodes the same shall not exceed 1 ohms.
- iii. Each Electrical System a system neutral earth.
- iv. At each earth provided for lightning conductor.
- v. At one point on each earthing system used to earth electrical equipment enclosure and each fence enclosing electrical equipment. The measurement shall be made before connection is made between the ground and the object to be grounded.

13.14 MODE OF MEASUREMENT:

- Each earth station complete with excavation, electrode, watering pipes, soil treatment, masonry chamber, CI / RCC Cover shall be treated as one unit for measurement.
- Earthing leads from earth station to switch board, motor, etc., shall be measured and paid per unit length. The cost will include end termination clamps, saddles, supports, excavation, refilling, etc., as required.
- The earthing of Lighting Fixtures, Switch Gear / Starter mounting in the Switch Board, Conduit Wiring, Cabling, etc., shall become part of cost of item and no separate payment shall be made.

Test Proforma

(INSTALLATION TESTING REPORT EARTHING INSTALLATIONS)

1. Earth System data

Type of Electrode	:
Total Number of electrodes	:
Main grid size	:

Material _____ :

2. **General Checks** Put Tick ☐ if O.K otherwise given details

Construction of earth electrodes as per Standard.

Size of earth conductor for various equipment O.K as per Standard.

Minimum distance kept between two electrodes.

Cleanliness and tightness of connection.

Inspect bolted and clamped connectors.

3. **Tests**

- 3.1 Measured earth resistance of each electrode in ohms.

No. 1

No. 2

No. 3

No. 4

No. 5

3.2 **Measurement of earth grid resistance (with all electrodes connected to grid)**

- a) At each electrical system earth or system neutral earth. _____ :
- b) At each point provided for structure lightning protection. _____ :
- c) At one point on each earthing systems used to earth electrical equipment enclosure. _____ :
- d) At one point on each earthing systems used to earth wiring system such as metal conduits etc., _____ :
- e) At one point on each fence enclosing electrical equipment. _____ :

4. **Remarks :**

PROJECT :		UNIT :	
Tested By	Witnessed By		Date
Contractor	CONSULTANT	OWNER	

TECHNICAL SPECIFICATION - SECTION- XIV**INSTALLATION****14.1 SCOPE:**

The specification covers the installation, testing and commissioning of all electrical equipment's and accessories required for the switchyard for efficient and trouble free operation.

14.2 STANDARDS:

The electrical installation work covered by this specification shall unless otherwise stated comply with the requirements of the latest edition of relevant Indian Standard, statutory regulations and codes of practices.

- Indian Electricity Rules –1956
- Tariff advisory committee - Approvals.
- IS – 10118 :Code of practice for selection, installation & maintenance of Switch gear and control gear
- IS – 6600 : Guide for loading of oil immersed transformers.
- IS – 3043 : Code of practice for earthing.
- IS – 2309 :Code of practice for protection of building and allied structures against lightning.
- IS – 2274 : Code of practice for electrical wiring installation.
- IS – 6665 : Code of practice for industrial lightning.

14.3 GENERAL REQUIREMENTS:

- The installation shall be carried out by an electrical contractor holding a valid license as required by the respective State Government. The contractor shall provide particulars of the license held by him or his subcontractor to the purchaser. The installation shall require approval of the Chief Electrical Inspector to the Government of Tamil Nadu and the contractor shall prepare all necessary drawings / documents and submit to CEIG and get safety certificates from CEIG. He shall also fully assist the purchaser in obtaining approval from any other statutory authorities for the successful commissioning of the Substation.
- Any modification in the equipment or installation that may be demanded by the inspector shall be carried out at no additional cost to the purchaser.
- In accordance with the specific installation instructions or as directed by the purchaser, the contractor shall unload, erect, assemble, install, wire, test and commission all electrical equipment's included in this contract. Equipment's shall be installed in a neat workman like manner with highest regard for safety.

- Erection materials, tools, testing instruments or any other machinery of any nature shall not be supplied by the purchaser. The contractor shall arrange for the same in a timely manner and he shall not be allowed to claim for any delay or extra cost of any nature.
- Consumable materials of any nature required for the erection job shall also have to be arranged by the contractor
- Clearing the site after completion of erection as well as regular clearance of unwanted, materials from site, returning all packing material and excess material and excess material shall also be covered under the scope of work.
- All equipment's and instruments of indoor and outdoor, shall be inscribed with number, nomenclature, danger boards and other instructions.
- The contractor shall touch up the surface for all equipment's, which are scratched and / or damaged during transportation and erection. The paint used shall match exactly the surface being touched up.
- The contractor shall employ skilled and semi-skilled laborers for erection, installation & testing as required. All electricians, cable jointers, wire men, welders and others employed shall be suitably qualified possessing valid certificates / licenses recognized by the competent authorities
- The contractor shall also furnish a list of Engineers/Supervisors and staff employed by him for erection and installation jobs, giving in brief, qualification and experience of such staff and indicating whether they hold such competency certificates / licenses to supervise the electrical installation jobs as required under Indian Electricity Rules & State electrical Inspectorate Rules.
- The contractor shall set up his own workshop and other facilities at site to undertake fabrication jobs, pipe bending, threading etc.
- The contractor shall be responsible for recording of all readings and observations during erection, testing and commissioning, in registers or on prescribed proforma. These shall be carried the presence of purchaser's representative. All such test data and records shall be duly signed by the contractor's Engineer / Purchaser's representative and shall be submitted to Purchaser in triplicate.
- The contractor shall carry out all tests at site for outdoor and indoor electrical equipment and commission the installation in the presence of Purchaser's representative. The contractor shall be responsible for final adjustment of relays, instruments, meters breakers etc., and also for submission of relay settings and calculations.

14.4 EQUIPMENT ERECTION:

- All support insulators, Circuit breakers, Isolators, Power Transformers, Instrument Transformers, etc. shall be properly handled and erected as per the relevant codes of practice and manufacturer's drawings and instruction manuals.
- Handling equipment, sling ropes etc. should be tested before erection and periodically for strength.
- For cleaning the inside and outside of hollow insulators, only muslin or leather clothes shall be used.
- Necessary Junction boxes for CT's and PT's shall be supplied and installed.

14.5 POWER CABLES:

- Contractor's scope of work includes unloading, laying, fixing, jointing, bending and terminating of cables. Contractor shall also supply all the necessary hardware's for jointing and terminating of cables. Cable shall be laid directly buried in earth, on cable racks in built up trenches, on cable trays and supports in conduits and ducts or bare on walls, ceiling etc. Where specific cable layouts are not shown in the drawing contractor shall route these cables as directed by Purchaser.
- It is the contractor's responsibility to ensure that he acquaints himself with the nature of the ground conditions of the project site.
- Markers shall be installed at all road crossing and joint positions. Their provision and installation shall be included in the areas.
- All necessary, care must be taken by the contractor while designing, excavating trenches and installation of cables.
- Contractor shall install, test and commission the cables specified in accordance with the latest revisions of Indian Standards.
- All cable work and the allied apparatus shall be designed and arranged to reduce the risk of fire and any damage that may cause in the event of fire. Wherever cables pass through any floor or wall opening suitable bushes supplied shall be sealed using fire-resisting materials to prevent fire spreading.
- Standard cable installation tools shall be utilized for cable pulling. Maximum pull tension shall not exceed the manufacturer's recommended value. Cable grips, reels or pulleys used shall be properly lubricated. The lubricant shall not injure the overall covering and shall not set up undesirable conditions of electrostatic stress.
- Sharp bending and kicking of cables shall be avoided. The bending radius for various types of cables shall not be less than those specified by manufacturer.
- Power and control cables shall be laid in separate cable trays. The order of laying of various cable in trenches shall be as specified below:
 - 33KV cables on top most tiers from top.
 - 415V cables on middle tier.

- Control cables in bottom most cable tier.
- Where cables cross roads and water, oil, gas or sewage pipes the cables shall be laid in reinforced spun concrete pipes. For road crossing the pipe for the cable shall be buried at not less than one-meter depth. Cable shall be protected at all times from mechanical injury and from absorption of moisture.
- Some extra length shall be kept in each cable run at a suitable point to enable one or two straight through joint to be made at a later date, if any fault occurs.
- Cable jointing shall be in accordance with relevant Indian Standard Codes of practice and Manufacture's special instruction. Materials and tools required for cable jointing work should be supplied by Contractor. Cable shall be firmly clamped on either sides of a straight through joint at not more than 300 mm away from the joints. Identification tags shall be provided at each joint and at all cable terminations. Before jointing insulation resistance of both sections of cables to be jointed shall be checked.
- Metal sheath and armour of the cable shall be bonded to the earthing system of the station.
- Each cable shall be identified with its designation number as indicated in the drawings.
- Cable clamps shall be of 3 mm thick galvanized M.S. spaced at every 1.5-M interval.

14.6 CABLE TRAYS, ACCESSORIES AND TRAY SUPPORTS:

- Cable trays shall either be run in concrete trenches or overhead supports from building steel, floor slab, etc.
- Cables shall be clamped to the cable trays in both horizontal runs and vertical runs by suitable prefabricated clamps.
- All cable trays and fitting will be ladder type and fabricated from M.S. sheet. They shall be hot dip galvanized.
- Cable trays shall be suitably supported at an interval of not more than one meter.

14.7 SWITCH GEAR, CONTROL AND RELAY PANEL:

- Switch gear, control and relay panel, etc. shall be installed in accordance with the latest edition of IS-10118 and manufacturer's instructions.
- The panels shall be installed on finished surfaces. The correct level shall be ascertained before final grouting.
- The contractor shall take utmost care in handling delicate equipment and mechanism like instruments, relays. Dragging shall be avoided as far as possible. Proper pipes shall be provided underneath when dragging for short distance. Wherever the instruments and relays are supplied separately, they shall be mounted only after the associated control panels have been erected and aligned. Any damage to relays and instruments shall be immediately reported to the purchaser. Contractor shall also make all necessary adjustments as specified by the manufacturer for proper functioning of the equipment.

- Contractor shall also carry out the following.
- Cleaning of internal parts like chambers, insulators etc. by vacuum cleaners.
- Prepare base for mounting the equipment including leveling.
- Drying of equipment in case of low insulation resistance.

14.8 TRANSFORMERS:

- The power transformers shall be tested and installed in accordance with the manufacturer's instructions.
- The level of rails on foundations shall be checked before erection and necessary stoppers shall be provided at the wheels after erection.
- Fresh silica gel shall be filled in the breather.
- Transformers for supplying to control circuits and lighting circuits shall be dry cast resin type. The requirements these transformers shall comply to IS: 12021.

14.9 CIRCUIT BREAKER:

- The vacuum circuit breaker shall be tested and installed in accordance with the manufacturer's inspection.
- The operating mechanism shall be installed and connected to the three poles and the control cubicle with compressor, air receiver, etc. shall be connected to the breather pole operating mechanism through pipes.
- Each pole of the circuit breaker and operating mechanism shall be accurately positioned not to cause any undue strain on the terminals and they shall be properly earthed.

14.10 ISOLATOR:

- The contractor shall inspect, clean, assemble and install the isolator on the fabricated base structure.
- The operating mechanism shall be installed on the structure and shall be tested by slowly bringing the isolator to the closed position and necessary adjustments shall be carried out as per manufacturer's instruction.
- Earth switches, frames, operating handles, etc. shall be earthed.

14.11 LIGHTING AND POWER PANELS:

- Lighting and Power Panels shall be located in the control room. The panel shall be made of 14/16 SWG CRCA sheet steel. The panels shall have hinged lockable doors with gaskets. Removable bottom gland plate shall be provided along with nickel plated single

compression glands,. An earth bus shall also be provided with two external grounding terminals.

14.12 MISCELLANEOUS ITEMS:

- The successful bidder shall supply and install the safety devices as required by the statutory authorities, but not limited to the following:
 - Danger boards.
 - Fire extinguishing appliances.
 - Rubber mats for switchgear panels, power distribution boards battery charges, control and relay panel, etc.
 - Rubber gloves, first aid charts, first aid box, etc.
 - Earthing rods.

TECHNICAL SPECIFICATION - SECTION- XV**TESTING & COMMISSIONING****15.1 GENERAL:**

The testing and commissioning for all electrical equipment at site shall be according to the procedure laid down below.

- All electrical equipment shall be installed, tested and commissioned in accordance with the latest relevant standards and codes of practices published by Indian standards, institution wherever applicable and stipulations made in relevant general specifications.
- The testing of all electrical equipment as well as the system as a whole shall be carried out to ensure that the equipment and its components are in satisfactory condition and will successfully perform its functional operation. The inspection of the equipment shall be carried out to ensure that all materials, workmanship and installation conform to the accepted design, engineering and construction standards, as well as accepted codes of practice and stipulations made in the relevant general specifications.
- The contractor using his own instruments, testing equipment as well as qualified testing personnel shall carry out all tests.
- The results of all tests shall be conform to the specification requirements as well as any specific performance data guaranteed during finalization of the contract. General
- At the completion of the work, the entire installation shall be subject to the following tests in presence of Employer/Consultant.
 - Wiring Continuity Test
 - Insulation Resistance Test
 - Earth Continuity Test
 - Earth Resistivity Test

15.2 PREPARATION OF THE ELECTRICAL SYSTEM FOR COMMISSIONING:

- After completion of the installation at site and for the preparation of Electrical system commissioning, the contractor shall carry out check and testing of all equipment and installation in accordance with the agreed standards, codes of practice of Indian Standards Institution and specific instructions furnished by the particular equipment suppliers.
- Checking required to be made on all equipment and installations at site shall comprise, but not be limited, to the following:

- The following checks shall be made on all equipment and installation at site:
 - Physical inspection for removal of any foreign bodies, external defects, such as damaged insulators, loose connecting bolts, loose foundation bolts etc.
 - Check for grease, insulating/lubricating oil leakage and its proper quantity.
 - Check for the free movement of mechanism for the circuit breakers, rotating part of the rotating machines and devices.
 - Check for tightness of all - cable, bus bars at termination/joints ends as well as earth connections in the main earthing network.
 - Check for Clearance of live bus bars and connectors from the metal enclosure.
 - Check the proper alignment of all draw out device like draw out type circuit breakers.
 - Continuity checks in case of power cables.
 - Checking of all mechanical and electrical interlocks including tripping of breakers using manual operation of relay.
 - Checking of alarm and annunciation circuits by manual actuation of relevant relays.
 - Check and calibrate devices requiring field adjustment/ calibration like adjustment of relay settings etc.,
 - Check proper connection to earth network of all non-current carrying parts of the equipment and installation.
 - Test reports for all meters are to be furnished.These tests shall be carried out on the equipment shall include but not be limited to the above.

15.3 H.T. BREAKER PANEL:

The following tests shall be carried out in accordance with IS 13118.

- Mechanical operation tests.
- Power frequency HV test.
- Insulation resistance test.
- Functional tests on control circuits.
- Relay operation tests by primary, secondary injection method.
- Checking of settings of all relays/releases as per single line diagram/specification.

15.4 DISTRIBUTION TRANSFORMER:

The following tests shall be carried out in accordance with IS

Measurement of insulation resistance test between:-

- HT & LT Windings
- Between HT & Earth
- Between LT & Earth

15.5 EARTH RESISTANCE TEST:

- Earth resistance of the Body earth Electrodes
- Earth resistance of the Neutral earth Electrodes
- Earth continuity

15.6 VOLTAGE RATIO TEST:

Ratio test on all taps of the transformers

15.7 CABLES:

- Insulation resistance test with 2,500V megger for high voltage power cables rated above 1.1KV grade and 1,000V megger for cables rated up to 1.1KV grade.
- All cables of 1.1KV and all H.V. cables shall be subjected to high voltage test after joining and terminating but before commissioning as per relevant standards.
- In each test, the metallic sheath/screen/armour should be connected to earth.
- Continuity of all the cores, correctness of all connections as per wiring diagram, correctness of polarity and phase of power cables and proper earth connection of cable glands, cable boxes, armour and metallic sheath, shall be checked.

15.8 EARTHING SYSTEM:

- Tests to ensure continuity of all earth connections.
- Tests to obtain earth resistance of the complete network by using earth tester. The test values obtained shall be within the limits.

All documents / records regarding test data, oscillographs and other measured values of important parameters finalized after site adjustment shall be handed over to the Owner in the form of test reports for their future use and reference.

15.9 ELECTRICAL:

-
1. The scope of work for testing and commissioning of the total installation shall be for the capital equipment's like switchgears, cables, etc., and also for the associated equipment like relays Cts, Pts cable etc.,
 2. The scope of work for testing and commissioning of electrical equipment for the above shall include but not limited to the following.
 3. Providing sufficient number of experienced Engineers, supervisors, Electricians, so that the installation can be commissioned in stipulated time.
 4. All the instruments, tools, and tackles required for carrying out the testing and the bidder shall provide commissioning.
 5. The testing of electrical equipment shall be carried but as per the relevant Indian standards/codes practices/Manufacturers instructions.

6. **Cleaning of Electrical equipment, contracts, cleaning and greasing etc., All the equipment and material required for above shall be supplied by the bidder.**
7. **Correcting the panel/equipment wiring for proper functioning for the schemes required /called etc.,**
8. **Installation and wiring of additional equipment on panels like auxiliary contractors, timers, etc., which may be additionally required for proper functioning of the schemes.**
9. **Checking of equipment earthing and system earthing as a whole.**
10. **Testing of all the cables.**
11. **Coordination with other contractors for testing and commissioning of interface cables.**

15.10

TEST TO BE CONDUCTED:

- a) **All tests shall be performed in the presence of the bidder and customer/consultant. For all types of visual inspections, checkings, pre-commissioning, commissioning test and acceptance tests, relevant IS for the tests given therein shall be followed in addition to the instructions in this technical specification the intention of giving the few test procedures, described below, is to provide a guideline for the bidder. However bidder shall not restrict themselves in carrying out only the tests described in this document.**
- b) **Bidder shall submit their proposed test procedures for approval and shall not commence testing without such approval is given.**
- c) **Bidder shall check and test all electrical equipment and system installed and supplied them, including equipment supplied by the owner.**
- d) **Bidder shall ensure that no tests are applied which may stress equipment above the limits for field-testing recommended by the manufacturer. Bidder shall be responsible for any damage to personnel or equipment resulting from improper test procedure including the equipment supplied by Client.**
- e) **All defective materials furnished by the bidder and defects due to poor workmanship revealed through field testing, shall be corrected at bidder expense without affecting the completion of the project.**
- f) **CLIENT reserves the right to interpret and approve all test results prior to energisation of circuits or apparatus.**
- g) **Bidder shall visually inspect all equipment for defects immediately results upon arrival at site including those supplied by CLIENT.**
- h) **Bidder shall test the buried earth grid and shall record the values. Bidder shall inspect and test all earthing work carried out by him, including all interconnections between ground loops, grounding of equipment and ensure all connections are permanent and that the earthing circuit is continuous.**
- i) **Bidder shall meggar and record earth resistance at various earth connection points.**
- j) **Switchgear rated 433 volts or more shall be tested with a 1000 volts megger.**
- k) **Auxiliary wiring rated less than 415 volts shall be tested with a 500 volts megger.**
- l) **ALL protective relays shall be tested at sufficient points to establish their proper functioning in accordance with manufacturer's specification and curves.**
- m) **Operation checks and functional checks on all switchgear panels.**

- n) **Bus bar, Wires and cables rated 433 volts or more shall be tested joints check with torque wrench for tightness.**
 - o) **Continuity testing of all wires and cables with a 1000 volts megger. Cables rated less than 433 volts shall be tested with a 500 volts megger.**
 - p) **No wires or cable having resistance between conductors or between conductors and ground if less than 100 mega-ohms shall be accepted.**
 - q) **All pre-commissioning test stated as per IS for respective items.**
-

NOTE: THE TENDERERS SHALL NOTE THAT ALL THE RATES QUOTED BY THEM ARE INCLUDING THE TESTING CHARGES FOR DOING THE ABOVE TESTS. SCOPE SHALL NOT REIMBURSE SEPARATELY ANY AMOUNT FOR ANY TESTING OF MATERILS.

TECHNICAL SPECIFICATION - SECTION-XVI**RISING MAIN (SANDWICH TYPE)****16.1 SCOPE**

This specification covers the Technical requirements of design, manufacture, test at works and Supply, Installation, Testing and Commissioning of LT Sandwich type Rising main.

16.2 STANDARDS

The equipment covered by this specification shall unless otherwise stated, be designed, constructed and tested in accordance with the latest revisions of relevant Indian / International Standards.

IS: 8623 (Part II)	:	Particular Requirements for Bus Trunking System.
IS: 2147	:	Degree of Protection provided by enclosures.
IS: 13947	:	Specification of Low Voltage Switchgear and Control Gear.
IEC: 60529	:	Degrees of protection provided by Enclosure.
IEC: 60439 - 1 / 2	:	Low Voltage Switchgear and Control gear Assemblies (Busway)

CONSTRUCTION:

The Rising Main shall be dust and Vermin Proof and the enclosure shall provide a degree of protection of not less than IP: 52.

The frame of the Rising Main shall be fabricated out of 1.6mm thick GS / Al. It shall be rigid & robust in construction & shall be treated with epoxy powder coating to prevent any possibility of corrosion.

The joints in enclosure shall be bolted and shall be provided with neoprene gaskets. All retaining catches, screws, bolts and nuts shall be zinc passivated.

Each length of Rising Main shall be provided with 2 Sets of spring loaded mounting brackets.

The Rising Main shall be provided with fire barrier wherever it crosses the wall.

16.3 BUS BARS:

The bus bars shall be made up of high conductivity Aluminium in sandwich configuration. The bus bars shall have a continuous current rating as mentioned in the Technical particulars and shall have fault withstand capacity of 35KA / 1Sec. The bus bars shall be 4W. The size of neutral bus bar shall be same as that of phase bus bars.

Bus bars shall be identified with red, yellow, blue and black colour tape bands at appropriate places and at terminations / Joints.

The bus bars shall be suitably supported at appropriate intervals to withstand the electro – mechanical stresses during short circuit. Bus bar shall be with full round edges.

Bolted type bus bar rigid joints shall be considered for connecting successive bus bars of its same phase. Bolted type splice plates, HTS zinc plated bolts and nuts with washers, shall be used to achieve an efficient joint.

The entire length of bus bar shall be tin coated and with epoxy resin insulation of flame retardant quality and prevent water and moisture that can cause reduction in dielectric resistance. The insulation of bus bars can also be glass - mica & polyester with number of layers. The insulation shall be class F (155°C).

All joints shall be maintenance free and single bolt operating system.

16.4 TAP OFF UNITS WITH SAFETY DEVICES

Tap off Units shall be provided in each floor, on the Rising Main. Maximum 3Nos. tap off units shall be provided per floor.

Each tap of point shall be provided with the necessary bus bar links with suitable support for connection to Tap off Boxes.

These points shall be provided with safety covers to prevent inadvertent touching of bus bars when no tap off unit is fitted.

A built in interlock system shall be provided to prevent opening of the cover when the control is ON.

Each tap of unit shall be provided with suitably rated MCCBs with interlocking facility with MCCB in the standby Rising Main and with necessary cable boxes for terminating the outgoing cable.

A safety interlocks to prevent the fixing or removal of plug in box when control is ON.

The Tap off boxes shall be rigidly fixed to the Rising Main using flange connectors.

The tap off box shall not be inserted with neutral on wrong side and polarity shall not be changed.

The tap off boxes shall not be located above 2.0m or below 0.5m above the floor level, sufficient gap shall be maintained between tap off points for ease of operation and maintenance.

Earth contact of plug in box shall make contact with bus trunking before phase and neutral contact is last to break when removed.

16.5 MOLDED CASE CIRCUIT BREAKER

The MCCB shall conform to IS:13947 / IEC947 in all respects. The MCCB shall comprise of switching mechanism, contact system, arc extinguishing device and the tripping unit all mounted in a moulded case made of high strength heat resistant and flame retardant thermosetting insulating material.

MCCB shall employ quick make and quick break switching mechanism independent of the speed of operation of the operating handle. The operating mechanism shall be trip free. The operating handle shall indicate the position of the MCCB in ON / OFF / TRIPPED. The operating handle shall have provision for door interlock and padlocking.

The MCCB shall be provided thermal magnetic trip unit with field adjustable over load protection and fixed short circuit protection.

MCCB shall be suitable for horizontal and vertical mounting and with line load reversibility.

All MCCBs shall be provided with 1NO + 1NC potential free contacts for BMS integration.

All MCCB shall be of current limiting type.

16.6 EARTHING

The Rising Main enclosure shall be earthed by two continuous copper earth bus of 25 x 3mm size, running on the sides, of the enclosure throughout the entire length of the Rising Main.

16.7 FIRE BARRIERS

Fire barriers provided shall be dust free and totally water and weatherproof resistant. It shall retain its properties under all temperature and atmospheric conditions and maintain its volume constancy. It shall have electric resistance and shall not contain asbestos or other toxic components. It shall give at least 4 hours of fire resistance.

16.8 TYPE TESTS

The bidder shall furnish two copies of type test certificate along with bid for the following,

- a. Short Time Current Test
- b. Temperature rise test

16.8.1 Inspection at Manufacturer's Works

Fully assembled Rising Main shall be offered for inspection at works with prior intimation.

Materials shall be dispatched to site only after issuing inspection clearance by the representative of purchaser / consultants. Following routing tests shall be conducted on the Rising Main.

1. Physical verification check
2. Megger Test
3. Power frequency HT test
4. Millivolt drop test on Rigid and expansion joints
5. Temperature Rise Test

16.8.2 Drawings

Bus bar size to be indicated along with the offer. After award of the order, the manufacturer shall submit the following documents for approval.

Design calculations.

- General arrangement showing the plan, elevation, different views.
- Detailed typical cross section of details of expansion joint, bus bar arrangements.
- Erection details with necessary supports.
- Test Certificates, if required.
- Complete technical particulars of the components.

16.8.3 TECHNICAL PARTICULARS- LT BUSDUCT (SANDWICH)

S. No.	Description		
1.	Service	Indoor / Outdoor	
2.	Type	Sandwich	
3.	Normal System Voltage KV rms	415V	
4.	Rated continuous current	A	As specified in BOQ
5.	Frequency	Hz	50 Hz

6.	System Earthing	Earthed
7.	No. of Phases	3 ϕ Neutral
8.	Max. Temp. Rise over design ambient of 50°C	40°C
9.	Insulation levels	
a.	Power frequency with-stand voltage dry	KV rms 2.5
10.	Short time current for a duration of 1 Second	KA rms As specified in BOQ
11.	Busbar	
a.	Material	Aluminium
12.	Enclosure	
a.	Material	GS / AL
b.	Degree of Protection	IP : 52 (Indoor)/ IP : 65(Outdoor)
13.	Bus Insulation	Class B Type with glass- mica & Polyester / Epoxy

16.8.4 TECHNICAL REQUIREMENTS FOR SANDWICH BUS DUCT RISING MAINS

1.	Application	:	Indoor
2.	Type	:	Sandwich
3.	Degree of Protection	:	IP 52
4.	Rated Voltage	:	415V
5.	Rated Continuous Current	:	As specified in BOQ
6.	Design Ambient Temperature	:	50°C
7.	Temperature rise over an ambient Temperature of 45°C	:	40°C
8.	Earthing		
a.	Material	:	Copper
b.	Size	:	25 x 3 mm
9.	Busbar		
a.	Number Phase / Wires	:	3 Ph, 4W
b.	Continuous Current	:	As specified in BOQ
c.	Frequency	:	50 Hz
d.	Fault Level	:	As specified in BOQ e. Material
		:	Aluminium
10.	Insulation Level		
a.	Power Frequency withstand Voltage	:	2.5kV
11.	Painting	:	Epoxy Powder Coated (Shade to be indicated later)

TECHNICAL SPECIFICATION

FIRE-FIGHTING WORKS

Sl.No	Details
a.	SECTION SECTION I: INTENT OF SPECIFICATION
b.	<u>SECTION II: BUILDING DETAILS AND DESIGN CONSIDERATION</u>
c.	<u>SECTION III: SPECIFICATION OF PIPING AND VALVES</u>
d.	<u>SECTION IV: SPECIFICATION FOR FIRE FIGHTING PUMPS</u>
e.	SECTION V: INSPECTION AND TESTING REQUIREMENT AND DOCUMENTS TO BE SUBMITTED FOR FIRE FIGHTING SYSTEM
f.	SECTION VI: COMMISSIONING OF FIRE FIGHTING SYSTEM
g.	SECTION VII: ALL PUMPS DATA TO BE FURNISHED BY MANUFACTURER
h.	<u>SECTION VIII: AUTOMATIC SPRINKLER SYSTEM</u>
i.	SECTION IX: SPECIFICATIONS FOR FIRE HYDRANT SYSTEM COMPONENTS
j.	SECTION XI: SIGNAGES
k.	SECTION X: FIRE EXTINGUISHER
l.	SECTION XI: AUTOMATIC DETECTION FIRE ALARM & PA SYSTEM
m.	SECTION XII: LIST OF APPROVED MAKES

SECTION - I

INTENT OF SPECIFICATION

This specification is intended to cover design, erection, testing, commissioning, performance demonstration at site and handing over of Fire Protection System as indicated in the schedule of requirement and scope of work.

This specification also includes complete earthwork, i.e. excavation and back filling for the entire buried piping for hydrant.

This specification also makes it obligatory for the contractor for arranging and obtaining necessary clearance / approval from all statutory authorities.

The equipment and installation shall confirm to high standard of engineering, design, and workmanship in all respect and shall be capable of performing continuous, satisfactory operation acceptable to the owner as well as to the various statutory authorities.

In case of any violation of the above contract, the owner reserves the right to change / reject / modify the equipment / system during detail engineering.

The entire system shall be designed and engineered based on the guidelines furnished in the specification, various codes / standards.

Items and quantities, which have been furnished in this specification, are tentative and indicative only.

Supplies and services to be covered under this tender specification and the conditions there off are detailed in the subsequent sections of the specifications.

In case of conflict among various sections, subsections, documents, drawings the same shall be referred to whose decision shall be final and binding to the firm. In all cases, the best advantages will go to the owner.

REFERENCES

IS 884-1985 Specification For First aid hose reel for Fire Fighting(First Revision)
(Reaffirmed November 1995.Amendment to1)

IS 901: 1988 Specification For coupling, Double male and double female instantaneous pattern for Fire Fighting (Third revision) (Reaffirmed in April 1993)

IS 902: 1992 Specification for Suction hose coupling for fire hose couplings for Fire Fighting Purposes (third revision)

IS 903: 1993 Specification For Fire hose delivery Couplings, Branch pipe, nozzles and nozzle spanner (fourth revision)

IS 904: 1983 Specification for two way and three way suction collecting heads for Fire Fighting purposes (Second revision)(Reaffirmed November 1995,amendment NO 1)

IS 905:1980 Specification for delivery breechings, dividing and collecting instantaneous pattern for Fire Fighting purposes (Second revision) (Reaffirmed February 1992)

IS 906:1988 Specification For Revolving branch pipe for Fire Fighting (Third revision)
(Reaffirmed November 1995)

IS 2189:1988 Code of Practice for Selection, Installation and maintenance of Automatic Fire Detection and Alarm System (Second Revision)Amendment No 1)

IS 2190:1992 Code Of Practice for selection, Installation and Maintenance of Portable First Aid fire Extinguishers (Third revision)

IS 636:1988 Non-Percolating Flexible fire fighting delivery hose (Third revision)

IS 933:1989 Specification For Portable fire Extinguisher Chemical foam (third revision)
Reaffirmed November 1995 Amendment No1)

IS 934:1989 Specification for portable fire extinguisher, Water type (soda acid)
(Fourth Revision) (Reaffirmed November 1995)

IS 937:1981 Specification for Washers for water fittings for fire fighting purposes (Second revision) (Reaffirmed February 1992 Amendment No 1)

IS 940: 1989 Specification for portable fire Extinguishers, Water type (Gas cartridge)
(Third revision) (Reaffirmed November 1995)

IS 941: 1985 Specification for Blowers and exhauster for fire fighting (Second revision)
(Reaffirmed November 1995 Amendment No 2)

IS 907: 1984 Specification for Suction strainers, cylindrical type for fire fighting (Second revision) (Reaffirmed November 1995 Amendment No 2)

- IS 908: 1975 Specification for Fire Hydrants ,Stand post type (Second revision) (Reaffirmed November 1995 Amendment No 2)
- IS 909: 1992 Specification for Under ground Fire Hydrant :sluice valve Type (Third revision)
- IS 952: 1986 Specification for Fog Nozzles, for Fire Brigade use (First revision) (Reaffirmed November 1995)
- IS 954: 1989 Functional requirements for Carbon di- for fire brigade use (Second revision) (Reaffirmed November 1995)
- IS 955: 1980 Functional requirements for Dry Powder tender for fire brigade use (150 kg Capacity (First revision) (Reaffirmed February 1992 Amendment No 3)
- IS 2097:1983 Specification for Foam making branch pipe (First revision) (Reaffirmed November 1995)
- IS 2171:1985 Specification for Fire Extinguishers, Dry powder (Cartridge type) (Third revision) (Reaffirmed November 1995 Amendment No 2)
- IS 2175:1988 Specification for Heat sensitive Fire Detectors for Fire Detectors for use in Automatic fire alarm System (Second revision) (Reaffirmed November 1995)
- IS 2546:1974 Specification for Galvanized mild steel Fire Bucket (First revision) (Reaffirmed November 1995 Amendment No1)
- IS 2871:1974 Specification for Branch pipe, Universal for Fire fighting Purposes (First revision) (Reaffirmed November 1995 Amendment No1)
- IS 2878:1986 Specification for Fire Extinguisher, Carbon-di-oxide type(Portable and trolley mounted) (Second revision) (Reaffirmed November 1995 Amendment No1)
- IS 4308:1982 Specification for Dry powder for fire fighting (First revision) (Reaffirmed January 1989 Amendment No3)
- IS 4861:1984 Specification for Dry powder for fire fighting in burning metals (First revision) (Reaffirmed January 1995 Amendment No1)
- IS 4928:1992 Specification for Unlined flax canvas hose for fire fighting (First revision)
- IS 4947:1985 Specification for Gas Cartridges for use in fire Extinguisher (Second revision) (Reaffirmed November 1995 Amendment No 4)
- IS 4989: Specification for foam concentrate (Compound) for Producing mechanical foam for fire fighting (Part 1 to 3)
- IS 5290: 1993 Specification for landing valves (Third Revision)
- IS 5714: 1981 Specification for Hydrant, Stand Pipe for fire fighting (First Revision) (Reaffirmed February 1992)

IS 6234: 1986 Specification for Portable extinguishers, Water type (Stored Pressure)First revision) (Reaffirmed November 1995,amendment No1)

IS 8090: 1992 Specification for Couplings, Branch pipe, Nozzle used in hose reel tubing for fire fighting (First revision) (Reaffirmed November 1995 (Amendment No1)

IS 8442: 1977 Specification for Stand post type water monitor for Fire fighting (Reaffirmed November 1995)(Amendment No3)

IS 9972: 1981 Specification for Automatic Sprinkler head (Reaffirmed February 1992 (Amendment No1)

IS 10204: 1982 Specification for Portable Extinguisher mechanical foam type (Reaffirmed November 1995)(Amendment No3)

IS 3034:1981 Code of Practice for Fire Safety of Industrial building electrical generating and distributing stations (First revision) (Reaffirmed November 1995)(Amendment No1)

IS 2189:1988 Code of Practice for Selection, Installation and maintenance of automatic fire detection and alarm System (Second revision)

IS 11360:1985 Specification for Smoke detectors for use in automatic electrical fire Alarm system (Reaffirmed November 1995)

IS 9668:1990 Code of Practice for Provision and maintenance of water supplies and fire stop (Reaffirmed November 1995)

IS 3844:1989 Code of Practice for Installation and maintenance of Internal Fire Hydrants and hose reel on premises (First Revision) (Reaffirmed November 1995)

All the selection of materials, Installation, Testing and commissioning shall be strictly in accordance with Indian Standard Specification and also Local Fire Force rules shall also be considered.

SECTION - II**BUILDING DETAILS AND DESIGN CONSIDERATION****1.0 TYPE OF SYSTEMS:**

Following are the various Fire Protection Systems proposed for protection under the present scope:

- 2.1 Fire Hydrant System
- 2.2 Automatic Sprinkler System.
- 2.3 Fire Extinguisher
- 2.4 Fire Suppression system
- 2.5 Automatic Smoke Detection and Alarm system
- 2.6 Public Address system
- 2.7 Applicable Codes & Standards: Unless specifically mentioned otherwise, all the applicable Codes and Standards published by the Bureau of Indian Standards and their subsequent revision shall govern in respect of design, workmanship, quality and properties of materials and method of testing.
NBC 2016 Part IV-Fire Protection System

2.0 BUILDING FIRE REQUIREMENTS:

PROPOSED TRAINING INSTITUTE & OFFICE BUILDING IN CHENNAI - FIRE SAFETY REQUIREMENTS AS PER NBC						
Types of Building Height	No. of floors	Fire Requirements	Terrace Tank	Pump Capacity	UG SUMP	
Training Institute & office Building (H=18.70m)	(Basement+ Ground+ 4Floors)	Fire Extinguisher Hose Reel Down Comer Sprinkler system for Basement Fire alarm & PA system - Automatic detection and alarm system	25,000 ltrs. (1 no.)	Terrace Booster Pump Capacity - 900 LPM (1 no.)	NA.	

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SECTION III

SPECIFICATION OF PIPING AND VALVES

Scope:-

The scope includes/ comprises the supply, installation, testing and commissioning of all pipes including pipe fittings, jointing, valves required as per B.O.Q. The drawings show the general layout of the piping and upon awarding of the contract, the contractor shall prepare his own detailed working drawing to suit site conditions and showing the details of fittings, location of valves etc.

Fire fighting system:

G.I piping:

All GI piping shall conform to IS 1239 for "Heavy" series (class C) with flanged joint. The pipe and fittings shall be able to withstand a hydrostatic test pressure of 5 Mpa (50 Kg/sq. cm) maintained for at least 3 seconds at manufacturing works (lab test). The flanged joints shall be provided on both sides of the valves. 3mm thick 3 ply rubber insertions conforming to IS 638 shall be provided between two flanges at the flanged joints. wet riser cum down comer for firefighting system, suction and delivery pipes of booster pumps, delivery pipes of domestic and flushing water pumps as well as pipes for priming of pumps shall made of GI.

B. General:

- i) All nuts and bolts shall be of MS hot dip galvanized.
- ii) All pipes shall have IS certification marks. The pipes and fittings shall be smooth, Sound, free from any imperfection and neatly dressed
- iii) GI pipes shall be properly supported with approved "U" clamps at proper interval.

Installation of Pipes :

The contractor, while making working drawings, must keep in view the openings in the building and other structural members through which the pipes are designed to pass or to be laid. All pipes shall be cut to accurate length at site.

Pipes shall be properly supported with suitable steel structures on walls/ roofs and PCC supports for the underground pipes. The tenderer shall adequately design all the brackets saddles, clamps, hangers, etc. and ensure their structural integrity.

The pipe-supports of steel shall have one coat of anticorrosive primer and two coats of rust preventive enamel paint.

The spacing of the pipe supports shall not exceed the following:

Pipe dia	Spacing
Upto 15mm dia	1200mm
20 to 25mm	1800mm
30 to 150mm	2400mm
150 and above	3000mm

Where pipe and clamps are of dissimilar materials, a gasket shall be provided in between. Pipe-hangers shall be fixed on walls and ceilings by means of raw plugs and approved

shear fasteners.

Vertical risers shall be parallel to wall and column lines and shall be straight. Risers passing through floor-slab shall be fixed at each floor by clamps or collars attached with a 12mm thick rubber pad or any resilient material.

Pipe sleeves of 50mm or larger dia shall be provided wherever pipe passes through walls and annular space filled with felt and finished with retaining rings.

All pipe-threading shall be in accordance to IS 554.

Installation of underground pipes:-

External underground Firefighting pipe lines shall be connected to the corresponding discharge headers of pumps as per the drawings. Wet Riser cum down comer for firefighting systems shall be connected to the correspondent underground pipelines as per the drawing.

Piping buried underground shall be properly laid & Back filling shall be done after all the required tests are carried out with excavated materials free from rubbish and large stones. It shall be tamped in layers.

5.0 Testing of Pipe:-

All pipes after installation shall be tested at a hydrostatic test pressure of at least 1.5 times the maximum operating pressure for a period of 2 hrs. All leaks and defects joints shall be rectified to the satisfaction of the engineer-in-charge. Pipes repaired subsequent to above pressure test shall be retested in the same manner. All tests shall be witnessed by the engineer-in-charge and record to be maintained for the same and to be submitted together with handing over documents.

6.0 Painting:-

After installation and testing, the pipes shall be painted as per B.O.Q.

7.0 Valves & Strainer

7.1 Ball Valve & Butterfly Valve:

The valves shall be of rating PN16 conforming to IS-14846. Valves shall be provided as shown in the applicable drawings conforming to the following specification:-

I) 15 to 50 mm gun metal Screwed

ii) 50mm and above body cast iron /ductile iron flanged end non-rising spindle.

The valves shall be tested hydraulically at 1.5 times of maximum system pressure.

Non-return valves:

The valves shall be of rating PN16 provided as shown in the applicable drawing and conform to the following specification.

i) (i) 15 to 65 mm- gun metal Screwed

ii) (ii) 80 and above- C.I./D.I- Flanged

Valves shall conform to IS 9338 or IS 5312 as required. The valve shall be tested hydraulically at 1.5 times of the max system pressure.

Pressure Release Valve:-

The valve shall be provided to release the excess pressure in the pipe line and pressure

release can be set as per site requirement. Body of the valve shall be cast iron.

Air Release Valve:-

The valve shall be provided for venting. The valve shall be double float type with C.I./G.M body, Valcanite balls, rubber seating etc. The air release valve shall be associated with an equal size peet valve.

Strainer:

All bronze, Y - type strainer with screwed ends, screwed cover and stainless steel screen for supply line.

SECTION IV

SPECIFICATION FOR FIRE FIGHTING PUMPS

Scope

The scope includes/comprises of the supply, erection, testing and commissioning of motor driven Firefighting pump set – 1 no. electrical driven booster pump on terrace – 1no along with base plate, foundation bolts, anti-vibration pads along with necessary accessories, complete in all respect as per following technical specifications and schedule of quantity for firefighting systems requirements.

The pumps shall be of horizontal and suction, top discharge centrifugal pump. The speed of motor driven pump shall be 2900 RPM and the Engine driven pump shall be 1800 to 2100 RPM.

Pump shall preferably be designed to have the best efficiency at the specified duty point. The pump shall be suitable for continuous operation at any point within the "Range of Operation" i.e 0% to 150% of rated capacity.

The pump shall be driven by drive unit directly coupled. A heavy duty coupling along with coupling guard shall be provide between the pump and drive unit.

The pump shall be capable of furnishing not less than 150% of rated capacity at a head of not less than 65% of the rated head.

The shut-off head shall not exceed 120% of rated head.

The drive unit power rating shall be the maximum of the following requirements.

15% margin over the pump shaft input power at the rated duty point.

5% margin over the pump shall input power required to drive the pump at 150% of the rated discharge.

The pump set shall be of securely mounted on a robust base frame and shall be free from vibration at all variations of load.

The pump shall also be linked to re-circulation pipeline required for minimum now through the pump during its operation without any discharge through headers.

The pump shall be provided with a nameplate indicating delivery head, capacity and RPM.

The material of construction of pump shall be as under:

- | | | |
|--------------------|---|--------------------------|
| i) Casing | : | Cast iron, IS:210, Gr 25 |
| ii) Impeller | : | Bronze IS:318, Gr LTB1 |
| iii) Wearing rings | : | Bronze |
| iv) Shaft sleeve | : | S/S Type AISI 410 |

v) Stuffing Box	:	FG 260 (Gr 25)
vi) Gland	:	FG 260 (Gr.25)
vii) Gland packing	:	Graphite Asbestos rope
viii) Shaft	:	Stainless Steel AISI 410
ix) Base frame	:	Fabricated M.S

MOTORS:

The motors for the wet riser & jockey pumps shall be of general purpose, constant speed, and sized for the maximum output at an ambient temperature of 40 Deg.C.

The motor for electrical driven pump sets shall be of TEFC type.

The motors shall be wound of class 'F' insulation.

The motors shall be rated for continuous duty. They shall also be suitable for long period of inactivity.

The operating speed for the Electrical pump / Jockey pump motor shall be of 2900 RPM.

The motor shall be suitable for electric supply of 400 v, 3 phase, 50 Hz and shall run continuously at rated input over the entire range of voltage and frequency variations as under.

RANGE OF VARIATION

i) Voltage	10%
ii) Frequency	5%
iii) Combined voltage frequency	19% (Absolute sum)

Motors shall be designed for direct on line starting

Motors shall be effectively grounded and shall be provided with two separate and distract grounding pads, each complete with tapped hole galvanized bolt and washer for connection to station ground conductors.

Motors shall confirm to IS : 325.

2. CONTROL PANEL FOR THE MOTOR DRIVEN PUMPSET

The starting switchgear for the motor driven pump set shall be suitable for direct on line starting.

The control circuit for the electrical driven fire fighting pump set shall be designed for automatic operation i.e., whenever the pressure reduces in the system, the pumps shall automatically starts. The stopping of the pumps is 'Manual'.

The control circuit for the pump set shall be designed that whenever pressure reduced the pump shall start automatically and when pressure reached rated system pressure then

it should automatically switched OFF.

Auto / Manual switch shall be provided for both the electrical driven fire fighting pump set so that the pump sets can be started / stopped manually also.

The panel shall be provided with the Voltmeter and ammeter with indicating 'R' 'Y' 'B' Lamps.

The control panel shall be of welded construction, fabricated from sheet metal having 2.03 (14 SWG) thickness and shall be dust and varnish proof.

The panel shall be completely factory wired absolutely ready in all respect for installation at site and incoming power cables to be laid by main electrical Contractor for which un drilled bottom gland plate shall be provided. The internal wirings of the panel shall be carried out with 650 v grade stranded copper wire of size rated for the current in the corresponding circuit. The minimum size of the wire shall not be less than 1.5 sq.mm stranded copper. The wiring termination shall be done using ferrules having indelible at the termination to reduce the possibility of short circuit between various wires. The Contractor shall be solely responsible for proper functioning of the equipment supplied. The internal control wiring shall be done with FRLS cable.

Neoprene or equivalent rubber gaskets shall be provided at all openings.

All the components used in the panel shall be of reputed makes.

Nameplate shall be provided at the panel.

MCCBs shall be provided for the panels. All the protection circuits shall be removed. The panels shall have 2 Nos. of incoming MCCBs with mechanical inter-locking facility.

2 (two) numbers earthing connection shall be provided for the panel. The earthing of the panel shall be done as per the rules and regulations.

Adequate number of potential free contacts shall be provided for connecting to the BAS system.

Submission of documents:

The supplier shall submit three copies of certified dimensional drawings, descriptive data on the equipment supplied, foundation details (design and drawing to be provided by the supplier), Dismantling operation, maintenance, spare parts list and any other relevant literature. The supplier shall also recommend list of spare parts that are necessary for maintenance purpose.

Packing: The pumps set shall be packed properly and perfectly in a rigid wooden casing.

SECTION – V**INSPECTION AND TESTING REQUIREMENT AND DOCUMENTS TO BE SUBMITTED FOR FIRE FIGHTING SYSTEM**

Physical and chemical test:-

The manufacturer shall furnish:

The actual results of chemical analysis and mechanical properties of the following:-

- a) Shaft
- b) Shaft sleeve and of other components as per standards
- c) Dynamic balance Certificate

Weldments :

Welding shall be done by qualified welder only. Weldment shall be examined by Dye Penetrant

Test in accordance with IS/ASTM

Submission of drawings

Following drawings shall be prepared and submitted for approval along with technical bid:

- General arrangement drawing indicating overall dimensions of all components
- Concrete foundation plan, base plate drawings and load details of all equipment.
- Complete electrical drawings.
- Instrumentation control circuit drawings and layout

Submission of Operation and Maintenance Manual :

The tenderer shall furnish 3 sets of operation and maintenance manuals, service & overhauling manuals and spare parts catalogue of the supplied equipment's and instruments.

SECTION – VI

COMMISSIONING OF FIRE FIGHTING SYSTEM

Pressure at Top most Hydrants: During operation of Fire fighting pump, the pressure at the top most hydrants shall not be less than 3.5Kg/sq.cm.

Auto/manual selection of FF pump:
Auto/Manual selection switches along with start/stop push buttons and indication lamps of FF pump shall be provided on the Fire Fighting pump Panel.

Level sensors

Level sensors will be installed in each tank. On seeing low water level in tanks, the respective pumps (jockey pump, electrical main pump and diesel main pump) will trip automatically.

Painting:

After complete installation and testing, pumps, accessories and fittings shall be given two coats of enamel paint of approved make over a coat of primer.

SECTION VII

All pumps data to be furnished by manufacturer

1. Terrace Booster Pump (Electric Driven)

Sr. No.	Description	To be filled by Tenderer
A	Pump	
1.	Pump make, model & type	
2.	Speed rpm	
3.	Duty parameters at 100% FLOW	
	3.1 Head (M) 3.2 Efficiency (%) 3.3 Break Horse Power (HP)	
4.	Duty parameters at 150% FLOW	
	4.1 Head (M) 4.2 Efficiency (%) 4.3 Break Horse Power (HP)	
5.	Shut off head (M)	
6.	Material of construction	
	6.1 Pump casing:- 6.2 Impeller:- 6.3 Shaft:-	
7.	Make of Bearing	
	7.1 Radial	
	7.2 Thrust	
8	Performance curve showing up to 150% of rate	To be enclosed

Sr. No.	Description	To be filled by Tenderer
	flow	
9	Diameter of Suction & Delivery	
B	Motor	
1	Make, Model & Type	
2	Rating (KW)	
3	Speed (RPM)	
4	Efficiency (%)	
5	Voltage (Volts)	415 V
6	Frequency (Hz)	50 Hz
7	No. of Phases	3 Phases
8	Class of Insulation	
9	Ingress Protection Class	
C	Overall Efficiency of Pump-Motor unit (%)	

SECTION –VIII

AUTOMATIC SPRINKLER SYSTEM

An automatic sprinkler system is considered for the building as per NBC standards. This would extinguish a fire in its early stages. The sprinkler system shall be provided as per the drawings inside the building. It consists of piping from down comer to distribute and convey water to the sprinklers in basement with ball valves & butterfly valves.

The Sprinkler senses temperature above 68 deg C and initiate sprinkling of water. An alarm valve fitted with down comer will sense fall in pressure due to sprinkler activation and sound an alarm. Each sprinkler will distribute over a definite area (approximately 12sq.m). The sprinklers are of K-80 factor as per standards.

SPRINKLER SYSTEM PHILOSOPHY:

1. A Sprinkler system has two functions to perform, i.e. to detect a fire and then provide an adequate distribution of water to control or extinguish it.
2. The suction of the pump is connected to a overhead tank.
3. The delivery of the pump is connected to the header pipe
4. Tapping will be taken from Vertical down comer and a network of piping will be kept under pressure and whenever Sprinkler bulb breaks, Water shall be sprinkled immediately and the reduction of pressure automatically sensed thereby switching on the Pump.
5. The pump can be either switched ON automatically or manually depending upon the status of the selector switch in the MCC / Auto start panel.
6. Flow switches shall be installed in the main branch of the sprinklers and these flow switches are connected to an annunciation panel kept in Fire panel room.
7. This annunciation panel will give indication whenever the sprinkler operates indicating the exact area.
8. Each flow switch will have 2 NO and 2 NC contacts so that 1 NO and 1 NC will be used for connecting to the annunciation panel and other 1 NO and 1NC will be used for connecting to the Building Management System.

SECTION – IX

SPECIFICATIONS FOR FIRE HYDRANT SYSTEM COMPONENTS & SPRINKLER:

GENERAL : The item includes supplying of C.I. Stand Post type Fire hydrant, C.I. sluice valve etc.

including fixing.

MATERIAL: Stand post column shall be fitted with 65 mm size instantaneous male coupling and 80 mm size C.I. duck-foot bend, C.I. sluice valve etc. Stand post hydrant shall conform to the relevant IS code. 80 mm socketed or flanged tailpiece shall be as per site requirements. Sluice valve shall conform to the relevant IS code with necessary flanged/lead joints.

FIXING: Hydrant and C.I. sluice valve shall be fixed in position as indicated in the drawing or as directed. They shall be fitted by means of flange joints on the pipe line.

TESTING: The Hydrant and the joints shall be tested under the testing clause of pipe line. The testing shall be done along with the testing of pipe line.

HYDRANT VALVES (Single/twin outlet landing valve)

GENERAL: The item includes supplying of Gun metal or stainless steel, single or double outlet (twin) hydrant landing valve with C.I./ S.S. hand wheel, detachable brass or stainless steel female coupling as specified in schedule of quantities.

MATERIAL: The hydrant landing valves shall be oblique Morris pattern GM or stainless steel conforming to IS 5290 /BS 5041 having instantaneous coupling having 100 / 150mm with blank cap and chain conforming to IS 901 to suit the hosepipes, including nuts, bolts, washers, gaskets with necessary fittings like 'tee' to main riser 150mm or 100 mm dia as specified.

FIXING: Twin outlet valves shall be fixed on 100 / 150mm nominal bore and single outlet shall be fixed on 80 /100mm nom. bore and court yard hydrant at 900 / 1200mm above ground level. The valve shall be flanged jointed.

PAINTING: - The hydrant valve unit shall be painted with three coats of signal red paint.

FIRE HOSE PIPES: -

GENERAL : The item include supplying of approved synthetic hose pipe of 63 mm dia x 15 m long with guaranteed bursting pressure of 35.7 kg Cm² fitted with 63 mm dia. Instantaneous male and female brass/ss coupling on each end as specified in schedule of quantities.

MATERIAL : The hose pipe shall be Unlined Flax Canvas for firefighting conforming to IS 4927, Fabric Reinforced Rubber Lined Woven-Jacketed (RRL) fire hose type A or Synthetic Fibre Woven-Jacketed (SFWJ)/ Elastomeric fire hose type B conforming to IS 636 or controlled percolations (CP) Hose conforming to IS 8423 all with ISI mark as specified in schedule of quantities. The instantaneous male and female coupling shall be of brass / SS conforming to IS 903 with ISI mark.

The coupling shall be bound and riveted to Hose pipe with copper rivets & 1.5mm copper wire.

PLACING: The fire hose shall be stored in the hose pipe box single or double either in flaked or rolled manner.

FIRE HOSE BOX:-

GENERAL : The item include supplying fabricated MS sheet metal / FRP hose pipe box shall be capable to accommodate 1 or 2 nos. hose pipe as specified in the schedule of quantities.

MATERIAL: The hose pipe box shall be fabricated from M.S. Sheet as per SOQ for accommodating 2 nos. of hose pipe and branch pipe with nozzle or hosing branch pipe with nozzle specified in schedule of quantities. The box shall have front glass door lock with angle iron frame.

FIXING: The hose pipe box shall be fixed with 4 nos. C.P. brass screws and rawal plugs on the wall as shown in the tender drawing and as shown by the Engr-in-Charge.

PAINTING: The MS box shall be painted with two coats of signal red enamel paint from outside and white enamel-paint from inside.

BRANCH PIPE:

GENERAL : The item include supplying of Gun metal / stainless steel branch pipe fitted with detachable hexagonal nozzle as specified in schedule of quantities.

MATERIAL: The branch pipe shall be GM or SS conforming to IS 903 or universal pattern as per IS 2871. The branch pipe shall be 63 mm dia fitted with detachable hexagonal nozzle of 19 mm dia. suitable to connect hose pipe.

PLACING: The branch pipe with detachable nozzle shall be placed in the hose pipe box.

FIRST AID HOSE REEL:-

GENERAL : The item include supplying of approved rubber hose reel of 20mm dia x 37 m (120') long with swinging drum with cut off outlet nozzle.

MATERIAL : The hose reel shall be high pressure rubber hose reel fitted on swinging wall mounted type drum of 20mm dia x 37 m (120') long conforming to IS 884 along with 8mm dia. Outlet cut off nozzle GM or chromium plated brass or stainless steel as specified in schedule of quantities.

FIXING: The swinging wall mounted drum fitted with rubber hose reel shall be fixed on necessary bracket on the wall with 4 nos. of C.P. brass screws. The hose reel shall be connected to wet/dry riser and shall be fixed near to riser.

PAINTING: The drum and brackets shall be painted with two coats of signal red enamel paint from outside and white enamel paint from inside.

FIRE BRIGADE SERVICE INLET SIAMESE CONNECTION:-

GENERAL: The item includes Siamese connections suitable to connect the mobile pump to wet riser for supplying water to the vertical riser in case of emergency.

MATERIAL: The siamese connections shall have 2, 3 or 4 nos. of 63 mm dia. GM make inlet (instantaneous male coupling) as specified in schedule of quantities. The double flange C.I. body swing type non-return valve conforming to IS5312 (part I) shall be flanged

jointed to the vertical pipe of siamese connection. This includes blank cap drain valve of 25mm dia and GI / MS pipe conforming to IS 1239 of necessary length and fittings required for connecting to main riser. The nom.bore of seamless pipe riser shall be 100mm or 150 mm as specified in schedule of quantities.

PAINTING: The exposed seamless pipe of siamese connection shall be painted with two coats of signal red anticorrosive paint over to coats of red oxide primer. The GM inlet housing box shall be painted with red enamel paint from outside and white enamel paint from inside. The underground pipe shall be given anticorrosive treatment as per IS 10221.

FIRE BRIGADE COLLECTIVE BREECHING INLET CONNECTION:

GENERAL: The item includes the inlet connection to the water tank with G.I. pipe for augmentation.

MATERIAL: The fire bridge inlet connection shall have 2, 3 or 4 nos. of 63 mm dia. GM make inlet (instantaneous male coupling) specified in schedule of quantities. The G.I. pipe for augmentation shall be confirming to IS 1239 (part I) of heavy quality and joints shall be with weld joint.

PAINTING: The exposed G.I. pipe of the connection shall be painted with two coats of signal red anticorrosive paint over two coats of red oxide primer. The underground pipe shall be given anticorrosive treatment as per IS 10221.

DEWATERING: The contract rate shall include bailing or pumping out all the water, if accumulated during the progress of the work while connecting the collective breeching to the water tank fire compartment, either from rain, seepage, springs or any other cause till completion of work.

TESTING: The Inlet connection shall be tested including fittings weld joints and shall be tested under the testing head (clause) of G.I. piping work.

SEAMLESS PIPE PIECE:

GENERAL: The 80 mm nominal MS seamless pipe of required length with MS flange at one end and other end welded with MS riser.

MATERIAL: MS seamless 80 mm dia pipe piece shall be conforming to IS 1978 of 300, 450, 600 mm length as specified in schedule of quantities. The pipe piece shall be with M.S. flange with holes welded or integral at one end for fixing Hydrant and other end to be welded with M.S. Riser in the shaft

FIXING: The pipe piece shall be welded with M.S. Riser in the shaft. The 10% of the welded joint shall be dye penetration (D.P.) tested to the relevant IS code. The flange at other end shall be welded or integral to the pipe piece.

PAINTING: The pipe piece shall be painted with two coats of signal real anticorrosive paint over a two coats of read oxide primer.

TESTING: The pipe piece shall be tested along with MS seamless pipe risers under the respective head (clause) testing.

ORIFICE PLATE (ORIFICE FLANGE):-

GENERAL : The item include providing and fixing GM / Brass or stainless steel orifice plate of not less than 6 mm thick or as specified in schedule of quantities, having suitable bore for reducing the pressure and to restrict the operating Pressure within 3.5 to 4.00 kg / sq cm at the hydrant outlet at every floor.

MATERIAL: The orifice plate shall be Gun Metal/Brass/Stainless Steel and bore size shall be properly designed to reduce the pressure at the fire hydrant outlet as per fire brigade requirement.

FIXING: The orifice plate shall be fixed in position as indicated in drawing or as directed. It shall be fitted by means of flanged or welded or screwed joint as specified in schedule of quantities.

TESTING: The orifice plate shall be tested or calibrated to give the required pressure. The bore shall be modified or orifice plate shall be replaced without any extra cost.

PENDANT / SIDE WALL SPRINKLER:

GENERAL: The item includes supplying and fixing sprinkler of specified dia to distribution pipes connected to riser pipes.

MATERIAL: The sprinkler shall be of automatic sprinkler as per IS 9972. It shall be fixed to the G.I heavy class distribution pipe line connected to riser pipe.

FIXING: The sprinkler shall be either welded or flange or screw jointed as specified in schedule of quantities.

TESTING: The sprinkler heads shall be tested for leak-proof joints along with piping system as per testing head (clause) of G.I heavy class piping work

PRESSURE GAUGE:-

GENERAL: The pressure gauge shall be of specified diameter and appropriate range and be complete with shut off gauge cocks.

MATERIAL: The pressure gauge shall be brass body siphon and cock dial type. The size of pressure gauge shall be 100 mm dia or 150 mm dia as specified in schedule of quantities. Dial range shall be adequate for the pressure encountered as specified. The Dial shall have the calibration in MKS and FPS units and shall be properly calibrated before installation.

Accuracy over the selected range shall be $\pm 1\%$.

FIXING: The pressure gauge shall be fixed & screwed in the pipe making or drilling necessary holes in pipe line etc.

TESTING: The pressure gauge shall be tested for the leak proofness of the joints along with pipeline laid under testing clause of relevant seamless piping work.

SECTION – X

FIRE EXTINGUISHERS:

PORTABLE FIRST-AID FIRE EXTINGUISHERS:

SYSTEM DESCRIPTION:

The extinguishers are used to put-off small fires. The extinguishers shall be used in the incipient stage of fire. Fire extinguishers are easy to handle. This is useful to put off the fire in the initial stage itself and thus avoiding major losses.

TYPES AND APPLICATIONS

Multipurpose ABC type fire extinguisher conforming to IS 15683-2006 shall be located in work station area, operations area & corridors.

Carbon dioxide type fire extinguishers (4.5 kg) conforming to IS 15683-2006 shall be located in workstation area, electrical panel room, pump room and lift machine room.

Mechanical foam type fire extinguishers (9 liters&50liters) shall be located at HSD Yard, D.G. Rooms, and substation and Firewater pump house and near oil filled Transformers.

FIRE EXTINGUISHERS:

Fire extinguishers are very valuable in dealing with fire situations, if used properly and efficiently in the early stages of a fire. The greatest advantage lies in its portability and mobility. In addition, their immediate availability and easy usage by one or two persons is another advantage.

The usefulness of these extinguishers is limited, as it is entirely dependent upon the presence of persons having knowledge to operate them. Further, their capacity is also limited and their operational values largely depend upon the initial charge being sufficient to overcome and extinguish the fire.

For example, depending upon the type and capacity of the extinguisher the discharge time may be 10 seconds to 90 seconds and the range of jet may be 3 to 10 meters.

Thus, these extinguishers are not expected to deal with a large fire, as all these are essentially first-aid fire fighting appliances. Routine inspection and periodical testing of fire extinguishers is necessary to ensure that these extinguishers are maintained in good working conditions.

As fire extinguishers are intended only for the use of incipient fires, they will be of little value if the fire is not extinguished or brought under control, in the early stages. Hence, all the employees shall be imparted periodical training in the operation and use of fire extinguishers.

The following types of fire extinguishers are to be deployed throughout the premises depending on the location:

- | | | |
|----------------------------------|---|----------------------------|
| 1) CO2 type extinguishers | : | 4.5 Kgs Cap |
| 2) ABC type extinguishers | : | 6 Kgs. Cap. |
| 3) DCP type extinguishers | : | 6 Kgs. Cap. |
| 4) Mechanical Foam extinguishers | : | 9 Lts. Cap. & 50 Lts. Cap. |

SL.NO.	BASIC TYPES OF FIRES	SUITABLE TYPE OF EXTINGUISHER
1.	Class A fire: Fire in ordinary combustibles like wood, paper, rubber, plastics etc.	Gas expelled water type
2.	Class B fire: Fire in flammable liquids, paints, grease, solvents etc.	Chemical extinguishers of CO2 and DCP type
3.	Class C fire: Fire in gaseous substances under pressure including liquefied gases	- do -
4.	Class D fire: Fire in reactive chemicals, active metals etc.	Special type of dry powder extinguishers

SECTION – XI

SIGNAGES

Signage's are provided as per Local fire force like exit signs & Floor indication (e.g. Ground floor, 1st floor) in case of fire /emergency, Staircase location indication etc. The location shall be each on landing of every staircase on each floor.

1. Signs shall be made out of 3mm thick PVC foam board with PVC reflective self-adhesive vinyl foam board or equivalent material with Mirror fasteners for fixing complete.
2. The exit sign with arrow indicating the way to the escape route is provided at suitable height from the floor level on the wall and will be illuminated by electric light
3. Signs are required at all exits, emergency exits and escape routes, which should comply with the graphic requirements of the relevant Indian Standards.
4. Exits shall be clearly visible and the route to reach the exits shall be clearly marked and signs posted to guide the occupants of the floor concerned.
5. Signs shall be illuminated and wired to an independent electrical circuit on an alternative source of supply.
6. The sizes and colours of the exit signs shall be in accordance with good practice
7. The colour of the exit signs shall be green.
8. Fire fighting equipment shall be suitably located and clearly marked by luminous signs

SECTION – XI

AUTOMATIC DETECTION FIRE ALARM & PA SYSTEM

1.FIRE DETECTION AND ALARM SYSTEM

All the selection of materials, Installation, Testing and commissioning shall be strictly in accordance with Indian Standard Specification and also Local Fire Force rules shall also be considered.

The Fire Detection System shall consist of Analogue addressable type smoke detection and heat detection system to cover the design area. These detectors shall be connected by fire retardant and heat resistant cables to main control panel(MCP). Total area shall be divided in to zones to suit the cabling and proper management during fire hazards.

Photocell and Ionization detectors shall be provided at spacing of 6.5mtrs. Maximum of 20 detectors shall be connected to one zone.

SMOKE DETECTORS

Smoke Detectors shall be of both Ionization type and Photocell type.

Ionization type smoke detectors shall be working on ionization principle. It shall be of dual chamber and dual source type. These shall be of solid state with L.E.D indication at the base. The radio active source used in the detector shall not emit beta or gamma rays and radio active material shall not be more than one Microcurie.

Photocell type smoke detectors shall be able to sense incipient fire by detecting the presence of visible and invisible products of combustion. The detectors shall be suitable for low voltage (24VDC) two wire supply,. The detectors shall be provided with response indicators and the sensitivity shall not vary with change in ambient temperature or its performance shall not be affected by air Velocity up to 5mtres per second. It shall have an in built arrangement so that time constant of about 4 seconds is provided so that puffs of smoke do not in advertently trigger the alarm. These detectors shall have indicating lamps that come 'ON' in alarm condition.

All detectors shall be identical in construction, design and characteristic to facilitate easy replacement. The detector housing shall be damage resistance made of polycarbonate or aluminum with plug in housing base of universal type suitable for either surface or recess mounting. The housing shall protect the detectors against dust accumulation/ingress and shall have ease in maintenance and functional tests at intervals.

The coverage area per smoke detector shall be specified and shall be proved by test. The detectors shall be estimated with spacing of 6.0mtrs and both types located alternatively with not more than 20 detectors forming one Zone for tendering purpose so as to keep all the parties participating at par. The sensitivity of detector shall be set and adjusted by the tender to suit site requirement.

Heat Detector

Heat Detector shall operate on rate of rise in temperature. The rate of rise sensing shall be by using a matched pair of thermistors to sense heat with one thermistor exposed to ambient while other is sealed. The imbalancing these two thermistors in event of fire shall trigger the alarm. This shall be re settable time and shall also trigger alarm in case the temperature reaches or rises beyond a set temperature point.

The operating parameters shall be as follows

Rate of rise in Temperature 9 degree Celsius or more per minute

Fixed Temperature 70 degree $\pm$ 5 degree Celsius

The detector shall incorporate response indicators Facility and to operate on 24 V Dc Two wire system

It shall be possible to loop the heat detectors with the smoke detectors and manual push points in the same circuit

The coverage area of heat detectors shall be specified and certified a testing agency like CBRI Roorkee

FIRE ALARM SYSTEM

The fire alarm system shall conform to relevant Indian Standards in respect to design and Installation and give audio /Visual alarm signals triggered by heat/Smoke detectors /Manual PB stations located suitably in an area to be monitored for fire. The system shall give pin point location of fire with warning system Zone wise as well as warning for any fault within the system.

The operating voltage system shall be 230 V AC/ 24 V DC with sealed lead acid battery backup and battery charger/ DC distribution board operating on 230 V AC $\pm$ 10%.

The system shall be capable to supervise all conditions with distinctive audio tones operated for system fault warning and fire alert conditions. All Audio alarms shall have except push button for cancellation of audible alarm reset future to ensure that no alarm warning is switched off inadvertently.

Following features shall be incorporated system fault warning Fire Alert and Annunciation System Fault warning /Fire Alert

Open Circuit in detector wiring

Short circuit in detector wiring

Normal conditions

Fire Conditions

AC Main Failure

DC Supply Failure

Battery low Alarm

1. ANNUNCIATION

'Flash 'on Fault

' Steady' On Accept

'OFF' on reset after fault clearance

'Flash', If reset with fault persisting

'Steady After accept even if fault cleared in the interim
'flash' Even if Fault disappears before accepting
Lamp test Push button shall be provided for testing the healthiness of all lamps.

The System shall be capable to switch – Off air conditioning or P & V System operating in the affected area in case of Fire.

2.HOOTERS

Dual tone hooters shall be electronic type and shall give discontinuous / Intermittent audible alarm automatically.

The sound level shall be 90 dB and above

3.MANUAL PUSH POINTS.

Minimum one manual push point shall be installed for each Zone in the manned area forming integral part of the Fire Detection System.

Manual push point shall be wall mounting break glass type units, completely encase in whether proof 16 SWG housing with hinged front cover with glass front window and housing push button.

The Front glass shall have words in bold letters " INCASE OF FIRE BREAK GLASS" a hammer with chain attached to the main housing shall be provided.

The push button housed inside shall be Red Mushroom head, normal in pressed condition and actuating when released by breaking glass in front of housing.

It shall have two sets of " NO" or " NC" potential free contacts suitable for 30 V DC, 3 Amps. The contact shall be wired up to terminal strip inside.

The housing shall accommodate minimum two cables and suitable glanding / conduit coupling arrangement shall be provided.

The metal housing shall be treated with one coat Iron Oxide before applying two coats of Fire Red enamel paint.

4.RESPONSE INDICATOR

Response indicator shall be provided in wall mounted 16 SWG M.S metal box duly painted as detailed for manual push points with provision for cable entry.

5.MAIN CONTROL PANEL

Main control panel shall conform to provisions of IS 2189. The panel shall be totally enclosed in dust and vermin proof type. Conforming to IP:54 and of 16 SWG dust inhibited sheet with oven baked finish. The panel shall be completely solid state design. Micro processor based panel also to be offered .

Main control panel shall be designed and installed so as to prevent ingress of sut and moisture and in inaccessible to Unauthorized persons. The main control shall be clearly labeled to indicate mode of operation and their function.

The panel shall have solid state circuits with restricted use of relays and shall have the following facilities.

A pair of red LED to indicate origin of faults and milk white jeweled lamp with Floor Numbers / Building /Area on it.

Distinct audible sounds for Fire and Fault Alarms.

Audible signal Acknowledge/ silence button and LED lamp to indicate its operation.

Voltmeter shall be provided

'System on' 'Standby on' test Key operation indication shall be provided.

'System off Switch, Its indication and floor isolation switches shall be provided. When the isolation switch its operated, fault signal must come on the respective floor indicators.

Failure of any indicator circuit shall not prevent the Fire Alarm from Sounding nor acknowledgement / Silencing of Alarm from one floor should prevent another alarm system from other Zone on the Same floor, or any other floor.

Test Facility to check Fire Alarm Circuit, Indicators shall be provided.

Fault warning in the Following events

Failure or disconnection of normal supply

Failure or disconnection of standby supply

Failure or Disconnection of Battery Charging equipment

Earth Faults

Any Fault on floor

Any Fault on floor as detailed earlier for system failure warning

The main Control shall work on 230 V normal supply. A standby power shall be immediately available in the event of failure of normal supply and shall be automatically connected as to maintain the Equipment in condition such that Fire Alarm originating from the operation of Fire Detector in separate Zone can be Subsequently given. The standby battery of storage type shall be of such type that when charged by associated battery charging equipment for a period of 24 hours shall have enough power supply to cope up with the additional load regulating in an alarm originating from two separate Zone for one hour. If Utilized send Emergency evacuation Alarm, It shall supply power for at least ten minutes. The batteries shall be lead acid sealed type with required all rating.

Suitable arrangements shall be incorporated to prevent batteries from discharging through the battery equipment in the event of its breakdown or a failure in the supply.

6.CONDUITS

These shall be 16 SWG M.S welded conduit and perfectly circular tubing and capable of being cleaned and with tight fitting joints.

All conduits shall be laid either on surface or in recess as required and as approved by consultants/ Purchaser and shall be protected from rust by one coat iron oxide and two coats of enamel paint.

7. ELECTRIC CABLES

The wiring for the fire detection and alarm system shall be Fire retardant & HR PVC insulated copper conductor having minimum 1.5 Sqmm cross sectional area. Conductor size for DC power shall not be less than 2.5 Sqmm copper.

The wiring for power supply to panel and between panels shall be PVC, HR aluminium conductor power cables of 1100 volts grade conforming to IS 1553.

All the Cables and wire shall be tagged for proper identification

The multi-core cables should not be shared for other low voltage or High voltage circuits.

All clamping of cables, detectors or any fixing of equipment shall be done with the use of anchor fasteners. Chipping of ceiling, floors, columns etc., shall not be permitted.

8.INSPECTION AND TESTING SCHEDULE

The bidder shall confirm to offer the above equipment for following inspection/ testing which may be witnessed by the purchaser's representative.

The Bidder shall make such arrangement as may be necessary for tests/ inspection. The Bidder shall be responsible for providing instrument of correct accuracy that may be required for carrying out these tests.

9.TESTOR

Workmanship

Operation and functional check-up

Simulation test by simulating fire Conditions for smoke and space heat detection.

Test for space covered by the detectors.

PUBLIC ADDRESS SYSTEM:

The Public Address System is designed to serve the multipurpose of playing the music or to make general announcement or to transit the fire tone under fire condition. These different signals are transmitted through the same set of speakers hence different levels of priorities are allotted to different signals. The music is with the least priority and the fire tone is having next priority and the emergency announcements are having the highest priority level.

The System shall have pre-recorded message to be announced in all the groups/zones in case of fire. A signal from fire alarm panel shall initiate such announcements. This announcement shall have highest priority. The system shall have a chime module. The chime module shall generate two types of signals consisting of either a two-note chime or a single gong tone to capture attention before paging. A DIP switch shall be provided on the P.C. board for selection of the chime or gong. Two LED indicators for the chime and paging function for the adjustment of the output level, two-note frequency and chiming interval. The chime shall be activated by a remote microphone and or through the amplifier. Normally background music shall be played in the lobbies, stair cases, smoke free lobbies, Toilets, Driver's Rest Room and other common areas paging will receive a higher priority and the highest priority shall be accorded to emergency announcements initiated by the fire control panel. The speakers are distributed in the entire facility and are configured in different zones. The announcement can be made in individual mode i.e. one zone at a time or to all the speakers simultaneously in ALL CALL mode. Public Addressing system comprising of

Wooden Cabinet / Metal Cabinet.

Ceiling ring speakers.

Public Addressing Amplifiers

Central Control Console

SPEAKERS:

Speakers are in wooden or metal enclosures. All the speakers are with its associated line matching transformers. The speakers are interconnected in the zone configurations. The types of speakers to be used in various areas are Common areas with false ceiling – 6 W flush mounting ceiling ring speakers Office areas – 6 W metallic/wooden cabinet speakers. These speakers shall have facilities for horizontal & vertical orientation and provision for

easy installation.

PUBLIC ADDRESSING AMPLIFIERS:

These amplifiers are designed to accept the input sources i.e., Microphone/CD player/Fire tone generator. The output is having a 100 Volts line to take care of line drop. The Amplifiers are stacked inside the rack system. Tone control circuit is provided to alter the quality of the output as per the user's requirement.

RACK ASSEMBLY:

Rack assembly is to stack the public addressing amplifiers. This should be factory prewired and have different channels for AC mains cable and Audio signal Cables. The rack should provided be with wheels for easy maintenance. One lockable rear door should be provided. The rack assembly is operating on 230 VAC.

CONTROL CONSOLE:

The heart of the entire public addressing system is the Console. One microphone is given for announcements. The selection of Music or Announcements is done through the console. The system shall make provision for dual channel output i.e both music and audio announcement shall be made for the entire system in which either music or audio announcement shall be selected through a hardware or software switching facility. The announcements can be made through the control console zone wise such as Music, Fire tone or Announcements. The console is provided with keypads on a sloped surface. The switches are having respective LED's. The different modes of transmission in P.A System are as follows:

MUSIC:

The music is through the CD/ Players can be fed to the amplifiers. The music transmission is having the least priority. That is if FIRE condition occurs, then the Alert tone overrides the music in the speaker. If any announcement is made, then the music will be stopped till the time of announcement and continues after pressing key. To play music a separate switch MUSIC is given and by operating this switch, the music is transmitted to all departments/zones.

ANNOUNCEMENTS:

Announcements can be made through the microphone and by selecting the required zones. Announcements can be made in following modes:

1. Individual mode
2. All Call mode In this mode announcement can be made to all the speakers simultaneously. This is useful when any common message to be passed to all. Also this is more convenient and fast to address the people during emergencies.

SPECIFICATIONS FOR CABLE & WIRING.**1.0 TYPE:**

Wiring shall be with plain annealed copper conductor, PVC insulated, PVC sheathed construction. The conductors of cable shall be solid circular type.

2.0 RATING:

The wires shall be rated for 1100 Volts.

3.0 CORE IDENTIFICATION:

Cores shall be provided with the standard color scheme of PVC insulation conforming to relevant standards.

4.0 CIRCUIT RATINGS:

The current rating shall be based on the following conditions: -

- a) Maximum conductor temperature : 70 Degree C.
- b) Ambient Air Temperature : 45 Degree C.
- c) Ground Temperature : 30 Degree C.
- d) Depth of Lying : 750 mm

5.0 STORING, LAYING, JOINTING AND TERMINATIONS:

5.1 STORING:

On receipt of cables/wires at site, the cables shall be inspected and stored in a safe location.

5.2 LAYING:

Cables/Wires shall be laid as per the specifications given below: -

Cables in Outdoor Trenches: Cables/Wires shall be laid in outdoor trenches wherever called for. The depth of the trenches shall not be less than 750 mm from the final ground level. The width of the trenches shall not be less than 300 mm. However, where more than one cable is laid an axial distance of not less than 150 mm shall be allowed between the cables. The trenches shall be cut square with vertical sidewalls and with uniform depth. Suitable shoring and propping may be done to avoid caving in of trench walls. The floor of the trench shall be rammed well. Wherever cables are bent, the minimum bending radius shall not be less than 12 times the diameter of the cable. After the cable/wire is laid and straightened, it shall be covered with 8cms thick layer of sand cushion. Over this, a course of cable/wire protection tiles or burnt brick shall be provided to cover the cables/wires by 50 mm on either side. Trenches shall be back filled with earth and consolidated and surplus earth to be disposed within the site as directed. Cables/Wires shall be laid in Hume pipes at all road crossings and in GI pipes at the wall entries. Approved cable markers made of CI indicating the voltage, number of cables/wires and the direction of run of the cables/wires shall be installed at regular intervals of 25 meters.

CABLE LAID INDOORS

Cable shall be laid in indoors wherever specified. Suitable clamps, hooks and saddles shall be used for securing the cables in position. Spacing between the cables shall not be less than 200 mm center to center. All concealed wiring shall be inside 25 mm HGMS conduit of reputed make.

JOINTING AND TERMINATIONS:

Cable jointing shall be done as per the recommendation of the cable manufacturer. Jointing shall be done by qualified cable jointers. Each termination shall be carried out using brass compression glands and cables sockets. Hydraulic crimping tool shall be used for making the end terminations. Cable gland shall be earthed by using suitable size G.I. wire/tape.

FIRE LIST OF RECOMMENDED MAKES		
SL.NO	ITEM DESCRIPTION	MAKE
1	FIRE PUMPS (DIESEL ENGINE)	KIRLOSKAR /GRUNDFOS
2	FIRE PUMPS (ELECTRIC DRIVEN)	KIRLOSKAR /GRUNDFOS
4	PIPES - G.I. / M.S.	TATA / JINDAL/ SAIL
5	PIPE FITTINGS - SOCKET WELDING	BHARATH FORGE / SANJAY FORGE
6	PIPE FITTINGS - BUTT WELDING	GOEL/MS C CLASS
7	BALL VALVE / TEST & DRAIN VALVE	TYCO/ZOLOTO
8	BUTTERFLY VALVES	TYCO/ZOLOTO
9	SLUICE VALVES	KIRLOSKAR / VB /HAWA
10	NON - RETURN VALVE/CHECK VALVE	RAPID /FIVALCO/TYCO
11	Y - STRAINER	UPADYAYA / SANT / H SARKER
12	AIR RELEASE VALVE	RB / TBS / VB / NEWAGE
13	HYDRANT VALVE	NEWAGE / WINCO/EVERSAFE
14	GUN METAL BRANCH PIPE	NEWAGE / WINCO/EVERSAFE
15	GUN METAL NOZZEL	NEWAGE / WINCO/EVERSAFE
16	RUBBER HOSE REEL	NEWAGE/CRC/ SAFEGAURD/EVERSAFE
17	SPRINKLER HEAD	NEWAGE / VIKING/RELIABLE/TYCO
18	DELUGE VALVE/ SPRINKLERS	HD FIRE/TYCO
19	SPRINKLER ICV	WORMALD / RELIABLE/HD/TYCO
20	PRESSURE GUAGE	H - GURU/HONEYWELL/INDFOS
21	PRESSURE SWITCH	DANFOSS / INDFOS/HONEYWELL
22	FIRE EXTINGUISHER	SAFEX / MINIMAX/CEASE FIRE
23	SUPPORTS	HITEC/FISCHER
24	FIRE SEALANT	HILTI / BIRLA 3 M
25	PAINT	ASIAN/BERGER/SHALIMAR
26	COATING & WRAPPING	PYPCOAT/ IWL/RUSHTECH
27	PVC INSULATED COPPER WIRES	POLYCAB /POWERFLEX/RR KABEL
28	CABLES	POLYCAB /POWERFLEX/RR KABEL
30	FIRE BUCKETS	SAFEX / MINIMAX/CEASE FIRE
31	FLOW SWITCH	NOTIFIER/HONEYWELL/SYSTEM SERVER
32	FLOW METER	GERAND/RAPIDROP/GVI
33	FOOT VALVE	UPADYAYA / SANT
34	SUBMERSIBLE PUMP	GRUNDFOS

35	CPVC PIPE	ASTRAL/ASHIRWARD/PRINCE
36	SIGNAGES	BRADY / PROLITE / AUTOGLOW
37	SPRINKLER ANNUNCIATION PANEL	HONEYWELL MORELY/SEIMENS
38	WELDING ROD	ESAB / COSMOS / ADOR
39	4 WAY BRIGADE INLET	NEWAGE / WINCO/EVERSAFE
40	FIRE ALARM PANEL	HONEYWELL MORELY/SEIMENS/EDWARD
41	CABLES	FINOLEX/POLYCAB/UNIVERSAL
42	SMOKE DETECTORS	EST/HONEYWELL/SEIMENS
43	HEAT DETECTORS	EST/HONEYWELL/SEIMENS
44	BEAM DETECTORS	EST/FIRERAY/SEIMENS/SYSTEN SENSOR
45	CONTROL MODULE	EST/HONEYWELL/SEIMENS
46	MONITOR MODULE	EST/HONEYWELL/SEIMENS
47	SPEAKER	HONEYWELL/SEIMENS/ BOSCH
48	2 X 1.5SQ MM FIRE ALARM ARMoured CABLE	POLYCAB/VARSHA/KEI/DATA FLEX
49	PSU FOR HOOTERS & AHU TRIPPING (INCLUSIVE OF BATTERY KIT & BATTERY	PANASONIC/EXIDE/AMARON